数理分析方法

老师课件证明过程汇编

原证明页保留 · 中文展开 · 页码索引

收录边界

仅以 sources/ 中老师原始课件是否实际展示论证为依据。收录显式 Proof、跨页续证、公式推导及带理由的例证；不因出现 “One can prove” 或 Questions/Prove 就收录。只陈述定理而没有推导的不补入。相同证明在不同课件重复出现时保留各自来源。 **阅读约定**

每项先给来源，再保留老师原页，随后展开中文步骤。原页是权威定位依据；补充与纠错是本汇编的批注，不假称为教师原文。伪证明保留作辨错，明确不作为有效证明。原课件PDF页码从封面计 1；另列已有 gpt 讲义对应页脚，避免把两套页码混淆。证明跨页时统一列出范围。 **重要说明**

教师出现了不完整论证时，依其原路线补足；原命题不成立时，说明具体错误，不把它包装为正确证明。108的伪定理条目属于原课件实际展示的错误论证。高维 Brouwer、Kakutani 仅陈述，故其证明不在本汇编。 2026 年 10 月 25 日

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1 000-Intro-Proof：无实际证明过程

000-PPT.pdf 为课程概览和人物介绍，没有符合本汇编范围的证明，因此不添加任何证明。

2 000-Proof（000-ZSM）

2.1 由分割公理证明上确界存在

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 19–20 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：23, 24。

原课件第 19 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded above. Let

$$B = \{x \in \mathbb{R} : x \text{ is an upper bound for } S\} \quad \text{and} \quad A = \mathbb{R} \setminus B.$$

Note that $B \neq \emptyset$; and if $x \in S$ then $x - 1$ is not an upper bound for S so $A \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then a would also be an upper bound for S, which contradicts the definition of A.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \sup S$.

If $c \in A$ then c is not an upper bound for S and so there exists $x \in S$ with $c < x$. But then $a = \dfrac{c + x}{2}$ is not an upper bound for S, i.e. $a \in A$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $a \leqslant c$ for all $a \in A$. Hence $c \in B$. But if $c \in B$ then $c \leqslant b$ for all $b \in B$; i.e. c is less than any upper bound for S.

原课件第 20 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then A is bounded above (by any element of B). Let $c = \sup A$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Now every member of B is an upper bound for A, from the first property of a Dedekind cut; hence $a \notin B$, as otherwise a would be an upper bound for A which is less than the least upper bound c. Hence $a \in A$.
- Next suppose $b > c$. Since c is an upper bound for A (in fact the least upper bound), it follows we cannot have $b \in A$, and thus $b \in B$.

沿原论证展开与批注

构造 令 B 为 S 的全部上界，$A = \mathbb{R} \setminus B$。有上界使 $B \neq \varnothing$；选 $s \in S$，$s-1$ 不是上界，故 $A \neq \varnothing$。

验证分割 若 $a \in A, b \in B$ 而 $a \geq b$，则所有 $s \in S$ 满足 $s \leq b \leq a$，使 a 也成上界，矛盾。因此 $a < b$，分割公理给分界点 c。

证明端点属于上界侧 若 $c \notin B$，有 $s \in S$ 满足 $s > c$。中点 $m = (c+s)/2$ 大于 c，应属于 B；但 $m < s$，又不是上界，矛盾。故 $c \in B$。所有 $b \in B$ 都有 $c \leq b$，所以 $c = \min B = \sup S$。关键技巧：用“全部上界”形成新集合，不是直接把原集合随意切开。

设 A, B 是一个分割。A 非空且任取 $b \in B$ 都是其上界，故可取 $c = \sup A$。

验证分界性质 若 $x < c$ 而 $x \in B$，则 x 是 A 的上界，违背 c 是最小上界，所以 $x \in A$。若 $x > c$，由于 c 是 A 的上界，x 不可能在 A，只能在 B。

唯一性补全 两个不同分界点之间的中点会被同时分入 A 和 B，矛盾。故确界公理能推出完整的分割公理。考试中“等价”必须写出两个方向，本页与前页缺一不可。

2.2 由分割公理证明下确界存在

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 21–22 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：25, 26。

原课件第 21 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded below. Let

$$A = \{x \in \mathbb{R} : x \text{ is a lower bound for } S\} \quad \text{and} \quad B = \mathbb{R} \setminus A.$$

Note that $A \neq \emptyset$; and if $x \in S$ then $x + 1$ is not a lower bound for S so $B \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then b would also be a lower bound for S, which contradicts the definition of B.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \inf S$.

If $c \in B$ then c is not a lower bound for S and so there exists $x \in S$ with $x < c$. But then $b = \dfrac{x + c}{2}$ is not a lower bound for S, i.e. $b \in B$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $c \leqslant b$ for all $b \in B$. Hence $c \in A$. But if $c \in A$ then $a \leqslant c$ for all $a \in A$; i.e. c is more than any lower bound for S.

原课件第 22 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then B is bounded below (by any element of A). Let $c = \inf B$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Since c is a lower bound for B (in fact the greatest lower bound), it follows we cannot have $a \in B$, and thus $a \in A$.
- Next suppose $c < b$. Now every member of A is a lower bound for B, from the first property of a Dedekind cut; hence $b \notin A$, as otherwise b would be a lower bound for B which is less than the greatest lower bound c. Hence $b \in B$.

沿原论证展开与批注 令 A 为 S 的全部下界，$B = \mathbb{R} \setminus A$。由下有界，$A \neq \varnothing$；取 $s \in S$，$s+1$ 不是下界，故 $B \neq \varnothing$。若 $a \in A, b \in B$ 且 $a \geq b$，则 $b \leq a \leq s$ 对所有 $s \in S$ 成立，矛盾，故 $a < b$。设分界点为 c。如果 $c \in B$，则有 $s \in S$ 满足 $s < c$。中点 $m = (s+c)/2$ 小于 c，应在 A，但 $m > s$ 又不可能是下界。因此 $c \in A$，且所有下界 $a \leq c$，故 $c = \inf S$。

记忆方法 上确界证明中选择“全部上界”的最小者；这里选择“全部下界”的最大者。中点总是用来挤出与分界方向冲突的数。

B 非空且任一 $a \in A$ 是其下界，取 $c = \inf B$。若 $x < c$，x 不可能属于 B，否则违反 c 是下界，所以 $x \in A$。若 $x > c$ 而 $x \in A$，则 x 是 B 的一个大于 c 的下界，违背 c 为最大下界，因此 $x \in B$。

原页勘误 末段英文写成 “less than the greatest lower bound”，应为 “greater than”。证明中的矛盾恰好来自找到一个更大的下界。

练习 把每个“不能”改写成反证的完整一句。这样能检查究竟使用了“c 是下界”还是“c 是最大下界”。

2.3 上确界逼近刻画的双向证明

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 24 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：28。

原课件第 24 页

Definition of Supremum

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) $b \leqslant u$ whenever u is any upper bound for S.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \leqslant b - \varepsilon$ for every $x \in S$, i.e. $b - \varepsilon$ is an upper bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $u < b$ then from $(2')$ it follows that u is not an upper bound of S. That is (2): $b \leqslant u$ whenever u is any upper bound for S.

沿原论证展开与批注

必要性 设 $b = \sup S$。它当然是上界。如果某个 $\varepsilon > 0$ 下找不到 $x > b - \varepsilon$，那么所有 $x \in S$ 都满足 $x \leq b - \varepsilon$，说明 $b - \varepsilon$ 是更小上界，矛盾。

充分性 设 b 是上界且有逼近性质。任取上界 u，要证 $b \leq u$。若 $u < b$，令 $\varepsilon = b - u > 0$。逼近性质给 $x \in S$ 满足 $x > b - \varepsilon = u$，违反 u 是上界。因此 $b \leq u$。

如何想到选精度 想让 $b - \varepsilon$ 恰好落在反证假设中的 u，所以解方程得到 $\varepsilon = b - u$。证明中不是“随便取一个小数”，而是围绕目标构造。

2.4 下确界逼近刻画的双向证明

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 25 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：29。

原课件第 25 页

Definition of Infimum

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) $l \geqslant d$ whenever d is any lower bound for S.

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose (2′) is not true. Then for some $\varepsilon > 0$ it follows that $x \geqslant l + \varepsilon$ for every $x \in S$, i.e. $l + \varepsilon$ is a lower bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $d > l$ then from (2′) it follows that d is not a lower bound of S. That is (2): $l \geqslant d$ whenever d is any lower bound for S.

沿原论证展开与批注

必要性 若 $\ell = \inf S$，它是下界。若某个 $\varepsilon > 0$ 使所有 $x \in S$ 都满足 $x \geq \ell + \varepsilon$，那么 $\ell + \varepsilon$ 是更大的下界，矛盾。

充分性 若 ℓ 是下界并能从上方逼近，任取下界 d。假设 $d > \ell$，令 $\varepsilon = d - \ell$，就有 $x \in S$ 满足 $x < \ell + \varepsilon = d$，违背下界性质。故每个下界 $d \leq \ell$。

一眼识别使用时机 需要构造 $x_n \in S$ 逼近下确界时，逐次取 $\varepsilon = 1/n$，得到 $\ell \leq x_n < \ell + 1/n$，于是 $x_n \to \ell$。存在逼近数列不意味着下确界属于 S。

2.5 递增情形：逐句拆解证明

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 28 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：32。

原课件第 28 页

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is increasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded above, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \leqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N > x - \varepsilon$, as otherwise $x - \varepsilon$ would be an upper bound. Choose such $N = N(\varepsilon)$. Since $x_N > x - \varepsilon$, then $x_n > x - \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x - \varepsilon < x_n \leqslant x$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

沿原论证展开与批注 设 $a = \sup\{x_n : n \in \mathbb{N}\}$，存在性来自非空和上有界。由上界定义，对所有 n 有 $x_n \le a$。任给 $\varepsilon > 0$，由于 $a - \varepsilon < a$，它不可能仍是上界。因此存在某个 K，使 $x_K > a - \varepsilon$。单调性告诉我们：只要 $n \ge K$，就有 $x_n \ge x_K > a - \varepsilon$。合起来：

$$a - \varepsilon < x_K \le x_n \le a, \qquad |x_n - a| = a - x_n < \varepsilon.$$

这正是极限定义，故 $x_n \to a$。

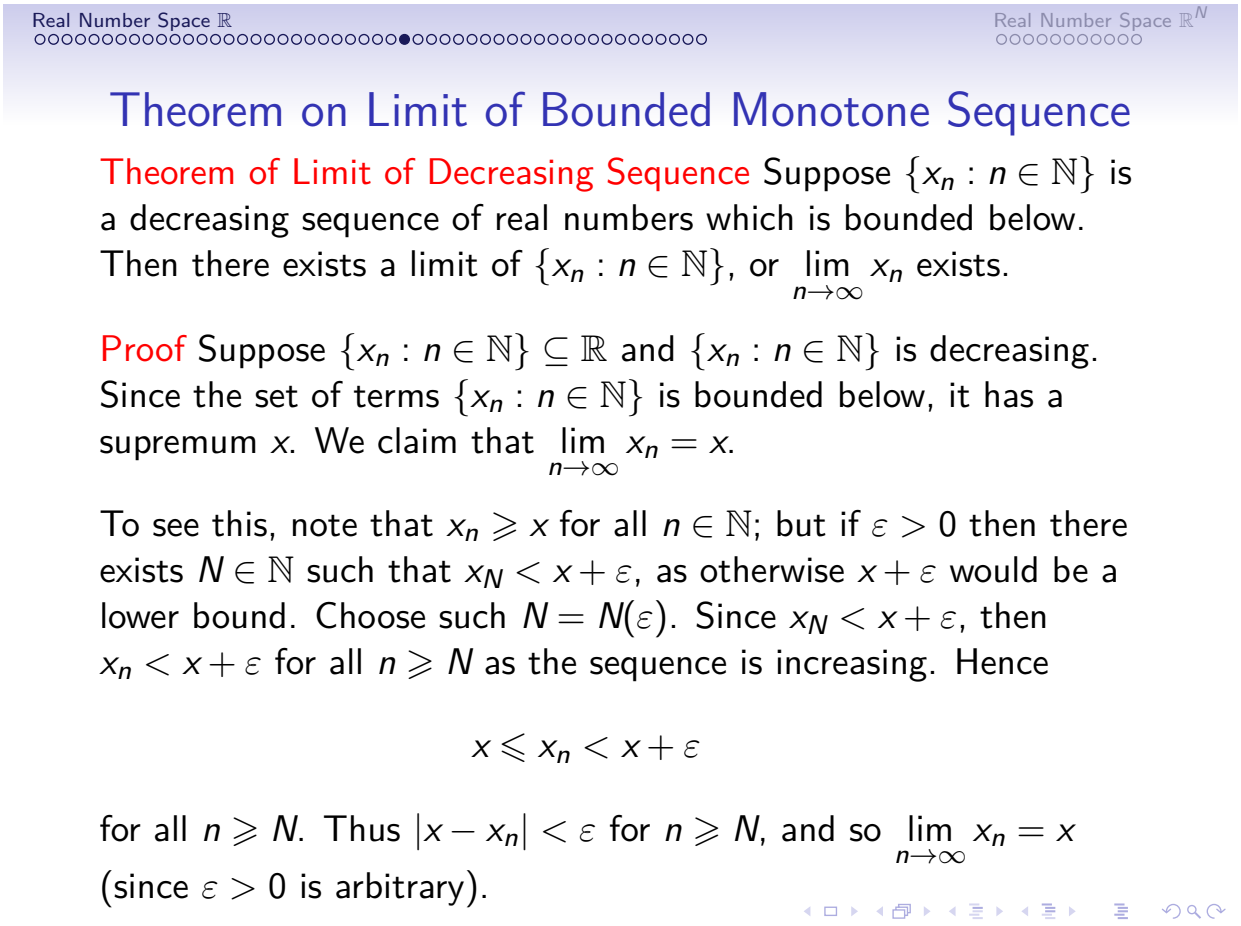
考试易漏 确界性质只给一个接近上界的项；如果没有单调性，后面的项可能又跑远。不能把这一步省成“由确界定义显然收敛”。

2.6 递减情形与课件笔误

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 29 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：33。

原课件第 29 页



沿原论证展开与批注

原页勘误 证明首段的 “supremum” 应改为 “infimum”；后面的 “increasing” 应改为 “decreasing”。

完整证明 令 $a = \inf\{x_n : n \in \mathbb{N}\}$。非空且下有界保证存在，并且 $a \le x_n$。任给 $\varepsilon > 0$，由下确界逼近性质，存在 K 使 $x_K < a + \varepsilon$。当 $n \ge K$，递减性给 $x_n \le x_K$，因此

$$a \le x_n \le x_K < a + \varepsilon, \qquad |x_n - a| = x_n - a < \varepsilon.$$

故 $x_n \to a$。

自检 以 $x_n = 1/n$ 检查：其极限是下确界 0，绝不是上确界 1。代入简单例子常能发现符号错误。

2.7 先把左端点的上确界放入每个区间

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 31 页。

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原课件第 31 页

Real Number Space $\mathbb{R}$ Real Number Space $\mathbb{R}^N$

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = \sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Proof We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_n \leqslant u_1$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $A = \{d_n : n \in \mathbb{N}\}$ that is bounded above by u_1. Thus this set has a supremum in the real numbers and denote it $D = \sup A = \lim_{n\to\infty} d_n$ so that $d_n \leqslant D$ for all $n \in \mathbb{N}$.
We now want to show that $D \leqslant u_n$ for all $n \in \mathbb{N}$. We will do this by showing that $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_n \leqslant u_{n'}$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_n \leqslant u_{n'}$.
So then $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $u_{n'}$ is an upper bound to the set A and so $D = \sup A \leqslant u_{n'}$ for all $n' \in \mathbb{N}$. Furthermore we have that $d_{n'} \leqslant D$ for all $n' \in \mathbb{N}$, and so $D \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $D \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

沿原论证展开与批注 所有 $d_n \le u_1$，因此 $D = \sup_n d_n$ 存在，并且 $d_m \le D$。还要证明 $D \le u_m$，才可说 $D \in I_m$。

关键交叉比较 固定任意 m，对任意 n：若 $n \le m$，则 $d_n \le d_m \le u_m$；若 $n > m$，则 $d_n \le u_n \le u_m$。所以 u_m 是全部左端点的上界，故 $D \le u_m$。于是每个 m 都有 $d_m \le D \le u_m$，即 $D \in \bigcap_m I_m$。

原页勘误 末尾英文 union 应为 intersection（交集）。只比较同一指标 $d_n \le u_n$ 不够，必须证明任意左右端点的交叉不等式。

2.8 再把右端点的下确界放入每个区间

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对应已有 `gpt-000-ZSM.pdf` 页脚：36。

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Cantor Nested Intervals Theorem

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Proof We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_1 \leqslant u_n$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $B = \{u_n : n \in \mathbb{N}\}$ that is bounded above by d_1. Thus this set has an infumum in the real numbers and denote it $U = \inf B = \lim_{n\to\infty} u_n$ so that $U \leqslant u_n$ for all $n \in \mathbb{N}$.
We now want to show that $l_n \leqslant U$ for all $n \in \mathbb{N}$. We will do this by showing that $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_{n'} \leqslant u_n$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_{n'} \leqslant u_n$.
So then $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $I_{n'}$ is a lower bound to the set B and so $u_{n'} \leqslant \sup B = U$ for all $n' \in \mathbb{N}$. Furthermore we have that $U \leqslant u_{n'}$ for all $n' \in \mathbb{N}$, and so $U \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $U \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

沿原论证展开与批注

纠正后证明 所有 $u_n \geq d_1$，所以右端点集合下有界，$U = \inf_n u_n$ 存在，且 $U \leq u_m$。上一页的交叉比较给固定 m 下，所有 n 满足 $d_m \leq u_n$；因此 d_m 是右端点集合的下界，故 $d_m \leq U$。所以 $d_m \leq U \leq u_m$，U 属于每个区间。

原页勘误 bounded above 应改为 bounded below；后面的 $\sup B$ 应为 $\inf B$；l_n、$l_{n'}$ 应统一为左端点 $d_n, d_{n'}$；关键结论是 $d_m \leq \inf B = U$，不是 $u_m \leq U$。本证明只用下确界定义，没有偷偷假设交集非空。

2.9 为什么交集恰好是一个闭区间

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 33 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：37。

原课件第 33 页

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = \sup\{d_n : n \in \mathbb{N}\} = \lim_{n\to\infty} d_n$ and $U = \inf\{u_n : n \in \mathbb{N}\} = \lim_{n\to\infty} u_n$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Proof We first prove that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$. Now let $x \in [D, U]$, then $D \leqslant x \leqslant U$. But we know that $d_n \leqslant D$ for all $n \in \mathbb{N}$ and $U \leqslant u_n$ for all $n \in \mathbb{N}$ and so $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Therefore $x \in [d_n, u_n]$ for all $n \in \mathbb{N}$ or in other words $x \in \bigcap_{n=1}^{\infty} I_n$. Therefore $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

We then prove that $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$. Let $x \in \bigcap_{n=1}^{\infty} I_n$. Then $d_n \leqslant x \leqslant u_n$ for all $n \in \mathbb{N}$. We also know that $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Suppose that x is such that $d_n \leqslant x < D$. Since D is the supremum of the set A then there exists an element $d_x \in \{d_n : n \in \mathbb{N}\}$ such that $x < d_x$ and so $x \notin [d_x, u_x]$ and therefore $x \notin \bigcap_{n=1}^{\infty} I_n$, a contradiction. Now suppose that x is such that $U < x \leqslant u_n$ for all $n \in \mathbb{N}$. Since U is the infimum of the set B then there exists an element $u_x \in \{u_n : n \in \mathbb{N}\}$ such that $u_x < x$ and so $x \notin [d_x, u_x]$ and so $x \notin \bigcap_{n=1}^{\infty} I_n$, once again, a contradiction. We must therefore have that $d_n \leqslant D \leqslant x \leqslant U \leqslant u_n$ and so $x \in [D, U]$ and so $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$.

沿原论证展开与批注 要证两个集合相等，标准做法是证明两个包含关系。

第一方向 若 $x \in [D, U]$，每个 n 都有 $d_n \le D \le x \le U \le u_n$，所以 $x \in I_n$，因此 x 属于交集。

第二方向 若 $x \in \bigcap_n I_n$，则每个 n 都有 $d_n \le x$，所以 x 是全体左端点的上界，从而 $D \le x$。同理 $x \le u_n$ 对所有 n 成立，使 x 成为右端点集合的下界，所以 $x \le U$。故 $x \in [D, U]$。这里的第二方向用确界定义直接完成，比课件的两次反证更短，但逻辑相同。$D \le U$ 已由左端点的上确界属于所有区间得到。

2.10 长度趋零如何产生唯一交点

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对应已有 `gpt-000-ZSM.pdf` 页脚：38。

原课件第 34 页

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

Proof The non-empty set $A = \{d_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above by u_1, then it has a limit $D = \lim_{n\to\infty} d_n$.

The non-empty set $B = \{u_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below by d_1, then it has a limit $U = \lim_{n\to\infty} u_n$.

Thus $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$, and

$$0 \leqslant U - D \leqslant u_n - d_n.$$

Note $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $D = U \in \bigcap_{n=1}^{\infty} I_n$.

沿原论证展开与批注 已有 $d_n \le D \le U \le u_n$，两端相减可得

$$0 \le U - D \le u_n - d_n.$$

右边趋于 0，所以固定非负数 $U - D$ 只能为 0，即 $D = U =: a$。交集 $[D, U] = \{a\}$，且单调收敛定理给 $d_n \to a, u_n \to a$。

不引用夹逼定理也能证明 若 $U - D = c > 0$，由长度趋零，某个 n 有 $u_n - d_n < c$，与 $c \le u_n - d_n$ 矛盾。

“唯一”的证明 若 a, b 都在每个区间，则 $|a - b| \le u_n - d_n$ 对所有 n 成立，同样推出 $a = b$。长度条件在这一步才真正用到。

2.11 Bolzano 定理：二分法抽子列

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对应已有 `gpt-000-ZSM.pdf` 页脚：39。

原课件第 35 页

Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a limit point.

Proof Let $\{x_n : n \in \mathbb{N}\}$ be a bounded sequence, we assume $\{x_n : n \in \mathbb{N}\} \subseteq [D, U]$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_1, U_1]$, and we take $x_{n_1} \in [D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts $\left[D_1, \frac{D_1+U_1}{2}\right]$ and $\left[\frac{D_1+U_1}{2}, U_1\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_2, U_2]$, and we take $x_{n_2} \in [D_2, U_2]$ with $n_2 > n_1$. We repeat this step and obtain a sequence $\{[D_k, U_k] : k \in \mathbb{N}\}$ of closed bounded nested intervals:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_k, U_k] \supseteq \cdots \quad \text{with } U_k - D_k = \frac{U - D}{2^k}$$

with a subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $x_{n_k} \in [D_k, U_k]$ for $k \in \mathbb{N}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{k=1}^{\infty} [D_k, U_k]$, and $|x_{n_k} - a| < U_k - D_k = \frac{U-D}{2^k}$, then $\lim_{k \to \infty} x_{n_k} = a$.

沿原论证展开与批注 把全部项放进 $[D, U]$。若 $D = U$，数列常值，直接收敛。否则把区间二等分，至少一半含无穷多个项的指标；否则两半加起来仅含有限项。保留这一半，重复二分，得到长度 $(U - D)/2^k$ 的闭区间套 I_k。每个 I_k 含无穷多个指标，所以可以逐次选 $n_k > n_{k-1}$ 且 $x_{n_k} \in I_k$。区间套定理给唯一交点 a。因为 a, x_{n_k} 都在 I_k，

$$|x_{n_k} - a| \le |I_k| = (U - D)2^{-k} \longrightarrow 0.$$

故选出的子列收敛。

细节 原页的严格 $<$ 一般应改为 $\le$，不影响极限。常值数列只有一个不同取值，却有无穷多个项；所以这里不能误说“无穷多个不同点”。

2.12 有界无限集合必有聚点：构造阶段

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 36–37 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：40, 41。

原课件第 36 页

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Proof. Note that E is a bounded infinite set in $\mathbb{R}$, there exists a constant M such that, for any $x \in E$, $|x| \leqslant M$. Define $I_0 = [-M, M]$, we divide R_0 into two small closed intervals: $[-M, 0]$ and $[0, M]$, then at least one small interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_1 with length M. Generally, we assume that there exists a sequence of closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_N, \quad I_N \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots, N$. We divide I_N into two small closed intervals, then at least one small closed interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_{N+1} with length $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_n \supseteq \cdots, \quad I_n \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots$.

原课件第 37 页

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $x \in \bigcap_{n=1}^{\infty} I_n$. We now check that this x is one of accumulation points of E.
Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large integer n_0, $\frac{M}{2^{n_0}} < \delta$. The length of I_{n_0+1} is strictly less than δ, then $I_{n_0+1} \subseteq N(x, \delta)$, and hence $E \cap I_{n_0+1} \subseteq E \cap N(x, \delta)$. x is an accumulation point of E.

沿原论证展开与批注 聚点 a 的定义是：每个 $\delta > 0$ 都能找到 $x \in E$ 满足 $0 < |x - a| < \delta$。允许 a 不属于 E，但找出的点必须不同于 a。把有界无限集合 E 放入 $[-M, M]$，取 $M > 0$。二分后至少一个闭半区间仍含 E 的无穷多个不同点，否则两边有限，其并也有限。反复保留无限的一半，得到 I_n 嵌套，长度 $2M/2^n$，且 $I_n \cap E$ 无限。

与上一页的区别 数列版本保留无限指标；集合版本保留无限不同点。这一区别保证下一页能排除交点本身，真正证明“聚点”。原页 R_0 应读作 I_0。

闭区间套给唯一交点 a。任给 $\delta > 0$，选择 n 使 $|I_n| < \delta$。对每个 $x \in I_n$，因 $a \in I_n$，有 $|x - a| \le |I_n| < \delta$，因此 $I_n \subset (a - \delta, a + \delta)$。又因为 $E \cap I_n$ 有无穷多个不同点，删去可能存在的点 a 后仍至少有一个 x。这个 x 满足 $x \in E$ 且 $0 < |x - a| < \delta$。由于 δ 任意，a 是聚点。

课件省略的关键句 只说邻域与集合相交尚不足以证明聚点，孤立点的邻域也会与集合相交。必须说“有不同于 a 的点”，无限性正是用来保证这一点。

2.13 收敛推出柯西：把误差分成两半

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 39–40 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：43, 44。

原课件第 39 页

Cauchy Convergence Criterion

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

Proof [Necessity] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a convergent sequence, and suppose $x = \lim_{n\to\infty} x_n$. For any $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that, when $n > N$, $|x_n - x| < \varepsilon$. It follows that for any $n_1 > N$ and $n_2 > N$,

$$|x_{n_1} - x_{n_2}| \leqslant |x_{n_1} - x| + |x_{n_2} - x| < 2\varepsilon.$$

Thus $\{x_n : n \in \mathbb{N}\}$ is a Cauchy sequence.

原课件第 40 页

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|x_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

From Bolzano-Weierstrass Theorem, the bounded sequence $\{x_n : n \in \mathbb{N}\}$ has a convergent subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $\lim_{k\to\infty} x_{n_k} = a$. For any $\varepsilon > 0$ there exists an integer K such that, when $k > K$, $|x_{n_k} - a| < \varepsilon$.
Take $k_0 = \max\{K+1, N+1\}$, then $n_{k_0} \geqslant n_{N+1} \geqslant N+1 > N$.
Then, when $n > N$, $|x_n - x_{n_{k_0}}| < \varepsilon$, and hence

$$|x_n - a| \leqslant |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - a| < 2\varepsilon$$

that is, $\lim_{n\to\infty} x_n = a$.

沿原论证展开与批注 设 $x_n \to a$。给定最终希望得到的 $\varepsilon > 0$，按收敛定义，对精度 $\varepsilon/2$ 选 K，使 $n \geq K$ 时 $|x_n - a| < \varepsilon/2$。若 $m, n \geq K$，利用三角不等式：

$$|x_n - x_m| = |(x_n - a) + (a - x_m)| \leq |x_n - a| + |x_m - a| < \varepsilon.$$

于是满足柯西定义。

为什么课件写 2ε 也能成立 因为 ε 是任意正数，可重新命名精度。但初学者考试最好从一开始取 $\varepsilon/2$，使结尾直接得到题目要求的 $< \varepsilon$，不留下量词解释负担。

先证明有界 在柯西条件中先取固定精度 1，得 K_0。固定 $m = K_0$，所有 $n \geq K_0$ 满足 $|x_n| \leq |x_n - x_{K_0}| + |x_{K_0}| < 1 + |x_{K_0}|$。再与前面有限项的绝对值取最大值，便得到全列的统一界。

找到候选极限 由 Bolzano 定理，有子列 $x_{n_j} \to a$。

证明整列趋于它 任给 $\varepsilon > 0$，柯西条件给 K，使 $m, n \geq K$ 时距离小于 $\varepsilon/2$。选一个足够大的 j，同时保证 $n_j \geq K$ 及 $|x_{n_j} - a| < \varepsilon/2$。以后每个 $n \geq K$ 都满足

$$|x_n - a| \leq |x_n - x_{n_j}| + |x_{n_j} - a| < \varepsilon.$$

这里先选一个共同的子列项作为桥梁，而不是对每个 n 随意更换未经控制的指标。

2.14 柯西充分性的第二证法：尾部下确界

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 41–42 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：45, 46。

原课件第 41 页

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \inf\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \geqslant y_n$ since y_{n+1} is the infimum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is an increasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

原课件第 42 页

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \leqslant x_n$ for each $n \in \mathbb{N}$, then $y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon$, then
$x_{n_2} \leqslant \inf\{x_{n_1} + \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} = \inf\{x_{n_1} : n_1 \geqslant N_2\} + \frac{1}{2}\varepsilon = y_{n_1} + \frac{1}{2}\varepsilon$.
Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim_{n\to\infty} x_n = a$.

沿原论证展开与批注 柯西列有界的证明同上一页。定义 $y_n = \inf\{x_k : k \geq n\}$。删掉第一项后尾集缩小，下确界只能上升：$y_{n+1} \geq y_n$。原数列的统一上下界也界住 y_n，所以 y_n 单调有界，存在极限 a。

几何解释 y_n 是从第 n 项起所有数的“下边界”。它不一定是实际某一项，但总有 $y_n \leq x_n$。柯西条件将迫使实际项与这条下边界越来越接近。

原页勘误 $|nn|$ 应为 $|x_n|$。不用把下确界视作最小值；即使尾集没有最小项，定义与证明仍然有效。

原页混用了取下确界的哑指标与自由指标。以下使用 j 表示被取遍的指标，避免混淆。给 $\varepsilon > 0$，取 K_1 使 $n \geq K_1$ 时 $|y_n - a| < \varepsilon/2$；再取 K_2 使 $n, j \geq K_2$ 时 $|x_n - x_j| < \varepsilon/2$。固定 $n \geq \max(K_1, K_2)$。对所有 $j \geq n$，有 $x_j > x_n - \varepsilon/2$，所以取下确界后

$$y_n \geq x_n - \varepsilon/2, \qquad 0 \leq x_n - y_n \leq \varepsilon/2.$$

从而 $|x_n - a| \leq |x_n - y_n| + |y_n - a| < \varepsilon$，故 $x_n \to a$。

细节 取确界后严格不等号可能变成非严格号，不能保留原页所有 $<$。由于另一误差严格小于 $\varepsilon/2$，最终仍得到严格 $< \varepsilon$。

2.15 第三证法：尾部上确界

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 43–44 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：47, 48。

原课件第 43 页

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \sup\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \leqslant y_n$ since y_{n+1} is the supremum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is a decreasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

原课件第 44 页

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \geqslant x_n$ for each $n \in \mathbb{N}$, then $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon \leqslant y_{n_1} + \frac{1}{2}\varepsilon$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$, then $y_{n_1} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} : n_1 \geqslant N_2\} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} - \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} \leqslant x_{n_2}$. Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim_{n\to\infty} x_n = a$.

沿原论证展开与批注 定义 $z_n = \sup\{x_k : k \geq n\}$。尾集合变小使上确界不增，故 z_n 递减；柯西列有界又保证 z_n 有下界。因此单调收敛定理给 $z_n \to a$。

对照下确界方法 下包络 y_n 从下方上升，上包络 z_n 从上方下降；柯西条件意味着两者之间的尾部振幅趋零。两种方法都是把不单调的数列转为单调边界。

原页勘误 同样应把 $|nn|$ 改为 $|x_n|$。因为 z_n 是上确界，所以 $x_n \leq z_n$，不能反过来。下一页用这一方向完成估计。

任给 $\varepsilon > 0$，取 K 同时保证 $n \geq K$ 时 $|z_n - a| < \varepsilon/2$，且 $n, j \geq K$ 时 $|x_n - x_j| < \varepsilon/2$。固定 $n \geq K$，对所有 $j \geq n$ 有 $x_j < x_n + \varepsilon/2$，所以

$$x_n \leq z_n = \sup_{j \geq n} x_j \leq x_n + \varepsilon/2.$$

于是 $|x_n - a| \leq |x_n - z_n| + |z_n - a| < \varepsilon$，整列收敛。

课件指标修正 不能把 $\sup_{j \geq K} x_j$ 写成任意的 z_{n_1}，两者的尾集起点不同。本页证明固定 n 后统一对 $j \geq n$ 取上确界，便消除了这个问题。上下包络证法任选一种熟练掌握即可，但应认识两种思路。

2.16 有限覆盖定理：把坏区间不断二分

来源： `000-ZSM-PPT.pdf`，原课件 PDF 第 46–47 页。

对应已有 `gpt-000-ZSM.pdf` 页脚：50, 51。

原课件第 46 页

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof (Reduction to Absurdity) Let $\mathscr{H}$ be an open covering of $[D, U]$. Assume that $[D, U]$ can not be covered by finite many open sets belonging to $\mathscr{H}$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts, then there exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_2, U_2]$. We repeat this step and obtain a sequence $\{[D_n, U_n] : n \in \mathbb{N}\}$ of closed bounded nested intervals that can not be covered by finite many open sets belonging to $\mathscr{H}$:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_n, U_n] \supseteq \cdots \quad \text{with } U_n - D_n = \frac{U - D}{2^n}.$$

原课件第 47 页

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{n=1}^{\infty} [D_n, U_n]$. Note that $\mathscr{H}$ is an open covering of $[D, U]$, then there exists $H_0 \in \mathscr{H}$ such that $a \in H_0$. Note that H_0 is an open set, then there exists $\delta > 0$ such that $(x-\delta, x+\delta) \subseteq H_0$.
Note $\lim_{n\to\infty} D_n = a = \lim_{n\to\infty} U_n$, then there exists an integer N such that, when $n > N$, $D_n \in (x-\delta, x+\delta)$ and $U_n \in (x-\delta, x+\delta)$.

$$[D_n, U_n] \subseteq (x-\delta, x+\delta) \subseteq H_0 \qquad \text{for } n > N.$$

which contradicts that $[D_n, U_n]$ can not be covered by finite many open sets belonging to $\mathscr{H}$.

沿原论证展开与批注 先证明闭区间 $[D,U]$。反设存在一个开覆盖，无法抽出有限子覆盖。二分成两个闭半区间，至少一个仍无法有限覆盖；否则两边各取有限个，合起来仍有限，便覆盖了整个区间，矛盾。如此递归，得到每个都“无法有限覆盖”的闭区间 I_n，且嵌套、长度 $(U-D)/2^n \to 0$。闭区间套定理给交点 a。

思路 我们暂时不直接挑覆盖集，而是把“失败的位置”压缩到一个点。开集在该点周围会留出余地，下一页正用这个余地制造矛盾。$D = U$ 的单点区间只需选一个覆盖它的开集，已经成立。

既然原族覆盖 $[D,U]$，有 $H_0 \in \mathcal{H}$ 含 a。H_0 开，故有 $\delta > 0$ 使 $(a-\delta, a+\delta) \subseteq H_0$。取 n 使 $|I_n| < \delta$。由于 $a \in I_n$，整个 I_n 都在该邻域内，于是仅一个 H_0 就覆盖 I_n，与构造矛盾。

补足课件未写的一步 定理针对任意闭有界 S，而非只有区间。取 $S \subseteq [D,U]$。因为 S 闭，$\mathbb{R} \setminus S$ 开，把它加入原开覆盖，便覆盖整个 $[D,U]$。用已证结论取有限子覆盖，再删除其中的 $\mathbb{R} \setminus S$（若有），剩下的原覆盖集仍覆盖 S。空集情形取空覆盖即可。

原页记号 邻域中心须统一写交点 a，不能在证明中突然换成未定义的 x。

3 101-Proof：无实际证明过程

101-ZSM-PPT.pdf 为数学章节引言，没有实际证明过程，故不添加证明。

4 102-Proof

4.1 结合律的展开式与排版纠正

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 33 页。

对应已有 gpt-102.pdf 页脚：37。

原课件第 33 页

Basic Operation: Symmetric Difference

$$
\begin{aligned}
& (A_1 \triangle A_2) \triangle A_3 = [(A_1 \setminus A_2) \cup (A_2 \setminus A_1)] \triangle A_3 \\
= \;& \{[(A_1 \setminus A_2) \cup (A_2 \setminus A_1)] \setminus A_3\} \cup \{A_3 \setminus [(A_1 \cup A_2) \setminus (A_1 \cap A_2)]\} \\
= \;& \{[(A_1 \setminus A_2) \setminus A_3] \cup [(A_2 \setminus A_1) \setminus A_3]\} \cup \{A_3 \cap [\mathbf{C}(A_1 \cup A_2) \cup (A_1 \cap A_2)]\} \\
= \;& [A_1 \cap \mathbf{C}(A_2 \cup A_3)] \cup [A_2 \cap \mathbf{C}(A_3 \cup A_1)] \cup [A_3 \cap \mathbf{C}(A_1 \cup A_2)] \cup [A_3 \cap (A_1 \cap A_2)] \\
= \;& [A_1 \setminus (A_2 \cup A_3)] \cup [A_2 \setminus (A_3 \cup A_1)] \cup [A_3 \setminus (A_1 \cup A_2)] \cup [A_1 \cap A_2 \cap A_3]
\end{aligned}
$$

$$
\begin{aligned}
& A_1 \triangle (A_2 \triangle A_3) = A_1 \triangle [(A_2 \setminus A_3) \cup (A_3 \setminus A_2)] \\
= \;& \{A_1 \setminus [(A_2 \cup A_3) \setminus (A_2 \cap A_3)]\} \cup \{[(A_2 \setminus A_3) \cup (A_3 \setminus A_2)] \setminus A_1\} \\
= \;& \{A_1 \cap [\mathbf{C}(A_2 \cup A_3) \cup (A_2 \cap A_3)]\} \cup \{[(A_2 \setminus A_3) \setminus A_1] \cup [(A_3 \setminus A_2) \setminus A_1]\} \\
= \;& [A_1 \cap \mathbf{C}(A_2 \cup A_3)] \cup [A_1 \cap (A_2 \cap A_3)] \cup [A_2 \cap \mathbf{C}(A_3 \cup A_1)] \cup [A_3 \cap (A_1 \cap A_2)] \\
= \;& [A_1 \setminus (A_2 \cup A_3)] \cup [A_1 \cap A_2 \cap A_3] \cup [A_2 \setminus (A_3 \cup A_1)] \cup [A_3 \setminus (A_1 \cup A_2)] \\
= \;& [A_1 \setminus (A_2 \cup A_3)] \cup [A_2 \setminus (A_3 \cup A_1)] \cup [A_3 \setminus (A_1 \cup A_2)] \cup [A_1 \cap A_2 \cap A_3]
\end{aligned}
$$

沿原论证展开与批注　目标与路线。证明对称差结合律。按老师的方法展开差集，把两种括号都化成同一个集合，而不把尚待证明的结合律当作已知。令 $A = A_1, B = A_2, C = A_3$。先展开

$$(A \triangle B) \triangle C = [(A \triangle B) \setminus C] \cup [C \setminus (A \triangle B)].$$

第一项由差对并的分配得到

$$[(A \setminus B) \cup (B \setminus A)] \setminus C = [A \setminus (B \cup C)] \cup [B \setminus (A \cup C)].$$

第二项用补集与德摩根律：

$$
\begin{aligned}
C \setminus (A \triangle B) &= C \cap [(A \cap B^c) \cup (B \cap A^c)]^c \\
&= C \cap (A^c \cup B) \cap (B^c \cup A) \\
&= [C \cap A^c \cap B^c] \cup [A \cap B \cap C].
\end{aligned}
$$

中间分配展开有四项，其中含 $A \cap A^c$ 或 $B \cap B^c$ 的项为空。于是整个结果为

$$[A \setminus (B \cup C)] \cup [B \setminus (A \cup C)] \cup [C \setminus (A \cup B)] \cup (A \cap B \cap C).$$

同样展开 $A \triangle (B \triangle C)$ 得到完全相同的四项，故两者相等。这一推导解释了为何三重共有部分也在三重对称差中。

4.2 抵消律与集合版三角不等式

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 34 页。

对应已有 gpt-102.pdf 页脚：38。

原课件第 34 页

Basic Operation: Symmetric Difference

From the property of the inverses in a Boolean group, it follows that the symmetric difference of two repeated symmetric differences is equivalent to the repeated symmetric difference of the join of the two multisets, where for each double set both can be removed. In particular:

$$(A_1 \triangle A_2) \triangle (A_2 \triangle A_3) = A_1 \triangle A_3.$$

This implies triangle inequality: the symmetric difference of A_1 and A_3 is contained in the union of the symmetric difference of A_1 and A_2 and that of A_2 and A_3.

$$(A_1 \triangle A_2) \triangle (A_2 \triangle A_3) = [(A_1 \triangle A_2) \triangle A_2] \triangle A_3 = A_1 \triangle A_3$$
$$(A_1 \triangle A_2) \triangle (A_2 \triangle A_3) = A_1 \triangle [A_2 \triangle (A_2 \triangle A_3)] = A_1 \triangle A_3$$

沿原论证展开与批注 老师利用结合律把重复的 A_2 放到一起，再用 $A_2 \triangle A_2 = \varnothing$ 消去：

$$\begin{aligned}(A_1 \triangle A_2) \triangle (A_2 \triangle A_3) &= A_1 \triangle [(A_2 \triangle A_2) \triangle A_3] \\ &= A_1 \triangle (\varnothing \triangle A_3) = A_1 \triangle A_3.\end{aligned}$$

第一步仅改变括号，不是分配律。再令 $X = A_1 \triangle A_2, Y = A_2 \triangle A_3$。由定义

$$X \triangle Y = (X \setminus Y) \cup (Y \setminus X) \subseteq X \cup Y,$$

代入前式得到 $A_1 \triangle A_3 \subseteq (A_1 \triangle A_2) \cup (A_2 \triangle A_3)$。

4.3 分配律的代数证明拆解

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 36 页。

对应已有 gpt-102.pdf 页脚：40。

原课件第 36 页

Basic Operation: Symmetric Difference

$$
\begin{aligned}
& A_1 \cap (A_2 \triangle A_3) = A_1 \cap [(A_2 \setminus A_3) \cup (A_3 \setminus A_2)] \\
= \;& [A_1 \cap (A_2 \setminus A_3)] \cup [A_1 \cap (A_3 \setminus A_2)] \\
= \;& \{A_1 \cap [A_2 \setminus (A_2 \cap A_3)]\} \cup \{A_1 \cap [A_3 \setminus (A_2 \cap A_3)]\} \\
= \;& [(A_1 \cap A_2) \setminus (A_1 \cap A_2 \cap A_3)] \cup [(A_1 \cap A_3) \setminus (A_1 \cap A_2 \cap A_3)] \\
= \;& [(A_1 \cap A_2) \cup (A_1 \cap A_3)] \setminus (A_1 \cap A_2 \cap A_3) \\
= \;& [A_1 \cap (A_2 \cup A_3)] \setminus (A_1 \cap A_2 \cap A_3)
\end{aligned}
$$

$$
\begin{aligned}
& (A_1 \cap A_2) \triangle (A_1 \cap A_3) = [(A_1 \cap A_2) \setminus (A_1 \cap A_3)] \cup [(A_1 \cap A_3) \setminus (A_1 \cap A_2)] \\
= \;& [(A_1 \cap A_2) \setminus (A_1 \cap A_2 \cap A_3)] \cup [(A_1 \cap A_3) \setminus (A_1 \cap A_2 \cap A_3)] \\
= \;& [(A_1 \cap A_2) \cup (A_1 \cap A_3)] \setminus (A_1 \cap A_2 \cap A_3) \\
= \;& [A_1 \cap (A_2 \cup A_3)] \setminus (A_1 \cap A_2 \cap A_3)
\end{aligned}
$$

沿原论证展开与批注 **老师路线：**展开对称差，把同一公式的左右两边都写成两个互斥部分。

$$
\begin{aligned}
A \cap (B \triangle C) &= A \cap [(B \cap C^c) \cup (C \cap B^c)] \\
&= (A \cap B \cap C^c) \cup (A \cap C \cap B^c).
\end{aligned}
$$

而

$$
\begin{aligned}
(A \cap B) \setminus (A \cap C) &= A \cap B \cap (A^c \cup C^c) = A \cap B \cap C^c, \\
(A \cap C) \setminus (A \cap B) &= A \cap C \cap B^c.
\end{aligned}
$$

将最后两项取并，正是 $(A \cap B) \triangle (A \cap C)$。每次删去的项都含 $A \cap A^c = \varnothing$。对称差不会给交集之外增加元素，因此也有 $A \cap (B \triangle C) \subseteq A \cap (B \cup C)$。

4.4 三个对称差推论的短证明与修正

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 37 页。

对应已有 gpt-102.pdf 页脚：41。

原课件第 37 页

Basic Operation: Symmetric Difference

Proof of (1).

$$\begin{aligned} \mathbf{C}A_1 \triangle \mathbf{C}A_2 &= (\mathbf{C}A_1 \setminus \mathbf{C}A_2) \cup (\mathbf{C}A_2 \setminus \mathbf{C}A_1) \\ &= (\mathbf{C}A_1 \cap A_2) \cup (\mathbf{C}A_2 \cap A_1) \\ &= (A_2 \setminus A_1) \cup (A_1 \setminus A_2) = A_1 \triangle A_2 \end{aligned}$$

Proof of (2). We first check $A_2 \subseteq A_1$. Otherwise, there exists $x_0 \in A_2 \setminus A_1$ and $A_1 \cup \{x_0\} \subseteq A_1 \triangle A_2 = A_1$, which is a contradiction!

If $A_2 \subseteq A_1$, then $A_1 = A_1 \triangle A_2 = A_1 \setminus A_2$ and hence $A_2 = \varnothing$.

Proof of (3).

$$E = E \triangle \varnothing = E \triangle (A \triangle A) = (E \triangle A) \triangle A = A' \triangle A.$$

沿原论证展开与批注 本页有三段证明。**(1) 两边同时取补不改变对称差。**

$$A^c \triangle B^c = (A^c \cap B) \cup (B^c \cap A) = (B \setminus A) \cup (A \setminus B) = A \triangle B.$$

用到 $X \setminus Y = X \cap Y^c$、双重补集及并的交换律。**(2)** $A \triangle B = A \iff B = \varnothing$。先假设等式。若存在 $x \in B \setminus A$，则 $x \in A \triangle B = A$，与 $x \notin A$ 矛盾。所以 $B \subseteq A$。因此 $B \setminus A = \varnothing$，对称差化为 $A \setminus B$。原假设便是 $A = A \setminus B$。若 $b \in B$，由于 $B \subseteq A$ 有 $b \in A$，但 $b \notin A \setminus B$，又矛盾；故 $B = \varnothing$。反向直接由 $A \triangle \varnothing = A$。原图中 $A \cup \{x\} \subseteq A \triangle B$ 的写法无须且一般不对，真正需要的只是 $\{x\} \subseteq A \triangle B$。**(3) 解对称差方程。**若 $E \triangle A = A'$，用已证结合律和 $A \triangle A = \varnothing$：

$$E = E \triangle \varnothing = E \triangle (A \triangle A) = (E \triangle A) \triangle A = A' \triangle A.$$

每步分别使用空集恒等元、相同集合自消、结合律及已知方程。反向代入 $E = A' \triangle A$，同样结合后消去两个 A，得 $E \triangle A = A'$。这说明“等式两边同时作 $\triangle A$”可以消掉一项，与普通加法移项的思路类似，但依据是对称差的结合与自消。

4.5 并集对称差的包含证明

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 38 页。

对应已有 gpt-102.pdf 页脚：42。

原课件第 38 页

Basic Operation: Symmetric Difference

Proof of (4).

$$
\begin{aligned}
& \left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) \triangle \left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right) \\
= & \left[\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) \setminus \left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right)\right] \cup \left[\left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right) \setminus \left(\bigcup_{\lambda\in\Lambda} A_\lambda\right)\right] \\
\subseteq & \left(\bigcup_{\lambda\in\Lambda} \left[A_\lambda \setminus A'_\lambda\right]\right) \cup \left(\bigcup_{\lambda\in\Lambda} \left[A'_\lambda \setminus A_\lambda\right]\right) \\
= & \bigcup_{\lambda\in\Lambda} \left[A_\lambda \setminus A'_\lambda\right] \cup \left[A'_\lambda \setminus A_\lambda\right] \\
= & \bigcup_{\lambda\in\Lambda} \left(A_\lambda \triangle A'_\lambda\right).
\end{aligned}
$$

沿原论证展开与批注 **目标：** $(\bigcup_\lambda A_\lambda)\triangle(\bigcup_\lambda A'_\lambda) \subseteq \bigcup_\lambda (A_\lambda \triangle A'_\lambda)$。老师先展开对称差，再分别估计两段差集。记 $U = \bigcup_\lambda A_\lambda, V = \bigcup_\lambda A'_\lambda$。任取 $x \in U \setminus V$。$x \in U$ 给出存在 λ_0 使 $x \in A_{\lambda_0}$；$x \notin V$ 则给出每个 λ 都有 $x \notin A'_\lambda$，特别有 $x \notin A'_{\lambda_0}$。于是

$$x \in A_{\lambda_0} \setminus A'_{\lambda_0} \subseteq \bigcup_\lambda (A_\lambda \setminus A'_\lambda).$$

这就证明 $U \setminus V$ 包含于右侧。交换两组集合得到

$$V \setminus U \subseteq \bigcup_\lambda (A'_\lambda \setminus A_\lambda).$$

将两个包含取并，再重排并集：

$$
\begin{aligned}
U\triangle V &= (U \setminus V) \cup (V \setminus U) \\
&\subseteq \bigcup_\lambda [(A_\lambda \setminus A'_\lambda) \cup (A'_\lambda \setminus A_\lambda)] \\
&= \bigcup_\lambda (A_\lambda \triangle A'_\lambda).
\end{aligned}
$$

这里只能保证包含：取 $A_1 = \{a\}, A_2 = \varnothing, A'_1 = \varnothing, A'_2 = \{a\}$，左边为空而右边为 $\{a\}$。

4.6 巧选特殊集合与补全双向证明

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 40 页。

对应已有 gpt-102.pdf 页脚：44。

原课件第 40 页

Basic Operation: Symmetric Difference

Example:

(1) Let A and A' be two subsets of the whole set S. If, for any $E \subseteq S$, $E \cap A = E \cup A'$, then $A = S$ and $A' = \varnothing$.

(2) Let A_1 and A_2 be two sets, then $A_1 = A_2$ if and only if there exists a set A_3 such that $A_1 \cap A_3 = A_2 \cap A_3$ and $A_1 \cup A_3 = A_2 \cup A_3$.

Proof. (1) If $E = S$, then $A = S$; Take $E = \mathbf{C}A$, then it is assumed that $\varnothing = \mathbf{C}A \cap A = \mathbf{C}A \cup A' \supseteq A'$, that is, $A' = \varnothing$.
(2) The necessity is obvious. Now we prove the sufficiency.

$$A_1 = A_1 \cup (A_3 \cap \mathbf{C}A_3) = (A_1 \cap A_3) \cup (A_1 \cap \mathbf{C}A_3)$$
$$A_2 = A_2 \cup (A_3 \cap \mathbf{C}A_3) = (A_2 \cap A_3) \cup (A_2 \cap \mathbf{C}A_3)$$

then $A_1 \cup A_3 = (A_1 \cap \mathbf{C}A_3) \cup A_3$ and $A_2 \cup A_3 = (A_2 \cap \mathbf{C}A_3) \cup A_3$.
Note that $A_3 \cap (A_1 \cap \mathbf{C}A_3) = \varnothing = A_3 \cap (A_2 \cap \mathbf{C}A_3)$, then $A_1 \cap \mathbf{C}A_3 = A_2 \cap \mathbf{C}A_3$. Thus
$A_1 = (A_1 \cap A_3) \cup (A_1 \cap \mathbf{C}A_3) = (A_2 \cap A_3) \cup (A_2 \cap \mathbf{C}A_3) = A_2$.

沿原论证展开与批注 题(1)假设对每个$E \subseteq S$均有$E \cap A = E \cup A'$。取$E = S$得$A = S$；取$E = \varnothing$得$A' = \varnothing$。这是“全称条件可以选特别方便的对象”的典型。 题(2)若$A_1 = A_2$，任意A_3都满足等式。反之已知与A_3的交相同、并相同，把并集等式两边与CA_3相交，得$A_1 \cap CA_3 = A_2 \cap CA_3$。再使用

$$A_i = (A_i \cap A_3) \cup (A_i \cap CA_3),$$

两块分别相同，故$A_1 = A_2$。这一分解用$A_3 \cup CA_3 = S$，不是靠把空集“分配”出两块。

4.7 练习1：先化简再谈充要条件

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 42 页。

对应已有 gpt-102.pdf 页脚：46。

原课件第 42 页

Basic Operation: Symmetric Difference

Proof of 1.

$$\begin{aligned} A' &= \mathbf{C}(E\cap A)\cap(\mathbf{C}E\cup A) = (\mathbf{C}E\cup\mathbf{C}A)\cap(\mathbf{C}E\cup A)\\ &= \mathbf{C}E\cup(\mathbf{C}A\cap\cup A) = \mathbf{C}E\cup\varnothing = \mathbf{C}E. \end{aligned}$$

If $\mathbf{C}A' = E$, then

$$\begin{aligned} &\mathbf{C}(E\cap A)\cap(\mathbf{C}E\cup A) = \mathbf{C}\left(\mathbf{C}A'\cap A\right)\cap\left(A'\cup A\right)\\ =\ &\left(A'\cup\mathbf{C}A\right)\cap\left(A'\cup A\right) = A'\cup(\mathbf{C}A\cap A) = A'\cup\varnothing = A'. \end{aligned}$$

Conversely,

$$\begin{aligned} \mathbf{C}A' &= (E\cap A)\cup\mathbf{C}(\mathbf{C}E\cup A) = (E\cap A)\cup(E\cap\mathbf{C}A)\\ &= E\cap(A\cup\mathbf{C}A) = E\cap S = E. \end{aligned}$$

沿原论证展开与批注 不论A是什么，

$$C(E\cap A)\cap(CE\cup A) = (CE\cup CA)\cap(CE\cup A) = CE\cup(CA\cap A) = CE.$$

于是题中等式就是$A' = CE$，它与$CA' = E$由双重补集律等价。原页的$CA\cap\cup A$是排版错误，应为$CA\cap A$。先化简可以避免把两个方向分别写成长式，却仍完整证明“当且仅当”。

4.8 练习4：不能漏掉反向包含

来源： 102-ZSM-PPT.pdf，原课件 PDF 第 43 页。

对应已有 gpt-102.pdf 页脚：47。

原课件第 43 页

Basic Operation: Symmetric Difference

Proof of 4 (1).

$$A = A \cap (A \cup A') = A \cap (E \cup E') = (A \cap E) \cup (A \cap E') = A \cap E$$

implies $A \subseteq E$, and

$$A' = A' \cap (A \cup A') = A' \cap (E \cup E') = (A' \cap E) \cup (A' \cap E') = A' \cap E'$$

implies $A' \subseteq E'$, then $A \cup A' \subseteq E \cup E' = A \cup A'$, thus $A = E$ and $A' = E'$.

Proof of 4 (2).

$$A_1 \cup A_2 = (A \cap E) \cup (A \cap E') = A \cap (E \cup E') = A \cap (A \cup A') = A.$$

沿原论证展开与批注 由$A \cup A' = E \cup E'$及$A \cap E' = \varnothing$，任取$x \in A$必在E或E'，后者被排除，故$A \subseteq E$。反向任取$x \in E$，共同并集保证$x \in A$或A'，由$A' \cap E = \varnothing$排除后者，故$E \subseteq A$。同理$A' \subseteq E'$且$E' \subseteq A'$。原页只写两个正向包含后直接宣布相等，需补上刚才的反向步骤。 第(2)题直接分配：$(A \cap E) \cup (A \cap E') = A \cap (E \cup E') = A \cap (A \cup A') = A$。

5 103-Proof

5.1 补集结论的证明

来源： 103-ZSM-PPT.pdf，原课件 PDF 第 18 页。

对应已有 gpt-103.pdf 页脚：21。

原课件第 18 页

Correspondences

Proof of Theorem 6. $y \in \mathbf{C}(\varphi(A))$, $y \notin \varphi(A) = \bigcup_{x \in A} \varphi(x)$, there isn't $x \in A$ such that $y \in \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$, and hence $y \in \varphi(\mathbf{C}A)$.

Conversely, $y \in \varphi(\mathbf{C}A) = \bigcup_{x \in \mathbf{C}A} \varphi(x)$, then $y \in \varphi(x)$ for some $x \in \mathbf{C}A$. If φ is injective, then $y \notin \varphi(x)$ for $x \in A$, and $y \in \varphi(x)$ for $x \notin A$, thus there isn't $x \in A$ such that $y \in \varphi(x)$, $y \notin \bigcup_{x \in A} \varphi(x) = \varphi(A)$, $y \in \mathbf{C}(\varphi(A))$.

沿原论证展开与批注 在 $T = \Phi(S)$ 下，若 $y \notin \Phi(A)$，从满值域选 $x \in S$ 使 $y \in \Phi(x)$；该 x 不在 A，故 $y \in \Phi(S \setminus A)$。反向需不同输入的输出集不相交，否则同一输出可同时由 A 内外产生。

5.2 常对应与恒等对应

来源： 103-ZSM-PPT.pdf，原课件 PDF 第 25 页。

对应已有 gpt-103.pdf 页脚：28。

原课件第 25 页

Operations on mappings

A mapping Δ is said to be constant if there exists a subset C of T such that $\Delta(x) = C$ for all S. A constant mapping satisfies the following property:

$$(\Delta \cap \varphi)(A) = \Delta(A) \cap \varphi(A).$$

Another important mapping is the identity mapping, which is the single-valued mapping I of S onto S defined by

$$I(x) = \{x\}.$$

$$\begin{aligned}(\Delta \cap \varphi)(A) &= \bigcup_{x\in A}(\Delta \cap \varphi)(x) = \bigcup_{x\in A}[\Delta(x) \cap \varphi(x)] \\ &= \bigcup_{x\in A}[C \cap \varphi(x)] = C \cap \left[\bigcup_{x\in A}\varphi(x)\right] \\ &= C \cap \varphi(A) = \Delta(A) \cap \varphi(A).\end{aligned}$$

沿原论证展开与批注 若 $\Delta(x) = C$，对非空 A 有 $\Delta(A) = C$，所以 $\bigcup_{x\in A}(C \cap \Phi(x)) = C \cap \Phi(A)$；空 A 时两边均为空。恒等对应的输出是单点集 $\{x\}$。

5.3 逆对应

来源： 103-ZSM-PPT.pdf，原课件 PDF 第 28 页。

对应已有 gpt-103.pdf 页脚：31。

原课件第 28 页

Upper and Lower Inverses

If φ is a mapping of S into T, its lower inverse is the mapping φ^- of T into S defined by

$$\varphi^-(y) = \{x \in S \mid y \in \varphi(x)\}.$$

This is a mapping whose domain is T^* and whose range is S^*.

Theorem 9. If φ is single-valued, φ^- is injective; if φ is injective, φ^- is single-valued; if φ is semi-single-valued, φ^- is semi-single-valued.

Proof. For $y \neq y'$ we want to prove $\varphi^-(y) \cap \varphi^-(y') = \varnothing$. Otherwise, there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is single-valued, then $y = \varphi(x) = y'$, which is a contradiction.

If $\varphi^-(y) \cap \varphi^-(y') \neq \varnothing$, then there exists $x \in \varphi^-(y) \cap \varphi^-(y')$, then $y \in \varphi(x)$ and $y' \in \varphi(x)$. If φ is injective, then $y = y'$ and hence $\varphi^-(y) = \varphi^-(y')$.

沿原论证展开与批注 记 $\Phi^-(y) = \{x : y \in \Phi(x)\}$，以下始终在有效定义域讨论非空值。**第一结论：单值对应的逆对应单射。**任取不同的 y, y'，若逆值集相交，取 $x \in \Phi^-(y) \cap \Phi^-(y')$。逆对应定义给 $y, y' \in \Phi(x)$；但 $\Phi(x)$ 恰含一个点，所以 $y = y'$，矛盾。于是不同输入的逆值集不交，这正是对应的单射定义。 **第二结论：单射对应的逆对应单值。**固定有效值域内的 y，其逆值非空。若有 $x, x' \in \Phi^-(y)$，则 $y \in \Phi(x) \cap \Phi(x')$，原对应单射使 $x = x'$。因此逆值恰含一个元素。注意必须固定同一个 y 比较两个原像；原页第二段固定同一个 x 比较两个输出，并没有证明此结论。 **第三句的额外纠错：**若 semi-single-valued 指每个值集“至多含一个元素”，则它本身不保证逆对应也半单值。反例 $\Phi(0) = \Phi(1) = \{a\}$，每个原值只有一点，但 $\Phi^-(a) = \{0, 1\}$。要保证逆值至多一点，仍需要不同输入的值集互不相交。故原页这句不能在没有补充定义或假设时当作已证定理。

5.4 下逆像

来源： 103-ZSM-PPT.pdf，原课件 PDF 第 29 页。

对应已有 gpt-103.pdf 页脚：32。

原课件第 29 页

Upper and Lower Inverses

For any non-empty subset B of T we have

$$\varphi^{-}(B) = \{x \in S \mid \varphi(x) \cap B \neq \varnothing\}$$

and we also write $\varphi^{-}(\varnothing) = \varnothing$. It is clear that the inverse of φ^{-} is $(\varphi^{-})^{-} = \varphi$ and that $y \in \varphi(x)$ is equivalent to $x \in \varphi^{-}(y)$.

$$\begin{aligned}
\left(\varphi^{-}\right)^{-}(A) &= \{y \in T \mid \varphi^{-}(y) \cap A \neq \varnothing\} \\
&= \{y \in T \mid \{x \in S \mid y \in \varphi(x)\} \cap A \neq \varnothing\} \\
&= \{y \in T \mid \{x \in A \mid y \in \varphi(x)\} \neq \varnothing\} \\
&= \{y \in T \mid y \in \varphi(A)\} = \varphi(A)
\end{aligned}$$

$$\left(\varphi^{-}\right)^{-}(x) = \{y \in T \mid x \in \varphi^{-}(y)\} = \{y \in T \mid y \in \varphi(x)\} = \varphi(x)$$

沿原论证展开与批注 $x \in \Phi^{-}(B) \iff \exists y \in B : y \in \Phi(x)$。交换有序对两坐标再交换回来不变，故逆对应的逆是原对应。集合逆像的“存在”含义对后面下半连续极重要。

5.5 补集对偶

来源： 103-ZSM-PPT.pdf，原课件 PDF 第 31–32 页。

对应已有 gpt-103.pdf 页脚：34, 35。

原课件第 31 页

Upper and Lower Inverses

Theorem 10. If B_1 and B_2 are subsets of T, then we have

(1) $\mathbf{C}\varphi^+(B) = \varphi^-(\mathbf{C}B)$ and $\mathbf{C}\varphi^-(B) = \varphi^+(\mathbf{C}B)$.

(2) $\varphi^+(B_1) \cup \varphi^+(B_2) \subseteq \varphi^+(B_1 \cup B_2)$.

(3) $\varphi^-(B_1) \cup \varphi^-(B_2) = \varphi^-(B_1 \cup B_2)$.

Proof.

$$\begin{aligned}\mathbf{C}\varphi^+(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \subseteq B\} \\ &= \mathbf{C}\{x \in S \mid \varphi(x) \cap \mathbf{C}B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap \mathbf{C}B \neq \varnothing\} = \varphi^-(\mathbf{C}B)\end{aligned}$$

$$\begin{aligned}\mathbf{C}\varphi^-(B) &= \mathbf{C}\{x \in S \mid \varphi(x) \cap B \neq \varnothing\} \\ &= \{x \in S \mid \varphi(x) \cap B = \varnothing\} \\ &= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}B\} = \varphi^+(\mathbf{C}B)\end{aligned}$$

原课件第 32 页

Upper and Lower Inverses

$$
\begin{aligned}
& \varphi^{+}(B_1) \cup \varphi^{+}(B_2) \\
= \; & \{x \in S \mid \varphi(x) \subseteq B_1\} \cup \{x \in S \mid \varphi(x) \subseteq B_2\} \\
\subseteq \; & \{x \in S \mid \varphi(x) \subseteq B_1 \cup B_2\} \\
= \; & \varphi^{+}(B_1 \cup B_2)
\end{aligned}
$$

$$
\begin{aligned}
& \varphi^{-}(B_1) \cup \varphi^{-}(B_2) \\
= \; & \{x \in S \mid \varphi(x) \cap B_1 \neq \varnothing\} \cup \{x \in S \mid \varphi(x) \cap B_2 \neq \varnothing\} \\
= \; & \{x \in S \mid (\varphi(x) \cap B_1) \cup (\varphi(x) \cap B_2) \neq \varnothing\} \\
= \; & \{x \in S \mid \varphi(x) \cap (B_1 \cup B_2) \neq \varnothing\} \\
= \; & \varphi^{-}(B_1 \cup B_2)
\end{aligned}
$$

沿原论证展开与批注 在有效定义域 S^* 内，非“全部在 B”等价于“有一个在 $T \setminus B$”。因此 $S^* \setminus \Phi^{+}(B) = \Phi^{-}(T \setminus B)$。允许空值却取整个 S 作补集会破坏课件公式。

下逆像保持并：有输出在 $B_1 \cup B_2$ 当且仅有输出在至少一个集合。上逆像仅有包含，因为所有输出可能分散在两集合。例值集 $\{0,1\}$，分别取 $B_1 = \{0\}, B_2 = \{1\}$ 即为严格包含。

5.6 Cantor 对角证明

来源： 103-ZSM-PPT.pdf，原课件 PDF 第 35 页。

对应已有 gpt-103.pdf 页脚：38。

原课件第 35 页

Countable Sets

Theorem 12. The set of subsets $\mathbb{N}$ of is not countable.

Proof: We show that there is no surjection from a set A to its power set 2^A. Consider any function $f : A \to 2^A$ and let

$$B = \{a \in A \mid a \notin f(a)\}$$

We claim that there is no $b \in A$ such that $f(b) = B$. Indeed, assume $f(b) = B$ for some $b \in A$. Then either $b \in B$ hence $b \notin f(b)$ which is a contradiction, or $b \notin B$ implying that $b \in B$ which is again a contradiction. Hence the map f is not surjective as claimed. □

沿原论证展开与批注 假设 $f : A \to \mathcal{P}(A)$ 满射。定义 $B = \{a \in A : a \notin f(a)\}$，由满射存在 b 使 $f(b) = B$。于是 $b \in B \iff b \notin B$，两种可能都矛盾。关键是 B 依赖于假设中的整张列表，必与每个列出集合在其对应位置不同。

6 104-Proof

6.1 负传递的等价形式

来源： 104-ZSM-PPT.pdf，原课件 PDF 第 17 页。

对应已有 gpt-104.pdf 页脚：19。

原课件第 17 页

Binary Relations

[Lemma 1.] A binary relation $\mathscr{R}$ is negatively transitive if and only if $x_1 \mathscr{R} x_2$ implies that, for all $x \in S$, $x_1 \mathscr{R} x$ or $x \mathscr{R} x_2$.

Proof.

$$
\begin{aligned}
& \text{Binary relation } \mathscr{R} \text{ is negatively transitive} \\
\Longleftrightarrow\ & \text{If } x_1 \mathscr{P} x \text{ and } x \mathscr{P} x_2 \text{ for some } x \in S, \text{ then } x_1 \mathscr{P} x_2 \\
\Longleftrightarrow\ & \text{If not } x_1 \mathscr{P} x_2, \text{ then not } [x_1 \mathscr{P} x \text{ and } x \mathscr{P} x_2] \text{ for all } x \in S \\
\Longleftrightarrow\ & \text{If not } x_1 \mathscr{P} x_2, \text{ then not } x_1 \mathscr{P} x \text{ or not } x \mathscr{P} x_2 \text{ for all } x \in S \\
\Longleftrightarrow\ & \text{If } x_1 \mathscr{R} x_2, \text{ then } x_1 \mathscr{R} x \text{ or } x \mathscr{R} x_2 \text{ for all } x \in S.
\end{aligned}
$$

沿原论证展开与批注 原式的逆否命题是 $x \succ z \Rightarrow (x \succ y \text{ 或 } y \succ z)$，对每个中间点 y 成立。推导用到 $\neg(P \wedge Q) = \neg P \vee \neg Q$，注意“或”允许两者同时成立。

6.2 传递与无循环的证明

来源： 104-ZSM-PPT.pdf，原课件 PDF 第 19 页。

对应已有 gpt-104.pdf 页脚：21。

原课件第 19 页

Preference Relations

[Proposition 1.] If $\succ$ is a strict preference relation, then it is irreflexive, transitive, and acyclic.

Proof. [1] Asymmetry directly implies irreflexivity.

[2] Suppose $x_1 \succ x_2$ and $x_2 \succ x_3$. By negative transitivity and Lemma 1, $x_1 \succ x_2$ implies either $x_1 \succ x_3$ or $x_3 \succ x_2$. But $x_3 \succ x_2$ is impossible because $x_2 \succ x_3$ is assumed and $\succ$ is asymmetric. Thus $x_1 \succ x_3$, which is transitivity.

Suppose $x_1 \succ x_2$ and $x_2 \succ x_3$. By negative transitivity and Lemma 1, $x_2 \succ x_3$ implies either $x_2 \succ x_1$ or $x_1 \succ x_3$. But $x_2 \succ x_1$ is impossible because $x_1 \succ x_2$ is assumed and $\succ$ is asymmetric. Thus $x_1 \succ x_3$, which is transitivity.

[3] If $x_1 \succ x_2, \cdots, x_{N-1} \succ x_N$, then by transitivity $x_1 \succ x_N$. Since $\succ$ is irreflexive, this implies $x_1 \neq x_N$. Thus $\succ$ is acyclic.

沿原论证展开与批注 非对称排除 $x \succ x$。若 $x \succ y \succ z$，由上页等价式，$x \succ y$ 推出 $x \succ z$ 或 $z \succ y$；后者与 $y \succ z$ 矛盾，故传递。沿有限循环反复传递得到 $x \succ x$，矛盾。

6.3 命题证明展开

来源： `104-ZSM-PPT.pdf`，原课件 PDF 第 22 页。

对应已有 `gpt-104.pdf` 页脚：24。

原课件第 22 页

Preference Relations

Proof. [1] follows from the definition of $\sim$ and the fact that $\succ$ is asymmetric.

[2] By the asymmetry of $\succ$, either $x_1 \not\succ x_2$ and $x_2 \not\succ x_1$ (or both) for all x_1 and x_2, thus $\succsim$ is complete. For the transitivity of $\succsim$, note that this follows immediately the negative transitivity of $\succ$.

[3] $\sim$ is reflexive because $\succ$ is irreflexive. $\sim$ is symmetric because the definition of $\sim$ is symmetric. For the transitivity, suppose $x_1 \sim x_2 \sim x_3$. Then $x_1 \not\succ x_2 \not\succ x_3$ and $x_3 \not\succ x_2 \not\succ x_1$. By negative transitivity of $\succ$, $x_1 \not\succ x_3 \not\succ x_1$, or $x_1 \sim x_3$.

[4] If $x_1 \succ x_2 \sim x_3$, then by [1] one of $x_1 \succ x_3$ or $x_3 \sim x_1$ or $x_1 \succ x_3$. But $x_3 \succ x_1$ is impossible, since then transitivity of $\succ$ would imply $x_3 \succ x_2$. And $x_3 \sim x_1$ is impossible, since then transitivity of $\sim$ would imply $x_2 \sim x_1$, contradicting $x_1 \succ x_2$. Thus $x_1 \succ x_3$ must hold.

If $x_2 \sim x_3 \succ x_4$, then by [1] one of $x_2 \succ x_4$ or $x_4 \sim x_2$ or $x_2 \succ x_4$. But $x_4 \succ x_2$ is impossible, since then transitivity of $\succ$ would imply $x_3 \succ x_2$. And $x_4 \sim x_2$ is impossible, since then transitivity of $\sim$ would imply $x_4 \sim x_3$, contradicting $x_3 \succ x_4$. Thus $x_2 \succ x_4$ must hold.

[5] $x_1 \succsim x_2$ if and only if $x_2 \not\succ x_1$ if and only if $x_1 \succ x_2$ or $x_1 \sim x_2$ (by part [1]).

[6] This is immediate from the definition of $\succsim$ and $\sim$.

沿原论证展开与批注 假设严格关系 $\succ$ 非对称且负传递，定义 $x \succeq y \iff \neg(y \succ x)$，$x \sim y \iff \neg(x \succ y) \wedge \neg(y \succ x)$。沿老师六小点依次证明。**[1] 三分性：**非对称排除两个严格方向同时成立；若两个方向均不成立，按定义就是无差异，所以三种情形恰一成立。**[2] 弱偏好完全与传递：**非对称保证 $\neg(x \succ y)$ 或 $\neg(y \succ x)$，故 $y \succeq x$ 或 $x \succeq y$。若 $x \succeq y, y \succeq z$，则 $\neg(y \succ x), \neg(z \succ y)$。对 z, y, x 用负传递，得 $\neg(z \succ x)$，即 $x \succeq z$。**[3] 无差异为等价关系：**非自反性给 $x \sim x$；交换两个否定条件得对称。若 $x \sim y, y \sim z$，负传递分别用于 x, y, z 和 z, y, x，得 $\neg(x \succ z)$ 与 $\neg(z \succ x)$，故 $x \sim z$。**[4] 严格与无差异可连接：**设 $x \succ y, y \sim z$。对 x, z 用三分性。如果 $z \succ x$，严格传递给 $z \succ y$，违反 $y \sim z$；如果 $z \sim x$，无差异传递给 $y \sim x$，违反 $x \succ y$。因此只能 $x \succ z$。对 $y \sim z, z \succ w$ 同理得 $y \succ w$。原图三分分支重复书写，应读为正向严格、反向严格、无差异这三种。**[5] 弱关系分解：**$x \succeq y$ 排除了 $y \succ x$，由三分性只剩 $x \succ y$ 或 $x \sim y$；这两种反过来也都排除 $y \succ x$，故等价。**[6] 双向弱关系：**$x \succeq y$ 且 $y \succeq x$，按定义恰是两个严格方向都不成立，也就是 $x \sim y$。每一项都须区分普通传递与负传递，不能仅凭文字相近交换使用。

6.4 还原弱关系

来源： 104-ZSM-PPT.pdf，原课件 PDF 第 28 页。

对应已有 gpt-104.pdf 页脚：30。

原课件第 28 页

Preference Relations

Proof. $\succ'$ is asymmetric because $x_1 \succ' x_2$ implies $x_2 \not\succeq' x_1$ from the definition of $\succ'$ and then $x_2 \not\succ' x_1$. $\succ'$ is negatively transitive because $x_1 \not\succ' x_2$ and $x_2 \not\succ' x_3$ implies $x_3 \succsim' x_2$ and $x_2 \succsim' x_1$ and then $x_3 \succsim' x_1$ from the completeness and transitivity of $\succsim'$, thus $x_1 \not\succ' x_3$ from the definition of $\succ'$.

$x_1 \succsim' x_2$ implies $x_2 \not\succ' x_1$ from the completeness of $\succsim'$, then $x_1 \succsim x_2$ from the definition of $\succsim$. $x_1 \succsim x_2$ implies $x_2 \not\succ' x_1$ from the definition of $\succsim$, then $x_1 \succsim' x_2$ from the completeness of $\succsim'$.

The proof of that $\sim'$ and $\sim$ are equivalent is from the definitions of the indifferences.

沿原论证展开与批注 原构造给出 $x \succeq y \iff \neg(y \succ' x) \iff \neg\neg(x \succeq' y) \iff x \succeq' y$。严格关系和无差异也由相同定义还原。这里的双重否定是逻辑步骤，不是经验判断。

7 105-Proof

7.1 由分割公理证明上确界存在

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 37–38 页。

对应已有 gpt-105.pdf 页脚：42, 43。

原课件第 37 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded above. Let

$$B = \{x \in \mathbb{R} : x \text{ is an upper bound for } S\} \quad \text{and} \quad A = \mathbb{R} \setminus B.$$

Note that $B \neq \emptyset$; and if $x \in S$ then $x - 1$ is not an upper bound for S so $A \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then a would also be an upper bound for S, which contradicts the definition of A.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \sup S$.

If $c \in A$ then c is not an upper bound for S and so there exists $x \in S$ with $c < x$. But then $a = \dfrac{c + x}{2}$ is not an upper bound for S, i.e. $a \in A$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $a \leqslant c$ for all $a \in A$. Hence $c \in B$. But if $c \in B$ then $c \leqslant b$ for all $b \in B$; i.e. c is less than any upper bound for S.

原课件第 38 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then A is bounded above (by any element of B). Let $c = \sup A$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Now every member of B is an upper bound for A, from the first property of a Dedekind cut; hence $a \notin B$, as otherwise a would be an upper bound for A which is less than the least upper bound c. Hence $a \in A$.
- Next suppose $b > c$. Since c is an upper bound for A (in fact the least upper bound), it follows we cannot have $b \in A$, and thus $b \in B$.

沿原论证展开与批注

构造 令 B 为 S 的全部上界，$A = \mathbb{R} \setminus B$。有上界使 $B \neq \varnothing$；选 $s \in S$，$s-1$ 不是上界，故 $A \neq \varnothing$。

验证分割 若 $a \in A, b \in B$ 而 $a \geq b$，则所有 $s \in S$ 满足 $s \leq b \leq a$，使 a 也成上界，矛盾。因此 $a < b$，分割公理给分界点 c。

证明端点属于上界侧 若 $c \notin B$，有 $s \in S$ 满足 $s > c$。中点 $m = (c+s)/2$ 大于 c，应属于 B；但 $m < s$，又不是上界，矛盾。故 $c \in B$。所有 $b \in B$ 都有 $c \leq b$，所以 $c = \min B = \sup S$。关键技巧：用“全部上界”形成新集合，不是直接把原集合随意切开。

设 A, B 是一个分割。A 非空且任取 $b \in B$ 都是其上界，故可取 $c = \sup A$。

验证分界性质 若 $x < c$ 而 $x \in B$，则 x 是 A 的上界，违背 c 是最小上界，所以 $x \in A$。若 $x > c$，由于 c 是 A 的上界，x 不可能在 A，只能在 B。

唯一性补全 两个不同分界点之间的中点会被同时分入 A 和 B，矛盾。故确界公理能推出完整的分割公理。考试中“等价”必须写出两个方向，本页与前页缺一不可。

7.2 由分割公理证明下确界存在

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 39–40 页。

对应已有 gpt-105.pdf 页脚：44, 45。

原课件第 39 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded below. Let

$$A = \{x \in \mathbb{R} : x \text{ is a lower bound for } S\} \quad \text{and} \quad B = \mathbb{R} \setminus A.$$

Note that $A \neq \emptyset$; and if $x \in S$ then $x + 1$ is not a lower bound for S so $B \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then b would also be a lower bound for S, which contradicts the definition of B.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \inf S$.

If $c \in B$ then c is not a lower bound for S and so there exists $x \in S$ with $x < c$. But then $b = \dfrac{x + c}{2}$ is not a lower bound for S, i.e. $b \in B$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $c \leqslant b$ for all $b \in B$. Hence $c \in A$. But if $c \in A$ then $a \leqslant c$ for all $a \in A$; i.e. c is more than any lower bound for S.

原课件第 40 页

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then B is bounded below (by any element of A). Let $c = \inf B$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Since c is a lower bound for B (in fact the greatest lower bound), it follows we cannot have $a \in B$, and thus $a \in A$.
- Next suppose $c < b$. Now every member of A is a lower bound for B, from the first property of a Dedekind cut; hence $b \notin A$, as otherwise b would be a lower bound for B which is less than the greatest lower bound c. Hence $b \in B$.

沿原论证展开与批注 令 A 为 S 的全部下界，$B = \mathbb{R} \setminus A$。由下有界，$A \neq \varnothing$；取 $s \in S$，$s+1$ 不是下界，故 $B \neq \varnothing$。若 $a \in A, b \in B$ 且 $a \geq b$，则 $b \leq a \leq s$ 对所有 $s \in S$ 成立，矛盾，故 $a < b$。设分界点为 c。如果 $c \in B$，则有 $s \in S$ 满足 $s < c$。中点 $m = (s+c)/2$ 小于 c，应在 A，但 $m > s$ 又不可能是下界。因此 $c \in A$，且所有下界 $a \leq c$，故 $c = \inf S$。

记忆方法 上确界证明中选择“全部上界”的最小者；这里选择“全部下界”的最大者。中点总是用来挤出与分界方向冲突的数。

B 非空且任一 $a \in A$ 是其下界，取 $c = \inf B$。若 $x < c$，x 不可能属于 B，否则违反 c 是下界，所以 $x \in A$。若 $x > c$ 而 $x \in A$，则 x 是 B 的一个大于 c 的下界，违背 c 为最大下界，因此 $x \in B$。

原页勘误 末段英文写成“less than the greatest lower bound”，应为“greater than”。证明中的矛盾恰好来自找到一个更大的下界。

练习 把每个“不能”改写成反证的完整一句。这样能检查究竟使用了“c 是下界”还是“c 是最大下界”。

7.3 上确界逼近刻画的双向证明

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 42 页。

对应已有 gpt-105.pdf 页脚：47。

原课件第 42 页

Definition of Supremum

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) $b \leqslant u$ whenever u is any upper bound for S.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose (2′) is not true. Then for some $\varepsilon > 0$ it follows that $x \leqslant b - \varepsilon$ for every $x \in S$, i.e. $b - \varepsilon$ is an upper bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $u < b$ then from (2′) it follows that u is not an upper bound of S. That is (2): $b \leqslant u$ whenever u is any upper bound for S.

沿原论证展开与批注

必要性 设 $b = \sup S$。它当然是上界。如果某个 $\varepsilon > 0$ 下找不到 $x > b - \varepsilon$，那么所有 $x \in S$ 都满足 $x \le b - \varepsilon$，说明 $b - \varepsilon$ 是更小上界，矛盾。

充分性 设 b 是上界且有逼近性质。任取上界 u，要证 $b \le u$。若 $u < b$，令 $\varepsilon = b - u > 0$。逼近性质给 $x \in S$ 满足 $x > b - \varepsilon = u$，违反 u 是上界。因此 $b \le u$。

如何想到选精度 想让 $b - \varepsilon$ 恰好落在反证假设中的 u，所以解方程得到 $\varepsilon = b - u$。证明中不是“随便取一个小数”，而是围绕目标构造。

7.4 下确界逼近刻画的双向证明

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 43 页。

对应已有 gpt-105.pdf 页脚：48。

原课件第 43 页

Definition of Infimum

Suppose S is a non-empty set of real numbers. Then $I = \inf S$ iff

(1) $x \geqslant I$ for all $x \in S$;

(2) $I \geqslant d$ whenever d is any lower bound for S.

Suppose S is a non-empty set of real numbers. Then $I = \inf S$ iff

(1) $x \geqslant I$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < I + \varepsilon$.

[(1) + (2) $\Rightarrow$ (2′)] Suppose (2′) is not true. Then for some $\varepsilon > 0$ it follows that $x \geqslant I + \varepsilon$ for every $x \in S$, i.e. $I + \varepsilon$ is a lower bound for S. This is a contradiction.

[(1) + (2′) $\Rightarrow$ (2)] If $d > I$ then from (2′) it follows that d is not a lower bound of S. That is (2): $I \geqslant d$ whenever d is any lower bound for S.

沿原论证展开与批注

必要性 若 $\ell = \inf S$，它是下界。若某个 $\varepsilon > 0$ 使所有 $x \in S$ 都满足 $x \geq \ell + \varepsilon$，那么 $\ell + \varepsilon$ 是更大的下界，矛盾。

充分性 若 ℓ 是下界并能从上方逼近，任取下界 d。假设 $d > \ell$，令 $\varepsilon = d - \ell$，就有 $x \in S$ 满足 $x < \ell + \varepsilon = d$，违背下界性质。故每个下界 $d \leq \ell$。

一眼识别使用时机 需要构造 $x_n \in S$ 逼近下确界时，逐次取 $\varepsilon = 1/n$，得到 $\ell \leq x_n < \ell + 1/n$，于是 $x_n \to \ell$。存在逼近数列不意味着下确界属于 S。

7.5 递增情形：逐句拆解证明

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 46 页。

对应已有 gpt-105.pdf 页脚：51。

原课件第 46 页

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof. Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is increasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded above, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \leqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N > x - \varepsilon$, as otherwise $x - \varepsilon$ would be an upper bound. Choose such $N = N(\varepsilon)$. Since $x_N > x - \varepsilon$, then $x_n > x - \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x - \varepsilon < x_n \leqslant x$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

沿原论证展开与批注 设 $a = \sup\{x_n : n \in \mathbb{N}\}$，存在性来自非空和上有界。由上界定义，对所有 n 有 $x_n \le a$。任给 $\varepsilon > 0$，由于 $a - \varepsilon < a$，它不可能仍是上界。因此存在某个 K，使 $x_K > a - \varepsilon$。单调性告诉我们：只要 $n \ge K$，就有 $x_n \ge x_K > a - \varepsilon$。合起来：

$$a - \varepsilon < x_K \le x_n \le a, \qquad |x_n - a| = a - x_n < \varepsilon.$$

这正是极限定义，故 $x_n \to a$。

考试易漏 确界性质只给一个接近上界的项；如果没有单调性，后面的项可能又跑远。不能把这一步省成“由确界定义显然收敛”。

7.6 递减情形与课件笔误

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 47 页。

对应已有 gpt-105.pdf 页脚：52。

原课件第 47 页

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof. Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is decreasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded below, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \geqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N < x + \varepsilon$, as otherwise $x + \varepsilon$ would be a lower bound. Choose such $N = N(\varepsilon)$. Since $x_N < x + \varepsilon$, then $x_n < x + \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x \leqslant x_n < x + \varepsilon$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

沿原论证展开与批注

原页勘误 证明首段的 “supremum” 应改为 “infimum”；后面的 “increasing” 应改为 “decreasing”。

完整证明 令 $a = \inf\{x_n : n \in \mathbb{N}\}$。非空且下有界保证存在，并且 $a \le x_n$。任给 $\varepsilon > 0$，由下确界逼近性质，存在 K 使 $x_K < a + \varepsilon$。当 $n \ge K$，递减性给 $x_n \le x_K$，因此

$$a \le x_n \le x_K < a + \varepsilon, \qquad |x_n - a| = x_n - a < \varepsilon.$$

故 $x_n \to a$。

自检 以 $x_n = 1/n$ 检查：其极限是下确界 0，绝不是上确界 1。代入简单例子常能发现符号错误。

7.7 先把左端点的上确界放入每个区间

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 49 页。

对应已有 gpt-105.pdf 页脚：54。

原课件第 49 页

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Proof. We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_n \leqslant u_1$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $A = \{d_n : n \in \mathbb{N}\}$ that is bounded above by u_1. Thus this set has a supremum in the real numbers and denote it $D = \sup A = \lim_{n\to\infty} d_n$ so that $d_n \leqslant D$ for all $n \in \mathbb{N}$.
We now want to show that $D \leqslant u_n$ for all $n \in \mathbb{N}$. We will do this by showing that $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_n \leqslant u_{n'}$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_n \leqslant u_{n'}$.
So then $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $u_{n'}$ is an upper bound to the set A and so $D = \sup A \leqslant u_{n'}$ for all $n' \in \mathbb{N}$. Furthermore we have that $d_{n'} \leqslant D$ for all $n' \in \mathbb{N}$, and so $D \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $D \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

沿原论证展开与批注 所有 $d_n \le u_1$，因此 $D = \sup_n d_n$ 存在，并且 $d_m \le D$。还要证明 $D \le u_m$，才可说 $D \in I_m$。

关键交叉比较 固定任意 m，对任意 n：若 $n \le m$，则 $d_n \le d_m \le u_m$；若 $n > m$，则 $d_n \le u_n \le u_m$。所以 u_m 是全部左端点的上界，故 $D \le u_m$。于是每个 m 都有 $d_m \le D \le u_m$，即 $D \in \bigcap_m I_m$。

原页勘误 末尾英文 union 应为 intersection（交集）。只比较同一指标 $d_n \le u_n$ 不够，必须证明任意左右端点的交叉不等式。

7.8 再把右端点的下确界放入每个区间

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 50 页。

对应已有 gpt-105.pdf 页脚：55。

原课件第 50 页



Cantor Nested Intervals Theorem

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Proof. We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_1 \leqslant u_n$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $B = \{u_n : n \in \mathbb{N}\}$ that is bounded above by d_1. Thus this set has an infumum in the real numbers and denote it $U = \inf B = \lim_{n\to\infty} u_n$ so that $U \leqslant u_n$ for all $n \in \mathbb{N}$.
We now want to show that $I_n \leqslant U$ for all $n \in \mathbb{N}$. We will do this by showing that $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_{n'} \leqslant u_n$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_{n'} \leqslant u_n$.
So then $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $I_{n'}$ is a lower bound to the set B and so $u_{n'} \leqslant \sup B = U$ for all $n' \in \mathbb{N}$. Furthermore we have that $U \leqslant u_{n'}$ for all $n' \in \mathbb{N}$, and so $U \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $U \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

沿原论证展开与批注

纠正后证明 所有 $u_n \geq d_1$，所以右端点集合下有界，$U = \inf_n u_n$ 存在，且 $U \leq u_m$。上一页的交叉比较给固定 m 下，所有 n 满足 $d_m \leq u_n$；因此 d_m 是右端点集合的下界，故 $d_m \leq U$。所以 $d_m \leq U \leq u_m$，U 属于每个区间。

原页勘误 bounded above 应改为 bounded below；后面的 $\sup B$ 应为 $\inf B$；l_n、$l_{n'}$ 应统一为左端点 $d_n, d_{n'}$；关键结论是 $d_m \leq \inf B = U$，不是 $u_m \leq U$。本证明只用下确界定义，没有偷偷假设交集非空。

7.9 为什么交集恰好是一个闭区间

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 51 页。

对应已有 gpt-105.pdf 页脚：56。

原课件第 51 页

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = sup\{d_n : n \in \mathbb{N}\} = \lim_{n\to\infty} d_n$ and $U = inf\{u_n : n \in \mathbb{N}\} = \lim_{n\to\infty} u_n$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Proof. We first prove that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$. Now let $x \in [D, U]$, then $D \leqslant x \leqslant U$. But we know that $d_n \leqslant D$ for all $n \in \mathbb{N}$ and $U \leqslant u_n$ for all $n \in \mathbb{N}$ and so $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Therefore $x \in [d_n, u_n]$ for all $n \in \mathbb{N}$ or in other words $x \in \bigcap_{n=1}^{\infty} I_n$.

Therefore $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

We then prove that $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$. Let $x \in \bigcap_{n=1}^{\infty} I_n$. Then $d_n \leqslant x \leqslant u_n$ for all $n \in \mathbb{N}$. We also know that $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Suppose that x is such that $d_n \leqslant x < D$. Since D is the supremum of the set A then there exists an element $d_x \in \{d_n : n \in \mathbb{N}\}$ such that $x < d_x$ and so $x \notin [d_x, u_x]$ and therefore $x \notin \bigcap_{n=1}^{\infty} I_n$, a contradiction. Now suppose that x is such that $U < x \leqslant u_n$ for all $n \in \mathbb{N}$. Since U is the infimum of the set B then there exists an element $u_x \in \{u_n : n \in \mathbb{N}\}$ such that $u_x < x$ and so $x \notin [d_x, u_x]$ and so $x \notin \bigcap_{n=1}^{\infty} I_n$, once again, a contradiction. We must therefore have that $d_n \leqslant D \leqslant x \leqslant U \leqslant u_n$ and so $x \in [D, U]$ and so $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$.

沿原论证展开与批注 要证两个集合相等，标准做法是证明两个包含关系。

第一方向 若 $x \in [D,U]$，每个 n 都有 $d_n \le D \le x \le U \le u_n$，所以 $x \in I_n$，因此 x 属于交集。

第二方向 若 $x \in \bigcap_n I_n$，则每个 n 都有 $d_n \le x$，所以 x 是全体左端点的上界，从而 $D \le x$。同理 $x \le u_n$ 对所有 n 成立，使 x 成为右端点集合的下界，所以 $x \le U$。故 $x \in [D,U]$。这里的第二方向用确界定义直接完成，比课件的两次反证更短，但逻辑相同。$D \le U$ 已由左端点的上确界属于所有区间得到。

7.10 长度趋零如何产生唯一交点

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 52 页。

对应已有 gpt-105.pdf 页脚：57。

原课件第 52 页

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

Proof. The non-empty set $A = \{d_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above by u_1, then it has a limit $D = \lim_{n\to\infty} d_n$.
The non-empty set $B = \{u_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below by d_1, then it has a limit $U = \lim_{n\to\infty} u_n$.
Thus $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$, and

$$0 \leqslant U - D \leqslant u_n - d_n.$$

Note $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $D = U \in \bigcap_{n=1}^{\infty} I_n$.

沿原论证展开与批注 已有 $d_n \le D \le U \le u_n$，两端相减可得

$$0 \le U - D \le u_n - d_n.$$

右边趋于 0，所以固定非负数 $U - D$ 只能为 0，即 $D = U =: a$。交集 $[D, U] = \{a\}$，且单调收敛定理给 $d_n \to a, u_n \to a$。

不引用夹逼定理也能证明 若 $U - D = c > 0$，由长度趋零，某个 n 有 $u_n - d_n < c$，与 $c \le u_n - d_n$ 矛盾。

“唯一”的证明 若 a, b 都在每个区间，则 $|a - b| \le u_n - d_n$ 对所有 n 成立，同样推出 $a = b$。长度条件在这一步才真正用到。

7.11 Bolzano 定理：二分法抽子列

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 53 页。

对应已有 gpt-105.pdf 页脚：58。

原课件第 53 页

Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.
Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a limit point.

Proof. Let $\{x_n : n \in \mathbb{N}\}$ be a bounded sequence, we assume $\{x_n : n \in \mathbb{N}\} \subseteq [D, U]$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_1, U_1]$, and we take $x_{n_1} \in [D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts $\left[D_1, \frac{D_1+U_1}{2}\right]$ and $\left[\frac{D_1+U_1}{2}, U_1\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_2, U_2]$, and we take $x_{n_2} \in [D_2, U_2]$ with $n_2 > n_1$. We repeat this step and obtain a sequence $\{[D_k, U_k] : k \in \mathbb{N}\}$ of closed bounded nested intervals:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_k, U_k] \supseteq \cdots \quad \text{with } U_k - D_k = \frac{U - D}{2^k}$$

with a subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $x_{n_k} \in [D_k, U_k]$ for $k \in \mathbb{N}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{k=1}^{\infty} [D_k, U_k]$, and $|x_{n_k} - a| < U_k - D_k = \frac{U-D}{2^k}$, then $\lim_{k\to\infty} x_{n_k} = a$.

沿原论证展开与批注 把全部项放进 $[D, U]$。若 $D = U$，数列常值，直接收敛。否则把区间二等分，至少一半含无穷多个项的指标；否则两半加起来仅含有限项。保留这一半，重复二分，得到长度 $(U - D)/2^k$ 的闭区间套 I_k。每个 I_k 含无穷多个指标，所以可以逐次选 $n_k > n_{k-1}$ 且 $x_{n_k} \in I_k$。区间套定理给唯一交点 a。因为 a, x_{n_k} 都在 I_k，

$$|x_{n_k} - a| \le |I_k| = (U - D)2^{-k} \longrightarrow 0.$$

故选出的子列收敛。

细节 原页的严格 $<$ 一般应改为 $\le$，不影响极限。常值数列只有一个不同取值，却有无穷多个项；所以这里不能误说“无穷多个不同点”。

7.12 有界无限集合必有聚点：构造阶段

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 54–55 页。

对应已有 gpt-105.pdf 页脚：59, 60。

原课件第 54 页

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Proof. Note that E is a bounded infinite set in $\mathbb{R}$, there exists a constant M such that, for any $x \in E$, $|x| \leqslant M$. Define $I_0 = [-M, M]$, we divide R_0 into two small closed intervals: $[-M, 0]$ and $[0, M]$, then at least one small interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_1 with length M. Generally, we assume that there exists a sequence of closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_N, \quad I_N \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots, N$. We divide I_N into two small closed intervals, then at least one small closed interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_{N+1} with length $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_n \supseteq \cdots, \quad I_n \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots$.

原课件第 55 页

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $x \in \bigcap_{n=1}^{\infty} I_n$. We now check that this x is one of accumulation points of E.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large integer n_0, $\frac{M}{2^{n_0}} < \delta$. The length of I_{n_0+1} is strictly less than δ, then $I_{n_0+1} \subseteq N(x,\delta)$, and hence $E \cap I_{n_0+1} \subseteq E \cap N(x,\delta)$. x is an accumulation point of E.

沿原论证展开与批注 聚点 a 的定义是：每个 $\delta > 0$ 都能找到 $x \in E$ 满足 $0 < |x - a| < \delta$。允许 a 不属于 E，但找出的点必须不同于 a。把有界无限集合 E 放入 $[-M, M]$，取 $M > 0$。二分后至少一个闭半区间仍含 E 的无穷多个不同点，否则两边有限，其并也有限。反复保留无限的一半，得到 I_n 嵌套，长度 $2M/2^n$，且 $I_n \cap E$ 无限。

与上一页的区别 数列版本保留无限指标；集合版本保留无限不同点。这一区别保证下一页能排除交点本身，真正证明“聚点”。原页 R_0 应读作 I_0。

闭区间套给唯一交点 a。任给 $\delta > 0$，选择 n 使 $|I_n| < \delta$。对每个 $x \in I_n$，因 $a \in I_n$，有 $|x - a| \le |I_n| < \delta$，因此 $I_n \subset (a - \delta, a + \delta)$。又因为 $E \cap I_n$ 有无穷多个不同点，删去可能存在的点 a 后仍至少有一个 x。这个 x 满足 $x \in E$ 且 $0 < |x - a| < \delta$。由于 δ 任意，a 是聚点。

课件省略的关键句 只说邻域与集合相交尚不足以证明聚点，孤立点的邻域也会与集合相交。必须说“有不同于 a 的点”，无限性正是用来保证这一点。

7.13 收敛推出柯西：把误差分成两半

来源： `105-ZSM-PPT.pdf`，原课件 PDF 第 57–58 页。

对应已有 `gpt-105.pdf` 页脚：62, 63。

原课件第 57 页

Cauchy Convergence Criterion

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

Proof [Necessity] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a convergent sequence, and suppose $x = \lim_{n\to\infty} x_n$. For any $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that, when $n > N$, $|x_n - x| < \varepsilon$. It follows that for any $n_1 > N$ and $n_2 > N$,

$$|x_{n_1} - x_{n_2}| \leqslant |x_{n_1} - x| + |x_{n_2} - x| < 2\varepsilon.$$

Thus $\{x_n : n \in \mathbb{N}\}$ is a Cauchy sequence.

原课件第 58 页

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|x_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

From Bolzano-Weierstrass Theorem, the bounded sequence $\{x_n : n \in \mathbb{N}\}$ has a convergent subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $\lim_{k\to\infty} x_{n_k} = a$. For any $\varepsilon > 0$ there exists an integer K such that, when $k > K$, $|x_{n_k} - a| < \varepsilon$.

Take $k_0 = \max\{K+1, N+1\}$, then $n_{k_0} \geqslant n_{N+1} \geqslant N+1 > N$. Then, when $n > N$, $|x_n - x_{n_{k_0}}| < \varepsilon$, and hence

$$|x_n - a| \leqslant |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - a| < 2\varepsilon$$

that is, $\lim x_n = a$.

沿原论证展开与批注 设 $x_n \to a$。给定最终希望得到的 $\varepsilon > 0$，按收敛定义，对精度 $\varepsilon/2$ 选 K，使 $n \geq K$ 时 $|x_n - a| < \varepsilon/2$。若 $m, n \geq K$，利用三角不等式：

$$|x_n - x_m| = |(x_n - a) + (a - x_m)| \leq |x_n - a| + |x_m - a| < \varepsilon.$$

于是满足柯西定义。

为什么课件写 2ε 也能成立 因为 ε 是任意正数，可重新命名精度。但初学者考试最好从一开始取 $\varepsilon/2$，使结尾直接得到题目要求的 $< \varepsilon$，不留下量词解释负担。

先证明有界 在柯西条件中先取固定精度 1，得 K_0。固定 $m = K_0$，所有 $n \geq K_0$ 满足 $|x_n| \leq |x_n - x_{K_0}| + |x_{K_0}| < 1 + |x_{K_0}|$。再与前面有限项的绝对值取最大值，便得到全列的统一界。

找到候选极限 由 Bolzano 定理，有子列 $x_{n_j} \to a$。

证明整列趋于它 任给 $\varepsilon > 0$，柯西条件给 K，使 $m, n \geq K$ 时距离小于 $\varepsilon/2$。选一个足够大的 j，同时保证 $n_j \geq K$ 及 $|x_{n_j} - a| < \varepsilon/2$。以后每个 $n \geq K$ 都满足

$$|x_n - a| \leq |x_n - x_{n_j}| + |x_{n_j} - a| < \varepsilon.$$

这里先选一个共同的子列项作为桥梁，而不是对每个 n 随意更换未经控制的指标。

7.14 柯西充分性的第二证法：尾部下确界

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 59–60 页。

对应已有 gpt-105.pdf 页脚：64, 65。

原课件第 59 页

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \inf\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \geqslant y_n$ since y_{n+1} is the infimum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is an increasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

原课件第 60 页

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \leqslant x_n$ for each $n \in \mathbb{N}$, then $y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon$, then
$x_{n_2} \leqslant \inf\{x_{n_1} + \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} = \inf\{x_{n_1} : n_1 \geqslant N_2\} + \frac{1}{2}\varepsilon = y_{n_1} + \frac{1}{2}\varepsilon$.
Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim_{n\to\infty} x_n = a$.

沿原论证展开与批注 柯西列有界的证明同上一页。定义 $y_n = \inf\{x_k : k \geq n\}$。删掉第一项后尾集缩小，下确界只能上升：$y_{n+1} \geq y_n$。原数列的统一上下界也界住 y_n，所以 y_n 单调有界，存在极限 a。

几何解释 y_n 是从第 n 项起所有数的“下边界”。它不一定是实际某一项，但总有 $y_n \leq x_n$。柯西条件将迫使实际项与这条下边界越来越接近。

原页勘误 $|nn|$ 应为 $|x_n|$。不用把下确界视作最小值；即使尾集没有最小项，定义与证明仍然有效。

原页混用了取下确界的哑指标与自由指标。以下使用 j 表示被取遍的指标，避免混淆。给 $\varepsilon > 0$，取 K_1 使 $n \geq K_1$ 时 $|y_n - a| < \varepsilon/2$；再取 K_2 使 $n, j \geq K_2$ 时 $|x_n - x_j| < \varepsilon/2$。固定 $n \geq \max(K_1, K_2)$。对所有 $j \geq n$，有 $x_j > x_n - \varepsilon/2$，所以取下确界后

$$y_n \geq x_n - \varepsilon/2, \qquad 0 \leq x_n - y_n \leq \varepsilon/2.$$

从而 $|x_n - a| \leq |x_n - y_n| + |y_n - a| < \varepsilon$，故 $x_n \to a$。

细节 取确界后严格不等号可能变成非严格号，不能保留原页所有 $<$。由于另一误差严格小于 $\varepsilon/2$，最终仍得到严格 $< \varepsilon$。

7.15 第三证法：尾部上确界

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 61–62 页。

对应已有 gpt-105.pdf 页脚：66, 67。

原课件第 61 页

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \sup\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \leqslant y_n$ since y_{n+1} is the supremum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is a decreasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

原课件第 62 页

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \geqslant x_n$ for each $n \in \mathbb{N}$, then $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon \leqslant y_{n_1} + \frac{1}{2}\varepsilon$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$, then
$y_{n_1} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} : n_1 \geqslant N_2\} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} - \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} \leqslant x_{n_2}$. Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim\limits_{n\to\infty} x_n = a$.

沿原论证展开与批注 定义 $z_n = \sup\{x_k : k \geq n\}$。尾集合变小使上确界不增，故 z_n 递减；柯西列有界又保证 z_n 有下界。因此单调收敛定理给 $z_n \to a$。

对照下确界方法 下包络 y_n 从下方上升，上包络 z_n 从上方下降；柯西条件意味着两者之间的尾部振幅趋零。两种方法都是把不单调的数列转为单调边界。

原页勘误 同样应把 $|nn|$ 改为 $|x_n|$。因为 z_n 是上确界，所以 $x_n \leq z_n$，不能反过来。下一页用这一方向完成估计。

任给 $\varepsilon > 0$，取 K 同时保证 $n \geq K$ 时 $|z_n - a| < \varepsilon/2$，且 $n, j \geq K$ 时 $|x_n - x_j| < \varepsilon/2$。固定 $n \geq K$，对所有 $j \geq n$ 有 $x_j < x_n + \varepsilon/2$，所以

$$x_n \leq z_n = \sup_{j \geq n} x_j \leq x_n + \varepsilon/2.$$

于是 $|x_n - a| \leq |x_n - z_n| + |z_n - a| < \varepsilon$，整列收敛。

课件指标修正 不能把 $\sup_{j \geq K} x_j$ 写成任意的 z_{n_1}，两者的尾集起点不同。本页证明固定 n 后统一对 $j \geq n$ 取上确界，便消除了这个问题。上下包络证法任选一种熟练掌握即可，但应认识两种思路。

7.16 有限覆盖定理：把坏区间不断二分

来源： 105-ZSM-PPT.pdf，原课件 PDF 第 64–65 页。

对应已有 gpt-105.pdf 页脚：69, 70。

原课件第 64 页

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof. (Reduction to Absurdity) Let $\mathscr{H}$ be an open covering of $[D, U]$. Assume that $[D, U]$ can not be covered by finite many open sets belonging to $\mathscr{H}$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts, then there exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_2, U_2]$. We repeat this step and obtain a sequence $\{[D_n, U_n] : n \in \mathbb{N}\}$ of closed bounded nested intervals that can not be covered by finite many open sets belonging to $\mathscr{H}$:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_n, U_n] \supseteq \cdots \quad \text{with } U_n - D_n = \frac{U - D}{2^n}.$$

原课件第 65 页

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

From Cantor Nested Intervals Theorem there exists a unique point $x \in \bigcap_{n=1}^{\infty} [D_n, U_n]$. Note that $\mathscr{H}$ is an open covering of $[D, U]$, then there exists $H_0 \in \mathscr{H}$ such that $x \in H_0$. Note that H_0 is an open set, then there exists $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq H_0$.

Note $\lim_{n\to\infty} D_n = x = \lim_{n\to\infty} U_n$, then there exists an integer N such that, when $n > N$, $D_n \in (x - \delta, x + \delta)$ and $U_n \in (x - \delta, x + \delta)$.

$$[D_n, U_n] \subseteq (x - \delta, x + \delta) \subseteq H_0 \qquad \text{for } n > N.$$

which contradicts that $[D_n, U_n]$ can not be covered by finite many open sets belonging to $\mathscr{H}$.

沿原论证展开与批注 先证明闭区间 $[D,U]$。反设存在一个开覆盖，无法抽出有限子覆盖。二分成两个闭半区间，至少一个仍无法有限覆盖；否则两边各取有限个，合起来仍有限，便覆盖了整个区间，矛盾。如此递归，得到每个都“无法有限覆盖”的闭区间 I_n，且嵌套、长度 $(U-D)/2^n \to 0$。闭区间套定理给交点 a。

思路 我们暂时不直接挑覆盖集，而是把“失败的位置”压缩到一个点。开集在该点周围会留出余地，下一页正用这个余地制造矛盾。$D = U$ 的单点区间只需选一个覆盖它的开集，已经成立。

既然原族覆盖 $[D,U]$，有 $H_0 \in \mathcal{H}$ 含 a。H_0 开，故有 $\delta > 0$ 使 $(a-\delta, a+\delta) \subseteq H_0$。取 n 使 $|I_n| < \delta$。由于 $a \in I_n$，整个 I_n 都在该邻域内，于是仅一个 H_0 就覆盖 I_n，与构造矛盾。

补足课件未写的一步 定理针对任意闭有界 S，而非只有区间。取 $S \subseteq [D,U]$。因为 S 闭，$\mathbb{R} \setminus S$ 开，把它加入原开覆盖，便覆盖整个 $[D,U]$。用已证结论取有限子覆盖，再删除其中的 $\mathbb{R} \setminus S$（若有），剩下的原覆盖集仍覆盖 S。空集情形取空覆盖即可。

原页记号 邻域中心须统一写交点 a，不能在证明中突然换成未定义的 x。

8 106-Proof

8.1 为什么 $p \to \infty$ 会得到最大坐标

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 8 页。

对应已有 gpt-106.pdf 页脚：13。

原课件第 8 页

Sets in $\mathbb{R}^N$

$$\lim_{p\to\infty} \rho_p(x^1,x^2) = \lim_{p\to\infty} \left\{ \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p \right\}^{\frac{1}{p}}$$

$$= \lim_{p\to\infty} e^{\frac{1}{p}\ln \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p} = e^{\lim_{p\to\infty} \frac{1}{p}\ln \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p}$$

$$= e^{\lim_{p\to\infty} \frac{\sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p \ln\left|x_n^1 - x_n^2\right|}{\sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p}} = e^{\sum_{n=1}^{N} \lim_{p\to\infty} \frac{\left|x_n^1 - x_n^2\right|^p}{\sum_{n'=1}^{N} \left|x_{n'}^1 - x_{n'}^2\right|^p} \ln\left|x_n^1 - x_n^2\right|}$$

$$= e^{\max_{1\leqslant n\leqslant N} \ln\left|x_n^1 - x_n^2\right|} = \max_{1\leqslant n\leqslant N} \left|x_n^1 - x_n^2\right|.$$

沿原论证展开与批注 课件用对数和洛必达法则计算；含有 $a_j = 0$ 时 $\log a_j$ 需要额外处理。以下夹逼证明更短且覆盖所有情形。令 $M = \max_j a_j$。因为某个 $a_j = M$，而每个 $a_j \le M$，有

$$M^p \le \sum_{j=1}^{N} a_j^p \le N M^p, \qquad M \le \rho_p(x,y) \le N^{1/p} M.$$

$N^{1/p} \to 1$，故夹逼得 $\rho_p \to M = \rho_\infty$。若 $M = 0$，左右本来全为0。证明模板是“最大项给下界，项数乘最大项给上界”。

8.2 聚点的数列判别：构造互不相同的点

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 18 页。

对应已有 gpt-106.pdf 页脚：23。

原课件第 18 页

Accumulation Points, Interior Points, and Boundary Points

Theorem 1. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E'$ if and only if P_0 is one of limit points of E, or equivalently, there exists a sequence of different points $P_n \in E$ such that $\lim_{n\to\infty} \rho(P_n, P_0) = 0$.

Proof. [Sufficiency] For any neighborhood $N(P_0, \delta)$ of center P_0 with $\delta > 0$, then, if N is large enough, $\rho(P_n, P_0) < \delta$ for $n \geqslant N$, then $P_n \in N(P_0, \delta)$, and there exist infinite such P_n.

[Necessity] Define $\delta_n = \dfrac{1}{n}$, then $\delta_n > 0$. $P_0 \in E'$, and hence infinite points in $N(P_0, \delta_1)$ belong to E. We can take P_1 arbitrarily. $P_0 \in E'$, and hence infinite points in $N(P_0, \delta_2)$ belong to E, thus $N(P_0, \delta_2) \setminus \{P_1\} \neq 0$. We can take $P_2 \in N(P_0, \delta_2) \setminus \{P_1\}$ arbitrarily, both P_1 and P_2 are different. Generally, we assume that we take different $P_1, P_2, \cdots, P_N$ with $P_n \in N(P_0, \delta_n)$ and $P_n \in E$ for $n = 1, \cdots, N$. Then infinite points in $N(P_0, \delta_{N+1})$ belong to E, thus $N(P_0, \delta_{N+1}) \setminus \{P_1, P_2, \cdots, P_N\} \neq 0$. We can take $P_{N+1} \in N(P_0, \delta_{N+1}) \setminus \{P_1, P_2, \cdots, P_N\}$ arbitrarily, P_{N+1} is different from $P_1, P_2, \cdots, P_N$. Therefore, we obtain a sequence of different points $P_n \in E$ for $n = 1, 2, \cdots$ such that $P_n \in N(P_0, \delta_n)$ and $P_n \in E$ for $n = 1, 2, \cdots$. Note $\lim_{n\to\infty} \delta_n = 0$, then $\lim_{n\to\infty} \rho(P_n, P_0) = 0$.

Q.E.D.

沿原论证展开与批注

充分性 若互不相同的 $x_k \in E$ 趋于 x，任何球 $B_r(x)$ 都包含全部足够晚的项，因此包含无穷多个不同的点，故 $x \in E'$。

必要性 假设 $x \in E'$。从 $E \cap B_1(x)$ 选 $x_1 \neq x$；已选 $x_1, \ldots, x_k$ 后，从 $E \cap B_{1/(k+1)}(x)$ 中排除有限集合 $\{x, x_1, \ldots, x_k\}$，仍有点可选，记为 x_{k+1}。于是点互不相同，且 $\|x_k - x\| < 1/k \to 0$。

容易漏的条件 若允许数列恒为 x，只能说明 $x \in \overline{E}$，不能说明 $x \in E'$。“互不相同”是本定理区别于闭包判别的关键。

8.3 导集的单调性与有限并公式

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 19 页。

对应已有 gpt-106.pdf 页脚：24。

原课件第 19 页

Accumulation Points, Interior Points, and Boundary Points

Theorem 2. If $A_1 \subseteq A_2$, then $A_1' \subseteq A_2'$ and $\overline{A}_1 \subseteq \overline{A}_2$.

Theorem 3. $(A_1 \cup A_2)' = A_1' \cup A_2'$.

Proof. Note $A_1 \subseteq A_1 \cup A_2$ and $A_2 \subseteq A_1 \cup A_2$, then $A_1' \subseteq (A_1 \cup A_2)'$ and $A_2' \subseteq (A_1 \cup A_2)'$ from Theorem 2, and hence $A_1' \cup A_2' \subseteq (A_1 \cup A_2)'$.

If $P \in (A_1 \cup A_2)'$, then there exists a sequence of different points $P_n \in A_1 \cup A_2$ such that $\lim_{n\to\infty} \rho(P_n, P) = 0$. If $P \in A_1'$, then $P \in A_1' \cup A_2'$. If $P \notin A_1'$, then at most finite P_n for $n = 1, 2, \cdots$ belong to A_1, the remaining infinite points belong to A_2, then $P \in A_2'$ from Theorem 1. Thus $P \in A_1' \cup A_2'$. Therefore $(A_1 \cup A_2)' \subseteq A_1' \cup A_2'$. Q.E.D.

Corollary 1. $\left(\bigcup_{n=1}^{N} A_n\right)' = \bigcup_{n=1}^{N} A_n'$.

沿原论证展开与批注 $A \subseteq B$ 时，任意球中无穷多个 A 点也都是 B 点，所以 $A' \subseteq B'$，进而 $\overline{A} \subseteq \overline{B}$。要证 $(A \cup B)' = A' \cup B'$，容易的一边来自单调性。对另一边用逆否命题更清楚：若 $x \notin A'$ 且 $x \notin B'$，分别有半径 r, s 的球只含有限多个 A, B 点。半径 $\min(r, s)$ 的球只含有限多个 $A \cup B$ 点，故 $x \notin (A \cup B)'$。重复这个论证得到任意有限并的公式；无限并不能照搬，例见第43页第3题。

8.4 内部的单调性与有限交公式

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 20 页。

对应已有 gpt-106.pdf 页脚：25。

原课件第 20 页

Accumulation Points, Interior Points, and Boundary Points

Theorem 4. If $A_1 \subseteq A_2$, then $A_1^\circ \subseteq A_2^\circ$.

Theorem 5. $(A_1 \cap A_2)^\circ = A_1^\circ \cap A_2^\circ$.

Proof. Note $A_1 \cap A_2 \subseteq A_1$ and $A_1 \cap A_2 \subseteq A_2$, then $(A_1 \cap A_2)^\circ \subseteq A_1^\circ$ and $(A_1 \cap A_2)^\circ \subseteq A_2^\circ$ from Theorem 2, and hence $(A_1 \cap A_2)^\circ \subseteq A_1^\circ \cap A_2^\circ$.

If $P \in A_1^\circ \cap A_2^\circ$, then $P \in A_k^\circ$ for $k = 1, 2$, and there exists a a neighborhood $N(P, \delta)$ of center P such that $N(P, \delta) \subseteq A_k$ for $k = 1, 2$, thus $N(P, \delta) \subseteq A_1 \cap A_2$. Therefore $P \in (A_1 \cap A_2)^\circ$, and hence $A_1^\circ \cap A_2^\circ \subseteq (A_1 \cap A_2)^\circ$. Q.E.D.

Corollary 2. $\left(\bigcap_{n=1}^{N} A_n\right)^\circ = \bigcap_{n=1}^{N} A_n^\circ$.

沿原论证展开与批注 若 $A \subseteq B$，包含于 A 的小球也包含于 B，所以 $A^\circ \subseteq B^\circ$。由此 $(A \cap B)^\circ \subseteq A^\circ \cap B^\circ$。反向取 $x \in A^\circ \cap B^\circ$，存在 $r_A, r_B > 0$ 分别使小球落入两集合；令 $r = \min(r_A, r_B) > 0$，就有 $B_r(x) \subseteq A \cap B$。因此等号成立。 课件略去的步骤是把两个可能不同的半径合并成一个。有限交可以取有限个正数的最小值；无限多个正数的下确界可能为0，这正是无限交不保开性的原因。

8.5 二分立方体证明：不断保留无限多个点

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 22–23 页。

对应已有 gpt-106.pdf 页脚：27, 28。

原课件第 22 页

Bolzano-Weierstrass Theorem

Proof. Note that E is a bounded infinite set, there exists a constant M such that, for any $x = (x_1, \cdots, x_N) \in E$, $|x_n| \leqslant M$ for $n = 1, \cdots, N$, and hence E includes in N-dimensional cube

$$R_0 = E[(x_1, \cdots, x_N); |x_n| \leqslant M \text{ for } n = 1, \cdots, N].$$

We use axises to divide R_0 into 2^N small closed cubes, then at least one small closed cube includes infinite points in E (otherwise E is a finite set!), we call such a small closed cube R_1 with edge M. Generally, we assume that there exists a sequence of closed cubes whose edges are parallel to axises.

$$R_0 \supseteq R_1 \supseteq \cdots \supseteq R_N, \quad R_N \cap E \text{ is an infinite set,}$$
$$\text{the edge of } R_n \text{ is } \frac{M}{2^{n-1}} \text{ for } n = 1, 2, \cdots, N$$

原课件第 23 页

Bolzano-Weierstrass Theorem

We use lines parallel to axises to divide R_N into 2^N small closed cubes, then at least one small closed cube includes infinite points in E (otherwise E is a finite set!), we call such a small closed cube R_{N+1} with edge $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed cubes whose edges are parallel to axises.

$$R_0 \supseteq R_1 \supseteq \cdots \supseteq R_n \supseteq \cdots, \quad R_n \cap E \text{ is an infinite set,}$$
$$\text{and the edge of } R_n \text{ is } \frac{M}{2^{n-1}} \text{ for } n = 1, 2, \cdots$$

Note $\lim\limits_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $P \in \bigcap\limits_{n=1}^{\infty} R_n$. We now check that this P is one of accumulation points of E.

Define $N(P, \delta)$ as any neighborhood of center P. Note $\lim\limits_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large natural number n_0, $\frac{M}{2^{n_0}} < \frac{\delta}{n_0^{\frac{1}{2}}}$, $n_0^{\frac{1}{2}} \frac{M}{2^{n_0}} < \delta$,. The diagonal length of R_{n_0+1} is strictly less than δ, then $R_{n_0+1} \subseteq N(P, \delta)$, and hence $E \cap R_{n_0+1} \subseteq E \cap N(P, \delta)$. There exist infinite points of $N(P, \delta)$ in E.
Q.E.D.

沿原论证展开与批注 把有界集 E 放进 $[-M, M]^N$。沿每个坐标二分，得到 2^N 个闭小方块。若每块只含有限个 E 点，则有限块合起来仍只有有限点，与 E 无限矛盾。因此至少一块仍含无限个点。选这一块再二分，如此构造

$$C_0 \supseteq C_1 \supseteq \cdots, \qquad \text{边长}(C_k) = 2M/2^k, \quad |C_k \cap E| = \infty.$$

小方块可以在边界上相交，不影响“有限个有限集之并仍有限”的推理。

每个 C_k 是 N 个闭区间的乘积；逐坐标应用闭区间套定理，得到唯一共同点 x。给定 $r > 0$，选 k 使 $\sqrt{N}(2M/2^k) < r$，则 C_k 的欧氏直径小于 r。因为 $x \in C_k$，故 $C_k \subset B_r(x)$。而 $C_k \cap E$ 无限，所以每个 $B_r(x)$ 都包含无限个 E 点，x 是聚点。 **课件修正：** 对角线系数应为固定维数的 $\sqrt{N}$，不能误写成不断变化的子块编号的平方根。最大范数下直径直接等于边长。

8.6 孤立点与可数性的证明起点

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 24–25 页。

对应已有 gpt-106.pdf 页脚：29, 30。

原课件第 24 页

Bolzano-Weierstrass Theorem

In mathematics, a point P is called an isolated point of a set E if P is an element of E but there exists a neighborhood of P which does not contain any other points of E.

Point P is an isolated point of set E if $P \in E$ and $P \notin D(P, \delta)$ for $\delta > 0$.

The boundary point which is not accumulation point is called to be an isolated point. The isolated point belongs to E. A set E is isolated if every point is isolated in E.

Each isolated set is a countable (finite or denumerable) set. In fact, if E is an isolated set, then, for any $P \in E$, there exists $\delta_P > 0$ such that $N(P, 2\delta_P) \cap E = \{P\}$. We pick a rational point R_P in $N(P, \delta_P)$. If $P_1 \neq P_2$, then $R_{P_1} \neq R_{P_2}$. Otherwise, $R_{P_1} = R_{P_2}$, then

$$\begin{aligned} \rho(P_1, P_2) &\leqslant \rho(P_1, R_{P_1}) + \rho(R_{P_1}, P_2) = \rho(P_1, R_{P_1}) + \rho(P_2, R_{P_2}) \\ &< \delta_{P_1} + \delta_{P_2} \leqslant 2 \max\{\delta_{P_1}, \delta_{P_2}\}. \end{aligned}$$

原课件第 25 页

Bolzano-Weierstrass Theorem

(1) If $\max\{\delta_{P_1}, \delta_{P_2}\} = \delta_{P_1}$, then $\rho(P_1, P_2) < 2\delta_{P_1}$, then $P_2 \in N(P_1, 2\delta_{P_1})$ and $N(P_1, 2\delta_{P_1}) \cap E = \{P_1, P_2\}$.

(2) If $\max\{\delta_{P_1}, \delta_{P_2}\} = \delta_{P_2}$, then $\rho(P_1, P_2) < 2\delta_{P_2}$, then $P_1 \in N(P_2, 2\delta_{P_2})$ and $N(P_2, 2\delta_{P_2}) \cap E = \{P_1, P_2\}$.

Therefore there exists an one - to -one correspondence from $P \in E$ to R_P. Note that the set of all rationals is denumerable, then E is a countable (finite or denumerable) set.

E is an isolated set if and only if $E \cap E' = \varnothing$.

E is a discrete set if $E' = \varnothing$. Any discrete set is an isolated set, but an isolated set may not be a discrete set. For example, the point set $\left\{\frac{1}{n}; n = 1, 2, \cdots\right\} \subseteq \mathbb{R}$ is an isolated set, but it is not a discrete set because 0 is an accumulation point.

沿原论证展开与批注 x 是 E 的孤立点，正确条件是 $x \in E$ 且存在 $r > 0$ 使 $B_r(x) \cap E = \{x\}$，等价于去心球与 E 无交。**课件修正：** $x \notin D(x, r)$ 总是成立，单独这一句不能定义孤立点。若所有点都孤立，对每个 $x \in E$ 选 $r_x > 0$ 使 $B_{2r_x}(x) \cap E = \{x\}$，再在 $B_{r_x}(x)$ 中选一个有理坐标点 $q_x \in \mathbb{Q}^N$。有理坐标点的存在来自每一坐标上有理数稠密；下一页证明这些 q_x 不会重复。

若不同的 x, y 选中了同一个 q，则 $\|x - y\| < r_x + r_y \le 2\max(r_x, r_y)$。若最大者是 r_x，则 $y \in B_{2r_x}(x) \cap E$，与该交集只有 x 矛盾；另一情形同理。故 $x \mapsto q_x$ 是单射。$\mathbb{Q}^N$ 是有限个可数集之积，仍可数，所以 E 可数。本课把 $E \cap E' = \varnothing$ 称为所有点孤立，把 $E' = \varnothing$ 称为 discrete。其他教材常把前者也称为离散子空间，答题应先写条件再写术语。$\{1/k\}$ 的每一点都孤立，但0仍是外部聚点。

8.7 聚点也可以用去心邻域定义

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 26–27 页。

对应已有 gpt-106.pdf 页脚：31, 32。

原课件第 26 页

Accumulation Points, Interior Points, and Boundary Points

Theorem 7. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E'$ if and only if, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P').

Proof. [Necessity] If $P_0 \in E'$, then, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P').

If $P_0 \in E'$, then for any $\delta > 0$ there exist infinite many points in $N(P_0, \delta) \cap E$. Take $P \in N(P_0, \delta)$, then $0 \leqslant d = \rho(P, P_0) < \delta$. Thus for any $P' \in N(P_0, \delta - d)$ we have
$\rho(P, P') \leqslant \rho(P, P_0) + \rho(P_0, P') < d + \delta - d = \delta$, and hence
$N(P_0, \delta - d) \subseteq N(P, \delta)$ and $N(P_0, \delta - d) \cap E \subseteq N(P, \delta) \cap E$.

Note that $N(P_0, \delta - d) \cap E$ has infinite many points, then there exists a point P', differing from P_0, in $N(P_0, \delta - d) \cap E \subseteq N(P, \delta) \cap E$ (there exist infinite such points P').

原课件第 27 页

Accumulation Points, Interior Points, and Boundary Points

[Sufficiency] If, in any neighborhood $N(P,\delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P'), then $P_0 \in E'$.

In any neighborhood $N(P,\delta)$ of point P_0, there exists a point P_1, differing from P_0, in E. Write $0 < \delta_1 = \rho(P_1, P_0) < \delta$. In δ_1 neighborhood $N(P_0,\delta_1)$ of center P_0, there exists a point P_2, differing from P_0, in E. Write $0 < \delta_2 = \rho(P_2, P_0) < \delta_1$, then $P_2 \in N(P_0,\delta_1)$. Generally, in δ_n neighborhood $N(P_0,\delta_n)$ of center P_0, there exists a point P_{n+1}, differing from P_0, in E. Write $0 < \delta_{n+1} = \rho(P_{n+1}, P_0) < \delta_n$, then $P_{n+1} \in N(P_0,\delta_n)$. From induction, there exist $\{\delta_n : n \in \mathbb{N}\}$ with $\delta_{n+1} < \delta_n$ for $n \in \mathbb{N}$ such that point P_{n+1}, differing from P_0, in E. Write $0 < \delta_{n+1} = \rho(P_{n+1}, P_0) < \delta_n$, then $P_{n+1} \in N(P_0,\delta_n)$. Therefore there exist infinitely many points in $N(P_0,\delta) \cap E$, and hence $P_0 \in E'$. Q.E.D.

沿原论证展开与批注 等价条件是 $x \in E' \iff \forall r > 0,\ (B_r(x) \setminus \{x\}) \cap E \neq \varnothing$。正向显然：无限多个点不可能只有中心一个。若课件说一个“含 x 的邻域”却以 y 为中心，只要 $x \in B_r(y)$，仍可找到以 x 为中心的小球 $B_{r-\|x-y\|}(x) \subset B_r(y)$。因为半径差严格为正，两种邻域表述等价。

逆向假设每个去心球都碰到 E。若某个 $B_r(x)$ 只含有限个异于 x 的点 $y_1,\dots,y_m$，取 $0 < s < \min(r, \|y_1 - x\|, \dots, \|y_m - x\|)$，则去心球 $B_s(x) \setminus \{x\}$ 不碰 E，矛盾。若原球根本没有异点，已经直接矛盾。因此每个球包含无限点。 课件递归选距离严格下降的点也能证明点数无限；但“严格下降的正数列”并不自动趋0，例如 $1+1/k$。若要同时构造收敛列，应额外把第 k 次半径限制到 $1/k$。

8.8 邻域换中心与内部的局部稳定性

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 28 页。

对应已有 gpt-106.pdf 页脚：33。

原课件第 28 页

Accumulation Points, Interior Points, and Boundary Points

Theorem 8. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E^\circ$ if and only if there exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$.

Proof. [Necessity] If $P_0 \in E^\circ$, then there exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$.
If $P_0 \in E^\circ$, then there exists $\delta > 0$ such that $N(P_0,\delta) \subseteq E$. Take $P \in N(P_0,\delta)$, then $0 \leqslant d = \rho(P,P_0) < \delta$. Thus for any $P' \in N(P,\delta-d)$ we have
$\rho(P_0,P') \leqslant \rho(P,P_0) + \rho(P,P') < d + \delta - d = \delta$, and hence $P' \in N(P_0,\delta)$. Thus $N(P,\delta-d) \subseteq N(P_0,\delta)$ and $N(P,\delta-d) \subseteq E$.

[Sufficiency] If there exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$, then $P_0 \in E^\circ$.
There exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$, then $N(P_0,\delta-\rho(P_0,P)) \subseteq N(P,\delta) \subseteq E$, and hence $P_0 \in E^\circ$. Q.E.D.

沿原论证展开与批注 若 $B_r(x) \subseteq E$，且 $y \in B_r(x)$，记 $d = \|x-y\| < r$。对 $z \in B_{r-d}(y)$，三角不等式给 $\|z-x\| < r-d+d = r$，故 $z \in E$。所以原球中的每个点也是内点。这一小计算将反复出现：可移动的余量等于原半径减去已走过的距离。若从一个非中心邻域反推中心邻域，也用同样的余量公式。

8.9 连续性与上下水平集闭：补上子列步骤

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 30 页。

对应已有 gpt-106.pdf 页脚：35。

原课件第 30 页

Closed Set

Example 2. If $f(x)$ is defined on $\mathbb{R}^N$, then the sufficient and necessary conditions for $f \in C\left(\mathbb{R}^N\right)$ are, for any $t \in \mathbb{R}$, the sets

$$E_1 = \left\{x \in \mathbb{R}^N \mid f(x) \geqslant t\right\} \qquad \text{and} \qquad E_2 = \left\{x \in \mathbb{R}^N \mid f(x) \leqslant t\right\}$$

are closed sets.

Proof [Necessity] For E_1, if $\{x_k; k \in \mathbb{N}\} \subseteq E_1$ and $\lim\limits_{k\to\infty} x_k = x_0$, from the continuity of $f(x)$ and $f\left(x_k\right) \geqslant t$, then $f\left(x_0\right) = \lim\limits_{k\to\infty} f\left(x_k\right) \geqslant t$, i.e. $x_0 \in E_1$. For E_2, if $\{x_k; k \in \mathbb{N}\} \subseteq E_2$ and $\lim\limits_{k\to\infty} x_k = x_0$, from the continuity of $f(x)$ and $f\left(x_k\right) \leqslant t$, then $f\left(x_0\right) = \lim\limits_{k\to\infty} f\left(x_k\right) \leqslant t$, i.e. $x_0 \in E_2$.

[Sufficiency] We adopt the reduction to absurdity. We assume that there exists $x_0 \in \mathbb{R}^N$ which is not a continuous point of $f(x)$, then there is $\varepsilon_0 > 0$ and point sequence $\{x_k; k \in \mathbb{N}\}$ such that $\lim\limits_{k\to\infty} x_k = x_0$. For every $k \in \mathbb{N}$,

$$f\left(x_k\right) \leqslant f\left(x_0\right) - \varepsilon_0 \qquad \text{or} \qquad f\left(x_k\right) \geqslant f\left(x_0\right) + \varepsilon_0.$$

If $f\left(x_k\right) \leqslant f\left(x_0\right) - \varepsilon_0$, then $x_k \in E_2[t = f\left(x_0\right) - \varepsilon_0]$, but $x_0 \notin E_2[t = f\left(x_0\right) - \varepsilon_0]$, which is a contradiction! If $f\left(x_k\right) \geqslant f\left(x_0\right) + \varepsilon_0$, then $x_k \in E_1[t = f\left(x_0\right) + \varepsilon_0]$, but $x_0 \notin E_1[t = f\left(x_0\right) + \varepsilon_0]$, which is a contradiction!

沿原论证展开与批注

正向 若 f 连续，且 $x_k \to x$、$f(x_k) \geq t$，则 $f(x) = \lim f(x_k) \geq t$，故 $\{f \geq t\}$ 闭；$\{f \leq t\}$ 同理。

反向 若在 x 不连续，则存在 $\varepsilon_0 > 0$，对每个 k 都能选 $\|x_k - x\| < 1/k$ 而 $|f(x_k) - f(x)| \geq \varepsilon_0$。每项至少属于以下一类：$f(x_k) \leq f(x) - \varepsilon_0$ 或 $f(x_k) \geq f(x) + \varepsilon_0$。两类中至少一类有无限多个下标，抽取该类子列。例如第一类得到闭集 $\{f \leq f(x) - \varepsilon_0\}$ 中的数列趋于 x，但 x 不在该闭集，矛盾。第二类同理。 **关键补足：** 不能假定所有项天然落在同一边；必须先用无限鸽巢原理抽子列。

8.10 开球的闭包与稠密

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 31 页。

对应已有 gpt-106.pdf 页脚：36。

原课件第 31 页

Closed Set

Example 3. The closure of the open ball $N(x_0, r)$ in $\mathbb{R}^N$ is the closed ball $B(x_0, r)$:

$$B(x_0, r) = \overline{N(x_0, r)} = \left\{x \in \mathbb{R}^N \mid |x - x_0| \leqslant r\right\}.$$

Proof Define $F = \left\{x \in \mathbb{R}^N \mid |x - x_0| \leqslant r\right\}$, then f is a closed set, so $\overline{N(x_0, r)} \subseteq F$. Conversely, if $x \in F$, define $x_k = \frac{1}{k}x_0 + \left(1 - \frac{1}{k}\right)x$, and

$$|x_0 - x_k| = \left(1 - \frac{1}{k}\right)|x - x_0| \leqslant \left(1 - \frac{1}{k}\right) r < r \text{ and } |x - x_k| = \frac{1}{k}|x - x_0| \leqslant \frac{1}{k} r < r.$$

This indicates that $\{x_k; k \in \mathbb{N}\} \subseteq N(x_0, r)$, and $\lim\limits_{k\to\infty} x_k = x$. Thus $x \in \overline{N(x_0, r)}$, that is, $F \subseteq \overline{N(x_0, r)}$.

Note: If $A \subseteq S$ and $\overline{A} = S$, then A is said to be dense in S, or A is a dense subset of S.

沿原论证展开与批注 设 $r > 0$。若 $x_k \in B_r(a)$ 且 $x_k \to x$，距离连续性给 $\|x - a\| \leq r$，故 $\overline{B_r(a)} \subseteq \overline{B}_r(a)$。反向对 $x \in \overline{B}_r(a)$，取 $x_k = a + (1 - 1/k)(x - a)$，则 $\|x_k - a\| = (1 - 1/k)\|x - a\| < r$，且 $x_k \to x$。因此等号成立。 A 在 S 中稠密，准确表述是 $A \subseteq S$ 且 $S \subseteq \overline{A}$，或相对闭包 $\overline{A}^S = S$。若直接写环境空间中的 $\overline{A} = S$，还暗中要求 S 闭。后面的第84–86页使用前一种一般定义。

8.11 导集与闭包本身都是闭集

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 33 页。

对应已有 gpt-106.pdf 页脚：38。

原课件第 33 页

Closed Set

Theorem 1. E' and $\overline{E}$ are closed sets.

Proof. If $P_0 \in (E')'$, then, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point $P_1 \in E'$ with $P_1 \neq P_0$ from Theorem 7 in Subsection 2, and hence, in any neighborhood $N(P', \delta')$ of point P_1, there exists a point $P_2 \in E$ with $P_2 \neq P_1$ and $P_2 \neq P_0$ from Theorem 7 in Subsection 2, thus $P_0 \in E'$. Therefore E' is a closed set.

Note $\overline{E} = E \cup E'$, then $\overline{E}' = E' \cup (E')' \subseteq E' \cup E' = E' \subseteq \overline{E}$, and hence $\overline{E}$ is a closed set. Q.E.D.

沿原论证展开与批注 证明 E' 闭：设 $x \in (E')'$。给定 $r > 0$，选 $y \in E'$ 满足 $0 < \|y - x\| < r/2$，再取 $s < \min(r - \|y - x\|, \|y - x\|)$。由于 y 是 E 的聚点，可选 $z \in E \cap B_s(y)$。这个小球不含 x，又包含于 $B_r(x)$，因此找到了 E 中异于 x 的近点。由第26–27页，$x \in E'$。再用有限并导集公式：$(\overline{E})' = (E \cup E')' = E' \cup (E')' \subseteq E' \subseteq \overline{E}$，所以 $\overline{E}$ 闭。课件略去的小球半径选择正是证明严密性所在。

8.12 有限个闭集之并仍闭：导集方法

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 34 页。

对应已有 gpt-106.pdf 页脚：39。

原课件第 34 页

Closed Set

Theorem 2. The union of finite closed sets is a closed set.

Proof. Define $F = \bigcup_{n=1}^{N} F_n$ where F_n is a closed set for $n = 1, \cdots, N$. Note that F_n is a closed set, $F_n' \subseteq F_n$, then

$$F' = \left(\bigcup_{n=1}^{N} F_n\right)' = \bigcup_{n=1}^{N} F_n' \subseteq \bigcup_{n=1}^{N} F_n = F.$$

Therefore F is a closed set. Q.E.D.

沿原论证展开与批注 设 $F = F_1 \cup \cdots \cup F_m$，每个 F_i 闭。由第19页的有限并公式，$F' = F_1' \cup \cdots \cup F_m' \subseteq F_1 \cup \cdots \cup F_m = F$，于是 F 闭。注意 “finite closed sets” 在这里表示有限个闭集，不是每个闭集本身只有有限个点。考试写明 $m < \infty$，防止把定理误用到无限并。

8.13 同一个有限并结论：闭包方法

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 35 页。

对应已有 gpt-106.pdf 页脚：40。

原课件第 35 页

Closed Set

Theorem 2. The union of finite closed sets is a closed set.

Proof. Define $F = \bigcup_{n=1}^{N} F_n$ where F_n is a closed set for $n = 1, \cdots, N$. Note that F_n is a closed set, $\overline{F}_n = F_n \cup F_n'$, then

$$\begin{aligned}
\overline{F} &= \overline{\bigcup_{n=1}^{N} F_n} = \left(\bigcup_{n=1}^{N} F_n\right) \cup \left(\bigcup_{n=1}^{N} F_n\right)' \\
&= \left(\bigcup_{n=1}^{N} F_n\right) \cup \left(\bigcup_{n=1}^{N} F_n'\right) \\
&= \bigcup_{n=1}^{N} \left(F_n \cup F_n'\right) = \bigcup_{n=1}^{N} \overline{F}_n = \bigcup_{n=1}^{N} F_n = F.
\end{aligned}$$

Therefore F is a closed set. Q.E.D.

沿原论证展开与批注 本页给另一种代数写法：

$$\overline{\bigcup_{i=1}^{m} F_i} = \bigcup_{i=1}^{m} (F_i \cup F_i') = \bigcup_{i=1}^{m} \overline{F_i} = \bigcup_{i=1}^{m} F_i.$$

第一个等号用闭包等于本集并导集及有限并导集公式；最后一个等号用每个 F_i 闭。这与上一页是一条证明的两种记号。复习时应能解释每个等号来自哪条定义，不能仅抄下一串等号。

8.14 可数个闭集之交仍闭：导集方法

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 36 页。

对应已有 gpt-106.pdf 页脚：41。

原课件第 36 页

Closed Set

Theorem 3. The intersection of multiple closed sets is a closed set.

Proof. Define $F = \bigcap_{n=1}^{\infty} F_n$ where F_n is a closed set for $n = 1, 2, \cdots$.

Note $F \subseteq F_n$, then $F' \subseteq F_n'$. Thus $F' \subseteq \bigcap_{n=1}^{\infty} F_n'$. Note that F_n is a closed set, $F_n' \subseteq F_n$, then

$$\left(\bigcap_{n=1}^{\infty} F_n\right)' = F' \subseteq \bigcap_{n=1}^{\infty} F_n' \subseteq \bigcap_{n=1}^{\infty} F_n = F.$$

Therefore F is a closed set. Q.E.D.

沿原论证展开与批注 设 $F = \bigcap_{k=1}^{\infty} F_k$。因为 $F \subseteq F_k$，单调性给 $F' \subseteq F_k' \subseteq F_k$，对每个 k 都成立，于是 $F' \subseteq \bigcap_k F_k = F$。这里不用交换“求导集”与“无限交”：一般只有包含，不应写成等号。这个证明实际上完全没有用到可数性，下一组页面将推广到任意指标集。

8.15 可数闭交的闭包写法

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 37 页。

对应已有 gpt-106.pdf 页脚：42。

原课件第 37 页

Closed Set

Theorem 3. The intersection of multiple closed sets is a closed set.

Proof. Define $F = \bigcap_{n=1}^{\infty} F_n$ where F_n is a closed set for $n = 1, 2, \cdots$. Note $F \subseteq F_n$, then $\overline{F} \subseteq \overline{F}_n$. Thus $\overline{F} \subseteq \bigcap_{n=1}^{\infty} \overline{F}_n$. Note that F_n is a closed set, $\overline{F}_n = F_n$, then

$$\overline{\bigcap_{n=1}^{\infty} F_n} = \overline{F} \subseteq \bigcap_{n=1}^{\infty} \overline{F}_n = \bigcap_{n=1}^{\infty} F_n = F.$$

Therefore F is a closed set. Q.E.D.

沿原论证展开与批注 对每个 k 有 $\overline{F} \subseteq \overline{F_k} = F_k$，所以 $\overline{F} \subseteq F$。再结合总成立的 $F \subseteq \overline{F}$，得到 $F = \overline{F}$。本页比上一页多用一次“双向包含推出相等”的动作。若 F 为空，也一样成立，不需要单独假设交集非空。

8.16 任意闭集族的交：不需要数得出来

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 38 页。

对应已有 gpt-106.pdf 页脚：43。

原课件第 38 页

Closed Set

Theorem 3. If $\{F_\gamma : \gamma \in \Gamma\}$ is a family of closed sets in $\mathbb{R}^N$, then its intersection $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ is a closed set.

Proof. Define $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ where F_γ is a closed set for $\gamma \in \Gamma$. Note $F \subseteq F_\gamma$, then $F' \subseteq F'_\gamma$. Thus $F' \subseteq \bigcap_{\gamma\in\Gamma} F'_\gamma$. Note that F_γ is a closed set, $F'_\gamma \subseteq F_\gamma$, then

$$\left(\bigcap_{n=1}^{\infty} F_n\right)' = F' \subseteq \bigcap_{\gamma\in\Gamma} F'_\gamma \subseteq \bigcap_{\gamma\in\Gamma} F_\gamma = F.$$

Therefore F is a closed set. Q.E.D.

沿原论证展开与批注 设 Γ 可以不可数，$F=\bigcap_{\gamma\in\Gamma}F_\gamma$。任取 $x\in F'$，因为 $F\subseteq F_\gamma$，可知 $x\in F'_\gamma\subseteq F_\gamma$，对所有 γ 成立，所以 $x\in F$。原页公式残留可数指标 n 的位置应统一改为 $\gamma\in\Gamma$。若指标集为空，约定交集为整个环境空间，仍为闭集。

8.17 任意闭交的闭包证明与笔误

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 39 页。

对应已有 gpt-106.pdf 页脚：44。

原课件第 39 页

Closed Set

Theorem 3. If $\{F_\gamma : \gamma \in \Gamma\}$ is a family of closed sets in $\mathbb{R}^N$, then its intersection $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ is a closed set.

Proof. Define $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ where F_γ is a closed set for $\gamma \in \Gamma$. Note $F \subseteq F_\gamma$, then $\overline{F} \subseteq \overline{F}_\gamma$. Thus $\overline{F} \subseteq \bigcap_{\gamma\in\Gamma} \overline{F}_\gamma$. Note that F_γ is a closed set, $\overline{F}_\gamma \subseteq \overline{F}$, then

$$\overline{\bigcap_{n=1}^{\infty} F_n} = \overline{F} \subseteq \bigcap_{\gamma\in\Gamma} \overline{F}_\gamma = \bigcap_{\gamma\in\Gamma} F_\gamma = F.$$

Therefore F is a closed set. Q.E.D.

沿原论证展开与批注 延续第37页，把 k 换成 γ：$\overline{F} \subseteq \overline{F_\gamma} = F_\gamma$，故 $\overline{F} \subseteq \bigcap_\gamma F_\gamma = F$。**课件修正：** 每个 F_γ 闭意味着 $\overline{F_\gamma} = F_\gamma$；不意味着 $\overline{F_\gamma} \subseteq F$，因为交集通常比单个集合小。不要把单调性方向倒过来。

8.18 为什么无限并会出现新极限点

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 40 页。

对应已有 gpt-106.pdf 页脚：45。

原课件第 40 页

Closed Set

Note that the union of an infinite number of closed sets is not necessarily a closed set. For example, let

$$F_k = \left[\frac{1}{k+1}, \frac{1}{k}\right] \subseteq \mathbb{R} \qquad (k = 1, 2, \cdots),$$

then $\bigcup_{k=1}^{\infty} F_k = (0, 1]$. Furthermore, $\overline{\bigcup_{k=1}^{\infty} F_k} \neq \bigcup_{k=1}^{\infty} \overline{F}_k$.
But we have the following simple facts: Let $E_\gamma \subseteq \mathbb{R}^N$ for $\gamma \in \Gamma$, then

$$\bigcup_{\gamma\in\Gamma} \overline{E}_\gamma \subseteq \overline{\bigcup_{\gamma\in\Gamma} E_\gamma} \qquad \text{and} \qquad \overline{\bigcap_{\gamma\in\Gamma} E_\gamma} \subseteq \bigcap_{\gamma\in\Gamma} \overline{E}_\gamma.$$

沿原论证展开与批注 $F_k = [1/(k+1), 1/k]$ 都闭，但 $\bigcup_k F_k = (0, 1]$ 不含聚点0，所以不闭。具体看 $x_k = 1/k$：每项落在某个闭区间，但没有一个固定的区间包含无限多个趋0的项。 一般总有

$$\bigcup_{\gamma} \overline{E_\gamma} \subseteq \overline{\bigcup_{\gamma} E_\gamma}, \qquad \overline{\bigcap_{\gamma} E_\gamma} \subseteq \bigcap_{\gamma} \overline{E_\gamma}.$$

第一式对每个 $E_\gamma \subseteq \bigcup E_\gamma$ 用单调性再取并；第二式对 $\bigcap E_\gamma \subseteq E_\gamma$ 用单调性再取交。等号一般不成立。

8.19 闭包是包含原集合的最小闭集

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 41 页。

对应已有 gpt-106.pdf 页脚：46。

原课件第 41 页

Closed Set

Theorem 4. The closure $\overline{A}$ of A is the intersection of all the closed sets which contain A.

Proof. Let E be the intersection of all the closed sets which contain A; by the preceding theorem, E is closed. Since $\overline{A}$ is a closed set which contains A, we have $E \subseteq \overline{A}$. But we also have $E = \overline{E} \supseteq \overline{A}$, by condition (2), whence $E = \overline{A}$. Q.E.D.

沿原论证展开与批注 令 $\mathcal{F} = \{F : F \text{ 闭且 } A \subseteq F\}$，$C = \bigcap_{F\in\mathcal{F}} F$。由于 $\overline{A}$ 本身闭且包含 A，它是被求交的成员之一，故 $C \subseteq \overline{A}$。另一方面每个 $F \in \mathcal{F}$ 满足 $\overline{A} \subseteq \overline{F} = F$，所以 $\overline{A} \subseteq C$。两边合并得 $C = \overline{A}$。“最小”按集合包含比较，不是按元素数量比较；$\mathcal{F}$ 至少含有整个空间，因此并非无从取交。

8.20 递减非空紧集的交非空：简洁完整证明

来源：106-ZSM-PPT.pdf，原课件 PDF 第 42 页。

对应已有 gpt-106.pdf 页脚：47。

原课件第 42 页

Closed Set

Theorem 5 (Cantor Nested Closed Sets Theorem) If $\{F_k; k \in \mathbb{N}\}$ is a sequence of non-empty bounded closed sets in $\mathbb{R}^N$ with $F_1 \supseteq F_2 \supseteq \cdots \supseteq F_k \supseteq \cdots$, then

$$\bigcap_{k=1}^{\infty} F_k \neq \varnothing.$$

Proof. If there are infinitely many identical sets in $\{F_k; k \in \mathbb{N}\}$, then there is a natural number $K \in \mathbb{N}$, define $F_k = F_K$ for $k \geqslant K$.

$$\bigcap_{k=1}^{\infty} = F_K \neq \varnothing.$$

Now let's assume that for all $k \in \mathbb{N}$, $F_{K+1} \subseteq F_k$, and F_{K+1} is the proper subset of F_k, i.e. $F_k \setminus F_{K+1} \neq \varnothing$ for $k \in \mathbb{N}$. We choose $x_k \in F_k \setminus F_{K+1}$ for $k \in \mathbb{N}$, then $\{x_k; k \in \mathbb{N}\}$ is a sequence of bounded distinct points in $\mathbb{R}^N$. According to Bolzano-Weierstrass theorem, there are $\{x_{k_q}; q \in \mathbb{N}\}$ and $x \in \mathbb{R}^N$ such that

$$\lim_{q \to \infty} |x_{k_q} - x| = 0.$$

Since each F_k is a closed set, then $x \in F_k$ for $k \in \mathbb{N}$, i.e., $x \in \bigcap_{k=1}^{\infty} F_k$.

Q.E.D.

沿原论证展开与批注 设 $F_1 \supseteq F_2 \supseteq \cdots$，各集非空、闭且 F_1 有界。从每个 F_k 任取 $x_k \in F_k$。整个数列都在 F_1 中，故有收敛子列 $x_{k_j} \to x$。任意固定 m，当 $k_j \geq m$ 时，$x_{k_j} \in F_{k_j} \subseteq F_m$。由 F_m 闭得到 $x \in F_m$。因为 m 任意，$x \in \bigcap_m F_m$。这个证明自动处理重复集合、重复点，不必按课件分情况。交集未必只有一点；加上直径趋0才唯一。若去掉有界性，$F_k = [k, \infty)$ 给出空交；去掉闭性，$F_k = (0, 1/k)$ 也给出空交。

8.21 内点与内部

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 46 页。

对应已有 gpt-106.pdf 页脚：51。

原课件第 46 页

Open Set

Theorem 1. If $E \subseteq \mathbb{R}^N$, then E° is an open set.

Proof. Let $E \subseteq \mathbb{R}^N$ and consider any $x_0 \in E^\circ$. Then $N_\delta(x_0) \subseteq E$ for some $\delta > 0$. We claim that $N_\delta(x_0) \subseteq E^\circ$, thus proving E° is open.

To establish the claim consider any $x \in N_\delta(x_0)$; we need to show that x is an interior point of E. Suppose $\rho(x, x_0) = r(< \delta)$. From the triangle inequality, $N_{\delta-r}(x) \subseteq N_\delta(x_0)$, and so $N_{\delta-r}(x) \subseteq E$, thus showing x is an interior point. Q.E.D.

沿原论证展开与批注 $x \in E^\circ$ 指存在 $r > 0$ 使 $B(x,r) \subseteq E$。内部不是边界。若 $y \in B(x,r)$，取 $s = r - \|x - y\| > 0$，三角不等式给 $B(y,s) \subseteq B(x,r)$。因此球中每点都是内点，E° 开。内部的单调性来自球的包含，幂等性来自内部本身是开集。

8.22 有限开集的交

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 47 页。

对应已有 gpt-106.pdf 页脚：52。

原课件第 47 页

Open Set

Theorem 2. The intersection of finite open sets is an open set.

Proof. Define $G = \bigcap_{n=1}^{N} G_n$ where G_n is an open set for $n = 1, \cdots, N$. Note that G_n is an open set, $G_n^\circ = G_n$, then

$$G^\circ = \left(\bigcap_{n=1}^{N} G_n\right)^\circ = \bigcap_{n=1}^{N} G_n^\circ = \bigcap_{n=1}^{N} G_n = G.$$

Therefore G is an open set. Q.E.D.

沿原论证展开与批注 任取 $x \in \bigcap_{i=1}^{m} G_i$，各取 $r_i > 0$ 使 $B(x, r_i) \subseteq G_i$。有限多个正数的最小值 r 仍正，故 $B(x, r) \subseteq \bigcap_i G_i$。无限交不能照搬：$\bigcap_{n\geq 1}(-1/n, 1/n) = \{0\}$ 不开。

8.23 有限开集的交

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 48 页。

对应已有 gpt-106.pdf 页脚：53。

原课件第 48 页

Open Set

Theorem 2. The intersection of finite open sets is an open set.

Proof. Define $G = \bigcap_{n=1}^{N} G_n$ where G_n is an open set for $n = 1, \cdots, N$. Suppose $x \in G$, then $x \in A_n$ for $n = 1, \cdots, N$, and for each n there exists $\delta_n > 0$ such that $N_{\delta_n}(x) \subseteq G_n$. Let $\delta = \min\{\delta_1, \cdots, \delta_N\}$. Then $\delta > 0$ and $N_\delta(x) \subseteq G$, implying G is open. Q.E.D.

沿原论证展开与批注 任取 $x \in \bigcap_{i=1}^{m} G_i$，各取 $r_i > 0$ 使 $B(x, r_i) \subseteq G_i$。有限多个正数的最小值 r 仍正，故 $B(x, r) \subseteq \bigcap_i G_i$。无限交不能照搬：$\bigcap_{n\geq 1}(-1/n, 1/n) = \{0\}$ 不开。

8.24 任意开集的并

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 49 页。

对应已有 gpt-106.pdf 页脚：54。

原课件第 49 页

Open Set

Theorem 3. The union of multiple open sets is an open set.

Proof. Define $G = \bigcup_{n=1}^{\infty} G_n$ where G_n is an open set for $n = 1, 2, \cdots$. Note $G_n \subseteq G$, then $G_n^\circ \subseteq G^\circ$. Thus $\bigcup_{n=1}^{\infty} G_n^\circ \subseteq G^\circ$. Note that G_n is an open set, $G_n^\circ = G_n$, then

$$G = \bigcup_{n=1}^{\infty} G_n = \bigcup_{n=1}^{\infty} G_n^\circ \subseteq G^\circ = \left(\bigcup_{n=1}^{\infty} G_n \right)^\circ .$$

Therefore G is an open set. Q.E.D.

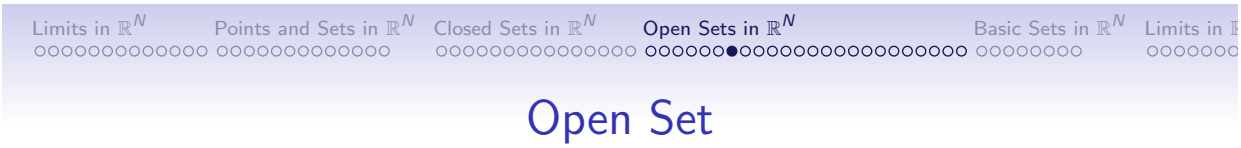
沿原论证展开与批注 若 $x \in \bigcup_i G_i$，只需选一个含 x 的 G_j；由其开性取球 $B(x,r) \subseteq G_j \subseteq \bigcup_i G_i$。这不需要所有 G_i 共同的正半径，所以任意甚至不可数并仍开。与有限交比较，关键是“存在一个”与“对所有”的区别。

8.25 任意开集的并

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 50 页。

对应已有 gpt-106.pdf 页脚：55。

原课件第 50 页



Open Set

Theorem 3. The union of multiple open sets is an open set.

Proof. Define $G = \bigcup_{n=1}^{\infty} G_n$ where G_n is an open set for $n = 1, 2, \cdots$. Suppose $x \in G$, then $x \in G_n$ for some n. For some such n choose $\delta_n > 0$ such that $N_{\delta_n}(x) \subseteq G_n$. Then certainly $N_{\delta_n}(x) \subseteq G$, and so G is open. Q.E.D.

沿原论证展开与批注 若 $x \in \bigcup_i G_i$，只需选一个含 x 的 G_j；由其开性取球 $B(x,r) \subseteq G_j \subseteq \bigcup_i G_i$。这不需要所有 G_i 共同的正半径，所以任意甚至不可数并仍开。与有限交比较，关键是“存在一个”与“对所有”的区别。

8.26 任意开集的并

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 51 页。

对应已有 gpt-106.pdf 页脚：56。

原课件第 51 页

Open Set

Theorem 3. If $\{G_\gamma : \gamma \in \Gamma\}$ is a family of open sets in $\mathbb{R}^N$, then its union $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ is an open set.

Proof. Define $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ where G_γ is an open set for $\gamma \in \Gamma$. Note $G_\gamma \subseteq G$, then $G_\gamma^\circ \subseteq G^\circ$. Thus $\bigcup_{\gamma\in\Gamma} G_\gamma^\circ \subseteq G^\circ$. Note that G_γ is an open set, $G_\gamma^\circ = G_\gamma$, then

$$G = \bigcup_{\gamma\in\Gamma} G_\gamma = \bigcup_{\gamma\in\Gamma} G_\gamma^\circ \subseteq G^\circ = \left(\bigcup_{\gamma\in\Gamma} G_\gamma\right)^\circ.$$

Therefore G is an open set. Q.E.D.

沿原论证展开与批注 若 $x \in \bigcup_i G_i$，只需选一个含 x 的 G_j；由其开性取球 $B(x,r) \subseteq G_j \subseteq \bigcup_i G_i$。这不需要所有 G_i 共同的正半径，所以任意甚至不可数并仍开。与有限交比较，关键是“存在一个”与“对所有”的区别。

8.27 任意开集的并

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 52 页。

对应已有 gpt-106.pdf 页脚：57。

原课件第 52 页

Open Set

Theorem 3. If $\{G_\gamma : \gamma \in \Gamma\}$ is a family of open sets in $\mathbb{R}^N$, then its union $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ is an open set.

Proof. Define $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ where G_γ is an open set for $\gamma \in \Gamma$.
Suppose $x \in G$, then $x \in G_\gamma$ for some $\gamma \in \Gamma$. For some such γ choose $\delta_\gamma > 0$ such that $N_{\delta_\gamma}(x) \subseteq G_\gamma$. Then certainly $N_{\delta_\gamma}(x) \subseteq G$, and so G is open. Q.E.D.

沿原论证展开与批注 若 $x \in \bigcup_i G_i$，只需选一个含 x 的 G_j；由其开性取球 $B(x,r) \subseteq G_j \subseteq \bigcup_i G_i$。这不需要所有 G_i 共同的正半径，所以任意甚至不可数并仍开。与有限交比较，关键是“存在一个”与“对所有”的区别。

8.28 实开集的区间分解

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 54 页。

对应已有 gpt-106.pdf 页脚：59。

原课件第 54 页

Open Set

Proof of (1). Let G be an open set in $\mathbb{R}$. For any point $a \in G$, since a is the interior point of G, there is $\delta > 0$ such that $(a-\delta, a+\delta) \subseteq G$. Now define

$$a' = \inf\{x : (x, a) \subseteq G\} \qquad \text{and} \qquad a'' = \sup\{x : (a, x) \subseteq G\},$$

(here, a' can be $-\infty$, and a'' can be $+\infty$). obviously, $a' < a < a''$ and $(a', a'') \subseteq G$. In fact, for any point z in the interval (a', a''), we assume $a' < z \leqslant a$, there must be x such that $a' < x < z$ and $(x, a) \subseteq G$, that is, $z \in G$. We call such an open interval (a', a'') the constituent interval I_a of G (about point a).

If $I_{a_1} = (a_1', a_1'')$, $I_{a_2} = (a_2', a_2'')$ is the constituent interval of G, then they are coincident or disjoint. For this purpose, we assume $a_1 < a_2$. If $I_{a_1} \cap I_{a_2} \neq \varnothing$, then there is $a_2' < a_1''$, then define $\min\{a_1', a_2'\} = a_0$ and $\max\{a_1'', a_2''\} = a^0$, then $(a_0, a^0) = (a_1', a_1'') \cup (a_2', a_2'')$. Take $x \in I_{a_1} \cap I_{a_2}$, then $I_x = (a_0, a^0)$ is a constituent interval and $(a_0, a^0) = (a_1', a_1'') = (a_2', a_2'')$.

Finally, we know that the family of disjoint intervals in $\mathbb{R}$ are countable.

沿原论证展开与批注 对 $a \in G$，把包含 a 且包含于 G 的开区间取并，得极大开区间 I_a。两个这样的区间相交，则并仍是区间，因此极大性使其相等。故不同 I_a 不交且覆盖 G。每个非空区间含有理数，选其中一个；不同区间对应不同有理数，故只有可数多个区间。端点允许无穷。

8.29 欧氏开集的方块分解

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 55–56 页。

对应已有 gpt-106.pdf 页脚：60, 61。

原课件第 55 页

Open Set

Proof of (2). Firstly, $\mathbb{R}^N$ is divided into countable semi-open and semi-closed cubes with side length of 1 by lattice points (coordinates are all integers), and the whole set is denoted as Γ_0. Each side of each cube in Γ_0 is bisected, then each cube can be divided into 2^N semi-open and semi-closed cubes with a side length of $\frac{1}{2}$, In Γ_0, all the sub-cubes thus created are denoted Γ_1. Continue to divide by this method to obtain the sequence $\{\Gamma_k; k \in \mathbb{N}\}$, where the side length of each cube in Γ_k is 2^{-k}, and this cube is the union of the corresponding 2^k disjoint cubes in Γ_{k+1}. We call the cube thus divided a dyadic cube.

原课件第 56 页

Open Set

Now put all cubes in Γ_0 included in G are taken out and written as $\mathscr{H}_0$. Then put all cubes in Γ_1 included in $G \setminus \bigcup_{H\in\mathscr{H}_0} H$ are taken out and written as $\mathscr{H}_1$ ($H \in \mathscr{H}_0$ is semi-open and semi-closed cube). And so on, $\mathscr{H}_K$ is the whole set of semi-open and semi-closed cubes in Γ_K included in $G \setminus \bigcup_{k=1}^{K-1} \bigcup_{H\in\mathscr{H}_k} H$. The whole set of semi-open and semi-closed cubes in $\mathscr{H}_K$ is countable for $K = 0, 1, 2, \cdots$. Since G is an open set, for any $x \in G$, there is $\delta > 0$ such that $B(x, \delta) \subseteq G$. The diameter of the cube in Γ_K approaches to zero as $K \to \infty$, so x will eventually fall into a certain cube in Γ_K. Thus

$$G = \bigcup_{k=0}^{\infty} \bigcup_{H\in\mathscr{H}_k} H, \quad H \text{ denotes semi-open and semi-closed dyadic cubes.}$$

沿原论证展开与批注 把空间用边长 2^{-k} 的半开方块分割，保留完全包含于 G 且未被更粗已选方块覆盖的方块。每层可数，总层数可数；嵌套网格保证所选方块两两不交。给定 $x \in G$，存在球包含于 G，足够细的含 x 方块直径小于该球半径，所以 x 最终被选中方块覆盖。半开约定避免公共边界重复。

8.30 可数拓扑基

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 57 页。

对应已有 gpt-106.pdf 页脚：62。

原课件第 57 页

Open Set

There exists another important fact about the open sets in $\mathbb{R}^N$, that is, there is an open set family Γ, which consists of countable open sets in $\mathbb{R}^N$ such that any open set in $\mathbb{R}^N$ is the union of some open sets in Γ. In fact, Γ can be taken as

$$\left\{ B\left(x, \frac{1}{K}\right) \mid x \text{ is a rational point in } \mathbb{R}^N, \text{ and } K \text{ is a natural number} \right\}$$

Firstly, Γ is a countable set. Secondly, for any point x in the open set G in $\mathbb{R}^N$, there must be $\delta > 0$ such that $B(x, \delta) \subseteq G$. Now take the rational point x' such that $d(x, x') < \frac{1}{K}$, where $K > \frac{2}{\delta}$ such that

$$x \in B\left(x', \frac{1}{K}\right) \subseteq B(x, \delta) \subseteq G.$$

Obviously, the union of such $B\left(x', \frac{1}{K}\right)$ is G. Q.E.D.

沿原论证展开与批注 所有有理中心、正有理半径的球构成可数族。若 $x \in G$ 开，选 $B(x, r) \subseteq G$，再选有理中心 q 足够近于 x，及有理半径 s 满足 $\|x - q\| < s < r - \|x - q\|$，便有 $x \in B(q, s) \subseteq G$。因此任何开集都是某些基球的并。

8.31 紧性与有限子覆盖

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 60 页。

对应已有 gpt-106.pdf 页脚：65。

原课件第 60 页

Open Set

Theorem 5 (Heine-Borel Finite Covering Theorem) An open covering of a bounded closed set in $\mathbb{R}^N$ contains a finite subcovering.

Proof Let F is a bounded closed set in $\mathbb{R}^N$, and Γ is an open covering of F. From the above Lemma, it can be assumed that Γ consists of countable open sets:

$$\Gamma = \{G_1, \cdots, G_K, \cdots\}.$$

Define

$$H_K = \bigcup_{k=1}^{K} G_k, \qquad L_K = F \cap \mathbf{C}H_K \quad k = 1, 2, \cdots.$$

Obviously, H_K is an open set, L_K is a closed set with $L_K \supseteq L_{K+1}$ for $k = 1, 2, \cdots$. There are two cases:

(1) there is K_0 such that L_{K_0} is an empty set, that is, $\mathbf{C}H_{K_0}$ does not contain any point of F, then $F \subseteq H_{K_0}$, the theorem is proved;

(2) If all L_K is not an empty set, then, from Cantor's closed set theorem, there is a point of $x_0 \in L_K$ for $k = 1, 2, \cdots$, i.e. $x_0 \in F$ and $x_0 \in \mathbf{C}H_K$ for $k = 1, 2, \cdots$. Thus there is a point $x_0 \in F$ does not belong to everything H_K, which is inconsistent with the original assumption, so case (2) does not exist. Q.E.D.

沿原论证展开与批注　老师的路线：先把覆盖缩成可数覆盖，再构造避开前面有限项的数列。先用有理球可数基，将任意开覆盖 $\mathcal{U}$ 缩为可数子覆盖：对每个包含于某个 $U \in \mathcal{U}$ 的基球 V_j，选择一个这样的 U_j。任意 $x \in F$ 都在某覆盖成员中，也在包含于该成员的基球中，因而被某个所选 U_j 覆盖。反设不存在有限子覆盖，则每个 n 可选

$$x_n \in F \setminus \bigcup_{j=1}^{n} U_j.$$

F 有界，Bolzano–Weierstrass 给子列 $x_{n_k} \to x$；F 闭，故 $x \in F$。可数覆盖给某个 j_0 使 $x \in U_{j_0}$。由于它开，收敛子列最终都在 U_{j_0}；同时子列指标最终满足 $n_k \geq j_0$，按选取方式 x_{n_k} 不在前 n_k 项的并，特别不在 U_{j_0}。矛盾，所以有限子覆盖存在。必须同时取到“子列靠近极限”和“子列指标超过 j_0”，不能漏掉其中一项。

8.32 紧性与有限子覆盖

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 61–63 页。

对应已有 gpt-106.pdf 页脚：66, 67, 68。

原课件第 61 页



Open Set

Theorem 5 (Heine-Borel Finite Covering Theorem) If $\mathscr{H}$ is an open covering of a closed and bounded subset E of $\mathbb{R}^N$ (for any x in E there exists a neighborhood $H \in \mathscr{H}$ such that $x \in H$); then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof. The proof method belongs to Lebesgue. We complete the proof by two steps.

原课件第 62 页

Open Set

(1) There exists $\delta > 0$ such that any δ neighborhood $N(x, \delta)$ of center of x in E is contained in some open neighborhood in $\mathscr{H}$. We use reduction to absurdity to check this setting, and assume that there isn't such $\delta > 0$. For any natural number n, we cannot take $\frac{1}{n}$ as δ, then there exists $x_n \in E$ such that $N\left(x_n, \frac{1}{n}\right)$ does not contain in any open neighborhood in $\mathscr{H}$. E is bounded, $\{x_n : n = 1, 2, \cdots\} \subseteq E$ is also bounded. If $\{x_n : n = 1, 2, \cdots\}$ has infinite different members, then there exists an accumulation points x_0 from Bolzano - Weierstrass Theorem. If $\{x_n : n = 1, 2, \cdots\}$ has only finite different members, then there exists at least a point x_0 presented infinite times. In a word, $\{x_n : n = 1, 2, \cdots\} \subseteq E$ has a convergent subsequence $\{x_{n_k} : n = 1, 2, \cdots\} \subseteq E$ with $\lim\limits_{k\to\infty} x_{n_k} = x_0$. Note that E is a closed set, then $x_0 \in E$. Since $\mathscr{H}$ is an open covering of a closed and bounded set E, then there exists $N \in \mathscr{H}$ such that $x_0 \in N$. Define $N = N(y_0, \eta)$, then there exists $\eta' = \min\{\eta, \rho(x_0, y_0)\} > 0$ such that $N(x_0, \eta') \subseteq N(y_0, \eta)$. Note $\lim\limits_{k\to\infty} x_{n_k} = x_0$, then we can take enough large natural number n_k such that $\rho(x_{n_k}, x_0) < \frac{\eta'}{2}$ and $\frac{1}{n_k} < \frac{\eta'}{2}$, then $N\left(x_{n_k}, \frac{1}{n_k}\right) \subseteq N(x_0, \eta') \subseteq N(y_0, \eta) \in \mathscr{H}$. This contradicts the definition of x_{n_k}. Therefore there exists such $\delta > 0$, which is called Lebesgue number.

原课件第 63 页

Open Set

(2) Since E is bounded, we can divide E into finite small cubes by using hyperplanes parallel axises such that the length in the each cube is strictly less than $\delta > 0$. Set these small cubes are $E_1, \cdots, E_K$ with $E = \sum_{k=1}^{K} E_k$, we take arbitrary $x_k \in F_k$ such that $F_k \subseteq N(x_k, \delta)$, then there exists $N_k \in \mathscr{H}$ such that $N(x_k, \delta) \subseteq N_k$. Thus we obtain finite open neighborhoods $N_1, \cdots, N_K$ with $E = \sum_{k=1}^{K} E_k \subseteq \bigcup_{k=1}^{K} N(x_k, \delta) \subseteq \bigcup_{k=1}^{K} N_k$. Q.E.D.

沿原论证展开与批注 开覆盖是开集族，其并包含目标集合；有限子覆盖必须从原覆盖取有限成员。有限维中闭有界等价于紧。证明闭有界给有限子覆盖：若没有，把含 E 的闭方块不断二分，每次保留不能有限覆盖的一个子方块。得到直径趋零的套方块及 $x_n \in E$，其极限 $x \in E$（用闭性）。某原覆盖开集含 x，故含 $B(x,r)$；足够小的整个所选方块都落入该球，可由一个开集覆盖，矛盾。另一证明的 Lebesgue 数步骤：若无统一半径，取 x_n 使 $B(x_n, 1/n)$ 不被单个覆盖成员包含；抽子列 $x_n \to x$，取 $B(x,r) \subset U$，当 $\|x_n - x\| < r/2, 1/n < r/2$ 时矛盾。闭性和有界性缺一不可：$(0,1)$ 可被 $(1/n, 1)$ 覆盖却不能有限覆盖，$\mathbb{N}$ 的局部小区间覆盖也不能有限覆盖。

8.33 紧性与有限子覆盖

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 64 页。

对应已有 gpt-106.pdf 页脚：69。

原课件第 64 页

Open Set

Note that in the above theorem, the boundedness condition cannot be absent. For example, in $\mathbb{R}$, there is no finite subcovering for open covering $\left\{\left(n-\frac{1}{2}, n+\frac{1}{2}\right)\right\}$ of natural number set. Similarly, the conditions of a closed set are also indispensable. For example, in $\mathbb{R}$, for the point set $\left\{1, \frac{1}{2}, \cdots, \frac{1}{n}, \cdots\right\}$, define an open covering

$$\left\{\left(\frac{1}{n}-\frac{1}{2n}, \frac{1}{n}+\frac{1}{2n}\right)=\left(\frac{1}{2n}, \frac{3}{2n}\right) ; n \in \mathbb{N}\right\},$$

there is no finite subcovering.

沿原论证展开与批注 开覆盖是开集族，其并包含目标集合；有限子覆盖必须从原覆盖取有限成员。有限维中闭有界等价于紧。证明闭有界给有限子覆盖：若没有，把含 E 的闭方块不断二分，每次保留不能有限覆盖的一个子方块。得到直径趋零的套方块及 $x_n \in E$，其极限 $x \in E$（用闭性）。某原覆盖开集含 x，故含 $B(x,r)$；足够小的整个所选方块都落入该球，可由一个开集覆盖，矛盾。另一证明的 Lebesgue 数步骤：若无统一半径，取 x_n 使 $B(x_n, 1/n)$ 不被单个覆盖成员包含；抽子列 $x_n \to x$，取 $B(x,r) \subset U$，当 $\|x_n - x\| < r/2, 1/n < r/2$ 时矛盾。闭性和有界性缺一不可：$(0,1)$ 可被 $(1/n, 1)$ 覆盖却不能有限覆盖，$\mathbb{N}$ 的局部小区间覆盖也不能有限覆盖。

8.34 紧集的邻域与逆定理

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 65 页。

对应已有 gpt-106.pdf 页脚：70。

原课件第 65 页

Open Set

Example: If F is a bounded closed set in $\mathbb{R}^N$, and G is an open set in $\mathbb{R}^N$ and $F \subseteq G$, then there is $\delta > 0$ such that when $|x| < \delta$,

$$F + \{x\} \equiv \{y + x \mid y \in F\} \subseteq G.$$

Proof. For any $y \in F$, as $y \in G$, there is $\delta_y > 0$ such that $B(y, \delta_y) \subseteq G$. As $\left\{ B\left(y, \frac{\delta_y}{2}\right) \mid y \in F \right\}$ constitutes an open covering of F, according to the finite sub-covering theorem, there is $y_1, \cdots, y_K \in F$ such that

$$F \subseteq \bigcup_{k=1}^{K} B\left(y_k, \frac{\delta_{y_k}}{2}\right).$$

Thus, each $y \in F$ belongs to at least one $B\left(y_k, \frac{\delta_{y_k}}{2}\right)$, and the distance between y and any point z in $\mathbf{C}G$ is

$$|y - z| \geqslant |z - y_k| - |y_k - y| > \delta_{y_k} - \frac{\delta_{y_k}}{2} = \frac{\delta_{y_k}}{2}.$$

Take $\delta = \frac{1}{2} \min\{\delta_{y_1}, \cdots, \delta_{y_K}\}$, then, when $|x| < \delta$, $y + x \in G$, that is, $F + \{x\} \subseteq G$.

沿原论证展开与批注 若紧 $F \subset G$ 开，给每个 $y \in F$ 选 $B(y, r_y) \subset G$，用半径 $r_y/2$ 的球有限覆盖 F，取有限 $r_y/2$ 的最小值 $\delta > 0$。任意 $\|h\| < \delta$ 都使 $F + h \subset G$。反之紧集必有界，因为球 $B(0, n)$ 覆盖它且有限子覆盖给统一半径；必闭，因为对 $z \notin E$，以 $x \in E$ 为中心取半径 $\|x - z\|/3$ 的球，有限覆盖后可在 z 周围找到避开 E 的正半径球。

8.35 紧集的邻域与逆定理

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 66 页。

对应已有 gpt-106.pdf 页脚：71。

原课件第 66 页

Open Set

Theorem 6. Let $E \subseteq \mathbb{R}^N$. If any open covering of E contains finite sub-coverings, then E is a bounded closed set.

Proof. If $y \in \mathbf{C}E$, then for every $x \in E$, there is $\delta_x > 0$ such that

$$B(x, \delta_x) \cap B(y, \delta_x) = \varnothing.$$

Obviously, $\{B(x, \delta_x) : x \in E\}$ is an open covering of E. Then there are a finite sub-covering, written as

$$B(x_1, \delta_{x_1}), \cdots, B(x_K, \delta_{x_K}),$$

then E is a bounded set. Now define $\delta_0 = \min\{\delta_{x_1}, \cdots, \delta_{x_K}\}$, then $B(y, \delta_0) \cap E = \varnothing$, that is, $y \notin E'$. Thus $E' \subseteq E$. E is a closed set and hence boundedness is obvious. Q.E.D.

Note that if any open covering of E contains a finite sub-covering, we call E a compact set. The above two theorems show that a compact set in $\mathbb{R}^N$ is a bounded closed set.

沿原论证展开与批注 若紧 $F \subset G$ 开，给每个 $y \in F$ 选 $B(y, r_y) \subset G$，用半径 $r_y/2$ 的球有限覆盖 F，取有限 $r_y/2$ 的最小值 $\delta > 0$。任意 $\|h\| < \delta$ 都使 $F + h \subset G$。反之紧集必有界，因为球 $B(0, n)$ 覆盖它且有限子覆盖给统一半径；必闭，因为对 $z \notin E$，以 $x \in E$ 为中心取半径 $\|x - z\|/3$ 的球，有限覆盖后可在 z 周围找到避开 E 的正半径球。

8.36 闭包与内部的抽象公理

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 70–71 页。

对应已有 gpt-106.pdf 页脚：75, 76。

原课件第 70 页

Close set and Open Set

Theorem 1. A one-one correspondence can be established between closure operations and interior operations by means of the formulae:

$$X^\circ = \mathbf{C}(\overline{\mathbf{C}X}) \qquad \text{and} \qquad \overline{X} = \mathbf{C}(\mathbf{C}X)^\circ$$

Proof. The correspondence $X \to X^\circ = \mathbf{C}(\overline{\mathbf{C}X})$ is clearly an interior operation, for

(1) $X^\circ = \mathbf{C}(\overline{\mathbf{C}X}) \subseteq \mathbf{C}(\mathbf{C}X) = X$,

(2) $X_1 \subseteq X_2 \Longrightarrow \mathbf{C}X_1 \supseteq \mathbf{C}X_2 \Longrightarrow (\overline{\mathbf{C}X_1}) \supseteq (\overline{\mathbf{C}X_2}) \Longrightarrow X_1^\circ \subseteq X_2^\circ$,

(3) $(X^\circ)^\circ = \mathbf{C}(\overline{\overline{\mathbf{C}X}}) = \mathbf{C}(\overline{\mathbf{C}X}) = X^\circ$,

(4) $S^\circ = \mathbf{C}\overline{\varnothing} = \mathbf{C}\varnothing = S$.

原课件第 71 页

Close set and Open Set

The correspondence $X \to \overline{X} = \mathbf{C}(\mathbf{C}X)^\circ$ is clearly a closure operation, for

(1) $\overline{X} = \mathbf{C}(\mathbf{C}X)^\circ \supseteq \mathbf{C}(\mathbf{C}X) = X$,

(2) $X_1 \subseteq X_2 \Longrightarrow \mathbf{C}X_1 \supseteq \mathbf{C}X_2 \Longrightarrow (\mathbf{C}X_1)^\circ \supseteq (\mathbf{C}X_2)^\circ \Longrightarrow \overline{X}_1 \subseteq \overline{X}_2$,

(3) $\overline{\overline{X}} = \mathbf{C}[(\mathbf{C}X)^\circ]^\circ = \mathbf{C}(\mathbf{C}X)^\circ = \overline{X}$,

(4) $\overline{\varnothing} = \mathbf{C}S^\circ = \mathbf{C}S = \varnothing$.

It is easily shown that the correspondence thus set up between the two kinds of operations is one-one. Q.E.D.

沿原论证展开与批注 闭包算子 c 满足扩张、单调、幂等及 $c(\varnothing)=\varnothing$；定义 $i(A)=S\setminus c(S\setminus A)$。补集反向包含把扩张变为收缩，单调仍为单调，幂等由两次取补消去得到，$i(S)=S$。反向公式 $c(A)=S\setminus i(S\setminus A)$，因此两构造互逆。闭集是闭包不动集：任意闭集交 F 满足 $c(F)\subseteq c(F_j)=F_j$ 对所有 j 成立，加扩张性得相等。若 $A\subset F$ 闭，则 $c(A)\subset F$；且 $c(A)$ 自己闭，所以它是所有含 A 闭集的交。注意这四条是较一般闭包系统公理，未必单独推出有限闭并性质，欧氏拓扑还有相应额外性质。

8.37 闭包与内部的抽象公理

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 72 页。

对应已有 gpt-106.pdf 页脚：77。

原课件第 72 页

Close set and Open Set

Remark. The complement of an open set is a closed set (and vice versa), for $\mathbf{C}F = \mathbf{C}\overline{F} = (\mathbf{C}F)^\circ$; in what follows, we shall study only properties of closure operations, those of interior operations being deducible immediately on appealing to duality, for $\mathbf{C}G = \mathbf{C}G^\circ = \overline{\mathbf{C}G}$.

沿原论证展开与批注 闭包算子 c 满足扩张、单调、幂等及 $c(\varnothing)=\varnothing$；定义 $i(A)=S\setminus c(S\setminus A)$。补集反向包含把扩张变为收缩，单调仍为单调，幂等由两次取补消去得到，$i(S)=S$。反向公式 $c(A)=S\setminus i(S\setminus A)$，因此两构造互逆。闭集是闭包不动集：任意闭集交 F 满足 $c(F)\subseteq c(F_j)=F_j$ 对所有 j 成立，加扩张性得相等。若 $A\subset F$ 闭，则 $c(A)\subset F$；且 $c(A)$ 自己闭，所以它是所有含 A 闭集的交。注意这四条是较一般闭包系统公理，未必单独推出有限闭并性质，欧氏拓扑还有相应额外性质。

8.38 闭包与内部的抽象公理

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 73 页。

对应已有 gpt-106.pdf 页脚：78。

原课件第 73 页

Close set and Open Set

Theorem 2. The intersection of a family $\mathscr{F} = (F_\gamma \mid \gamma \in \Gamma)$ of closed sets is a closed set.

Proof. Writing $F = \bigcap_{\gamma\in\Gamma} F_\gamma$, we have, by condition (2) for a closure operation,

$$\overline{F} \subseteq \overline{F}_\gamma = F_\gamma$$

for each γ. Therefore $\overline{F} \subseteq F$; hence, by condition (1), $\overline{F} = F$. Q.E.D.

沿原论证展开与批注 闭包算子 c 满足扩张、单调、幂等及 $c(\varnothing) = \varnothing$；定义 $i(A) = S \setminus c(S \setminus A)$。补集反向包含把扩张变为收缩，单调仍为单调，幂等由两次取补消去得到，$i(S) = S$。反向公式 $c(A) = S \setminus i(S \setminus A)$，因此两构造互逆。闭集是闭包不动集：任意闭集交 F 满足 $c(F) \subseteq c(F_j) = F_j$ 对所有 j 成立，加扩张性得相等。若 $A \subset F$ 闭，则 $c(A) \subset F$；且 $c(A)$ 自己闭，所以它是所有含 A 闭集的交。注意这四条是较一般闭包系统公理，未必单独推出有限闭并性质，欧氏拓扑还有相应额外性质。

8.39 闭包与内部的抽象公理

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 74 页。

对应已有 gpt-106.pdf 页脚：79。

原课件第 74 页

Close set and Open Set

Theorem 2. The union of a family $\mathscr{G} = (G_\gamma \mid \gamma \in \Gamma)$ of open sets is an open set.

Proof. Writing $G = \bigcup_{\gamma\in\Gamma} G_\gamma$, we have, by condition (2) for a interior operation,

$$G_\gamma = G_\gamma^\circ \subseteq G^\circ$$

for each γ. Therefore $G \subseteq G^\circ$; hence, by condition (1), $G^\circ = G$. Q.E.D.

沿原论证展开与批注 闭包算子 c 满足扩张、单调、幂等及 $c(\varnothing)=\varnothing$；定义 $i(A)=S\setminus c(S\setminus A)$。补集反向包含把扩张变为收缩，单调仍为单调，幂等由两次取补消去得到，$i(S)=S$。反向公式 $c(A)=S\setminus i(S\setminus A)$，因此两构造互逆。闭集是闭包不动集：任意闭集交 F 满足 $c(F)\subseteq c(F_j)=F_j$ 对所有 j 成立，加扩张性得相等。若 $A\subset F$ 闭，则 $c(A)\subset F$；且 $c(A)$ 自己闭，所以它是所有含 A 闭集的交。注意这四条是较一般闭包系统公理，未必单独推出有限闭并性质，欧氏拓扑还有相应额外性质。

8.40 闭包与内部的抽象公理

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 75 页。

对应已有 gpt-106.pdf 页脚：80。

原课件第 75 页

Close set and Open Set

Theorem 3. The closure $\overline{X}$ of X is the intersection of all the closed sets which contain X.

Proof. Let E be the intersection of all the closed sets which contain X; by the preceding theorem, E is closed. Since $\overline{X}$ is a closed set which contains X, we have $E \subseteq \overline{X}$. But we also have $E = \overline{E} \supseteq \overline{X}$, by condition (2), whence $E = \overline{X}$. Q.E.D.

沿原论证展开与批注 闭包算子 c 满足扩张、单调、幂等及 $c(\varnothing) = \varnothing$；定义 $i(A) = S \setminus c(S \setminus A)$。补集反向包含把扩张变为收缩，单调仍为单调，幂等由两次取补消去得到，$i(S) = S$。反向公式 $c(A) = S \setminus i(S \setminus A)$，因此两构造互逆。闭集是闭包不动集：任意闭集交 F 满足 $c(F) \subseteq c(F_j) = F_j$ 对所有 j 成立，加扩张性得相等。若 $A \subset F$ 闭，则 $c(A) \subset F$；且 $c(A)$ 自己闭，所以它是所有含 A 闭集的交。注意这四条是较一般闭包系统公理，未必单独推出有限闭并性质，欧氏拓扑还有相应额外性质。

8.41 闭包的序列判据与相对闭性

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 79 页。

对应已有 gpt-106.pdf 页脚：84。

原课件第 79 页

Limits in $\mathbb{R}^N$

One can prove:

(3) If X is a subset of $\mathbb{R}^N$, its adherence $\overline{X}$ is closed.

Proof: For any $\varepsilon > 0$ and any $x \in \overline{\overline{X}}$, there exists $x_\varepsilon \in \overline{X}$ such that $d(x, x_\varepsilon) < \frac{\varepsilon}{2}$ and then there exists $x'_\varepsilon \in X$ such that $d(x_\varepsilon, x'_\varepsilon) < \frac{\varepsilon}{2}$, hence $d(x, x'_\varepsilon) \leqslant d(x, x_\varepsilon) + d(x_\varepsilon, x'_\varepsilon) < \varepsilon$, which implies $x \in \overline{X}$. Hence $\overline{\overline{X}} \subseteq \overline{X}$. $\overline{X}$ is closed. $\square$

That is, $\overline{X} = \overline{\overline{X}}$. Since every closed set containing X clearly contains $\overline{X}$, the adherence of X can be characterized as the smallest closed set containing X.

沿原论证展开与批注 $x \in \overline{A}$ 当且仅对任意 n 可选 $a_n \in A$ 使 $\|x - a_n\| < 1/n$。闭集即任何集内收敛序列的极限仍在集内。闭包幂等可用二段逼近：先在 $\overline{A}$ 找距 x 小于 $\varepsilon/2$ 的点，再在 A 找距它小于 $\varepsilon/2$ 的点。任意闭交用序列同时属于每个因子；有限闭并则用有限抽屉原理找到无限多项属于某一个闭集的子列。相对闭包为 $S \cap \overline{A}$，只保留属于背景空间 S 的极限；例如 $(0,1)$ 在自身中闭，却不在 $\mathbb{R}$ 中闭。

8.42 闭包的序列判据与相对闭性

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 80 页。

对应已有 gpt-106.pdf 页脚：85。

原课件第 80 页

Limits in $\mathbb{R}^N$

It is easy to prove that:

(4) If $\mathscr{X}$ is a set of closed subsets of $\mathbb{R}^N$, their intersection, $\bigcap_{X\in\mathscr{X}} X$, is a closed subset of $\mathbb{R}^N$.

Proof: For any $x \in \overline{\bigcap_{X\in\mathscr{X}} X}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq \bigcap_{X\in\mathscr{X}} X$ such that $x^q \to x$. And for any $X \in \mathscr{X}$, $\{x^q; q \in \mathbb{N}\} \subseteq X$, which implies $x \in \overline{X}$ for any $X \in \mathscr{X}$. As X is closed, $x \in X$ for any $X \in \mathscr{X}$. Then $x \in \bigcap_{X\in\mathscr{X}} X$, which implies $\overline{\bigcap_{X\in\mathscr{X}} X} \subseteq \bigcap_{X\in\mathscr{X}} X$. So $\bigcap_{X\in\mathscr{X}} X$ is a closed subset of $\mathbb{R}^N$. □

沿原论证展开与批注 $x \in \overline{A}$ 当且仅对任意 n 可选 $a_n \in A$ 使 $\|x - a_n\| < 1/n$。闭集即任何集内收敛序列的极限仍在集内。闭包幂等可用二段逼近：先在 $\overline{A}$ 找距 x 小于 $\varepsilon/2$ 的点，再在 A 找距它小于 $\varepsilon/2$ 的点。任意闭交用序列同时属于每个因子；有限闭并则用有限抽屉原理找到无限多项属于某一个闭集的子列。相对闭包为 $S \cap \overline{A}$，只保留属于背景空间 S 的极限；例如 $(0,1)$ 在自身中闭，却不在 $\mathbb{R}$ 中闭。

8.43 闭包的序列判据与相对闭性

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 81 页。

对应已有 gpt-106.pdf 页脚：86。

原课件第 81 页

Limits in $\mathbb{R}^N$

It is a little more difficult to prove that:

(5) If $X_1, \cdots, X_n, \cdots, X_N$ ore N closed subsets of $\mathbb{R}^N$, their union, $\bigcup_{n=1}^{N} X_n$, is a closed subset of $\mathbb{R}^N$.

Proof: For any $x \in \overline{\bigcup_{n=1}^{N} X_n}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq \bigcup_{n=1}^{N} X_n$ such that $x^q \to x$. So there exists $n_0 \in \{1, \cdots, N\}$ such that there are infinite elements $\{x^{q_k}; q_k \in \mathbb{N}\}$ of $\{x^q; q \in \mathbb{N}\}$ in X_{n_0}, hence $x^{q_k} \to x$, which implies $x \in \overline{X}_{n_0}$. As X_{n_0} is closed, $x \in X_{n_0}$. Hence $x \in \bigcup_{n=1}^{N} X_n$. $\bigcup_{n=1}^{N} X_n$ is a closed subset of $\mathbb{R}^N$. □

沿原论证展开与批注 $x \in \overline{A}$ 当且仅对任意 n 可选 $a_n \in A$ 使 $\|x - a_n\| < 1/n$。闭集即任何集内收敛序列的极限仍在集内。闭包幂等可用二段逼近：先在 $\overline{A}$ 找距 x 小于 $\varepsilon/2$ 的点，再在 A 找距它小于 $\varepsilon/2$ 的点。任意闭交用序列同时属于每个因子；有限闭并则用有限抽屉原理找到无限多项属于某一个闭集的子列。相对闭包为 $S \cap \overline{A}$，只保留属于背景空间 S 的极限；例如 $(0,1)$ 在自身中闭，却不在 $\mathbb{R}$ 中闭。

8.44 任意欧氏子集都可分

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 84–85 页。

对应已有 gpt-106.pdf 页脚：89, 90。

原课件第 84 页

Limits in $\mathbb{R}^N$

One can prove:

(6) Every subset S of $\mathbb{R}^N$ contains a countable subset X such that $S \subseteq \overline{X}$.

Proof: As $\overline{\mathbb{Q}^N} = \mathbb{R}^N$ and $\mathbb{Q}^N$ is countable, let $\mathbb{Q}^N = \{q^1, q^2, \cdots\}$ and

$$T = \left\{(n,k) \in \mathbb{N}^2 : S \cap B\left(q^n, \frac{1}{k}\right) \neq \varnothing\right\}.$$

For every $(n,k) \in T$, let

$$y_{n,k} \in S \cap B\left(q^n, \frac{1}{k}\right),$$

then

$$X = \{y_{n,k} : (n,k) \in T\}$$

is a countable subset of S. Then we need to show $S \subseteq \overline{X}$.

原课件第 85 页

Limits in $\mathbb{R}^N$

Suppose any $x \in S$ and any $\varepsilon > 0$, there exists $k \in \mathbb{N}$ such that $\frac{2}{k} < \varepsilon$. Since $\overline{\mathbb{Q}^N} = \mathbb{R}^N$, there exists n so that

$$q^n \in B\left(x, \frac{1}{k}\right)$$

This implies $x \in B\left(q^n, \frac{1}{k}\right)$. So $(n, k) \in T$ and $y_{n,k} \in X$ and

$$d\left(x, y_{n,k}\right) < \frac{2}{k} < \varepsilon.$$

Hence $S \subseteq \overline{X}$. $\square$

沿原论证展开与批注 枚举有理中心有理半径的球，对每个与 S 相交的球选择一个点 $s_j \in S$，形成可数集合 D。给定 $x \in S, \varepsilon > 0$，由可数基选球 B 满足 $x \in B \subset B(x, \varepsilon)$，其选点 s_j 与 x 距离小于 ε。故 $D \subset S \subset \overline{D}$。不能直接取 $S \cap \mathbb{Q}^N$，它可能为空，例如 $S = \{\sqrt{2}\}$。

8.45 有界、紧与有限积

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 92 页。

对应已有 gpt-106.pdf 页脚：97。

原课件第 92 页

Limits in $\mathbb{R}^N$

A subset S of $\mathbb{R}^N$ is said to be compact if it is closed and bounded. It is easily proved that:

(7) If $S_1, \cdots, S_k, \cdots, S_K$ are closed (resp. compact) subsets of $\mathbb{R}^{N_1}, \cdots, \mathbb{R}^{N_k}, \cdots, \mathbb{R}^{N_K}$, then $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is a closed (resp. compact) subset of $\mathbb{R}^{\sum_{k=1}^{K} N_k}$.

Proof: Let $x = (x_1, \cdots, x_k, \cdots, x_K) \in S_1 \times \cdots \times S_k \times \cdots \times S_K$ where $x_k \in S_k$ for $k = 1, \cdots, K$.

For any $x \in \overline{S_1 \times \cdots \times S_k \times \cdots \times S_K}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq S_1 \times \cdots \times S_k \times \cdots \times S_K$ such that $x^q \to x$. Hence for any $k = 1, \cdots, K$, we have $x_k^q \to x_k$. As S_k is closed, $x_k \in S_k$. Hence $x \in S_1 \times \cdots \times S_k \times \cdots \times S_K$. So $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is a closed.

If $S_1, \cdots, S_k, \cdots, S_K$ are compact, then for $k = 1, \cdots, K$, any $x_k \in S_k$, there exists $M_k \geqslant 0$ such that $\|x_k\| \leqslant M_k$. Let $M = \max\{M_1, \cdots, M_k, \cdots, M_K\}$, then for any $x = (x_1, \cdots, x_k, \cdots, x_K) \in S_1 \times \cdots \times S_k \times \cdots \times S_K$, we have $\|x\| \leqslant M$, i.e. $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is bounded. As $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is also closed, we have that $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is compact. □

沿原论证展开与批注 有界是存在统一 M 使每个点的范数不大于 M。有限积的闭性由坐标收敛和各因子闭性给出；有界性由各因子界的平方和给出。因此有限个紧集的积紧。有限维条件很关键，无限维中闭且有界一般不保证紧。

8.46 连通的多种判据

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 94 页。

对应已有 gpt-106.pdf 页脚：99。

原课件第 94 页

Limits in $\mathbb{R}^N$

Proof: Let A is connected. Assume A is not an interval, then we can take three real numbers a_1, a_2, a such that $a_1 < a < a_2$, $a_1 \in A$, $a_2 \in A$ and $a \notin A$. Let $A_1 = A \cap (-\infty, a]$, $A_2 = A \cap [a, \infty)$, then $A = A_1 \cup A_2$ and $A_1 \cap A_2 = \varnothing$. Notice that $(-\infty, a]$ and $[a, \infty)$ are both non-empty closed subsets of $\mathbb{R}$, hence A_1 and A_2 are closed in A, hence A is therefore not connected. Contradiction!

Conversely let A be an interval of $\mathbb{R}$. To derive a contradiction, suppose we could write $A = A_1 \cup A_2$ for nonempty and disjoint closed subsets A_1 and A_2 of A. Pick any $a_1 \in A_1$ and $a_2 \in A_2$ and let $a_1 < a_2$ without loss of generality. Define $a = \sup\{t \in A_1 : t < a_2\}$ [or $a = \inf\{t \in A_2 : a_1 < t\}$], and note that $a_1 \leqslant a \leqslant a_2$ and hence $a \in A$ (because A is an interval).

If $a \in A_1$, then $a \neq a_2$ (since $A_1 \cap A_2 = \varnothing$), so $a < a_2$. As $a \notin A_2$ and A_2 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_2$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a, a + \varepsilon)$, $a_1 < x < a_2$, hence $x \notin A_1 \cup A_2$, i.e. $x \notin A$. Contradiction!

If $a \in A_2$, then $a \neq a_1$ (since $A_1 \cap A_2 = \varnothing$), so $a_1 < a$. As $a \notin A_1$ and A_1 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_1$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a - \varepsilon, a)$, $a_1 < x < a_2$, hence $x \notin A_1 \cup A_2$, i.e. $x \notin A$. Contradiction! □

沿原论证展开与批注 连通指不能分成两个非空不交的相对开集；取补立刻等价于不能分成两个非空不交的相对闭集，也等价于没有非平凡既开又闭子集。$\partial A = \overline{A} \setminus A^\circ = \varnothing$ 等价于 A 既开又闭，故得边界判据。若不交集合 A, B 覆盖空间且互不碰对方闭包，它们在空间中均闭也均开，给分离判据。最后，非平凡 clopen 集 A 的示性函数 $1_A : X \to \{0, 1\}$ 连续，因为每个离散开集的逆像均开；反之非常值连续函数到 $\{0, 1\}$ 的两个纤维给分离。实数连通集必是区间：若中间缺点 c，两侧相对开集分离它。区间必连通：若分离且 a 在一侧、$b > a$ 在另一侧，取 $s = \sup\{t \in [a, b] : t$ 在第一侧$\}$，利用任一侧的相对开性都矛盾。

8.47 连通的多种判据

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 96 页。

对应已有 gpt-106.pdf 页脚：101。

原课件第 96 页

Connectedness

(8) A subset A of $\mathbb{R}$ is connected if and only if it is an interval.

Proof. Let A is connected. Assume A is not an interval, then we can take three real numbers a_1, a_2, a such that $a_1 < a < a_2$, $a_1 \in A$, $a_2 \in A$ and $a \notin A$. Let $A_1 = A \cap (-\infty, a]$, $A_2 = A \cap [a, \infty)$, then $A = A_1 \cup A_2$ and $A_1 \cap A_2 = \varnothing$. Notice that $(-\infty, a]$ and $[a, \infty)$ are both non-empty closed subsets of $\mathbb{R}$, hence A_1 and A_2 are closed in A, hence A is therefore not connected. Contradiction!
Conversely let A be an interval of $\mathbb{R}$. To derive a contradiction, suppose we could write $A = A_1 \cup A_2$ for nonempty and disjoint closed subsets A_1 and A_2 of A. Pick any $a_1 \in A_1$ and $a_2 \in A_2$ and let $a_1 < a_2$ without loss of generality. Define $a = \sup\{t \in A_1 : t < a_2\}$, and note that $a_1 \leqslant a \leqslant a_2$ and hence $a \in A$ (because A is an interval).
If $a \in A_1$, then $a \neq a_2$ (since $A_1 \cap A_2 = \varnothing$), so $a < a_2$. As $a \notin A_2$ and A_2 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_2$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a, a + \varepsilon)$, $a_1 < a < a_2$, hence $x \notin A_1 \cap A_2$, i.e. $x \notin A$. Contradiction!
If $a \in A_2$, then $a > a_1$. Using similar method, we can also get to the contradiction! □

沿原论证展开与批注 连通指不能分成两个非空不交的相对开集；取补立刻等价于不能分成两个非空不交的相对闭集，也等价于没有非平凡既开又闭子集。$\partial A = \overline{A} \setminus A^\circ = \varnothing$ 等价于 A 既开又闭，故得边界判据。若不交集合 A, B 覆盖空间且互不碰对方闭包，它们在空间中均闭也均开，给分离判据。最后，非平凡 clopen 集 A 的示性函数 $1_A : X \to \{0,1\}$ 连续，因为每个离散开集的逆像均开；反之非常值连续函数到 $\{0,1\}$ 的两个纤维给分离。实数连通集必是区间：若中间缺点 c，两侧相对开集分离它。区间必连通：若分离且 a 在一侧、$b > a$ 在另一侧，取 $s = \sup\{t \in [a,b] : t$ 在第一侧$\}$，利用任一侧的相对开性都矛盾。

8.48 连通的多种判据

来源： 106-ZSM-PPT.pdf，原课件 PDF 第 98–99 页。

对应已有 gpt-106.pdf 页脚：103, 104。

原课件第 98 页

Connectedness

Proof of (1) $\Longrightarrow$ (2): If (2) is false, then there exists non-empty, disjoint and closed sets A_1 and A_2 such that $X = A_1 \cup A_2$. Then $\varnothing = A_1 \cap A_2$, hence $X = \mathbf{C}\varnothing = \mathbf{C}A_1 \cup \mathbf{C}A_2$ where $\mathbf{C}A_1$ and $\mathbf{C}A_2$ are disjoint, non-empty and open sets. Thus (1) is false.
Proof of (2) $\Longrightarrow$ (1): If (1) is false, then there exists non-empty, disjoint and open sets A_1 and A_2 such that $X = A_1 \cup A_2$. Then $\varnothing = A_1 \cap A_2$, hence $X = \mathbf{C}\varnothing = \mathbf{C}A_1 \cup \mathbf{C}A_2$ where $\mathbf{C}A_1$ and $\mathbf{C}A_2$ are disjoint, non-empty and closed sets. Thus (2) is false.
Proof of (1) $\Longrightarrow$ (3) and (2) $\Longrightarrow$ (3): If the clopen subset S of X is a proper subset ($S \notin \{\varnothing, X\}$), then $X = S \cup \mathbf{C}S$, where $\mathbf{C}S$ is a clopen proper subset of X, thus X is not connected.
Proof of (3) $\Longrightarrow$ (1): If (1) is false, then there exist non-empty, disjoint and open sets A_1 and A_2 such that $X = A_1 \cup A_2$, hence $A_2 = \mathbf{C}A_1$ is closed. Thus A_2 a clopen proper subset of X, that is, (3) is false.
Proof of (3) $\Longrightarrow$ (2): If (2) is false, then there exist non-empty, disjoint and closed sets A_1 and A_2 such that $X = A_1 \cup A_2$, hence $A_2 = \mathbf{C}A_1$ is open. Thus A_2 a clopen proper subset of X, that is, (3) is false.

原课件第 99 页

Connectedness

Proof of (3) $\Longleftrightarrow$ (4): $A \subseteq X$ with $\partial A = \varnothing$ if and only if A is a clopen subset of X.
$A \subseteq X$ with $\partial A = \varnothing$ if and only if $\overline{A} = \overset{\circ}{A} \cup \partial(A)$ with $\partial A = \varnothing$ if and only if $\overline{A} = \overset{\circ}{A}$ if and only if $\overline{A} = A = \overset{\circ}{A}$ if and only if A is a clopen subset of X.
Proof of (2) $\Longrightarrow$ (5): If (5) is false, then there exist non-empty separated sets A_1 and A_2 such that $X = A_1 \cup A_2$. Since $\overline{A_1} \cap A_2 = \varnothing$ and $\overline{A_2} \cap A_1 = \varnothing$, then $\overline{A_1} \subseteq A_1$ and $\overline{A_2} \subseteq A_2$, i.e. A_1 and A_2 are non-empty closed. Thus (2) is false.
Proof of (5) $\Longrightarrow$ (3): If (3) is false, then there exist a non-empty clopen proper subset A of X, hence $X = A \cup \mathbf{C}A$ where $\mathbf{C}A$ is a non-empty clopen proper subset of X. Thus A and $\mathbf{C}A$ are non-empty separated sets, that is, (5) is false.
Proof of (3) $\Longrightarrow$ (6): As $\{0\}$ and $\{1\}$ are clopen proper subsets in the space $\{0,1\}$, two-point space endowed with the discrete metric. Then if f is continuous, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are clopen subsets, then $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are either X or $\varnothing$, which implies f is constant.
Proof of (6) $\Longrightarrow$ (3): As $\{0\}$ and $\{1\}$ are clopen proper subsets in the space $\{0,1\}$, two-point space endowed with the discrete metric. Then if f is continuous, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are clopen subsets. Then if f is constant, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are either X or $\varnothing$. Therefore the only clopen subsets of X are X and the empty set. □

沿原论证展开与批注 连通指不能分成两个非空不交的相对开集；取补立刻等价于不能分成两个非空不交的相对闭集，也等价于没有非平凡既开又闭子集。$\partial A = \overline{A} \setminus A^\circ = \varnothing$ 等价于 A 既开又闭，故得边界判据。若不交集合 A, B 覆盖空间且互不碰对方闭包，它们在空间中均闭也均开，给分离判据。最后，非平凡 clopen 集 A 的示性函数 $1_A : X \to \{0,1\}$ 连续，因为每个离散开集的逆像均开；反之非常值连续函数到 $\{0,1\}$ 的两个纤维给分离。实数连通集必是区间：若中间缺点 c，两侧相对开集分离它。区间必连通：若分离且 a 在一侧、$b > a$ 在另一侧，取 $s = \sup\{t \in [a,b] : t \text{ 在第一侧}\}$，利用任一侧的相对开性都矛盾。

9 107-Proof

9.1 连续性的五个等价条件

来源： `107-ZSM-PPT.pdf`，原课件 PDF 第 9–11 页。

对应已有 `gpt-107.pdf` 页脚：12, 13, 14。

原课件第 9 页

Continuous Functions

Proof: The function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|x - x_0\| < \delta \qquad \text{implies} \qquad \|f(x) - f(x_0)\| < \varepsilon,$$

or, the function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in N_\delta(x_0) \qquad \text{implies} \qquad f(x) \in N_\varepsilon(f(x_0)),$$

that is, $f(N_\delta(x_0)) \subseteq N_\varepsilon(f(x_0))$, equivalently, the function f is continuous at the point x_0 if for any neighborhood V of $f(x_0)$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$.

原课件第 10 页

Continuous Functions

Proof of (1) $\Longrightarrow$ (2): For each open set G in Y, we want to check that set $f^{-1}(G)$ is an open set in X.
For any $x_0 \in f^{-1}(G)$, $f(x_0) \in G$. For any neighborhood V of $f(x_0)$ with $f(x_0) \in V \subseteq G$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$, then $x_0 \in U \subseteq f^{-1}(V) \subseteq f^{-1}(G)$.
Proof of (2) $\Longrightarrow$ (1): For any neighborhood V of $f(x_0)$, there exists neighborhood V' of $f(x_0)$ such that $f(x_0) \in V' \subseteq V$, then $x_0 \in f^{-1}(V') \subseteq f^{-1}(V)$ and $f^{-1}(V')$ is an open set.
Proof of (2) $\Longrightarrow$ (3): For each closed set F in Y, $G \equiv Y \setminus F$ is open, then
$f^{-1}(G) = f^{-1}(Y \setminus F) = f^{-1}(Y) \setminus f^{-1}(F) = X \setminus f^{-1}(F)$ is open, then the set $f^{-1}(F)$ is an closed set in X.
Proof of (3) $\Longrightarrow$ (2): For each open set G in Y, $F \equiv Y \setminus G$ is closed, then
$f^{-1}(F) = f^{-1}(Y \setminus G) = f^{-1}(Y) \setminus f^{-1}(G) = X \setminus f^{-1}(G)$ is closed, then the set $f^{-1}(G)$ is an open set in X.

原课件第 11 页

Continuous Functions

Proof of (3) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq \overline{f(A)}$, then $A \subseteq f^{-1}\left(\overline{f(A)}\right)$ and $f^{-1}\left(\overline{f(A)}\right)$ is a closed set in X, hence $\overline{A} \subseteq f^{-1}\left(\overline{f(A)}\right)$. Thus $f\left(\overline{A}\right) \subseteq \overline{f(A)}$.

Proof of (4) $\Longrightarrow$ (3): For each closed set F in Y, define $A = f^{-1}(F)$, then $f(A) = f\left(f^{-1}(F)\right) \subseteq F$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)} \subseteq \overline{F} = F$. Thus $\overline{A} \subseteq f^{-1}(F) = A$, $A = \overline{A}$ is closed, that is, the set $f^{-1}(F)$ is an closed set in X.

Proof of (3) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq f^{-1}\left(\overline{B}\right)$ and $f^{-1}\left(\overline{B}\right)$ is a closed set in X, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (3): For each closed set F in Y, $\overline{f^{-1}(F)} \subseteq f^{-1}\left(\overline{F}\right) = f^{-1}(F)$, then the set $f^{-1}(F)$ is an closed set in X.

Proof of (4) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq X$, then $f\left(\overline{f^{-1}(B)}\right) \subseteq \overline{f\left(f^{-1}(B)\right)} \subseteq \overline{B}$, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq Y$, then $\overline{A} \subseteq \overline{f^{-1}(f(A))} \subseteq f^{-1}\left(\overline{f(A)}\right)$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)}$. $\square$

沿原论证展开与批注 连续的 ε–δ 形式为：对每个 $\varepsilon > 0$ 存在 $\delta > 0$，所有定义域内距 x 小于 δ 的点，其像距 $f(x)$ 小于 ε。否定它可逐个取 $\delta = 1/n$ 构造 $x_n \to x$ 而像距始终至少 ε_0，所以与数列定义等价。连续推出开集逆像开：$f(x) \in G$ 先取像球包含于 G，再用连续性拉回一个输入球。开逆像与闭逆像用补集对偶。闭逆像推出 $f(\overline{A}) \subseteq \overline{f(A)}$：后者的逆像是含 A 的闭集，故含闭包。反向若有闭包包含性质，取 $A = f^{-1}(F)$ 且 F 闭，则 $f(\overline{A}) \subseteq F$，故 $\overline{A} \subseteq A$，逆像闭。不要把像闭包上的横线漏掉。

9.2 实值连续函数的水平集

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 13 页。

对应已有 gpt-107.pdf 页脚：16。

原课件第 13 页

Continuous Functions

Proof: If A function from S to $\mathbb{R}$ is continuous on S, as every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in $\mathbb{R}$, then the inverse image is also closed in S.

Conversely, if the inverse image of every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in S. Assume the function is $f : S \to \mathbb{R}$. So for any open interval of $\mathbb{R}$, the form (a, b), (a, ∞), $(-\infty, a)$, we have

$$f^{-1}(-\infty, a) = f^{-1}(\mathbb{R}) \setminus f^{-1}[a, \infty)$$
$$f^{-1}(a, \infty) = f^{-1}(\mathbb{R}) \setminus f^{-1}(-\infty, a]$$
$$f^{-1}(a, b) = f^{-1}(-\infty, b) \setminus f^{-1}(-\infty, a]$$

are all open in S.

We also have following lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals. So for any open set $O \subseteq \mathbb{R}$, there exists countable open interval $\{I_n; n \in \mathbb{N}\}$ such that $O = \bigcup_{n=1}^{\infty} I_n$. So

$$f^{-1}(O) = \bigcup_{n=1}^{\infty} f^{-1}(I_n)$$

are also open in S. Hence f is continuous. □

沿原论证展开与批注 实值函数连续当且仅所有 $\{f \le a\}$ 与 $\{f \ge a\}$ 闭。必要性是闭集逆像；充分性先取补得 $\{f > a\}, \{f < b\}$ 开，再相交得 $f^{-1}((a,b))$ 开，任意实开集为开区间并，所以逆像开。实开集的极大区间分解见106第53页：不交区间各含不同有理数，因此可数。

9.3 实值连续函数的水平集

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 14 页。

对应已有 gpt-107.pdf 页脚：17。

原课件第 14 页

Continuous Functions

Lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals.

Proof: Let G be a non-empty open set in $\mathbb{R}$. Write $a \sim b$ if the closed interval $[a, b]$ if $a < b$ or $[b, a]$ if $b < a$, lies in G. This is an equivalence relation, in particular $a \sim b$ since $\{a\}$ is itself a closed interval. G is therefore the union of disjoint equivalence classes. Let $C(a)$ be the equivalence class containing a. Let $G = \bigcup_{a \in I} C(a)$ where $C(a) \neq C(a')$ if $a \neq a'$. $C(a)$ is clearly an interval. Also $C(a)$ is open, for if $k \in C(a)$, then $(k - \varepsilon, k + \varepsilon) \subseteq G$ for sufficiently small ε. But then $(k - \varepsilon, k + \varepsilon) \subseteq C(a)$; so G is the union of disjoint intervals. For each $C(a)$ choose a rational q_a, this map $I \to \mathbb{Q} : G(a) \longmapsto q_a$ is injective, so I is countable. □

沿原论证展开与批注 实值函数连续当且仅所有 $\{f \le a\}$ 与 $\{f \ge a\}$ 闭。必要性是闭集逆像；充分性先取补得 $\{f > a\}, \{f < b\}$ 开，再相交得 $f^{-1}((a,b))$ 开，任意实开集为开区间并，所以逆像开。实开集的极大区间分解见106第53页：不交区间各含不同有理数，因此可数。

9.4 紧集的连续像与极值

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 15 页。

对应已有 gpt-107.pdf 页脚：18。

原课件第 15 页

Continuous Functions

One can prove:

(4) Let f be a function from S to T. If f is continuous on S, and if S is compact, then $f(S)$ is compact.

Proof: Let $\bigcup_{\gamma\in\Gamma} U_\gamma$ be an open covering of $f(S)$ on T. Then the sets $f^{-1}(U_\gamma)$ are open and $\left\{f^{-1}(U_\gamma);\gamma\in\Gamma\right\}$ form an open covering of S since we have show

$$f^{-1}\left(\bigcup_{\gamma\in\Gamma} U_\gamma\right)=\bigcup_{\gamma\in\Gamma} f^{-1}(U_\gamma)$$

Since S is compact, there exists a finite subset $\Gamma'\subseteq\Gamma$ such that $\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)$ still forms a covering of S, then

$$f(S)=f\left(\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)\right)=f\left(f^{-1}\left(\bigcup_{\gamma\in\Gamma'} U_\gamma\right)\right)\subseteq\bigcup_{\gamma\in\Gamma'} U_\gamma$$

沿原论证展开与批注 设 K 非空紧，$\{U_i\}$ 覆盖 $f(K)$，则 $\{f^{-1}(U_i)\}$ 为 K 的开覆盖。取有限子覆盖再取像得 $f(K)$ 的有限子覆盖，故像紧。实值时像非空闭有界，令 $M=\sup f(K)$，选像中数 $y_n>M-1/n$，闭性给 $M\in f(K)$，故有 $x_M\in K$ 达到最大。最小同理。M 是值，x_M 是达到该值的点；最优点可以不唯一。

9.5 紧集的连续像与极值

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 16 页。

对应已有 gpt-107.pdf 页脚：19。

原课件第 16 页

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately:

(4′) (Weierstrass' Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is non-empty compact, then $f(S)$ has a maximum and a minimum.

Proof: $f(S)$ is a compact subset of $\mathbb{R}$, hence closed and bounded. Now, any bounded set $A \subseteq \mathbb{R}$ has a least upper bound $\sup A$ and a greatest lower bound $\inf A$, and these two points belong to the closure A. But applying this to $A = f(S)$, which is closed, we conclude that $\sup f(S)$ and $\inf f(S)$ belong to $f(S)$ itself, which is exactly what is being claimed. □

沿原论证展开与批注 设 K 非空紧，$\{U_i\}$ 覆盖 $f(K)$，则 $\{f^{-1}(U_i)\}$ 为 K 的开覆盖。取有限子覆盖再取像得 $f(K)$ 的有限子覆盖，故像紧。实值时像非空闭有界，令 $M = \sup f(K)$，选像中数 $y_n > M - 1/n$，闭性给 $M \in f(K)$，故有 $x_M \in K$ 达到最大。最小同理。M 是值，x_M 是达到该值的点；最优点可以不唯一。

9.6 连通像与介值

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 18 页。

对应已有 gpt-107.pdf 页脚：21。

原课件第 18 页

Continuous Functions

It is easy to prove (using (3)),

(5) Let f be a function from S to T. If f is continuous on S, and if S is connected, then $f(S)$ is connected.

Proof: If $f(S)$ is not connected in Y, then we can find two non-empty and disjoint open subsets V_1 and V_2 of $f(X)$ such that $V_1 \cup V_2 = f(S)$. But then, $f^{-1}(V_1)$ and $f^{-1}(V_2)$ are nonempty open subsets of S. Moreover, these sets are disjoint, and we have $f^{-1}(V_1) \cup f^{-1}(V_2) = f^{-1}(V_1 \cup V_2) = S$. Thus S is not connected. Contradiction! $\square$

沿原论证展开与批注 若 $f(K) = A \cup B$ 是两个非空不交相对开集的分离，则逆像为 K 的非空不交相对开分离，矛盾。实数连通集是区间，所以连续函数的两个函数值之间的每个数也是某个函数值。在 $[a,b]$ 上，若 $f(a)f(b) < 0$ 就存在零点。仅能得存在，不保证唯一；唯一通常还需严格单调等条件。

9.7 连通像与介值

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 19 页。

对应已有 gpt-107.pdf 页脚：22。

原课件第 19 页

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately (see (8) of 1.6.u):

(5′) (Bolzano's Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is connected, then $f(S)$ is an interval.

Thus, if x^1, x^2 are two points of S, and y is a real number such that $f\left(x^1\right) \leqq y \leqq f\left(x^2\right)$, then there is an $x \in S$ such that $f(x) = y$. In other words, f takes on all values between $f\left(x^1\right)$ and $f\left(x^2\right)$. This result is often applied to the case where S is an interval of $\mathbb{R}$.

The Intermediate Value Theorem. Let X be a connected metric space and $f : X \to \mathbb{R}$ a continuous function. If $f(x_1) \leqslant \alpha \leqslant f(x_2)$ for some $x_1 \in X$ and $x_2 \in X$, then there exists a $x \in X$ such that $f(x) = \alpha$.

Proof. $f(X)$ is connected in $\mathbb{R}$. Hence $f(X)$ must be an interval. □

沿原论证展开与批注 若 $f(K) = A \cup B$ 是两个非空不交相对开集的分离，则逆像为 K 的非空不交相对开分离，矛盾。实数连通集是区间，所以连续函数的两个函数值之间的每个数也是某个函数值。在 $[a,b]$ 上，若 $f(a)f(b) < 0$ 就存在零点。仅能得存在，不保证唯一；唯一通常还需严格单调等条件。

9.8 实值半连续的定义与等价判据

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 26 页。

对应已有 gpt-107.pdf 页脚：29。

原课件第 26 页

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open.

Proof. The family $\mathscr{D}^-$ of intervals

$$\{(\lambda, +\infty) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of lower semi-continuity that f is lower semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^-)$. Since $S_\lambda^- = f^{-1}((\lambda, +\infty))$, the theorem stated is equivalent to this result. □

Corollary 1. A numerical function f is lower semi-continuous if and only if, for each number λ, the lower contour set $T_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

沿原论证展开与批注 下半连续指 $f(x) \le \liminf f(x_n)$，允许向下的低点；上半连续指 $f(x) \ge \limsup f(x_n)$，允许向上的高点。下半连续等价于 $\{f > a\}$ 开，也等价于 $\{f \le a\}$ 闭：若 $a < f(x)$，定义给邻域中 $f > a$；反向对任意 $a < f(x)$ 用该开集取邻域。上半连续用 $-f$ 转化，得到 $\{f < a\}$ 开及 $\{f \ge a\}$ 闭。两者同时成立即连续。例 $f(0) = 0, f(x) = 1$ 对 $x \neq 0$ 下半连续但不连续，其负函数上半连续。

9.9 实值半连续的定义与等价判据

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 27 页。

对应已有 gpt-107.pdf 页脚：30。

原课件第 27 页

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open.

Proof. The family $\mathscr{D}^+$ of intervals

$$\{(-\infty, \lambda) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of upper semi-continuity that f is upper semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^+)$. Since $S_\lambda^+ = f^{-1}((-\infty, \lambda))$, the theorem stated is equivalent to this result. □

Corollary 1. A numerical function f is upper semi-continuous if and only if, for each number λ, the upper contour set $T_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

沿原论证展开与批注 下半连续指 $f(x) \le \liminf f(x_n)$，允许向下的低点；上半连续指 $f(x) \ge \limsup f(x_n)$，允许向上的高点。下半连续等价于 $\{f > a\}$ 开，也等价于 $\{f \le a\}$ 闭：若 $a < f(x)$，定义给邻域中 $f > a$；反向对任意 $a < f(x)$ 用该开集取邻域。上半连续用 $-f$ 转化，得到 $\{f < a\}$ 开及 $\{f \ge a\}$ 闭。两者同时成立即连续。例 $f(0) = 0, f(x) = 1$ 对 $x \neq 0$ 下半连续但不连续，其负函数上半连续。

9.10 实值半连续的定义与等价判据

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 28 页。

对应已有 gpt-107.pdf 页脚：31。

原课件第 28 页

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open, i.e. the lower contour set $T_\lambda^- = \mathbf{C}_X S_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)-\epsilon}^- = \{x \in X \mid f(x) > f(x_0) - \epsilon\}$ is an open set and $x \in S_{f(x_0)-\epsilon}^-$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)-\epsilon}^-$, then $f(x) > f(x_0) - \epsilon$ for all $x \in U$. Hence f is lower semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is lower semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$. For any $x_0 \in S_\lambda^-$, $f(x_0) > \lambda$, then let $\epsilon = \dfrac{f(x_0) - \lambda}{2}$, there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $f(x_0) - \epsilon = \dfrac{f(x_0) + \lambda}{2} > \lambda$, $U \subseteq S_\lambda^-$, then S_λ^- is open. □

沿原论证展开与批注 下半连续指 $f(x) \le \liminf f(x_n)$，允许向下的低点；上半连续指 $f(x) \ge \limsup f(x_n)$，允许向上的高点。下半连续等价于 $\{f > a\}$ 开，也等价于 $\{f \le a\}$ 闭：若 $a < f(x)$，定义给邻域中 $f > a$；反向对任意 $a < f(x)$ 用该开集取邻域。上半连续用 $-f$ 转化，得到 $\{f < a\}$ 开及 $\{f \ge a\}$ 闭。两者同时成立即连续。例 $f(0) = 0, f(x) = 1$ 对 $x \neq 0$ 下半连续但不连续，其负函数上半连续。

9.11 实值半连续的定义与等价判据

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 29 页。

对应已有 gpt-107.pdf 页脚：32。

原课件第 29 页

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open, i.e. the upper contour set $T_\lambda^+ = \mathbf{C}_X S_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)+\epsilon}^+ = \{x \in X \mid f(x) < f(x_0) + \epsilon\}$ is an open set and $x \in S_{f(x_0)+\epsilon}^+$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)+\epsilon}^+$, then $f(x) < f(x_0) + \epsilon$ for all $x \in U$. Hence f is upper semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is upper semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$. For any $x_0 \in S_\lambda^+$, $f(x_0) < \lambda$, then let $\epsilon = \dfrac{\lambda - f(x_0)}{2}$, there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $f(x_0) + \epsilon = \dfrac{f(x_0) + \lambda}{2} < \lambda$, $U \subseteq S_\lambda^+$, then S_λ^+ is open. □

沿原论证展开与批注 下半连续指 $f(x) \leq \liminf f(x_n)$，允许向下的低点；上半连续指 $f(x) \geq \limsup f(x_n)$，允许向上的高点。下半连续等价于 $\{f > a\}$ 开，也等价于 $\{f \leq a\}$ 闭：若 $a < f(x)$，定义给邻域中 $f > a$；反向对任意 $a < f(x)$ 用该开集取邻域。上半连续用 $-f$ 转化，得到 $\{f < a\}$ 开及 $\{f \geq a\}$ 闭。两者同时成立即连续。例 $f(0) = 0, f(x) = 1$ 对 $x \neq 0$ 下半连续但不连续，其负函数上半连续。

9.12 半连续函数的极值证明

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 30 页。

对应已有 gpt-107.pdf 页脚：33。

原课件第 30 页

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a lower semi-continuous function f attains in K the value $m = \inf_{x \in K} f(x)$ (in other words, $\inf_{x \in K} f(x)$ is a minimum).

Proof. Let μ be a number such that $\mu > m$. Then $S_\mu^- = \{x \in K \mid f(x) \leqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu > m} S_\mu^- \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = m$. □

沿原论证展开与批注　老师使用闭子集的有限交性质，不是直接用最小化序列。假设 K 非空紧，f 为有限实值下半连续函数。先证明有下界：对每个 $x \in K$，下半连续给邻域 U_x，使 $f(z) > f(x) - 1$ 对 $z \in U_x$ 成立。取有限覆盖 $U_{x_1}, \dots, U_{x_m}$，则 $f(z) > \min_i (f(x_i) - 1)$ 对所有 $z \in K$ 成立。因此 $m = \inf_K f$ 有限。对每个 $\mu > m$，定义 $F_\mu = \{x \in K : f(x) \leq \mu\}$。由确界定义，存在 x 使 $f(x) < \mu$，所以 F_μ 非空；下半连续使它闭。任意有限多个这样的集合的交就是最小阈值对应的一项，仍非空。紧性的有限交性质给 $\bigcap_{\mu > m} F_\mu \neq \varnothing$。取其中 x_0，则 $f(x_0) \leq \mu$ 对每个 $\mu > m$ 成立，故 $f(x_0) \leq m$；又 m 是下界，故 $f(x_0) = m$。

9.13 半连续函数的极值证明

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 31–32 页。

对应已有 gpt-107.pdf 页脚：34, 35。

原课件第 31 页

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a lower semi-continuous function on a compact metric space X. Then f is bounded below and attains its minimum.

Proof. Let $U_\lambda^- = (\lambda, +\infty)$. $\left\{U_\lambda^-; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^-\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^-\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^-\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^-\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^-$$

hence $f(x) \geqq \min\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded below.

原课件第 32 页

Numerical Functions Defined on a Topological Space

Let $m = \inf_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is lower semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_0) - \epsilon < f(x_{n_k})$$

let $k \to \infty$, we have

$$f(x_0) - \epsilon \leqq m.$$

As ϵ is arbitrary, we have $f(x_0) \leqq m$, and by definition of m, we have $f(x_0) \geqq m$, hence x_0 is a minimizer. □

沿原论证展开与批注 非空紧 K 上有限实值下半连续函数先必有下界：若无，取 $f(x_n) < -n$，紧性给子列趋于 $x \in K$，下半连续将推出有限 $f(x) \leq -\infty$，矛盾。设有限 $m = \inf_K f$，选 $f(x_n) < m + 1/n$，再取收敛子列，得到 $f(x) \leq \liminf f(x_n) = m$，而下界给反向，故取到最小值。对 $-f$ 应用此结论得上半连续函数有最大值。原页35趋近的值应为 M 而非 m。不能只从紧集直接说任意函数有界。

9.14 半连续函数的极值证明

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 33 页。

对应已有 gpt-107.pdf 页脚：36。

原课件第 33 页

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a upper semi-continuous function f attains in K the value $M = \sup_{x \in K} f(x)$ (in other words, $\sup_{x \in K} f(x)$ is a maximum).

Proof. Let μ be a number such that $\mu < M$. Then $S_\mu^+ = \{x \in K \mid f(x) \geqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu < M} S_\mu^+ \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = M$. □

沿原论证展开与批注 老师用上水平闭集的有限交性质证明取最大值。先由上半连续性对每个 x 取邻域使附近 $f(z) < f(x) + 1$，紧性给有限子覆盖，因而全域有共同上界。令 $M = \sup_K f$。对每个 $\mu < M$，$F_\mu = \{x \in K : f(x) \geq \mu\}$ 非空且闭；任意有限交是最大阈值对应的一项，仍非空。故紧性给一点 x_0 属于全部 F_μ。于是 $f(x_0) \geq \mu$ 对每个 $\mu < M$ 成立，迫使 $f(x_0) \geq M$；而上界定义给 $f(x_0) \leq M$，所以达到最大值。条件 $K \neq \varnothing$ 及函数实值用于保证有限的 M 有意义。

9.15 半连续函数的极值证明

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 34–35 页。

对应已有 gpt-107.pdf 页脚：37, 38。

原课件第 34 页

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a upper semi-continuous function on a compact metric space X. Then f is bounded above and attains its maximum.

Proof. Let $U_\lambda^+ = (-\infty, \lambda)$. $\left\{U_\lambda^+; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^+\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^+\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^+\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^+\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^+$$

hence $f(x) \leqq \max\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded above.

原课件第 35 页

Numerical Functions Defined on a Topological Space

Let $M = \sup_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is upper semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_{n_k}) < f(x_0) + \epsilon$$

let $k \to \infty$, we have

$$M \leqq f(x_0) + \epsilon.$$

As ϵ is arbitrary, we have $M \leqq f(x_0)$, and by definition of M, we have $f(x_0) \leqq M$, hence x_0 is a maximizer. □

沿原论证展开与批注 非空紧 K 上有限实值下半连续函数先必有下界：若无，取 $f(x_n) < -n$，紧性给子列趋于 $x \in K$，下半连续将推出有限 $f(x) \leq -\infty$，矛盾。设有限 $m = \inf_K f$，选 $f(x_n) < m + 1/n$，再取收敛子列，得到 $f(x) \leq \liminf f(x_n) = m$，而下界给反向，故取到最小值。对 $-f$ 应用此结论得上半连续函数有最大值。原页35趋近的值应为 M 而非 m。不能只从紧集直接说任意函数有界。

9.16 上确界、下确界与有限运算

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 36 页。

对应已有 gpt-107.pdf 页脚：39。

原课件第 36 页

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of lower semi-continuous functions, the function g such that $g(x) = \sup_{i \in I} f_i(x)$ is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) > g(x_0) - \varepsilon.$$

Since f_i is lower semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \implies f_i(x) > f_i(x_0) - \varepsilon > g(x_0) - 2\varepsilon$$

and therefore

$$x \in U(x_0) \implies g(x) \geqq g(x_0) - 2\varepsilon.$$

Hence g is lower semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. □

沿原论证展开与批注　沿老师的确界逼近法。令 $g = \sup_i f_i$，假设 $g(x_0)$ 有限。给 $\varepsilon > 0$，由上确界定义选一个 i 使 $f_i(x_0) > g(x_0) - \varepsilon/2$。下半连续给邻域 U 使 $f_i(x) > f_i(x_0) - \varepsilon/2$。于是对所有 $x \in U$，

$$g(x) \geq f_i(x) > f_i(x_0) - \varepsilon/2 > g(x_0) - \varepsilon.$$

这正是下半连续。原文的 2ε 可通过预先分配 $\varepsilon/2$ 改写成标准精度。不需要所有 i 同时有同一个邻域，只需逼近上确界的那个 i。

9.17 上确界、下确界与有限运算

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 37 页。

对应已有 gpt-107.pdf 页脚：40。

原课件第 37 页

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of upper semi-continuous functions, the function g such that $g(x) = \inf_{i \in I} f_i(x)$ is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) < g(x_0) + \varepsilon.$$

Since f_i is upper semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \implies f_i(x) < f_i(x_0) + \varepsilon < g(x_0) + 2\varepsilon$$

and therefore

$$x \in U(x_0) \implies g(x) \leqq g(x_0) + 2\varepsilon.$$

Hence g is upper semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. □

沿原论证展开与批注 令 $g = \inf_i f_i$，给 $\varepsilon > 0$，先选 i 使 $f_i(x_0) < g(x_0) + \varepsilon/2$，再由 f_i 上半连续取邻域使 $f_i(x) < f_i(x_0) + \varepsilon/2$。于是 $g(x) \le f_i(x) < g(x_0) + \varepsilon$，故上半连续。选近似最优项在先，控制该项的邻域在后，量词顺序不能倒过来。

9.18 上确界、下确界与有限运算

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 38 页。

对应已有 gpt-107.pdf 页脚：41。

原课件第 38 页

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are lower semi-continuous functions then the function $g(x) = \inf_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) > f_n(x_0) - \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) > f_n(x_0) - \varepsilon,$$

whence

$$g(x) = \inf_{n \in I} f_n(x) \geqq \inf_{n \in I} \{f_n(x_0) - \varepsilon\} = g(x_0) - \varepsilon$$

and so g is lower semi-continuous. □

沿原论证展开与批注 若 $g = \sup_i f_i$ 且各 f_i 下半连续，则 $\{g > a\} = \bigcup_i \{f_i > a\}$ 开，故 g 下半连续。上半连续函数的任意下确界对偶。有限下半连续函数的最小值满足 $\{\min_i f_i > a\} = \bigcap_i \{f_i > a\}$，有限交开；上半连续函数有限最大值对偶。无限交不保证开，故不能删掉“有限”。若确界可能无穷，应注明把函数视为扩展实值。

9.19 上确界、下确界与有限运算

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 39 页。

对应已有 gpt-107.pdf 页脚：42。

原课件第 39 页

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are upper semi-continuous functions then the function $g(x) = \sup_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) < f_n(x_0) + \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) < f_n(x_0) + \varepsilon,$$

whence

$$g(x) = \sup_{n \in I} f_n(x) \leqq \sup_{n \in I} \{f_n(x_0) + \varepsilon\} = g(x_0) + \varepsilon$$

and so g is upper semi-continuous. □

沿原论证展开与批注 若 $g = \sup_i f_i$ 且各 f_i 下半连续，则 $\{g > a\} = \bigcup_i \{f_i > a\}$ 开，故 g 下半连续。上半连续函数的任意下确界对偶。有限下半连续函数的最小值满足 $\{\min_i f_i > a\} = \bigcap_i \{f_i > a\}$，有限交开；上半连续函数有限最大值对偶。无限交不保证开，故不能删掉“有限”。若确界可能无穷，应注明把函数视为扩展实值。

9.20 半连续函数的加法与正乘法

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 40 页。

对应已有 gpt-107.pdf 页脚：43。

原课件第 40 页

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two lower semi-continuous numerical functions, the function $f_1 + f_2$ is lower semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) > a\} = \bigcup_{\lambda} \{x \mid f_1(x) > a - \lambda\} \cap \{x \mid f_2(x) > \lambda\}$$

is a union of open sets and so is open. □

沿原论证展开与批注 老师将和函数的水平集拆成开集的并。以 $f+g>a$ 为目标，有

$$\{x : f(x)+g(x) > a\} = \bigcup_{\lambda \in \mathbb{R}} \big(\{x : f(x) > a-\lambda\} \cap \{x : g(x) > \lambda\}\big).$$

右边属于某项时，两个不等式相加立即得左边。反向若和严格大于 a，就在 $g(x)$ 与 $a-f(x)$ 之间选一个实数 λ，使两个所需严格不等式同时成立。每项是两个开水平集的交，因此开；任意并仍开，所以和下半连续。并集中的 λ 可以随 x 而变，不要求全域共用同一个值。

9.21 半连续函数的加法与正乘法

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 41 页。

对应已有 gpt-107.pdf 页脚：44。

原课件第 41 页

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two upper semi-continuous numerical functions, the function $f_1 + f_2$ is upper semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) < a\} = \bigcup_{\lambda} \{x \mid f_1(x) < a - \lambda\} \cap \{x \mid f_2(x) < \lambda\}$$

is a union of open sets and so is open. □

沿原论证展开与批注 老师将和函数的水平集拆成开集的并。以 $f+g<a$ 为目标，有

$$\{x : f(x)+g(x)<a\} = \bigcup_{\lambda\in\mathbb{R}} \big(\{x : f(x)<a-\lambda\} \cap \{x : g(x)<\lambda\}\big).$$

右边属于某项时，两个不等式相加立即得左边。反向若和严格小于 a，就在 $g(x)$ 与 $a-f(x)$ 之间选一个实数 λ，使两个所需严格不等式同时成立。每项是两个开水平集的交，因此开；任意并仍开，所以和上半连续。并集中的 λ 可以随 x 而变，不要求全域共用同一个值。

9.22 半连续函数的加法与正乘法

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 42 页。

对应已有 gpt-107.pdf 页脚：45。

原课件第 42 页

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive lower semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is lower semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) > \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) > \lambda\} \cap \left\{x \mid f_2(x) > \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. □

沿原论证展开与批注 假设 f, g 严格正，老师按乘积水平集证明。对 $a > 0$，

$$\{x : f(x)g(x) > a\} = \bigcup_{\lambda>0} \big(\{x : f(x) > \lambda\} \cap \{x : g(x) > a/\lambda\}\big).$$

右边到左边由正数不等式相乘。左边到右边，在正数 $f(x)$ 与 $a/g(x)$ 之间选 λ：若乘积大于 a，选 $a/g(x) < \lambda < f(x)$；若乘积小于 a，选 $f(x) < \lambda < a/g(x)$。这样严格验证集合等式。每项开，故并开。还须处理 $a \le 0$：乘积严格正，所以大于 a 的水平集是全域，小于 a 的水平集为空，均开。因此乘积下半连续。正性用于除法及保持不等号方向。

9.23 半连续函数的加法与正乘法

来源： 107-ZSM-PPT.pdf，原课件 PDF 第 43 页。

对应已有 gpt-107.pdf 页脚：46。

原课件第 43 页

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive upper semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is upper semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) < \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) < \lambda\} \cap \left\{x \mid f_2(x) < \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. □

沿原论证展开与批注 假设 f, g 严格正，老师按乘积水平集证明。对 $a > 0$，

$$\{x : f(x)g(x) < a\} = \bigcup_{\lambda>0} \big(\{x : f(x) < \lambda\} \cap \{x : g(x) < a/\lambda\}\big).$$

右边到左边由正数不等式相乘。左边到右边，在正数 $f(x)$ 与 $a/g(x)$ 之间选 λ：若乘积大于 a，选 $a/g(x) < \lambda < f(x)$；若乘积小于 a，选 $f(x) < \lambda < a/g(x)$。这样严格验证集合等式。每项开，故并开。还须处理 $a \le 0$：乘积严格正，所以大于 a 的水平集是全域，小于 a 的水平集为空，均开。因此乘积上半连续。正性用于除法及保持不等号方向。

10 108-Proof

10.1 上半连续与闭图：双向证明

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 11 页。

对应已有 gpt-108.pdf 页脚：17。

原课件第 11 页

Continuous Correspondences

h. Semicontinuities and continuity on S are defined as semicontinuities and continuity at every point of S. It is clear that:

(1) The correspondence φ is upper semicontinuous on S if and only if its graph is closed in $S \times T$.

Proof. If the graph $G(\varphi)$ is closed in $S \times T$, then the correspondence φ is upper semicontinuous on S. For any $x^0 \in S$, if $x^q \to x^0$, $y^q \in \varphi(x^q)$, and $y^q \to y^0$, then $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$. As $G(\varphi)$ is closed, then $(x^0, y^0) \in G(\varphi)$ and hence $y^0 \in \varphi(x^0)$.

Proof. If the correspondence φ is upper semicontinuous on S, then its graph is closed in $S \times T$. If $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$, then $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q)$. As φ is upper semicontinuous on x^0, then $y^0 \in \varphi(x^0)$ and hence $(x^0, y^0) \in G(\varphi)$. □

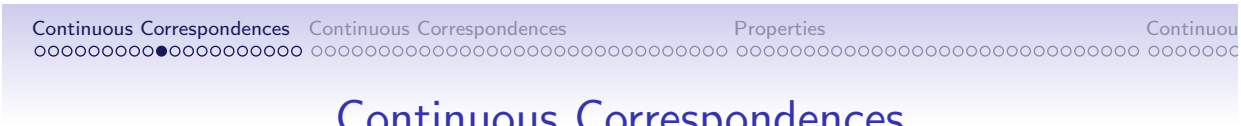
沿原论证展开与批注 本页先采用老师前面给出的闭图式上半连续定义：$x_n \to x, y_n \to y, y_n \in \Phi(x_n) \Rightarrow y \in \Phi(x)$。图为 $G = \{(x, y) : y \in \Phi(x)\}$。**闭图推出该序列条件：**任取满足条件的两列，$(x_n, y_n) \in G$ 且在乘积空间趋于 (x, y)。由 G 闭得 $(x, y) \in G$，按图的定义即 $y \in \Phi(x)$。**反向：**任取图中收敛点列 $(x_n, y_n) \to (x, y)$。乘积收敛给两个坐标收敛，图中归属给 $y_n \in \Phi(x_n)$，于是上半连续定义给 $y \in \Phi(x)$，即极限仍在图中，故图闭。注意不能删除结果列收敛这个前提。在本节共同紧陪域的背景下，这个闭图判据与后面开集式上半连续相协调；离开该背景时应检查紧性及闭值条件。

10.2 连续函数与对应的复合

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 12 页。

对应已有 gpt-108.pdf 页脚：18。

原课件第 12 页



Continuous Correspondences

i. As in 1.7.c let S_1, S_2, T be subsets of $\mathbb{R}^{K_1}$, $\mathbb{R}^{K_2}$, $\mathbb{R}^{L}$ respectively, f be a function from S_1 to S_2, and φ be a correspondence from S_2 to T. Define a correspondence ψ from S_1 to T by $\psi(x) = \varphi(f(x))$ for every x in S_1. It is immediate that:

(2) If f is continuous at the point x of S_1, and φ is upper semicontinuous (resp. lower semicontinuous) at the point $f(x)$ of S_2, then ψ is upper semicontinuous (resp. lower semicontinuous) at x.

Proof. If $x^q \to x$, $y^q \in \psi(x^q)$, and $y^q \to y$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$,
$y^q \in \psi(x^q) = \varphi(f(x^q)) \equiv \varphi(z^q)$. As φ is upper semicontinuous at the point $z \equiv f(x)$, then $y \in \varphi(z) = \varphi(f(x)) = \psi(x)$. Hence ψ is upper semicontinuous at x.
Proof. If $x^q \to x$, and $y \in \varphi(x)$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$, $y \in \psi(x) = \varphi(f(x)) \equiv \varphi(z)$. As φ is lower semicontinuous at the point $z \equiv f(x)$, then there is (y^q) such that $y^q \in \varphi(z^q) \equiv \varphi(f(x^q)) = \psi(x^q)$ and $y^q \to y$. Hence ψ is lower semicontinuous at x. □

沿原论证展开与批注 令 $\Psi = \Phi \circ f$，即 $\Psi(x) = \Phi(f(x))$。**上半连续的闭图式证明：** 给定 $x_n \to x, y_n \to y, y_n \in \Psi(x_n)$，先由 f 连续令 $z_n = f(x_n) \to f(x) = z$。再把归属重写为 $y_n \in \Phi(z_n)$；由 Φ 在 z 的条件得 $y \in \Phi(z) = \Psi(x)$。 **下半连续：** 给定 $x_n \to x$ 及固定 $y \in \Psi(x)$，于是 $y \in \Phi(f(x))$。由 f 连续及 Φ 下半连续，能选 $y_n \in \Phi(f(x_n)) = \Psi(x_n)$ 且 $y_n \to y$，正是所需。原页下半连续段开头的 $y \in \Phi(x)$ 应改为 $y \in \Psi(x)$，否则定义域可能不匹配。上半连续是验证任意已有的收敛选择，下半连续则需要主动构造逼近给定旧选择的输出列。

10.3 积对应的证明逐步写

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 14 页。

对应已有 gpt-108.pdf 页脚：20。

原课件第 14 页

Continuous Correspondences

Proof. If $x^q \to x$, $y^q \in \varphi(x^q)$, and $y^q \to y$, then $(y_1^q, \cdots, y_N^q) = y^q \in \varphi(x^q) = \prod_{n=1}^{N} \varphi_n(x^q)$ and $(y_1^q, \cdots, y_N^q) = y^q \to y = (y_1, \cdots, y_N)$, that is, $y_n^q \in \varphi_n(x^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. As φ_n is upper semicontinuous at x, then $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. Therefore $y = (y_1, \cdots, y_N) \in \prod_{n=1}^{N} \varphi_n(x) = \varphi(x)$. Hence φ is upper semicontinuous at x.

Proof. If $x^q \to x$, and $y \in \varphi(x)$, then $(y_1, \cdots, y_N) = y \in \varphi(x) = \prod_{n=1}^{N} \varphi_n(x)$, that is, $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. As φ_n is lower semicontinuous at the point x, then there is (y_n^q) such that $y_n^q \in \varphi_n(z^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. Thus $y^q = (y_1^q, \cdots, y_N^q) \in \prod_{n=1}^{N} \varphi_n(x^q) = \varphi(x^q)$ and $y^q = (y_1^q, \cdots, y_N^q) \to (y_1, \cdots, y_N) = y$. Hence φ is lower semicontinuous at x. □

沿原论证展开与批注 上半连续：任取 $x_q \to x$、$y^q \in \varphi(x_q)$、$y^q \to y$。对每个 i，有 $y_i^q \in \varphi_i(x_q)$ 且 $y_i^q \to y_i$；上半连续得 $y_i \in \varphi_i(x)$，全部坐标同时成立即 $y \in \prod_i \varphi_i(x)$。

下半连续：给定 $y = (y_1, \ldots, y_N) \in \varphi(x)$。对每个 i，下半连续允许选 $y_i^q \in \varphi_i(x_q)$ 趋于 y_i；合成 $y^q = (y_1^q, \ldots, y_N^q)$，它可行，且 $\|y^q - y\|^2 = \sum_i |y_i^q - y_i|^2 \to 0$。**纠正：** 课件的 $\varphi_n(z^q)$ 应为 $\varphi_n(x^q)$。

10.4 最优值连续的完整证明与原页纠错

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 19 页。

对应已有 gpt-108.pdf 页脚：25。

原课件第 19 页

Continuous Correspondences

Proof. Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is lower semi-continuous continuous (l.s.c.) at $x_0 \in S$, for any $y_0 \in \varphi(x_0)$, there exists $(y_n) \subseteq T$ such that $y_n \in \varphi(x_n)$ and $y_n \to y^0$. By the continuity of f, we have $f(x_n, y_n) \to f(x_0, y_0)$. Since $f(x_n, y_n) \leqq g(x_n)$, then

$$f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \leqq \underline{\lim}_{n\to\infty} g(x_n)$$

and hence $g(x_0) \leqq \underline{\lim}_{n\to\infty} g(x_n)$. Therefore g is lower semi-continuous continuous (l.s.c.) at x_0.

Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, for any sequence $(y_n \in \varphi(x_n))$, the exists a subsequence $(y_{n_k} \in \varphi(x_{n_k}))$ such that $y_{n_k} \to y_0$ and $y^0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then

$$g(x_0) = f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \geqq \overline{\lim}_{n\to\infty} g(x_n).$$

Therefore g is upper semi-continuous continuous (u.s.c.) at x_0.

沿原论证展开与批注 固定任意 $x_n \to x_0$，记 $g_0 = g(x_0)$。**先证明下界**。取一个最优点 $y_0 \in \mu(x_0)$。由 φ 下半连续，可选 $z_n \in \varphi(x_n)$ 使 $z_n \to y_0$。连续性给

$$f(x_n, z_n) \to f(x_0, y_0) = g_0, \qquad g(x_n) \geq f(x_n, z_n),$$

故 $\liminf_n g(x_n) \geq g_0$。这里用 $\liminf$，因为还没有证明 $g(x_n)$ 的极限存在。

再证明上界。若 $\limsup_n g(x_n) > g_0$，则存在 $\varepsilon > 0$ 和子列，使 $g(x_{n_k}) \geq g_0 + \varepsilon$。取最优点 $y_k \in \mu(x_{n_k})$，所以 $f(x_{n_k}, y_k) = g(x_{n_k})$。共同紧 T 可抽子列 $y_{k_j} \to y$，闭图给 $y \in \varphi(x_0)$。取极限得

$$f(x_0, y) = \lim_j f(x_{n_{k_j}}, y_{k_j}) \geq g_0 + \varepsilon,$$

但可行性要求 $f(x_0, y) \leq g_0$，矛盾。于是 $\limsup_n g(x_n) \leq g_0$，上下夹住得 $g(x_n) \to g_0$。一般紧值 u.s.c. 情形用第87页的子列结论代替共同紧 T。

纠正： 原页把未知是否存在的 $\lim g(x_n)$ 直接写出来，并把任意可行 y_n 当成最优点；还过早写 $g(x_0) = f(x_0, y_0)$。以上证明逐项补上所需理由。

10.5 最优解对应为何紧且上半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 20 页。

对应已有 gpt-108.pdf 页脚：26。

原课件第 20 页

Continuous Correspondences

Proof. For each x_0, f is continuous on $S \times T$, and if φ is compact-valued continuous at $x_0 \in S$, then $\mu(x_0)$ is non-empty and compact-valued from the Extreme Value Theorem.

Let $x_n \to x_0$, $y_n \in \mu(x_n)$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, then there exists $(y_{n_k}) \subseteq T$ such that $y_{n_k} \in \varphi(x_{n_k})$ and $y_{n_k} \to y_0$ and $y_0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. By the continuity of g, we have $g(x_{n_k}) \to g(x_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then $f(x_0, y_0) = g(x_0)$, so $y_0 \in \mu(x_0)$. Thus μ is upper semicontinuous (u.s.c.) at x_0. □

沿原论证展开与批注 对固定 x，极值定理给最优点存在。$\mu(x) = \varphi(x) \cap \{y : f(x,y) = g(x)\}$ 是紧集的闭子集，所以紧。接着任取 $x_n \to x_0$、$y_n \in \mu(x_n)$、$y_n \to y_0$。因为 $\mu(x_n) \subseteq \varphi(x_n)$，闭图给 $y \in \varphi(x_0)$；又

$$f(x_n, y_n) = g(x_n) \implies f(x_0, y) = g(x_0),$$

故 $y \in \mu(x_0)$，最优图闭。

在共同紧 T 中闭图即上半连续。一般情形：若某开集 $G \supseteq \mu(x_0)$ 不能包含附近全部最优点，就可选 $x_n \to x_0$、$y_n \in \mu(x_n) \setminus G$；紧值 u.s.c. 的 φ 给收敛子列，刚才论证使其极限在 $\mu(x_0) \subseteq G$，但各点在闭集 $T \setminus G$ 中，矛盾。只证明“某个好子列存在”而不说明怎样排除坏序列，容易留下逻辑缺口。

10.6 最优值连续而最优解不下半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 21 页。

对应已有 gpt-108.pdf 页脚：27。

原课件第 21 页

Continuous Correspondences

Example: Let S and T be the real interval $[0,1]$. Define φ by $\varphi(x) = [0,1]$ for all $x \in S$, and f by $f(x,y) = xy$. For $x \neq 0$, $\mu(x)$ consists of the single element 1; for $x = 0$, $\mu(x) = [0,1]$.

$$\mu(x) = \begin{cases} [0,1], & \text{for} \quad x = 0, \\ \{1\}, & \text{for} \quad x \neq 0. \end{cases}$$

(It will be helpful to draw the graph of μ.) For all x, $g(x) = x$.

$$g(x) = \max\{xy \mid y \in [0,1]\} = x \max\{y \mid y \in [0,1]\} = x.$$

沿原论证展开与批注 $S = T = [0,1]$，$\varphi(x) = [0,1]$，$f(x,y) = xy$。$x > 0$ 时乘以正数保序，所以唯一最优点为1；$x = 0$ 时所有收益都为0，故 $\mu(0) = [0,1]$。$g(x) = x$ 连续。

μ 在0上半连续，因为任何附近最优点的极限都在 $[0,1]$。但不下半连续：取 $x_n = 1/n$ 和目标 $y_0 = 0 \in \mu(0)$，每个 $y_n \in \mu(x_n)$ 都只能等于1，不能趋于0。这个反例说明最大值定理的结论已经是恰当强度，不能擅自把 u.s.c. 改为连续。

10.7 逆像刻画的证明

来源： `108-ZSM-PPT.pdf`，原课件 PDF 第 29–30 页。

对应已有 `gpt-108.pdf` 页脚：35, 36。

原课件第 29 页

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^{-}(G)$ is a open set in S. If $x_0 \in \varphi^{-}(G)$, then $\varphi(x_0) \cap G \neq \varnothing$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

i.e. $U(x_0) \subseteq \varphi^{-}(G)$. Therefore $\varphi^{-}(G)$ is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^{-}(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^{-}(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^{-}(G)$, that is, if G meets $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$, then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, thus φ is lower semi-continuous (l.s.c.) at x_0.

原课件第 30 页

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq F\} \\
&= \mathbf{C}_S\left(\varphi^+(F)\right)
\end{aligned}$$

is open, that is, $\varphi^+(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^-(G)$ is open. □

沿原论证展开与批注 下逆像 $\Phi^-(V) = \{x : \Phi(x) \cap V \neq \varnothing\}$。下半连续当且仅开集的下逆像开：若某点属于该逆像，定义给整个邻域也属于它；反向取逆像本身为邻域。上逆像 $\Phi^+(V) = \{x : \Phi(x) \subseteq V\}$，同样论证得到上半连续的开逆像判据。非空值下的补集对偶把两种开集判据换成闭集判据；允许空值时应先限制到有效定义域。

10.8 逆像刻画的证明

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 32–33 页。

对应已有 gpt-108.pdf 页脚：38, 39。

原课件第 32 页

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^+(G)$ is a open set in S. If $x_0 \in \varphi^+(G)$, then $\varphi(x_0) \subseteq G$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

i.e. $U(x_0) \subseteq \varphi^+(G)$. Therefore $\varphi^+(G)$ is a neighbourhood of each of its points and so is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^+(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^+(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$, that is, if G contains $\varphi(x_0)$ [$\varphi(x_0) \subseteq G$], then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, thus φ is upper semi-continuous (u.s.c.) at x_0.

原课件第 33 页

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(F)\right)
\end{aligned}$$

is open, that is, $\varphi^-(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq G\} \\
&= \mathbf{C}_S\left(\varphi^+(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^+(G)$ is open. □

沿原论证展开与批注 下逆像 $\Phi^-(V) = \{x : \Phi(x) \cap V \neq \varnothing\}$。下半连续当且仅开集的下逆像开：若某点属于该逆像，定义给整个邻域也属于它；反向取逆像本身为邻域。上逆像 $\Phi^+(V) = \{x : \Phi(x) \subseteq V\}$，同样论证得到上半连续的开逆像判据。非空值下的补集对偶把两种开集判据换成闭集判据；允许空值时应先限制到有效定义域。

10.9 紧值上半连续对应的紧像

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 34 页。

对应已有 gpt-108.pdf 页脚：40。

原课件第 34 页

Semi-continuous Mappings

Upper semi-continuous (u.s.c.) correspondences of S into T have the following fundamental property:

Theorem 1.3. If $\varphi : S \to T$ is a compact-valued u.s.c. correspondence, then the image $\varphi(K)$ of a compact subset K of S is also compact.

Proof. Let $\{G_\lambda \mid \lambda \in \Lambda\}$ be an open covering of $\varphi(K)$. If $x \in K$, the set $\varphi(x)$, which is compact, can be covered by a finite number of the $G_\lambda(x)$ for $\lambda \in \Lambda$, $\varphi(x) \subseteq \bigcup_{\lambda\in\Lambda} G_\lambda(x)$; let G_x denote the union of the sets in such a finite family, $G_x = \bigcup_{\lambda\in\Lambda} G_\lambda(x)$, then $\varphi(K) \subseteq \bigcup_{x\in K} \varphi(x) \subseteq \bigcup_{x\in K} G_x$. Then $\{\varphi^+(G_x) \mid x \in K\}$ is an open covering of K and so it contains a finite covering $\varphi^+(G_{x_1}), \cdots, \varphi^+(G_{x_n}), \cdots, \varphi^+(G_{x_N})$. $K \subseteq \bigcup_{n=1}^{N} \varphi^+(G_{x_n})$. The sets $G_{x_1}, \cdots, G_{x_n}, \cdots, G_{x_N}$ cover $\varphi(K)$ and so $\varphi(K)$ can be covered by a finite number of the G_{x_n} for $n = 1, \cdots, N$. That is, $\varphi(K) \subseteq \bigcup_{n=1}^{N} G_{x_n}$.

□

沿原论证展开与批注 设 K 紧且 Φ 紧值上半连续。给 $\Phi(K)$ 的任意开覆盖，对每个 $x \in K$ 先有限覆盖紧集 $\Phi(x)$，把对应有限个开集的并记作 V_x。上半连续给 U_x 使 $\Phi(U_x) \subset V_x$。再用 K 紧，从 U_x 取有限子覆盖。两次有限选择的总开集数仍有限，覆盖 $\Phi(K)$。两个紧性分别作用于“每个值”和“输入集”。

10.10 闭对应与闭图

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 39 页。

对应已有 gpt-108.pdf 页脚：45。

原课件第 39 页

Semi-continuous Mappings

Proposition 1.2. Let S and T be two metric spaces. A mapping $\varphi : S \rightrightarrows T$ is a closed correspondence if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Theorem 1.4. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Proof. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $(x_n, y_n) \longrightarrow (x_0, y_0)$ and $(x_n, y_n) \in G(\varphi)$, then $(x_0, y_0) \in G(\varphi)$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$. □

沿原论证展开与批注 闭图指若 $x_n \to x, y_n \to y, y_n \in \Phi(x_n)$，则 $y \in \Phi(x)$。等价于图在乘积空间闭；也等价于图外每个 (x, y) 可找到乘积开邻域与图不交。固定 x 用常数输入得闭图蕴含闭值。连续 $f(x, y)$ 的弱不等式定义对应有闭图，因为不等式保极限。任意闭图对应的逐点交为图的任意闭交，有限并为有限闭并，有限积由坐标判据闭。闭值本身不推出闭图，例如单值跳跃函数。

10.11 闭对应与闭图

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 40 页。

对应已有 gpt-108.pdf 页脚：46。

原课件第 40 页

Properties of Semi-continuous Mappings

Proposition 1.3. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n : S \rightrightarrows \prod_{n=1}^{N} T_n$ is a closed correspondence.

Proof 1. If $\left(y_1^0, \cdots, y_N^0\right) = y^0 \notin \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then there exists $n_0 = 1, \cdots, N$ such that $y_{n_0}^0 \notin \varphi_{n_0}\left(x^0\right)$. Therefore there exist neighbourhoods $U_{n_0}\left(x^0\right)$ and $V_{n_0}\left(y_{n_0}^0\right)$ such that $x \in U_{n_0}\left(x^0\right)$ implies $\varphi_{n_0}(x) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$, that is, $\varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $U\left(x^0\right) = \bigcap_{n=1}^{N} U_n\left(x^0\right)$, then $x \in U\left(x_0\right)$ implies

$$\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) \subseteq \varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$$

and hence $\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $V\left(y^0\right) = \prod_{n=1}^{N} V_n\left(y_n^0\right)$, then

$$\varphi\left(U\left(x^0\right)\right) \cap V\left(y^0\right) = \prod_{n=1}^{N}\left[\varphi_n\left(U\left(x^0\right)\right) \cap V_n\left(y_n^0\right)\right] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $\left(y_1^q, \cdots, y_N^q\right) = y^q \to y^0 = \left(y_1^0, \cdots, y_N^0\right)$, and $\left(y_1^q, \cdots, y_N^q\right) = y^q \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$, then $y_n^q \to y_n^0$ and $y_n^q \in \varphi_n\left(x^q\right)$ for $n = 1, \cdots, N$. If every φ_n is closed at the point x_0 of S, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$, and hence $y^0 = \left(y_1^0, \cdots, y_N^0\right) \in \prod_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \prod_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

沿原论证展开与批注 闭图指若 $x_n \to x, y_n \to y, y_n \in \Phi(x_n)$，则 $y \in \Phi(x)$。等价于图在乘积空间闭；也等价于图外每个 (x, y) 可找到乘积开邻域与图不交。固定 x 用常数输入得闭图蕴含闭值。连续 $f(x, y)$ 的弱不等式定义对应有闭图，因为不等式保极限。任意闭图对应的逐点交为图的任意闭交，有限并为有限闭并，有限积由坐标判据闭。闭值本身不推出闭图，例如单值跳跃函数。

10.12 闭对应与闭图

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 41 页。

对应已有 gpt-108.pdf 页脚：47。

原课件第 41 页

Semi-continuous Mappings

Proposition 1.4. Let S and T be two metric spaces. If $\varphi : S \rightrightarrows T$ is a closed correspondence at x, then the set $\varphi(x)$ is closed in T.

Proof: Let y_n be a sequence in $\varphi(x)$ with $y_n \to y$. Define $x_n = x$ for all n, so $x_n \to x$ and $y_n \in \varphi(x_n)$. Since φ is a closed correspondence at x, we have $y \in \varphi(x)$. This shows that $\varphi(x)$ is a closed set. □

沿原论证展开与批注 闭图指若 $x_n \to x, y_n \to y, y_n \in \Phi(x_n)$，则 $y \in \Phi(x)$。等价于图在乘积空间闭；也等价于图外每个 (x, y) 可找到乘积开邻域与图不交。固定 x 用常数输入得闭图蕴含闭值。连续 $f(x, y)$ 的弱不等式定义对应有闭图，因为不等式保极限。任意闭图对应的逐点交为图的任意闭交，有限并为有限闭并，有限积由坐标判据闭。闭值本身不推出闭图，例如单值跳跃函数。

10.13 闭对应与闭图

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 42 页。

对应已有 gpt-108.pdf 页脚：48。

原课件第 42 页

Semi-continuous Mappings

Example 1.1. If f is a continuous numerical function in $S \times T$, the correspondence $\varphi : S \rightrightarrows T$ defined by $\varphi(x) = \{y \in T \mid f(x, y) \leqq 0\}$ is a closed correspondence of S into T, for, in $S \times T$, the graphical representation

$$G(\varphi) = \{(x, y) \in S \times T \mid f(x, y) \leqq 0\}$$

is a closed set.

In particular, if λ is a continuous numerical function in a metric space (X, d), the mapping

$$\varphi(x) = B_{\lambda(x)}(x) = \{y \in X \mid d(x, y) - \lambda(x) \leqq 0\}$$

is closed.

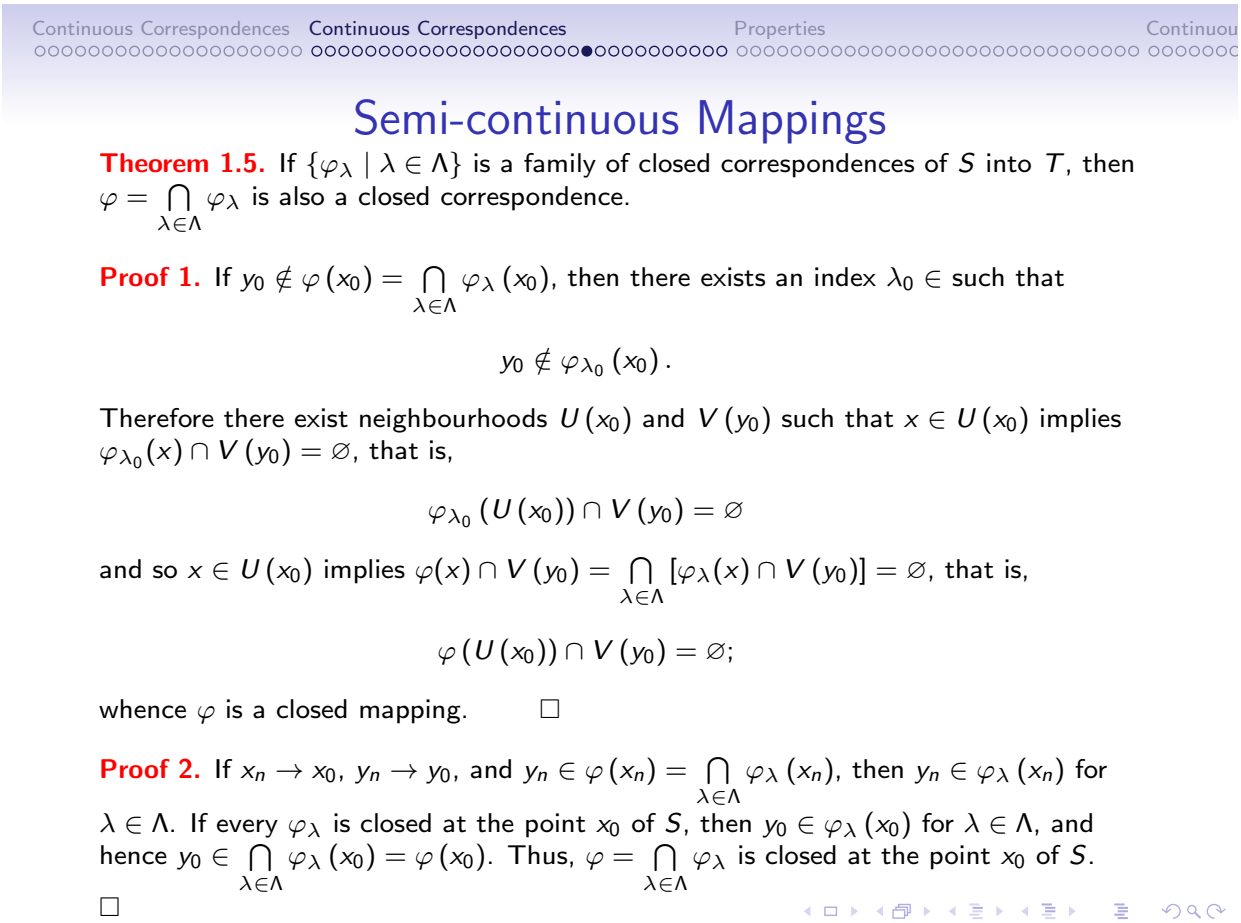
沿原论证展开与批注 闭图指若 $x_n \to x, y_n \to y, y_n \in \Phi(x_n)$，则 $y \in \Phi(x)$。等价于图在乘积空间闭；也等价于图外每个 (x, y) 可找到乘积开邻域与图不交。固定 x 用常数输入得闭图蕴含闭值。连续 $f(x, y)$ 的弱不等式定义对应有闭图，因为不等式保极限。任意闭图对应的逐点交为图的任意闭交，有限并为有限闭并，有限积由坐标判据闭。闭值本身不推出闭图，例如单值跳跃函数。

10.14 闭对应与闭图

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 43 页。

对应已有 gpt-108.pdf 页脚：49。

原课件第 43 页



沿原论证展开与批注 闭图指若 $x_n \to x, y_n \to y, y_n \in \Phi(x_n)$，则 $y \in \Phi(x)$。等价于图在乘积空间闭；也等价于图外每个 (x,y) 可找到乘积开邻域与图不交。固定 x 用常数输入得闭图蕴含闭值。连续 $f(x,y)$ 的弱不等式定义对应有闭图，因为不等式保极限。任意闭图对应的逐点交为图的任意闭交，有限并为有限闭并，有限积由坐标判据闭。闭值本身不推出闭图，例如单值跳跃函数。

10.15 闭对应与闭图

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 44 页。

对应已有 gpt-108.pdf 页脚：50。

原课件第 44 页

Properties of Semi-continuous Mappings

Proposition 1.5. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is a closed correspondence.

Proof 1. If $y_0 \notin \varphi(x_0) = \bigcup_{n=1}^{N} \varphi_n(x_0)$, then, for all $n = 1, \cdots, N$,

$$y_0 \notin \varphi_n(x_0).$$

Therefore there exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi_n(x) \cap V(y_0) = \varnothing$, that is,

$$\varphi_n(U(x_0)) \cap V(y_0) = \varnothing$$

and so $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(x) \cap V(y_0)] = \varnothing$, that is,

$$\varphi(U(x_0)) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(U(x_0)) \cap V(y_0)] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q) = \bigcup_{n=1}^{N} \varphi_n(x^q)$, then there exists $n_0 \in \{1, \cdots, N\}$ such that $y^q \in \varphi_{n_0}(x^q)$. If every φ_n is closed at the point x_0 of S, then $y^0 \in \varphi_{n_0}\left(x^0\right)$, and hence $y^0 \in \bigcup_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

沿原论证展开与批注 闭图指若 $x_n \to x, y_n \to y, y_n \in \Phi(x_n)$，则 $y \in \Phi(x)$。等价于图在乘积空间闭；也等价于图外每个 (x, y) 可找到乘积开邻域与图不交。固定 x 用常数输入得闭图蕴含闭值。连续 $f(x, y)$ 的弱不等式定义对应有闭图，因为不等式保极限。任意闭图对应的逐点交为图的任意闭交，有限并为有限闭并，有限积由坐标判据闭。闭值本身不推出闭图，例如单值跳跃函数。

10.16 闭图与上半连续之间的紧性桥梁

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 45 页。

对应已有 gpt-108.pdf 页脚：51。

原课件第 45 页

Semi-continuous Mappings

Theorem 1.6. Every compact-valued u.s.c. correspondence is closed.

Proof. Let φ be a upper semi-continuous (u.s.c.) correspondence of S into T and suppose that $y_0 \notin \varphi(x_0)$. Since $\varphi(x_0)$ is compact, there exists an open set G in T containing $\varphi(x_0)$ and a neighbourhood $V(y_0)$ such that

$$G \cap V(y_0) = \varnothing.$$

Since φ is a upper semi-continuous (u.s.c.) correspondence, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G.$$

Then

$$x \in U(x_0) \Longrightarrow \varphi(x) \cap V(y_0) = \varnothing$$

and φ is therefore closed. □

沿原论证展开与批注 紧值上半连续对应有闭图：若 $y \notin \Phi(x)$，紧值闭且 $\operatorname{dist}(y, \Phi(x)) > 0$，取分离开邻域，再用上半连续，附近图不能靠近 (x, y)。若 Ψ 闭图，Φ 紧值上半连续，则 $\Phi \cap \Psi$ 紧值上半连续。证明反证：若有 $x_n \to x$ 和交值 y_n 逃出包含交值的开集 G，紧像性质给子列 $y_n \to y$，两图闭使 y 属于交值，开性使子列最终进 G，矛盾。值域空间紧时，闭图亦推出上半连续：逃出开集的选择总能抽取收敛子列。

10.17 闭图与上半连续之间的紧性桥梁

来源： `108-ZSM-PPT.pdf`，原课件 PDF 第 47–49 页。

对应已有 `gpt-108.pdf` 页脚：53, 54, 55。

原课件第 47 页

Semi-continuous Mappings

Proof. For each x, the set $\varphi(x)$ is compact, because it is a closed set contained in a compact set $\varphi_2(x)$. Thus φ is compact-valued.
Let G be an open set such that $\varphi(x_0) = \varphi_1(x_0) \cap \varphi_2(x_0) \subseteq G$. We shall prove that there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G$$

that is, $\varphi(U(x_0)) \subseteq G$.
(1) If $\varphi_2(x_0) \subseteq G$, there is nothing to prove. φ_2 is upper semi-continuous (u.s.c.), there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi_2(x) \subseteq G$$

that is, $\varphi_2(U(x_0)) \subseteq G$ and hence $\varphi(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G$.
(2) Suppose that $\varphi_2(x_0) \cap (\mathbf{C}G) = K \neq \varnothing$. Let y be a point of K, $y \notin \varphi_1(x_0)$; then there exist neighbourhoods $V(y)$ and $U_y(x_0)$ such that

$$x \in U_y(x_0) \Longrightarrow \varphi_1(x) \cap V(y) = \varnothing,$$

that is, $\varphi_1(U_y(x_0)) \cap V(y) = \varnothing$.

原课件第 48 页

Semi-continuous Mappings

Since the set K is compact, there exist elements $y_1, y_2, \cdots, y_N$ in K such that $V(y_1), V(y_2), \cdots, V(y_N)$ cover K. Write $K \subseteq \bigcup_{n=1}^{N} V(y_n) \equiv V(K)$, then $\varphi_2(x_0) \subseteq G \cup V(K)$.
If φ_2 is upper semi-continuous (u.s.c.), then there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi_2(x) \in G \cup V(K),$$

that is, $\varphi_2(U'(x_0)) \subseteq G \cup V(K)$.
Putting $U(x_0) = U_{y_1}(x_0) \cap U_{y_2}(x_0) \cap \cdots \cap U_{y_N}(x_0) \cap U'(x_0)$, we have

$$\varphi_1(U(x_0)) \cap V(y_n) \subseteq \varphi_1(U_{y_n}(x_0)) \cap V(y_n) = \varnothing,$$

that is, $\varphi_1(U(x_0)) \cap V(y_n) = \varnothing$, and hence

$$\varphi_1(U(x_0)) \cap V(K) = \varphi_1(U(x_0)) \cap \bigcup_{n=1}^{N} V(y_n) = \bigcup_{n=1}^{N} [\varphi_1(U(x_0)) \cap V(y_n)] = \varnothing$$

原课件第 49 页

Semi-continuous Mappings

Therefore

$$\begin{cases} (\varphi_1 \cap \varphi_2)(U(x_0)) \cap V(K) \subseteq \varphi_1(U(x_0)) \cap V(K) = \varnothing, \\ (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G \cup V(K), \end{cases}$$

whence

$$\varphi(U(x_0)) = (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq G,$$

that is,

$$x \in U(x_0) \Longrightarrow (\varphi_1 \cap \varphi_2)(x) \subseteq G.$$

Therefore φ is upper semi-continuous (u.s.c.). □

沿原论证展开与批注　老师路线：用闭图分离，再以紧性把逐点邻域变成有限个共同邻域。令 $H = \Phi_1 \cap \Phi_2$，Φ_1 闭图，Φ_2 紧值上半连续。先证每个 $H(x)$ 紧：闭图给 $\Phi_1(x)$ 闭，故交值是紧 $\Phi_2(x)$ 的闭子集。固定 x_0 及开集 $G \supset H(x_0)$，令 $K = \Phi_2(x_0) \setminus G$。这是紧集。每个 $y \in K$ 都不属于 $\Phi_1(x_0)$，否则属于 $H(x_0) \subset G$，矛盾。闭图分离给邻域 U_y 和 V_y，使 $\Phi_1(U_y) \cap V_y = \varnothing$。以有限 V_{y_i} 覆盖紧 K，令 $V = \bigcup_i V_{y_i}$，$U_1 = \bigcap_i U_{y_i}$；有限交保证 U_1 是邻域，且 $\Phi_1(U_1) \cap V = \varnothing$。又 $\Phi_2(x_0) \subset G \cup V$，由上半连续取 U_2 使 $\Phi_2(U_2) \subset G \cup V$。对 $x \in U_1 \cap U_2$，$H(x)$ 既在 $G \cup V$ 中又不碰 V，所以 $H(x) \subset G$。这正是上半连续。若 K 为空，直接对 $\Phi_2(x_0) \subset G$ 用上半连续即可。

10.18 闭图与上半连续之间的紧性桥梁

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 51 页。

对应已有 gpt-108.pdf 页脚：57。

原课件第 51 页

Semi-continuous Mappings

Corollary 1.1. If T is a compact space, a correspondence $\varphi : S \rightrightarrows T$ is closed-valued, then the correspondence φ is closed if and only if it is upper semi-continuous (u.s.c.).

Proof. If T is a compact space and φ is a closed-valued mapping, $\varphi(x)$ is a closed set for each $x \in S$, thus $\varphi(x)$ is a compact set and hence φ is compact-valued. If φ is u.s.c., then it is closed. Conversely, $\Delta : S \to \{T\}$ is the mapping such that $\Delta(x) = T$ for each x, Δ is a compact-valued u.s.c. correspondence. If φ is closed, then $\varphi = \varphi \cap \Delta$ is u.s.c. □

沿原论证展开与批注 紧值上半连续对应有闭图：若 $y \notin \Phi(x)$，紧值闭且 $\operatorname{dist}(y, \Phi(x)) > 0$，取分离开邻域，再用上半连续，附近图不能靠近 (x, y)。若 Ψ 闭图，Φ 紧值上半连续，则 $\Phi \cap \Psi$ 紧值上半连续。证明反证：若有 $x_n \to x$ 和交值 y_n 逃出包含交值的开集 G，紧像性质给子列 $y_n \to y$，两图闭使 y 属于交值，开性使子列最终进 G，矛盾。值域空间紧时，闭图亦推出上半连续：逃出开集的选择总能抽取收敛子列。

10.19 紧空间上的不变紧集

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 52–53 页。

对应已有 gpt-108.pdf 页脚：58, 59。

原课件第 52 页

Semi-continuous Mappings

Theorem 1.8. If S is a compact space and φ is a upper semi-continuous (u.s.c.) correspondence of S into itself such that, for each x, $\varphi(x) \neq \varnothing$, then there exists a non-empty compact subset K of S such that $\varphi(K) = K$.

Proof. The sets $S, \varphi(S), \varphi^2(S), \cdots$ form a sequence of non-empty compact sets; if any of these is equal to its successor, it is a set of the required kind and so the desired result is proved. We therefore assume that the sets are distinct; since $S \supseteq \varphi(S)$, we have $\varphi(S) \supseteq \varphi^2(S)$, $\varphi^2(S) \supseteq \varphi^3(S)$ and so on. Thus the sequence of sets $\varphi^n(S)$ is decreasing. By the finite intersection axiom, we have

$$K = \bigcap_{n=1}^{\infty} \varphi^n(S) \neq \varnothing.$$

For each n, we have $K \subseteq \varphi^{n-1}(S)$ and so $\varphi(K) \subseteq \varphi^n(S)$, whence $\varphi(K) \subseteq K$.

原课件第 53 页

Semi-continuous Mappings

We now prove that $K \subseteq \varphi(K)$. Let a be a point of K; then, for each n, there exists a point x_n in $\varphi^n(S)$ such that $a \in \varphi(x_n)$. The sequence $\{x_n : n \in \mathbb{N}\}$ admits a cluster point x_0 and a sub-sequence $\{x_{k_n} : n \in \mathbb{N}\}$ converges to x_0. Since at most $n-1$ points of this sub-sequence are outside $\varphi^n(S)$ and $\varphi^n(S)$ is compact, we have $x_0 \in \varphi^n(S)$ and so $x_0 \in K$. Also $(x_{k_n}, a) \longrightarrow (x_0, a)$ and $(x_{k_n}, a) \in G(\varphi) = \sum_{x \in S} \varphi(x)$ and so, since φ is a closed mapping, $a \in \varphi(x_0)$; therefore $a \in \varphi(K)$. It follows that $K \subseteq \varphi(K)$ and hence that $\varphi(K) = K$. $\square$

沿原论证展开与批注 假设非空紧 S、非空紧值上半连续自对应。令 $K_n = \Phi^n(S)$，紧像定理使它们非空紧，单调性使 $K_{n+1} \subseteq K_n$，故交 K 非空紧。单调性给 $\Phi(K) \subseteq K$。对 $a \in K$，可取 $x_n \in K_n$ 使 $a \in \Phi(x_n)$，抽子列 $x_n \to x$，套集闭性给 $x \in K$，闭图给 $a \in \Phi(x)$。故 $K = \Phi(K)$。这不保证某点 $x \in \Phi(x)$，不要把不变集合误当不动点。

10.20 复合与下半连续并

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 55 页。

对应已有 gpt-108.pdf 页脚：61。

原课件第 55 页

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \cap \varphi_2^-(G) \neq \varnothing\right\} \\
&= \varphi_1^-\left(\varphi_2^-(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^-(G)$ is open and so φ is lower semi-continuous (l.s.c.) □

沿原论证展开与批注 复合的下逆像满足 $(\Psi \circ \Phi)^-(V) = \Phi^-(\Psi^-(V))$，故两次下半连续使开集逐次变开；上逆像的全称量词也能逐层写出，得上半连续复合。任意并的下逆像为各下逆像之并，因此任意并保下半连续。每个等式都来自“存在中间选择”或“所有中间选择”，而不是把对应当单值函数计算。

10.21 复合与下半连续并

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 56 页。

对应已有 gpt-108.pdf 页脚：62。

原课件第 56 页

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set in Z, then

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq F\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq F\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(F)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(F)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^+(F)$ is closed and so φ is lower semi-continuous (l.s.c.) □

沿原论证展开与批注 复合的下逆像满足 $(\Psi \circ \Phi)^-(V) = \Phi^-(\Psi^-(V))$，故两次下半连续使开集逐次变开；上逆像的全称量词也能逐层写出，得上半连续复合。任意并的下逆像为各下逆像之并，因此任意并保下半连续。每个等式都来自“存在中间选择”或“所有中间选择”，而不是把对应当单值函数计算。

10.22 复合与下半连续并

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 57 页。

对应已有 gpt-108.pdf 页脚：63。

原课件第 57 页

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq G\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq G\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(G)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^+(G)$ is open and so φ is upper semi-continuous (u.s.c.) □

沿原论证展开与批注 复合的下逆像满足 $(\Psi \circ \Phi)^-(V) = \Phi^-(\Psi^-(V))$，故两次下半连续使开集逐次变开；上逆像的全称量词也能逐层写出，得上半连续复合。任意并的下逆像为各下逆像之并，因此任意并保下半连续。每个等式都来自“存在中间选择”或“所有中间选择”，而不是把对应当单值函数计算。

10.23 复合与下半连续并

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 58 页。

对应已有 gpt-108.pdf 页脚：64。

原课件第 58 页

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set of Z, then

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap F \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^{-}(F) \neq \varnothing\right\} \\
&= \varphi_1^{-}\left(\varphi_2^{-}(F)\right).
\end{aligned}$$

Therefore if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^{-}(F)$ is closed and so φ is upper semi-continuous (u.s.c.) □

沿原论证展开与批注 复合的下逆像满足 $(\Psi \circ \Phi)^{-}(V) = \Phi^{-}(\Psi^{-}(V))$，故两次下半连续使开集逐次变开；上逆像的全称量词也能逐层写出，得上半连续复合。任意并的下逆像为各下逆像之并，因此任意并保下半连续。每个等式都来自“存在中间选择”或“所有中间选择”，而不是把对应当单值函数计算。

10.24 复合与下半连续并

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 59 页。

对应已有 gpt-108.pdf 页脚：65。

原课件第 59 页

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} [\varphi_\lambda(x) \cap G] \neq \varnothing\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap G \neq \varnothing\} \\
&= \bigcup_{\lambda\in\Lambda} \varphi_\lambda^-(G).
\end{aligned}$$

Therefore the set $\varphi^-(G)$ is open and so φ is l.s.c. □

沿原论证展开与批注 复合的下逆像满足 $(\Psi\circ\Phi)^-(V)=\Phi^-(\Psi^-(V))$，故两次下半连续使开集逐次变开；上逆像的全称量词也能逐层写出，得上半连续复合。任意并的下逆像为各下逆像之并，因此任意并保下半连续。每个等式都来自“存在中间选择”或“所有中间选择”，而不是把对应当单值函数计算。

10.25 复合与下半连续并

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 60 页。

对应已有 gpt-108.pdf 页脚：66。

原课件第 60 页

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}F \subseteq \bigcap_{\lambda\in\Lambda} \mathbf{C}(\varphi_\lambda(x))\right\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}F \subseteq \mathbf{C}(\varphi_\lambda(x))\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq F\} \\
&= \bigcap_{\lambda\in\Lambda} \varphi_\lambda^+(F).
\end{aligned}$$

Therefore the set $\varphi^+(F)$ is closed and so φ is l.s.c. □

沿原论证展开与批注 复合的下逆像满足 $(\Psi\circ\Phi)^-(V)=\Phi^-(\Psi^-(V))$，故两次下半连续使开集逐次变开；上逆像的全称量词也能逐层写出，得上半连续复合。任意并的下逆像为各下逆像之并，因此任意并保下半连续。每个等式都来自“存在中间选择”或“所有中间选择”，而不是把对应当单值函数计算。

10.26 对应交集的条件与伪定理

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 62 页。

对应已有 gpt-108.pdf 页脚：68。

原课件第 62 页

Properties of Semi-continuous Mappings

Proof. By Theorem 1.5, the mapping φ_λ is compact-valued u.s.c., then the mapping $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed. Suppose that $y_0 \notin \varphi(x_0) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0)$, then there exists $\lambda_0 \in$ such that $y_0 \notin \varphi_{\lambda_0}(x_0)$. There exist neighbourhood $U(x_0)$ and neighbourhood $V(y_0)$ such that

$$x \in U(x_0) \qquad \Longrightarrow \qquad \varphi_{\lambda_0}(x) \cap V(y_0) = \varnothing,$$

that is, $\varphi_{\lambda_0}(U(x_0)) \cap V(y_0) = \varnothing$, then $\bigcap_{\lambda\in\Lambda} \varphi_\lambda(U(x_0)) \cap V(y_0) = \varnothing$ and hence $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed.

The intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ of a family of compact-valued mappings φ_λ is compact-valued from Non-empty Finite-intersection Theorem.

On the other hand, $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda = \varphi_{\gamma'} \cap \bigcap_{\gamma\neq\gamma'} \varphi_\lambda$. The mapping $\varphi_{\gamma'}$ is compact-valued u.s.c., and the mapping $\bigcap_{\gamma\neq\gamma'} \varphi_\lambda$ is closed, therefore, by Theorem 1.7, the mapping φ is u.s.c. □

沿原论证展开与批注 紧值上半连续对应的任意交是闭图，固定一个紧值上半连续因子作容器，运用第45页交集定理得紧值上半连续（交值可能为空）。第63–64页删掉紧值后的证明不能沿用，尤其 $(\bigcap_i \Phi_i)^-(F)$ 一般不等于 $\bigcap_i \Phi_i^-(F)$：各因子与 F 相交可能在不同点。第65–66页明确为伪定理：下半连续的有限交不必下半连续。反例 $\Phi_1(x) = \{0\}$，$\Phi_2(x) = \{x\}$，交仅在0非空；若要求非空值，可改 $\Phi_1(x) = \{0,1\}, \Phi_2(x) = \{x,1\}$，在0交含0与1，非零附近只含1，不下半连续。

10.27 对应交集的条件与伪定理

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 63 页。

对应已有 gpt-108.pdf 页脚：69。

原课件第 63 页

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T\left(\varphi_\lambda(x)\right)\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_\lambda(x)\right)\} = \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq G\} = \bigcup_{\lambda\in\Lambda} \varphi_\lambda^+(G)
\end{aligned}$$

Therefore the set $\varphi^+(G)$ is open and so φ is u.s.c. □

沿原论证展开与批注　本页展示的论证有错误，不能作为定理的有效证明。老师试图证明 $\Phi^+(G) = \bigcup_\lambda \Phi_\lambda^+(G)$，从而借开集的并得到开性。但从

$$T \setminus G \subseteq \bigcup_\lambda (T \setminus \Phi_\lambda(x))$$

不能推出存在单个 λ 使 $T \setminus G \subseteq T \setminus \Phi_\lambda(x)$。一个集合被许多集合的并覆盖，不意味着被其中某一个覆盖。例 $E = \{0,1\}, E_1 = \{0\}, E_2 = \{1\}$ 即否定此推理。 在对应层面，取某点值 $\Phi_1 = \{0,1\}, \Phi_2 = \{0,2\}, G = (-1/2, 1/2)$，交值 $\{0\}$ 在 G 内，却没有任何一个原值集整体在 G 内，直接否定老师使用的逆像等式。紧值上半连续交集的正确证明应沿前面第47–49页的闭图加紧容器定理；此处不能因错误等式最后得到开集就宣布证明完成。

10.28 对应交集的条件与伪定理

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 64 页。

对应已有 gpt-108.pdf 页脚：70。

原课件第 64 页

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid F \subseteq \mathbf{C}_T \left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid F \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T(\varphi_\lambda(x))\right\} = \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid F \subseteq \mathbf{C}_T(\varphi_\lambda(x))\}\right) \\
&= \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F = \varnothing\}\right) \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F \neq \varnothing\} = \bigcap_{\lambda\in\Lambda} \varphi_\lambda^-(F).
\end{aligned}$$

Therefore the set $\varphi^-(F)$ is closed and so φ is u.s.c. □

沿原论证展开与批注　本页展示的论证有错误，不能作为定理的有效证明。老师试图证明 $\Phi^+(G) = \bigcup_\lambda \Phi_\lambda^+(G)$，从而借开集的并得到开性。但从

$$T \setminus G \subseteq \bigcup_\lambda (T \setminus \Phi_\lambda(x))$$

不能推出存在单个 λ 使 $T \setminus G \subseteq T \setminus \Phi_\lambda(x)$。一个集合被许多集合的并覆盖，不意味着被其中某一个覆盖。例 $E = \{0,1\}, E_1 = \{0\}, E_2 = \{1\}$ 即否定此推理。在对应层面，取某点值 $\Phi_1 = \{0,1\}, \Phi_2 = \{0,2\}, G = (-1/2, 1/2)$，交值 $\{0\}$ 在 G 内，却没有任何一个原值集整体在 G 内，直接否定老师使用的逆像等式。紧值上半连续交集的正确证明应沿前面第47–49页的闭图加紧容器定理；此处不能因错误等式最后得到开集就宣布证明完成。本页换成闭集和下逆像也不能回避同一问题：每个值集都与同一个闭集相交，不保证它们的共同交还与该闭集相交，因为各自的交点可能不同。

10.29 对应交集的条件与伪定理

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 65 页。

对应已有 gpt-108.pdf 页脚：71。

原课件第 65 页

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T G\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid G \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T(\varphi_n(x))\right\} = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid G \subseteq \mathbf{C}_T(\varphi_n(x))\}\right) \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T G\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G = \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G \neq \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^-(G).
\end{aligned}$$

Therefore $\varphi^-(G)$ is an open set and so φ is l.s.c. □

沿原论证展开与批注　原页已标为 Pseudo-Theorem（伪定理），这里收录是为了拆解错误。核心错误为

$$(\Phi_1 \cap \Phi_2)^-(G) = \Phi_1^-(G) \cap \Phi_2^-(G).$$

左侧要求同一个点同时在两个值集中且在 G 中；右侧允许两个不同交点。取常值 $\Phi_1(x) = \{0,1\}, \Phi_2(x) = \{1,2\}$ 和开 $G = (-1/2, 1/2) \cup (3/2, 5/2)$，右侧为全域而左侧为空，故等式错。 命题本身也假：取 $\Phi_1(x) = \{0,1\}$、$\Phi_2(x) = \{x,1\}$，两者均为非空紧值连续对应。交值在 $x = 0$ 为 $\{0,1\}$，在非零 x 只有 $\{1\}$。开邻域 $(-1/4, 1/4)$ 遇到交值在0的值集，却不遇到任何非零附近输入的交值，违反下半连续定义。

10.30 对应交集的条件与伪定理

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 66 页。

对应已有 gpt-108.pdf 页脚：72。

原课件第 66 页

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T \left(\bigcap_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq F\} = \bigcup_{n=1}^{N} \varphi_n^+(F).
\end{aligned}$$

Therefore $\varphi^+(F)$ is a closed set and so φ is l.s.c. □

沿原论证展开与批注　原页已标为 Pseudo-Theorem（伪定理），这里收录是为了拆解错误。核心错误为

$$(\Phi_1 \cap \Phi_2)^-(G) = \Phi_1^-(G) \cap \Phi_2^-(G).$$

左侧要求同一个点同时在两个值集中且在 G 中；右侧允许两个不同交点。取常值 $\Phi_1(x) = \{0,1\}, \Phi_2(x) = \{1,2\}$ 和开 $G = (-1/2, 1/2) \cup (3/2, 5/2)$，右侧为全域而左侧为空，故等式错。 命题本身也假：取 $\Phi_1(x) = \{0,1\}$、$\Phi_2(x) = \{x,1\}$，两者均为非空紧值连续对应。交值在 $x = 0$ 为 $\{0,1\}$，在非零 x 只有 $\{1\}$。开邻域 $(-1/4, 1/4)$ 遇到交值在0的值集，却不遇到任何非零附近输入的交值，违反下半连续定义。本页第二种证明把“一个集合包含于并集”误换成“包含于某个成员”，同样犯了存在见证不统一的问题。第一种错在把分别存在当共同存在，第二种错在把逐点可选成员当全体共用成员。

10.31 有限并保持上半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 67 页。

对应已有 gpt-108.pdf 页脚：73。

原课件第 67 页

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 1. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} = \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

沿原论证展开与批注 若开集 G 包含 $\bigcup_{i=1}^{m} \Phi_i(x)$，它包含每个 $\Phi_i(x)$。为每个因子取邻域 U_i 使其值在 G 内，有限交 U 仍为邻域，故并值在 G 内。紧值时有限并亦紧。原页逆像公式须注意：并对应的上逆像是各上逆像的交，下逆像是各下逆像的并。不能把两者写成同一种交并。

10.32 有限并保持上半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 68 页。

对应已有 gpt-108.pdf 页脚：74。

原课件第 68 页

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 2. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if F is a closed set in T, we have

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(F).
\end{aligned}$$

Therefore $\varphi^{-}(F)$ is a closed set and so φ is u.s.c. □

沿原论证展开与批注 若开集 G 包含 $\bigcup_{i=1}^{m} \Phi_i(x)$，它包含每个 $\Phi_i(x)$。为每个因子取邻域 U_i 使其值在 G 内，有限交 U 仍为邻域，故并值在 G 内。紧值时有限并亦紧。原页逆像公式须注意：并对应的上逆像是各上逆像的交，下逆像是各下逆像的并。不能把两者写成同一种交并。

10.33 有限并保持上半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 69 页。

对应已有 gpt-108.pdf 页脚：75。

原课件第 69 页

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

沿原论证展开与批注 若开集 G 包含 $\bigcup_{i=1}^{m} \Phi_i(x)$，它包含每个 $\Phi_i(x)$。为每个因子取邻域 U_i 使其值在 G 内，有限交 U 仍为邻域，故并值在 G 内。紧值时有限并亦紧。原页逆像公式须注意：并对应的上逆像是各上逆像的交，下逆像是各下逆像的并。不能把两者写成同一种交并。

10.34 有限并保持上半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 70 页。

对应已有 gpt-108.pdf 页脚：76。

原课件第 70 页

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^-(F).
\end{aligned}$$

Therefore $\varphi^-(F)$ is a closed set and so φ is u.s.c. □

沿原论证展开与批注 若开集 G 包含 $\bigcup_{i=1}^{m} \Phi_i(x)$，它包含每个 $\Phi_i(x)$。为每个因子取邻域 U_i 使其值在 G 内，有限交 U 仍为邻域，故并值在 G 内。紧值时有限并亦紧。原页逆像公式须注意：并对应的上逆像是各上逆像的交，下逆像是各下逆像的并。不能把两者写成同一种交并。

10.35 有限积对应的半连续性

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 71–72 页。

对应已有 gpt-108.pdf 页脚：77, 78。

原课件第 71 页

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned}\varphi^- \left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} G_n^k \neq \varnothing\right\} \\ &= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap G_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap G_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^- \left(G_n^k\right).\end{aligned}$$

Then $\varphi^- \left(E^k\right)$ is an open set and therefore so is $\varphi^-(G) = \bigcup_k \varphi^- \left(E^k\right)$.

$$\begin{aligned}\varphi^-(G) &= \varphi^- \left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcup_k E^k\right) \neq \varnothing\right\} \\ &= \left\{x \in S \mid \bigcup_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\ &= \bigcup_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcup_k \varphi^- \left(E^k\right).\end{aligned}$$

Hence φ is l.s.c. □

原课件第 72 页

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} F_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq F_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(F_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is a closed set and therefore so is $\varphi^+(F) = \bigcap_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(F) &= \varphi^+\left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcap_k E^k\right\} \\ &= \bigcap_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcap_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is l.s.c. □

沿原论证展开与批注 下半连续积：给定值点 $y = (y_i)$ 和含它的开集，取乘积开邻域 $\prod_i V_i$ 包含于其中。各因子的下半连续保证附近可分别选 $y_i' \in \Phi_i(x') \cap V_i$，于是组成积中选择。紧值上半连续积：给任意输入收敛序列及积值选择，依次抽每个因子的收敛子列，有限次后所有坐标同时收敛且极限在积值中；若不满足上半连续，选逃离开邻域的序列即矛盾。没有紧值的第73–74页不能用上述紧性证明，也不能把任意开集当单个开矩形；本讲使用有紧值条件的正确版本。

10.36 有限积对应的半连续性

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 73–74 页。

对应已有 gpt-108.pdf 页脚：79, 80。

原课件第 73 页

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} Y_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} G_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq G_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(G_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is an open set and therefore so is $\varphi^+(G) = \bigcup_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(G) &= \varphi^+\left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcup_k E^k\right\} \\ &= \bigcup_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcup_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is u.s.c. □

原课件第 74 页

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned}\varphi^-\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} F_n^k \neq \varnothing\right\} \\ &= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap F_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap F_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^-\left(F_n^k\right).\end{aligned}$$

Then $\varphi^-\left(E^k\right)$ is a closed set and therefore so is $\varphi^-(F) = \bigcap_k \varphi^-\left(E^k\right)$.

$$\begin{aligned}\varphi^-(F) &= \varphi^-\left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcap_k E^k\right) \neq \varnothing\right\} \\ &= \left\{x \in S \mid \bigcap_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\ &= \bigcap_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcap_k \varphi^-\left(E^k\right).\end{aligned}$$

Hence φ is u.s.c. □

沿原论证展开与批注 下半连续积：给定值点 $y = (y_i)$ 和含它的开集，取乘积开邻域 $\prod_i V_i$ 包含于其中。各因子的下半连续保证附近可分别选 $y_i' \in \Phi_i(x') \cap V_i$，于是组成积中选择。紧值上半连续积：给任意输入收敛序列及积值选择，依次抽每个因子的收敛子列，有限次后所有坐标同时收敛且极限在积值中；若不满足上半连续，选逃离开邻域的序列即矛盾。没有紧值的第73–74页不能用上述紧性证明，也不能把任意开集当单个开矩形；本讲使用有紧值条件的正确版本。

10.37 有限积对应的半连续性

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 75–76 页。

对应已有 gpt-108.pdf 页脚：81, 82。

原课件第 75 页

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a compact-valued u.s.c. correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a compact-valued u.s.c. correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof. To fix our ideas, we consider the case $N = 2$. Since $\varphi_1(x)$ and $\varphi_2(x)$ are compact, so is $\varphi(x) = \varphi_1(x) \times \varphi_2(x)$, by Tychonoff's Theorem.

原课件第 76 页

Properties of Semi-continuous Mappings

Let G be an open set in T and. let a be an element of $\varphi^+(G)$. Since $\varphi(a)$ is compact and is contained in G, it can be covered by a finite number of elementary open sets contained in G; let $E^1, \cdots, E^K$ be such a finite covering. Suppose that $(y_1, y_2) \in \varphi_1(a) \times \varphi_2(a)$; let $E(y_1)$ denote the union of the E^k meeting the set $\{y_1\} \times \varphi_2(a)$ and let $E(y_2)$ be the union of the E^k meeting the set $\varphi_1(a) \times \{y_2\}$. The projection $\pi_2 E(y_1)$ of the set $E(y_1)$ on the space Y_2 is open, by Theorem 2 of §9, Chapter IV. As y_1 varies the sets. $\pi_2 E(y_1)$ are finite in number and therefore their intersection is open; similarly the intersection of the sets $\pi_1 E(y_2)$ is open. We write

$$G_1 = \bigcap_{y_2 \in \varphi_2(a)} \pi_1 E(y_2) \qquad \text{and} \qquad G_2 = \bigcap_{y_1 \in \varphi_1(a)} \pi_2 E(y_1).$$

Then $E = G_1 \times G_2$ is an elementary open set of T and

$$\varphi^+(E) = \{x \mid \varphi_1(x) \times \varphi_2(x) \subseteq G_1 \times G_2\} = \left[\varphi_1^+(G_1) \cap \varphi_2^+(G_2)\right] \cup \varphi_1^+(\varnothing) \cup \varphi_2^+(\varnothing).$$

Therefore $\varphi^+(E)$ is an open set of S; since $\varphi(a) \subseteq E \subseteq G$, we also have

$$a \in \varphi^+(E) \subseteq \varphi^+(G)$$

Therefore the set $\varphi^+(G)$ is a neighbourhood of each of its points and so is open; hence φ is compact-valued u.s.c. □

沿原论证展开与批注 下半连续积：给定值点 $y = (y_i)$ 和含它的开集，取乘积开邻域 $\prod_i V_i$ 包含于其中。各因子的下半连续保证附近可分别选 $y_i' \in \Phi_i(x') \cap V_i$，于是组成积中选择。紧值上半连续积：给任意输入收敛序列及积值选择，依次抽每个因子的收敛子列，有限次后所有坐标同时收敛且极限在积值中；若不满足上半连续，选逃离开邻域的序列即矛盾。没有紧值的第73–74页不能用上述紧性证明，也不能把任意开集当单个开矩形；本讲使用有紧值条件的正确版本。

10.38 和对应与凸包

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 78–79 页。

对应已有 gpt-108.pdf 页脚：84, 85。

原课件第 78 页

Properties of Semi-continuous Mappings

Proof. We use mathematical induction to verify this theorem.

(1) If $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $U(x_0)$ such that $x \in U(x_0)$ implies $\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq G$.

(1) Let T be a topological vector space. Define $f : T^N \to T$ by $f(y_1, \cdots, y_N) = \sum_{n=1}^{N} y_n$, then f is a continuous function, $f(\varphi_1 x_0, \cdots, \varphi_n(x_0)) = \sum_{n=1}^{N} \varphi_n(x_0)$. As you know, $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

原课件第 79 页

Properties of Semi-continuous Mappings

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $V = U(0)$ of origin such that $\sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$, there exist N neighborhoods $V_1, \cdots, V_N$ with $\sum_{n=1}^{N} V_n \subseteq V$ such that $\sum_{n=1}^{N} [\varphi_n(x_0) + V_n] = \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$. $\varphi_n : X \to Y$ is u.s.c. for $n = 1, \cdots, N$, then $\varphi_n(x_0) \subseteq \varphi_n(x_0) + V_n$, there exists a neighborhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad x \in \varphi_n(x_0) + V_n.$$

For any $x \in N(x_0) \equiv \bigcap_{n=1}^{N} U_n(x_0)$,

$$\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq \sum_{n=1}^{N} [\varphi_n(x_0) + V_n] \subseteq \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G.$$

Thus $\varphi = \sum_{n=1}^{N} \varphi_n$ is compact-valued and u.s.c. □

沿原论证展开与批注 有限和是积对应经连续加法映射的像，故紧值上半连续性由积与复合得到。下半连续凸包：任取值中一点 $y = \sum_{i=1}^{m} \lambda_i y_i$，对每个 y_i 用下半连续选取随输入趋近的 y_{in}，保持同一组权重，便得凸组合趋于 y。紧值上半连续凸包在 $\mathbb{R}^N$ 也成立：Carathéodory 定理把每点表示为至多 $N+1$ 个点的凸组合，权重在紧单纯形中；对权重和值点抽子列得极限仍在凸包。有限维假设不可遗漏。

10.39 和对应与凸包

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 80 页。

对应已有 gpt-108.pdf 页脚：86。

原课件第 80 页

Properties of Semi-continuous Mappings

Theorem 2.6. Let T be a linear vector space. If $\varphi : S \to \mathbb{R}^N$ is a lower semi-continuous (l.s.c.) correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_0 \in \mathbf{CE}(\varphi)(x_0) = \mathbf{CE}(\varphi(x_0))$,

$$y_0 = \sum_{k=1}^{K} \lambda^k z_0^k, \qquad z_0^k \in \varphi(x_0), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a sequence $z_n^k \to z_0^k$ such that $z_n^k \in \varphi(x_n)$, $y_n = \sum_{k=1}^{K} \lambda^k z_n^k \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_n))$. Therefore $\mathbf{CE}(\varphi)$ is lower semi-continuous (l.s.c.). □

沿原论证展开与批注 有限和是积对应经连续加法映射的像，故紧值上半连续性由积与复合得到。下半连续凸包：任取值中一点 $y = \sum_{i=1}^{m} \lambda_i y_i$，对每个 y_i 用下半连续选取随输入趋近的 y_{in}，保持同一组权重，便得凸组合趋于 y。紧值上半连续凸包在 $\mathbb{R}^N$ 也成立：Carathéodory 定理把每点表示为至多 $N+1$ 个点的凸组合，权重在紧单纯形中；对权重和值点抽子列得极限仍在凸包。有限维假设不可遗漏。

10.40 和对应与凸包

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 81 页。

对应已有 gpt-108.pdf 页脚：87。

原课件第 81 页

Properties of Semi-continuous Mappings

Theorem 2.6′. If $\varphi : S \to \mathbb{R}^N$ is a compact-valued u.s.c. correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a compact-valued u.s.c. correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_n \in \mathbf{CE}(\varphi)(x_n) = \mathbf{CE}(\varphi(x_n))$,

$$y_n = \sum_{k=1}^{K} \lambda_n^k z_n^k, \qquad z_n^k \in \varphi(x_n), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a convergent subsequence $\{z_{n'}^k, n' \in \mathbb{N}\} \subseteq \{z_n^k, n \in \mathbb{N}\}$ such that $\lim_{n'} z_{n'}^k \to z_0^k \in \varphi(x_0)$, then there corresponds a subsequence $\{y_{n'}, n' \in \mathbb{N}\} \subseteq \{y_n, n \in \mathbb{N}\}$ such that $y_{n'} = \sum_{k=1}^{K} \lambda_{n'}^k z_{n'}^k \to \sum_{k=1}^{K} \lambda_n^k z_0^k = y_0 \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_0))$. Therefore $\mathbf{CE}(\varphi)$ is compact-valued and u.s.c. □

沿原论证展开与批注 有限和是积对应经连续加法映射的像，故紧值上半连续性由积与复合得到。下半连续凸包：任取值中一点 $y = \sum_{i=1}^{m} \lambda_i y_i$，对每个 y_i 用下半连续选取随输入趋近的 y_{in}，保持同一组权重，便得凸组合趋于 y。紧值上半连续凸包在 $\mathbb{R}^N$ 也成立：Carathéodory 定理把每点表示为至多 $N+1$ 个点的凸组合，权重在紧单纯形中；对权重和值点抽子列得极限仍在凸包。有限维假设不可遗漏。

10.41 闭包对应与内半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 82 页。

对应已有 gpt-108.pdf 页脚：88。

原课件第 82 页

Properties of Semi-continuous Mappings

Theorem 2.7. $\varphi : S \to T$ is a lower semi-continuous (l.s.c.) correspondence if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. For any open set $G \subseteq T$ with $\varphi(x_0) \subseteq G$ equivalently $\overline{\varphi}(x_0) \subseteq G$, there exists a neighbourhood $U(x_0)$ such that, for any $x \in U(x_0)$, $\varphi(x) \subseteq G$ equivalently $\overline{\varphi}(x) \subseteq G$. Thus φ is l.s.c. if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is l.s.c. $\square$

沿原论证展开与批注 本页横线记号应理解为逐点闭包 $\overline{\Phi}(x) = \overline{\Phi(x)}$，不是凸包。且下半连续必须用“遇到开集”的下逆像，不能用本页误写的“整个值集包含于开集”。修正后仍沿老师的开集判据：对每个开 G，

$$\overline{\Phi(x)} \cap G \neq \varnothing \iff \Phi(x) \cap G \neq \varnothing.$$

右到左由 $\Phi(x) \subseteq \overline{\Phi(x)}$。左到右取 $y \in \overline{\Phi(x)} \cap G$，由 G 开选 $B(y, r) \subset G$，闭包定义保证该球遇到 $\Phi(x)$，故 G 也遇到它。因此两对应的下逆像对每个开集都完全相同，开性也相同，由下半连续的开逆像判据完成等价证明。原图有两个术语/符号错误，应先修正再使用。

10.42 闭包对应与内半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 83 页。

对应已有 gpt-108.pdf 页脚：89。

原课件第 83 页

Properties of Semi-continuous Mappings

Theorem 2.7′. Let T be a regular metric space. If $\varphi : S \to T$ is a u.s.c. correspondence, then its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a u.s.c. correspondence.

Proof. For any open set $G \subseteq T$ with $\overline{\varphi}(x_0) \subseteq G$, $\overline{\varphi}(x_0) \cap (\mathbf{C}_T G) = \varnothing$. According to the regularity of metric space, there exists an open set G' and O such that

$$\overline{\varphi}(x_0) \subseteq G', \qquad \mathbf{C}_T G \subseteq O, \qquad G' \cap O = \varnothing$$

As φ is u.s.c., there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G',$$

that is, $\varphi(U(x_0)) \subseteq G'$. Since $G' \cap (\mathbf{C}_T G) = \varnothing$, then $\overline{\varphi}(U(x_0)) \subseteq G$. Therefore $\overline{\varphi}$ is u.s.c. $\square$

沿原论证展开与批注 仍按逐点闭包理解本页的横线，不能称作凸包。固定 x_0 与开集 G 包含 $\overline{\Phi(x_0)}$。在度量空间中，两不交闭集 $\overline{\Phi(x_0)}$ 和 $T \setminus G$ 可取不交开邻域 V, W 分别包含它们（度量空间正规性）；这使 $\overline{V} \subset T \setminus W \subset G$。由原对应上半连续，取输入邻域 U 使 $\Phi(x) \subset V$ 对 $x \in U$ 成立。取闭包便得 $\overline{\Phi(x)} \subset \overline{V} \subset G$。这正是闭包对应上半连续。老师的路线是“分离开邻域—上半连续—取闭包”，关键补充是确保选到的 V 的闭包仍在 G，只写 $V \subset G$ 不足以推出这一点。

10.43 闭包对应与内半连续

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 85–86 页。

对应已有 gpt-108.pdf 页脚：91, 92。

原课件第 85 页

Semi-continuous Correspondences

Theorem 3.1 (Lower Semi-continuity and Inner Semi-continuity) Let X and Y be two metric spaces. A correspondence $\varphi : X \rightrightarrows Y$. φ is l.s.c. at x if and only if it is i.s.c. at x.

Proof. Suppose φ is not l.s.c. at some $x_0 \in X$. Then there exists an open subset G of Y such that $\varphi(x_0) \cap G \neq \varnothing$ and, for every $n \in \mathbb{N}$, there exists an $x_n \in B_{\frac{1}{n}}(x_0)$ with $\varphi(x_n) \cap G = \varnothing$. Note that $x_n \to x_0$, and pick any $y_0 \in \varphi(x_0) \cap G$. By hypothesis, there must exist a sequence $\{y_n : n \in \mathbb{N}\} \subseteq Y$ such that $y_n \to y_0$ and $y_n \in \varphi(x_n)$ for each n. But since $y_0 \in G$ and G is open, $y_n \in G$ for n large enough, contradicting that $\varphi(x_n) \cap G = \varnothing$ for all n.

原课件第 86 页

Semi-continuous Correspondences

Conversely, if $x_n \to x_0$ and $y_0 \in \varphi(x_0)$, then $y_0 \in \varphi(x_0) \cap B_{\frac{1}{n}}(y_0)$. By lower semi-continuity, there exists a $\varepsilon_n > 0$ with $\lim_{n\to\infty} \varepsilon_n = 0$ such that

$$x \in B_{\varepsilon_n}(x_0) \quad \Longrightarrow \quad \varphi(x) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing.$$

Since $x_n \to x_0$, so there exists k_n such that for all $n \geqslant k_n$, $x_n \in B_{\varepsilon_n}(x_0)$, then $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$. We may assume k_n is increasing.

For $n < k_n$, we pick $y_n \in \varphi(x_n)$. For any $k_n \leqslant n < k_{n+1}$, we have $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$, pick $y_n \in \varphi(x_n) \cap B_{\frac{1}{n}}(y_0)$. Then $d(y_n, y_0) < \dfrac{1}{n}$. $y_n \to y_0$. Hence φ is i.s.c. at x_0. $\square$

沿原论证展开与批注 第82–83页的横线运算是闭包，不应标为 convex hull。Φ 与逐点闭包的下逆像对开集相同，因为开集遇到闭包就遇到原集合，故下半连续等价。上半连续的闭包版本用正则性：给开集 G 包含闭包值，先取开邻域 V 使 $\Phi(x) \subset V \subset \overline{V} \subset G$（紧值时可有限选取），再用上半连续。度量空间中下半连续等价于：任意 $x_n \to x, y \in \Phi(x)$ 可选 $y_n \in \Phi(x_n)$ 趋于 y。证明依次对球 $B(y, 1/k)$ 取输入邻域，再用足够大的指标分段选择。

10.44 外半连续的术语与序列选取

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 87–89 页。

对应已有 gpt-108.pdf 页脚：93, 94, 95。

原课件第 87 页

Semi-continuous Correspondences

Theorem 3.1′ (Upper Semi-continuity and outer Semi-continuity) Let X and Y be two metric spaces. Mapping $\varphi : X \rightrightarrows Y$ is a correspondence.

(1) If φ is compact-valued u.s.c. at x, then it is o.s.c. at x.

(2) If φ is o.s.c. at x, then φ is u.s.c. at x.

Proof of (1). Suppose φ is u.s.c. at x_0, $x_n \to x_0$ and $y_n \in \varphi(x_n)$. Since φ is compact-valued, $\varphi(x_0)$ has a bounded neighborhood U. Then $S = \{x_0, x_1, x_2, \cdots\}$ is sequentially compact in X, hence S is compact in X. By Proposition 1, $\varphi(S)$ is sequentially compact in Y. So, if $y_n \in \varphi(x_n)$ for each n, then $\{y_n : n \in \mathbb{N}\}$ — being a sequence in $\varphi(S)$ – must possess a subsequence that converges to some $y_0 \in \varphi(S)$. With an innocent abuse of notation, we denote this subsequence also by $\{y_n : n \in \mathbb{N}\}$, and write $y_n \to y_0$.

原课件第 88 页

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$, and observe that this implies $\varepsilon = d_Y(y_0, \varphi(x_0)) > 0$ because $\varphi(x_0)$ is a compact (hence closed) set. Now define

$$T = \left\{ z \in Y : d_Y(z, \varphi(x_0)) \leqslant \frac{\varepsilon}{2} \right\}$$

Since $d_Y(z, \varphi(x_0))$ is a continuous map on z, T is a closed set. Moreover, by definition, we have

$$\varphi(x_0) \subseteq T^\circ$$

and $y_0 \notin T$. But, since φ is u.s.c. at x_0, there exists $\delta > 0$ such that

$$\varphi(B_\delta(x_0)) \subseteq T^\circ.$$

Thus there is $n \in \mathbb{N}$ such that $y_n \in \varphi(x_n) \subseteq T^\circ$ for all $n \geqslant N$. But then, since T is a closed set and $y_n \to y_0$, we must have $y_0 \in T$, a contradiction.

原课件第 89 页

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$. Note that $\varphi(x_0)$ is compact, there exist two neighbourhoods $V(y_0)$ and $V'(\varphi(x_0))$ such that $V(y_0) \cap V'(\varphi(x_0)) = \varnothing$, then $y_0 \notin \overline{V'}(\varphi(x_0))$.

If φ is u.s.c. at x_0 and $x_n \to x_0$, then there exists a $N \in \mathbb{N}$ such that $\varphi(x_n) \subseteq V'(\varphi(x_0))$ for $n \geqslant N$, thus $y_n \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, and $y_0 \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, which contradicts $y_0 \notin \overline{V'}(\varphi(x_0))$.

沿原论证展开与批注 本课此处的 o.s.c. 要求：任意 $x_n \to x$ 及 $y_n \in \Phi(x_n)$，有收敛子列且极限属于 $\Phi(x)$。它比文献中仅要求“已有收敛选择的极限不逃出”的闭图式 outer semicontinuity 更强。紧值上半连续给该性质：集合 $\{x, x_1, x_2, \ldots\}$ 紧，紧像定理给子列，闭图给极限归属。反向若上半连续失效，选开 $G \supset \Phi(x)$ 及逃出 G 的 y_n，由假设抽子列极限在 $\Phi(x)$，与开性矛盾。复合用先抽中间值再抽末端值，有限积用有限次子列抽取；不能无限次抽取后还把未构造的子列当作存在。

10.45 外半连续的术语与序列选取

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 90 页。

对应已有 gpt-108.pdf 页脚：96。

原课件第 90 页

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then there exists an open subset G of Y with $\varphi(x_0) \subseteq G$ and

$$\varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G \neq \varnothing$$

for any $n \in \mathbb{N}$. But then there exist $(x_n) \subseteq X$ and $(y_n) \subseteq Y$ such that $x_n \to x_0$ and $y_n \in \varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G$ for each n. Since $\varphi(x_0) \subseteq G$ and each y_n belongs to the closed set $Y \setminus G$, no subsequence of (y_n) can converge to a point in $\varphi(x_0)$. □

沿原论证展开与批注 本课此处的 o.s.c. 要求：任意 $x_n \to x$ 及 $y_n \in \Phi(x_n)$，有收敛子列且极限属于 $\Phi(x)$。它比文献中仅要求“已有收敛选择的极限不逃出”的闭图式 outer semicontinuity 更强。紧值上半连续给该性质：集合 $\{x, x_1, x_2, \ldots\}$ 紧，紧像定理给子列，闭图给极限归属。反向若上半连续失效，选开 $G \supset \Phi(x)$ 及逃出 G 的 y_n，由假设抽子列极限在 $\Phi(x)$，与开性矛盾。复合用先抽中间值再抽末端值，有限积用有限次子列抽取；不能无限次抽取后还把未构造的子列当作存在。

10.46 外半连续的术语与序列选取

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 91 页。

对应已有 gpt-108.pdf 页脚：97。

原课件第 91 页

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then for any open set $G \subseteq Y$ with $\varphi(x_0) \subseteq G$ and any neighbourhood $U_n(x_0)$, there exist $x_n \in U_n(x_0)$ such that $\varphi(x_n) \nsubseteq G$. As $x_n \to x_0$, there exists $y_n \in \varphi(x_n)$ and $y_n \notin G$. Then there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $y_{n'} \to \varphi(x_0)$. However, $y_n \notin G$, there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $\lim_{n' \to \infty} y_{n'} \notin G$, then $\lim_{n' \to \infty} y_{n'} \notin \varphi(x_0)$. This is a contradiction! □

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10.47 外半连续的术语与序列选取

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 92 页。

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原课件第 92 页

Properties of Semi-continuous Mappings

Theorem 3.2. (Inner Semi-continuity) If φ_1 is an i.s.c. mapping of X into Y and φ_2 is an i.s.c. mapping of Y into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an i.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_0 \in \varphi(x_0) = (\varphi_2 \cdot \varphi_1)(x_0) = \varphi_2(\varphi_1(x_0))$, then there exists $y_0 \in \varphi_1(x_0)$ such that $z_0 \in \varphi_2(y_0)$.

Whenever $x_n \to x_0$ and $y_0 \in \varphi_1(x_0)$, by inner semi-continuity of φ_1, there exists a sequence $y_n \in \varphi_1(x_n)$ such that $y_n \to y_0$. On the other hand, $z_0 \in \varphi_2 y_0$, by inner semi-continuity of φ_2, there exists a sequence $z_n \in \varphi_2(y_n)$ such that $z_n \to z_0$.

$[y_n \in \varphi_1(x_n)$ and $z_n \in \varphi_2(y_n)]$ imply $z_n \in \varphi_2(\varphi_1(x_n)) = \varphi(x_n)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is i.s.c. $\square$

沿原论证展开与批注 本课此处的 o.s.c. 要求：任意 $x_n \to x$ 及 $y_n \in \Phi(x_n)$，有收敛子列且极限属于 $\Phi(x)$。它比文献中仅要求“已有收敛选择的极限不逃出”的闭图式 outer semicontinuity 更强。紧值上半连续给该性质：集合 $\{x, x_1, x_2, \ldots\}$ 紧，紧像定理给子列，闭图给极限归属。反向若上半连续失效，选开 $G \supset \Phi(x)$ 及逃出 G 的 y_n，由假设抽子列极限在 $\Phi(x)$，与开性矛盾。复合用先抽中间值再抽末端值，有限积用有限次子列抽取；不能无限次抽取后还把未构造的子列当作存在。

10.48 外半连续的术语与序列选取

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 93 页。

对应已有 gpt-108.pdf 页脚：99。

原课件第 93 页

Properties of Semi-continuous Mappings

Theorem 3.2′ (Outer Semi-continuity) If φ_1 is an o.s.c. mapping of X into Y and φ_2 is an o.s.c. mapping of Y into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an o.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_n \in \varphi(x_n) = (\varphi_2 \cdot \varphi_1)(x_n) = \varphi_2(\varphi_1(x_n))$, then there exists $y_n \in \varphi_1(x_n)$ such that $z_n \in \varphi_2(y_n)$.

Whenever $x_n \to x_0$ and $y_n \in \varphi_1(x_n)$, by outer semi-continuity of φ_1, there exists a convergent subsequence $y_{n_1} \to y_0$ with $y_0 \in \varphi_1(x_0)$. On the other hand, $z_n \in \varphi_2(y_n)$, then $z_{n_1} \in \varphi_2(y_{n_1})$. By outer semi-continuity of φ_2, there exists a convergent subsequence $z_{n_2} \to z_0$ with $z_0 \in \varphi_2(y_0)$.

$[y_0 \in \varphi_1(x_0)$ and $z_0 \in \varphi_2(y_0)]$ imply $z_0 \in \varphi_2(\varphi_1(x_0)) = \varphi(x_0)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is o.s.c. □

沿原论证展开与批注 本课此处的 o.s.c. 要求：任意 $x_n \to x$ 及 $y_n \in \Phi(x_n)$，有收敛子列且极限属于 $\Phi(x)$。它比文献中仅要求“已有收敛选择的极限不逃出”的闭图式 outer semicontinuity 更强。紧值上半连续给该性质：集合 $\{x, x_1, x_2, \ldots\}$ 紧，紧像定理给子列，闭图给极限归属。反向若上半连续失效，选开 $G \supset \Phi(x)$ 及逃出 G 的 y_n，由假设抽子列极限在 $\Phi(x)$，与开性矛盾。复合用先抽中间值再抽末端值，有限积用有限次子列抽取；不能无限次抽取后还把未构造的子列当作存在。

10.49 外半连续的术语与序列选取

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 94 页。

对应已有 gpt-108.pdf 页脚：100。

原课件第 94 页

Semi-continuous Correspondences

Theorem 3.3 (Inner Semi-continuity) The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of i.s.c. mappings φ_n of X into Y_n is an i.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $\left(y_1^0, \cdots, y_N^0\right) = y^0 \in \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$. If every φ_n is i.s.c., then there is a sequence $y_n^q \in \varphi_n\left(x^q\right)$ for each q such that $y_n^q \to y_n^0$. Thus $y^q = \left(y_1^q, \cdots, y_N^q\right) \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$ such that $y^q = \left(y_1^q, \cdots, y_N^q\right) \to \left(y_1^0, \cdots, y_N^0\right) = y^0$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is i.s.c. □

沿原论证展开与批注 本课此处的 o.s.c. 要求：任意 $x_n \to x$ 及 $y_n \in \Phi(x_n)$，有收敛子列且极限属于 $\Phi(x)$。它比文献中仅要求“已有收敛选择的极限不逃出”的闭图式 outer semicontinuity 更强。紧值上半连续给该性质：集合 $\{x, x_1, x_2, \ldots\}$ 紧，紧像定理给子列，闭图给极限归属。反向若上半连续失效，选开 $G \supset \Phi(x)$ 及逃出 G 的 y_n，由假设抽子列极限在 $\Phi(x)$，与开性矛盾。复合用先抽中间值再抽末端值，有限积用有限次子列抽取；不能无限次抽取后还把未构造的子列当作存在。

10.50 外半连续的术语与序列选取

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 95 页。

对应已有 gpt-108.pdf 页脚：101。

原课件第 95 页

Semi-continuous Correspondences

Theorem 3.3′ **(Outer Semi-continuity)** The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of o.s.c. mappings φ_n of X into Y_n is an o.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $\left(y_1^q, \cdots, y_N^q\right) = y^q \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$, then $y_n^q \in \varphi_n\left(x^q\right)$ for $n = 1, \cdots, N$. If φ_1 is o.s.c., then there is a convergent subsequence $y_1^{q_1} \to y_1^0$ with $y_1^0 \in \varphi_1 x^0$. In this case, $y_n^{q_1} \in \varphi_n x^{q_1}$ for $n = 1, \cdots, N$. If φ_2 is o.s.c., then there is a convergent subsequence $y_n^{q_2} \to y_n^0$ with $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, 2$. In this case, $y_n^{q_2} \in \varphi_n x^{q_2}$ for $n = 1, \cdots, N$. Follow these steps, we can obtain that, if φ_N is o.s.c., then there is a convergent subsequence $y_n^{q_N} \to y_n^0$ with $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$. Hence $y^{q_N} = \left(y_1^{q_N}, \cdots, y_N^{q_N}\right) \to \left(y_1^0, \cdots, y_N^0\right) = y^0$ and $y^0 = \left(y_1^0, \cdots, y_N^0\right) \in \prod_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is o.s.c. □

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10.51 最优值的两种半连续性

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 96 页。

对应已有 gpt-108.pdf 页脚：102。

原课件第 96 页

Berge's Maximum Theorem

Theorem 4.1. If f is a l.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \sup\{f(x,y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. Suppose that $x_0 \in S$ and let y_0 be such that

$$y_0 \in \varphi(x_0); \qquad f(x_0, y_0) \geqq M(x_0) - \varepsilon.$$

There exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that

$$(x,y) \in U(x_0) \times V(y_0) \Longrightarrow f(x,y) \geqq f(x_0,y_0) - \varepsilon \geqq M(x_0) - 2\varepsilon;$$

there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi(x) \cap V(y_0) \neq \varnothing.$$

Therefore

$$x \in U(x_0) \cap U'(x_0) \Longrightarrow M(x) \geqq M(x_0) - 2\varepsilon.$$

□

沿原论证展开与批注 设 $M(x) = \sup_{y \in \Phi(x)} f(x,y)$。若 f 下半连续且 Φ 下半连续，任取 $a < M(x)$，选可行 y 满足 $f(x,y) > a$。目标下半连续给乘积邻域使函数值仍大于 a，可行对应下半连续给附近可行点落入该邻域，故附近 $M > a$。若 f 上半连续，Φ 非空紧值上半连续，则每点最大值存在。取输入序列及最优点，由紧像抽子列趋于可行极限，上半连续给 $\limsup M(x_n) \leq M(x)$。对 limsup 严格写法需先选逼近它的子列，再抽最优点子列。

10.52 最优值的两种半连续性

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 98 页。

对应已有 gpt-108.pdf 页脚：104。

原课件第 98 页

Berge's Maximum Theorem

Proof. Suppose that $x_0 \in S$; to each y in $\varphi(x_0)$ there correspond neighbourhoods $U_y(x_0)$ and $V(y)$ such that

$$(x,z) \in U_y(x_0) \times V(y) \quad \Longrightarrow \quad f(x,z) \leqq f(x_0,y) + \varepsilon.$$

Since $\varphi(x_0)$ is compact, it can be covered by a finite number of neighbourhoods of the form $V(y)$, say $V(y_1), V(y_2), \cdots, V(y_N)$. Putting $U'(x_0) = \bigcap_{n=1}^{N} U_{y_n}(x_0)$ and $V(\varphi(x_0)) = \bigcup_{n=1}^{N} V(y_n)$, we have

$$x \in U'(x_0), y \in V(\varphi(x_0)) \Longrightarrow f(x,y) \leqq \max_n f(x_0,y_n) + \varepsilon \leqq M(x_0) + \varepsilon.$$

Moreover there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq V(\varphi(x_0))$$

and so

$$x \in U(x_0) \cap U'(x_0) \quad \Longrightarrow \quad M(x) = \max_{y \in \varphi(x)} f(x,y) \leqq M(x_0) + \varepsilon.$$

□

沿原论证展开与批注　保留老师的有限覆盖思路。固定 x_0，令 $M_0 = \max_{y\in\Phi(x_0)} f(x_0,y)$；非空紧值与目标上半连续保证最大值存在。给 $\varepsilon > 0$。对每个 $y \in \Phi(x_0)$，联合上半连续给乘积邻域 $U_y \times V_y$，使其中 $f(x,z) < f(x_0,y) + \varepsilon \le M_0 + \varepsilon$。有限个 V_{y_i} 覆盖紧值 $\Phi(x_0)$。上半连续可行对应给 U_0，使 $\Phi(x) \subset \bigcup_i V_{y_i}$ 对 $x \in U_0$ 成立。取 $U = U_0 \cap \bigcap_i U_{y_i}$。任意 $x \in U, z \in \Phi(x)$，选覆盖它的 V_{y_i}，同时 $x \in U_{y_i}$，于是 $f(x,z) < M_0 + \varepsilon$。对所有可行 z 取最大值得 $M(x) \le M_0 + \varepsilon$，故上半连续。这里紧性只用于 x_0 的值集上取得有限覆盖，随后才能对输入邻域取有限交。

10.53 Berge 最大值与唯一选择

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 99 页。

对应已有 gpt-108.pdf 页脚：105。

原课件第 99 页

Berge's Maximum Theorem

Theorem 4.3 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \max\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $f(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $f = \varphi \cap \Delta$ is u.s.c. $\square$

沿原论证展开与批注 必须补充非空紧值：Φ 连续且非空紧值，f 连续，则第96页的两个方向给 M 连续。最优对应 $A(x) = \{y \in \Phi(x) : f(x,y) = M(x)\}$ 非空紧；等式由连续性保极限，故闭图，再以 Φ 为紧容器得上半连续。若最优点唯一，单点值上半连续等价普通连续，故唯一最优选择连续。实数紧值取 $f(y) = y$，最大值唯一，得到连续选择；没有紧值时例如 $(0,1)$ 根本没有最大点。原页100的球应理解为非空闭球，开球一般不取到坐标最大值。

10.54 Berge 最大值与唯一选择

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 101 页。

对应已有 gpt-108.pdf 页脚：107。

原课件第 101 页

Berge's Maximum Theorem

Theorem 4.4. Let φ be a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$. If the family of sets $\{\varphi(x) \mid x \in S\}$ is selective, there is a single-valued continuous mapping σ of S into T such that, for each x, $\sigma(x) \in \varphi(x)$.

Proof. Let f be a continuous numerical function in T for which $\{\varphi(x) \mid x \in S\}$ is selective; if $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ then ϕ is a single-valued mapping of S into T. Moreover, it is u.s.c. by the preceding theorem and so is continuous (since it is single-valued). The mapping $\sigma(x) = \phi(x)$ satisfies

$$\sigma(x) \in \varphi(x) \quad \text{for all} \quad x \in S$$

as required. $\square$

沿原论证展开与批注 必须补充非空紧值：Φ 连续且非空紧值，f 连续，则第96页的两个方向给 M 连续。最优对应 $A(x) = \{y \in \Phi(x) : f(x,y) = M(x)\}$ 非空紧；等式由连续性保极限，故闭图，再以 Φ 为紧容器得上半连续。若最优点唯一，单点值上半连续等价普通连续，故唯一最优选择连续。实数紧值取 $f(y) = y$，最大值唯一，得到连续选择；没有紧值时例如 $(0,1)$ 根本没有最大点。原页100的球应理解为非空闭球，开球一般不取到坐标最大值。

10.55 Berge 最大值与唯一选择

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 102 页。

对应已有 gpt-108.pdf 页脚：108。

原课件第 102 页

Berge's Maximum Theorem

Corollary 4.1. If φ is a continuous mapping of X into $\mathbb{R}$ such that, for each x, $\varphi(x) \neq \varnothing$, there exists a continuous single-valued mapping σ such that, for each x, $\sigma(x) \in \varphi(x)$

Proof. It is sufficient to take $f(y) = y$, whence $\sigma(x) = \max \varphi(x)$. □

沿原论证展开与批注 必须补充非空紧值：Φ 连续且非空紧值，f 连续，则第96页的两个方向给 M 连续。最优对应 $A(x) = \{y \in \Phi(x) : f(x,y) = M(x)\}$ 非空紧；等式由连续性保极限，故闭图，再以 Φ 为紧容器得上半连续。若最优点唯一，单点值上半连续等价普通连续，故唯一最优选择连续。实数紧值取 $f(y) = y$，最大值唯一，得到连续选择；没有紧值时例如 $(0,1)$ 根本没有最大点。原页100的球应理解为非空闭球，开球一般不取到坐标最大值。

10.56 最小化的对偶证明

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 103–104 页。

对应已有 gpt-108.pdf 页脚：109, 110。

原课件第 103 页

Berge's Maximum Theorem

Theorem 4.5. If f is a l.s.c. numerical function in $S \times T$ and φ is a compact-valued u.s.c. correspondence of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x_0) - \varepsilon \leqslant M(x).$$

原课件第 104 页

Berge's Maximum Theorem

If φ is compact-valued, then there exist $y_1, \cdots, y_K \in \varphi(x_0)$ such that $\varphi(x_0) \subseteq \bigcup_{k=1}^{K} V_k(y_k)$. If φ is u.s.c., then there exists a neighbourhood $U_0(x_0)$ such that

$$x \in U_0(x_0) \Longrightarrow \varphi(x) \subseteq \bigcup_{k=1}^{K} V_k(y_k).$$

If f is l.s.c., then, for any $\varepsilon > 0$, $y_k \in \varphi(x_0)$, there exists a neighbourhood $U_{y_k}(x_0)$ such that, for any $x \in U_{y_k}(x_0)$ and any $y \in V_k(y_k)$, $f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Define $U(x_0) = U_0(x_0) \cap \bigcap_{k=1}^{K} U_{y_k}(x_0)$. Note that $y_1, \cdots, y_K \in \varphi(x_0)$, then, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y_k \in \varphi(x_0)} f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Therefore, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y \in \varphi(x)} f(x, y) = M(x_0)$. □

沿原论证展开与批注 对最小化使用 $-f$ 的最大化，或直接证明：紧值上半连续可行集及下半连续目标给最小值函数下半连续；下半连续可行集及上半连续目标给下确界函数上半连续。后一证明先选 y_0 使 $f(x_0, y_0) < m(x_0) + \varepsilon/2$，再使附近某可行选择的目标小于 $f(x_0, y_0) + \varepsilon/2$。两项合成 $m(x) < m(x_0) + \varepsilon$。第104页末尾应是 $M(x)$，不是 $M(x_0)$；第106页错误的集合包含应改成存在 $y \in \Phi(x) \cap V$。第107页最小对应的等式最安全写为 $f(y) = m(x)$，原图弱不等式方向不能把所有较大值也选入。

10.57 最小化的对偶证明

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 105–106 页。

对应已有 gpt-108.pdf 页脚：111, 112。

原课件第 105 页

Berge's Maximum Theorem

Theorem 4.6. If f is a u.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is u.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x) \leqslant M(x_0) + \varepsilon.$$

原课件第 106 页

Berge's Maximum Theorem

From the definition of infimum, for any $\varepsilon > 0$, there exists $y_0 \in \varphi(x_0)$ such that $f(x_0, y_0) \leqslant M(x_0) + \frac{\varepsilon}{2}$.
If f is u.s.c., then there exist two neighbourhoods $U_1(x_0)$ and $V(y_0)$ such that, for any $x \in U_1(x_0)$ and any $y \in V(y_0)$, $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2}$.
Therefore $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon$.
If φ is l.s.c., for $\varphi(x_0) \cap V(y_0) \neq \varnothing$, then there exists a neighbourhood $U_2(x_0)$ such that

$$x \in U_2(x_0) \Longrightarrow \varphi(x) \subseteq \varphi(x) \cap V(y_0) \neq \varnothing.$$

Define $U(x_0) = U_1(x_0) \cap U_2(x_0)$. for any $x \in U(x_0)$, there exists $y \in V(y_0)$, such that

$$M(x) = \inf_{y \in \varphi(x)} f(x, y) \leqslant f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon.$$

Therefore M is u.s.c. □

沿原论证展开与批注 对最小化使用 $-f$ 的最大化，或直接证明：紧值上半连续可行集及下半连续目标给最小值函数下半连续；下半连续可行集及上半连续目标给下确界函数上半连续。后一证明先选 y_0 使 $f(x_0, y_0) < m(x_0) + \varepsilon/2$，再使附近某可行选择的目标小于 $f(x_0, y_0) + \varepsilon/2$。两项合成 $m(x) < m(x_0) + \varepsilon$。第104页末尾应是 $M(x)$，不是 $M(x_0)$；第106页错误的集合包含应改成存在 $y \in \Phi(x) \cap V$。第107页最小对应的等式最安全写为 $f(y) = m(x)$，原图弱不等式方向不能把所有较大值也选入。

10.58 最小化的对偶证明

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 107 页。

对应已有 gpt-108.pdf 页脚：113。

原课件第 107 页

Berge's Maximum Theorem

Theorem 4.7 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \min\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $\phi = \varphi \cap \Delta$ is u.s.c. $\square$

沿原论证展开与批注 对最小化使用 $-f$ 的最大化，或直接证明：紧值上半连续可行集及下半连续目标给最小值函数下半连续；下半连续可行集及上半连续目标给下确界函数上半连续。后一证明先选 y_0 使 $f(x_0, y_0) < m(x_0) + \varepsilon/2$，再使附近某可行选择的目标小于 $f(x_0, y_0) + \varepsilon/2$。两项合成 $m(x) < m(x_0) + \varepsilon$。第104页末尾应是 $M(x)$，不是 $M(x_0)$；第106页错误的集合包含应改成存在 $y \in \Phi(x) \cap V$。第107页最小对应的等式最安全写为 $f(y) = m(x)$，原图弱不等式方向不能把所有较大值也选入。

10.59 环境依赖偏好与最大元

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 110–111 页。

对应已有 gpt-108.pdf 页脚：116, 117。

原课件第 110 页

Berge's Maximum Theorem

Proof. If $\varphi(x) = \varnothing$, then for any $y \in \beta(x)$, there exists $y' \in \beta(x)$ such that $y \prec_x y'$. Define $r(z) = \{y \in \beta(x) \mid y \prec_x z\}$, then $\beta(x) \subseteq \bigcup_{z\in\beta(z)} \{r(z)\}$. Note that $\beta(x)$ is compact and $\bigcup_{z\in\beta(z)} \{r(z)\}$ is its open covering, then there exist $r(z_1), \cdots, r(z_K)$, such that $\beta(x) \subseteq \bigcup_{k=1}^{K} r(z_k)$. $\{z_1, \cdots, z_K\}$ in finite, then there is the maximal to be z_K, as $\prec_x$ is transitive, then for $k = 1, \cdots, K-1$, $z_K \notin r(z_k)$. But $\prec_x$ is not reflexive, then $z_K \notin r(z_K)$, and hence $z_K \notin \beta(x)$, which is a contradiction. Therefore $\varphi(x) \neq \varnothing$.

原课件第 111 页

Berge's Maximum Theorem

According to the continuity of $\prec_x$, $\{y \in \beta(x) \mid y \not\prec_x z\}$ is closed, then

$$\varphi(x) = \{y \in T \mid z \in \beta(x), y \not\prec_x z\} = \bigcap_{z \in \beta(x)} \{y \in \beta(x) \mid y \not\prec_x z\}$$

is closed, $\varphi(x) \neq \varnothing$ and $\varphi(x) \subseteq \beta(x)$, $\beta(x)$ is compact, then $\varphi(x)$ is compact.

In order to check that φ is u.s.c.at x_0, we want to prove that φ is closed-valued,

$$\text{for any } (x_n, y_n) \to (x, y),\ y_n \in \varphi(x_n) \quad \Longrightarrow \quad y \in \varphi(x).$$

Since β is l.s.c., then, for $z \in \beta(x)$, there exists $z_n \to z$ such that, for any n, $z_n \in \varphi(x_n)$, then $y_n \not\prec_{x_n} z_n$. From the continuity of $\prec_x$, $x_n \to x$, $y_n \to y$, $z_n \to z$, then $y \not\prec_x z$, and hence $y = \max_{\prec_x} \beta(x)$, that is, $y \in \varphi(x)$, thus φ is u.s.c. □.

沿原论证展开与批注 假设 β 非空紧值连续，每个严格关系传递且非自反，且“可行且非严格优于”的三元组集合闭。若无极大元，每个 y 被某个 z 严格压过，这些严格下截集构成 $\beta(x)$ 的相对开覆盖。有限子覆盖的有限个 z_i 中存在极大者（传递且无循环），它不被任一 z_i 压过，矛盾。极大元集为所有非优于截集的交，故闭且紧。若 $x_n \to x, y_n$ 为极大元且 $y_n \to y$，先由可行图闭得 $y \in \beta(x)$，再对任意 $z \in \beta(x)$ 用下半连续选 $z_n \in \beta(x_n) \to z$。由 y_n 不差于 z_n 及三元组闭性得极限不被 z 压过。第111页 z_n 应属于可行集 $\beta(x_n)$，不必属于最优集。以连续效用定义严格偏好即可回到 Berge 定理；第112页最优对应内目标须写成 $f(x, y)$，并补紧值条件。

10.60 环境依赖偏好与最大元

来源： 108-ZSM-PPT.pdf，原课件 PDF 第 113 页。

对应已有 gpt-108.pdf 页脚：119。

原课件第 113 页

Continuous Correspondences Continuous Correspondences Properties Continuou

Berge's Maximum Theorem

Proof. Define

$$y \prec_x z \qquad \Longleftrightarrow \qquad f(x,y) \prec f(x,z)$$

Since f is continuous, then $\{(x,y,z) \in S \times T \times T \mid f(x,y) \not\prec f(x,z)\}$ is closed, then for any $x \in S$ and $y \in Y$, $\{z \in Y \mid f(x,y) \prec_x f(x,z)\}$ is closed, then f is u.s.c.

We now check that M is continuous. $\{x\}$ is a single-point mapping of correspondence, the product mapping $(x, r(x))$ is u.s.c., f is continuous, then $f(x, r(x))$ is u.s.c., but $f(x, r(x)) = \{M(x)\}$, then $\alpha : S \to \mathbb{R}$ is continuous. $\square$

沿原论证展开与批注 假设 β 非空紧值连续，每个严格关系传递且非自反，且“可行且非严格优于”的三元组集合闭。若无极大元，每个 y 被某个 z 严格压过，这些严格下截集构成 $\beta(x)$ 的相对开覆盖。有限子覆盖的有限个 z_i 中存在极大者（传递且无循环），它不被任一 z_i 压过，矛盾。极大元集为所有非优于截集的交，故闭且紧。若 $x_n \to x, y_n$ 为极大元且 $y_n \to y$，先由可行图闭得 $y \in \beta(x)$，再对任意 $z \in \beta(x)$ 用下半连续选 $z_n \in \beta(x_n) \to z$。由 y_n 不差于 z_n 及三元组闭性得极限不被 z 压过。第111页 z_n 应属于可行集 $\beta(x_n)$，不必属于最优集。以连续效用定义严格偏好即可回到 Berge 定理；第112页最优对应内目标须写成 $f(x,y)$，并补紧值条件。

11 109-Proof

11.1 集合取负、作差与闭包的和

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 10 页。

对应已有 gpt-109.pdf 页脚：16。

原课件第 10 页

Vectors in $\mathbb{R}^N$

One defines $-X$ as the set of elements of $\mathbb{R}^N$ of the form $-x$ where $x \in X$. One writes $X^1 - X^2$ for $X^1 + \left(-X^2\right)$. If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, their sum is denoted $\sum_{k=1}^{K} X_k$. It is easy to prove :

(1) If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, then $\sum_{k=1}^{K} \overline{X_k} \subseteq \overline{\sum_{k=1}^{K} X_k}$.

That is, the sum of their adherences is contained in the adherence of their sum.

Proof. If $X_1, \cdots, X_k, \cdots, X_K$ are K subsets of $\mathbb{R}^N$, then for any $k = 1, \cdots, K$ and any $x_k \in \overline{X_k}$, there exists $\left\{x_k^q; q \in \mathbb{N}\right\} \subseteq X_k$ such that $\lim_{q\to\infty} x_k^q = x_k$. Then $\left\{\sum_{k=1}^{K} x_k^q; q \in \mathbb{N}\right\} \subseteq \overline{\sum_{k=1}^{K} X_k}$ and $\lim_{q\to\infty} \sum_{k=1}^{K} x_k^q = \sum_{k=1}^{K} x_k \in \overline{\sum_{k=1}^{K} X_k}$, hence $\sum_{k=1}^{K} \overline{X_k} \subseteq \overline{\sum_{k=1}^{K} X_k}$. □

沿原论证展开与批注 $-A = \{-a : a \in A\}$，$A - B = A + (-B)$，并非集合差 $A \setminus B$。本页命题正确读作

$$\overline{A_1} + \cdots + \overline{A_K} \subseteq \overline{A_1 + \cdots + A_K}.$$

任取左边的 $z = \sum_k a_k$。因为 $a_k \in \overline{A_k}$，对每个 k 选 $a_{k,q} \in A_k$ 使 $a_{k,q} \to a_k$。有限个极限可以相加，故 $z_q = \sum_k a_{k,q} \to z$，且 z_q 在和集内。按闭包定义，z 在右边。横线所在位置至关重要；这并没有说闭集之和一定闭。

11.2 紧集之和紧：先看有界性

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 11 页。

对应已有 gpt-109.pdf 页脚：17。

原课件第 11 页

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. In $\mathbb{R}^N$, a set is compact if and only if it is closed and bounded.

If $X_1, \cdots, X_k, \cdots, X_K$ are K compact subsets of $\mathbb{R}^N$, then they are bounded, there exists a constant M_k such that $\|x_k\| \leqq M_k$ for $x_k \in X_k$, $k = 1, \cdots, K$. For the sum $\sum_{k=1}^{K} X_k$, any element x can be written as $x = \sum_{k=1}^{K} x_k$, where $x_k \in X_k$, thus

$$\|x\| = \left|\sum_{k=1}^{K} x_k\right| \leqq \sum_{k=1}^{K} |x_k| \leqq \sum_{k=1}^{K} M_k.$$

Therefore, the sum $\sum_{k=1}^{K} X_k$ is bounded.

沿原论证展开与批注 若 A_k 紧，则有常数 M_k 使 $\|a_k\| \le M_k$。任取 $z \in \sum_k A_k$，有 $z = \sum_k a_k$，从而 $\|z\| \le \sum_k \|a_k\| \le \sum_k M_k$。上界不依赖所选 z，故整个和集有界。如果有一个 $A_k = \varnothing$，和集为空，紧性自动成立；下面默认各集合非空。此页只完成“有界”，还不能宣告紧。

11.3 紧集之和紧：连续像的短证明

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 12 页。

对应已有 gpt-109.pdf 页脚：18。

原课件第 12 页

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. The sum of closed sets is not necessarily closed. However, in $\mathbb{R}^N$, the sum of compact (which are closed and bounded) sets is closed. This is because the sum of compact sets can be viewed as the image of the compact set $\prod_{k=1}^{K} X_k$ under the continuous function $(x_1, \cdots, x_K) \mapsto \sum_{k=1}^{K} x_k$. Since the continuous image of a compact set is compact, the sum $\sum_{k=1}^{K} X_k$ is closed.

Since the sum of K compact subsets of $\mathbb{R}^N$ is both bounded and closed, it is compact.

沿原论证展开与批注 有限乘积 $P = A_1 \times \cdots \times A_K$ 紧，映射 $F(a_1, \ldots, a_K) = \sum_k a_k$ 连续，而 $F(P) = \sum_k A_k$。连续映射把紧集送到紧集，所以结论成立。这条论证其实已同时得到闭性和有界性。

为什么不能把紧换成闭。 在 $\mathbb{R}^2$ 中取 $A = \{(u, 1/u) : u > 0\}$、$B = \{(v, 0) : v \in \mathbb{R}\}$，两者都闭，然而 $A + B = \mathbb{R} \times (0, \infty)$ 不闭。A 的闭性来自：若它的点列有有限极限，则第一坐标不可能趋于 0 或无穷，极限仍满足乘积为 1 且正。

11.4 紧集之和紧：逐层抽取子列

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 13 页。

对应已有 gpt-109.pdf 页脚：19。

原课件第 13 页

Vectors in $\mathbb{R}^N$

One can also prove:

(2) The sum of K compact subsets of $\mathbb{R}^N$ is compact.

Proof. If $X_1, \cdots, X_k, \cdots, X_K$ are K compact subsets of $\mathbb{R}^N$, for any sequence $\{x^q; q \in \mathbb{N}\} \subseteq \sum_{k=1}^{K} X_k$, we assume $x^q = \sum_{k=1}^{K} x_k^q$ such that $x_k^q \in X_k$ for all $k = 1, \cdots, K$. For compact set X_1, there exists a subsequence $\{x_1^{q_1}; q_1 \in \mathbb{N}\}$ of sequence $\{x_1^q; q \in \mathbb{N}\}$ such that $\lim_{q_1 \to \infty} x_1^{q_1} = x_1^0 \in X_1$. For compact set X_2, there exists a subsequence $\{x_2^{q_2}; q_2 \in \mathbb{N}\}$ of sequence $\{x_2^{q_1}; q_1 \in \mathbb{N}\}$ such that $\lim_{q_2 \to \infty} x_2^{q_2} = x_2^0 \in X_2$ with $\lim_{q_2 \to \infty} x_1^{q_2} = x_1^0 \in X_1$. Continuing this procedure, we have that, for compact set X_K, there exists a subsequence $\{x_K^{q_K}; q_K \in \mathbb{N}\}$ of sequence $\{x_K^{q_{K-1}}; q_{K-1} \in \mathbb{N}\}$ such that $\lim_{q_K \to \infty} x_K^{q_K} = x_K^0 \in X_K$ with $\lim_{k_K \to \infty} x_k^{q_K} = x_k^0 \in X_k$ for $k = 1, \cdots, K-1$.

沿原论证展开与批注 不用连续像定理也能证明。给定 $z_q = \sum_{k=1}^{K} a_{k,q}$，先利用 A_1 紧，挑索引子列，使第一个分量收敛；只在这条子列中再挑，使第二个分量收敛。第一个分量仍然收敛，因为收敛序列的子列保留原极限。重复有限次，得到同一条索引子列 q_j，使每个 $a_{k,q_j} \to a_k \in A_k$。不能分别随意选 K 条不同索引子列后直接相加。

11.5 对应之和半连续：补全漏掉的抽子列步骤

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 17 页。

对应已有 gpt-109.pdf 页脚：23。

原课件第 17 页

Vectors in $\mathbb{R}^N$

Proof. If $x^q \to x^0$, $y^q \in \varphi(x^q)$ and $y^q \to y^0$, then $y^q = \sum_{k=1}^{K} y_k^q$ with $y_k^q \in \varphi_k(x^q)$ and $y^0 = \sum_{k=1}^{K} y_k^0$ with $y_k^q \to y_k^0$. If every φ_k is upper semicontinuous at the point $x^0 \in S$, then $y_k^0 \in \varphi_k\left(x^0\right)$. Thus

$$y^0 = \sum_{k=1}^{K} y_k^0 \in \sum_{k=1}^{K} \varphi_k\left(x^0\right) = \varphi\left(x^0\right)$$

φ is upper semicontinuous at x^0.

If $x^q \to x^0$ and $y^0 \in \varphi\left(x^0\right)$, then $y^0 = \sum_{k=1}^{K} y_k^0$ with $y_k^0 \in \varphi_k\left(x^0\right)$. If every φ_k is lower semicontinuous at the point $x^0 \in S$, then there is $\left(y_k^q\right)$ such that $y_k^q \in \varphi_k(x^q)$ and $y_k^q \to y_k^0$. Thus

$$y^q = \sum_{k=1}^{K} y_k^q \in \sum_{k=1}^{K} \varphi_k(x^q) = \varphi(x^q), \qquad y^q = \sum_{k=1}^{K} y_k^q \to \sum_{k=1}^{K} y_k^0 = y^0$$

φ is lower semicontinuous at x^0. □

沿原论证展开与批注

上半连续。 设 $x_q \to x_0$，$y_q \in \varphi(x_q)$，$y_q \to y_0$。分解 $y_q = \sum_k y_{k,q}$，其中 $y_{k,q} \in \varphi_k(x_q) \subset T_k$。由各 T_k 紧，取共同子列使 $y_{k,q_j} \to y_k$。各对应在 x_0 的闭图性质给 $y_k \in \varphi_k(x_0)$，故 $y = \sum_k y_k \in \varphi(x_0)$。总值域紧，因此这推出所需上半连续性。原页直接写各分量收敛，省略了关键依据。

下半连续。 给定 $y = \sum_k y_k \in \varphi(x_0)$ 和任意 $x_q \to x_0$。各 φ_k 下半连续，存在 $y_{k,q} \in \varphi_k(x_q)$ 且 $y_{k,q} \to y_k$；取和便得 $y_q \in \varphi(x_q)$ 且 $y_q \to y_0$。这里要逼近的是指定的旧选择 y_0。

11.6 正半独立：方向不能抵消

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 23 页。

对应已有 gpt-109.pdf 页脚：29。

原课件第 23 页

Vectors in $\mathbb{R}^N$

m. Given K cones $C_1, \cdots, C_k, \cdots, C_K$ with vertex 0, they are said to be positively semi-independent if "$x_k \in C_k$ for every k, and $\sum_{k=1}^{K} x_k = 0$" implies "$x_k = 0$ for every k," i.e., if it is impossible to take a vector in each cone so that their sum is 0, unless they are all equal to 0. It is clear that two cones C_1, C_2 with vertex 0 are positively semi-independent if and only if $C_1 \cap (-C_2) = \{0\}$.

Proof. If $C_1 \cap (-C_2) = \{0\}$ and for $x_1 \in C_1$ and $x_2 \in C_2$ with $x_1 + x_2 = 0$, then $x_1 = -x_2 \in C_1 \cap (-C_2) = \{0\}$, and hence $x_1 = x_2 = 0$. Thus, C_1 and C_2 are positively semi-independent.

If C_1 and C_2 with vertex 0 are positively semi-independent, and if $x \in C_1 \cap (-C_2)$, then $x \in C_1$ and $-x \in C_2$ with $x + (-x) = 0$, then $x = 0$. Hence $C_1 \cap (-C_2) = \{0\}$. □

沿原论证展开与批注 锥 $C_1, \ldots, C_K$ 正半独立，意思是 $x_k \in C_k$ 且 $\sum_k x_k = 0$ 只能发生在全部 $x_k = 0$ 时。两个锥时等价于 $C_1 \cap (-C_2) = \{0\}$。

两方向证明。 若交仅含 0，由 $x_1 + x_2 = 0$ 得 $x_1 = -x_2 \in C_1 \cap (-C_2)$，所以两者均零。反过来，若正半独立，任取 $z \in C_1 \cap (-C_2)$，则 $z \in C_1$、$-z \in C_2$，两者和为零，故 $z = 0$。两个同向非负射线正半独立，两个相反射线则不满足。

11.7 为什么截尾锥递减；闭包不能漏

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 25 页。

对应已有 gpt-109.pdf 页脚：31。

原课件第 25 页

Vectors in $\mathbb{R}^N$

Remark. If $M_1 \leqq M_2$, then $S^{M_1} \supseteq S^{M_2}$ and $\Gamma\left(S^{M_1}\right) \supseteq \Gamma\left(S^{M_2}\right)$. $\Gamma\left(S^M\right)$ is a monotonically decreasing set-sequence. Hence

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right) = \lim_{M \to \infty} \Gamma\left(S^M\right).$$

Therefore $\mathbf{A}S$ is the cone generated by the points in S which are at infinity.

As $\Gamma\left(S^M\right)$ is the least closed cone with vertex 0 containing S_M, then we can see that

$$\begin{aligned} \Gamma\left(S^M\right) &= \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}} \\ &= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S^M}\right\} \end{aligned}$$

沿原论证展开与批注 若 $M_1 \leq M_2$，距离门槛提高，故 $S^{M_2} \subset S^{M_1}$，进而 $\Gamma(S^{M_2}) \subset \Gamma(S^{M_1})$。所谓极限就是递减集合族的交。准确的公式是

$$\Gamma(E) = \overline{\{\alpha x : \alpha \geq 0,\ x \in E\} \cup \{0\}}.$$

闭包负责加入仅能逼近的边界方向；$\{0\}$ 负责空集情形。去掉闭包、或把“存在任意接近的点”误写成“恰好存在同方向的点”，都会使后面的证明失效。

11.8 有界当且仅当没有非零渐近方向

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 29 页。

对应已有 gpt-109.pdf 页脚：35。

原课件第 29 页

Vectors in $\mathbb{R}^N$

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

Proof. It is clear that C_∞ cannot contain any non-zero direction if C is bounded. As C is bounded, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = 0$. Hence $C_\infty = \{0\}$.

Conversely, if C is unbounded, then there exists a sequence $\{x_k; k \in \mathbb{N}\} \subseteq C$ with $x_k \neq 0$ for all $k \in \mathbb{N}$ such that $\|x_k\| \to \infty$ and thus the vectors $d_k = \frac{x_k}{\|x_k\|} \in \{d \in \mathbb{R}^N : \|d\| = 1\}$.
Therefore, we can extract a subsequence of $\{d_k\}$ such that $\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k = d$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|d\| = 1$. This non-zero vector d is an element of C_∞, a contradiction! □

沿原论证展开与批注 若 $\|x\| \le M$ 对所有 $x \in C$ 成立，那么任意 x_k/t_k 的范数至多 $M/t_k \to 0$，故只可能得到 0；非空性保证 0 确实存在。反向用逆否命题：若 C 无界，对每个整数 k 选 $x_k \in C$ 使 $\|x_k\| \ge k$。单位向量 $u_k = x_k/\|x_k\|$ 位于紧的单位球面，所以有子列趋向 u，且 $\|u\| = 1$。取 $t_j = \|x_{k_j}\| \to \infty$，就得到 $u \in C_\infty$，是非零方向。关键是“无界—归一化—球面紧—非零极限”。

11.9 单调性、平移不变性与乘积的预告

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 30 页。

对应已有 gpt-109.pdf 页脚：36。

原课件第 30 页

Vectors in $\mathbb{R}^N$

o. It is obvious that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$".

Proof. If $S_1 \subseteq S_2$, then $S_1^M \subseteq S_2^M$ for $M \geqq 0$, thus $\Gamma\left(S_1^M\right) \subseteq \Gamma\left(S_2^M\right)$ for $M \geqq 0$, and hence $\mathbf{A}S_1 \subseteq \mathbf{A}S_2$.

One can prove that:

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

In other words, a translation of S does not alter its asymptotic cone. It is then easy to prove that:

(6) If $T \neq \varnothing$ and S are two subsets of $\mathbb{R}^N$, then $\mathbf{A}S \subseteq \mathbf{A}(S + T)$.

One can also prove that:

(7) If, for every $k = 1, \cdots, K$, S_k is a subset of $\mathbb{R}^{N_k}$, then $\mathbf{A}\left(\prod_{k=1}^{K} S_k\right) \subseteq \prod_{k=1}^{K} \mathbf{A}S_k$.

Proof of (6). Let $x \in T$. Then $\mathbf{A}S = \mathbf{A}(S + \{x\}) \subseteq \mathbf{A}(S + T)$. □

沿原论证展开与批注 $S_1 \subset S_2$ 时，在 S_1 中见证某渐近方向的点列也在 S_2 中，所以 $AS_1 \subset AS_2$。若 $T \neq \varnothing$，取固定 $a \in T$，则 $S + a \subset S + T$；结合平移不变性，$AS = A(S + a) \subset A(S + T)$。这里 a 必须固定，不能随序列任意变化。乘积的结论一般只是包含，原因是各分量需要共同的缩放尺度，见第33页。

11.10 平移不改变渐近锥：逐步证明

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 31 页。

对应已有 gpt-109.pdf 页脚：37。

原课件第 31 页

Vectors in $\mathbb{R}^N$

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S+\{x\}) = \mathbf{A}S$.

Proof. We just need to prove $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$, then to prove $\mathbf{A}S \subseteq \mathbf{A}(S+\{x\})$, we just need to let $S = (S+\{x\})+\{-x\}$ and we arrive at the conclusion. Let $T = S+\{x\}$.

For any $d \in \mathbf{A}(S+\{x\})$, $d \in (S+\{x\})_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq S+\{x\}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Let $\{x'_k = x_k - x\} \subseteq S$ and

$$\left\| \frac{x'_k}{t_k} - d \right\| \leqq \left\| \frac{x'_k}{t_k} - \frac{x_k}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| = \left\| \frac{x}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| \to 0$$

hence $\lim\limits_{k\to\infty} \dfrac{x'_k}{t_k} = d$, i.e., $d \in S_\infty$, hence $d \in \mathbf{A}S$. Hence $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$. □

沿原论证展开与批注 设 $d \in (S+a)_\infty$，存在 $z_k \in S+a$、$t_k \to \infty$ 使 $z_k/t_k \to d$。写 $z_k = x_k + a$，$x_k \in S$，则

$$\|x_k/t_k - d\| \leq \|z_k/t_k - d\| + \|a\|/t_k \to 0.$$

故 $(S+a)_\infty \subset S_\infty$。把 $S+a$ 当作新集合，再平移 $-a$，得到反向包含。直观是固定距离的搬动除以越来越大的尺度后消失。

11.11 平移不变性的等价链写法

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 32 页。

对应已有 gpt-109.pdf 页脚：38。

原课件第 32 页

Vectors in $\mathbb{R}^N$

(5) If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

Proof.

$$
\begin{aligned}
& d \in \mathbf{A}(S) \Longleftrightarrow d \in S_\infty \\
\Longleftrightarrow \quad & \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \\
\Longleftrightarrow \quad & \text{there exist } \{x_k + x; k \in \mathbb{N}\} \subseteq S + \{x\} \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k + x}{t_k} = d \\
\Longleftrightarrow \quad & d \in (S + \{x\})_\infty \Longleftrightarrow d \in \mathbf{A}(S + \{x\})
\end{aligned}
$$

□

In other words, a translation of S does not alter its asymptotic cone.

沿原论证展开与批注 本页将第31页的双包含压缩为等价关系。核心等式是 $(x_k + a)/t_k = x_k/t_k + a/t_k$，最后一项趋于 0，所以两边具有同一个极限。练习时先自行写出 x_k 属于哪个集合，再写极限；不必背每一行排版。若平移量变为 a_k 且 a_k/t_k 不趋零，结论便可能改变。

11.12 乘积的渐近锥包含于各渐近锥的乘积

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 33 页。

对应已有 gpt-109.pdf 页脚：39。

原课件第 33 页

Vectors in $\mathbb{R}^N$

(7) If, for every $k = 1, \cdots, K$, S_k is a subset of $\mathbb{R}^{N_k}$, then $\mathbf{A}\left(\prod_{k=1}^{K} S_k\right) \subseteq \prod_{k=1}^{K} \mathbf{A}S_k$.

Proof. For $d \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$. If $d = 0$, then $d \in \prod_{n=1}^{N} \mathbf{A}S_n$ is trivial. If $d \neq 0$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \prod_{n=1}^{N} S_n$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let

$$x_k = \left(x_k^1, \cdots, x_k^n, \cdots, x_k^N\right), d = \left(d^1, \cdots, d^n, \cdots, d^N\right)$$

where $x_k^n \in S_n$ and $d^n \in \mathbb{R}$ for $n = 1, \cdots, N$. For each $n = 1, \cdots, N$, we have $\lim_{k\to\infty} \frac{x_k^n}{t_k} = d^n$. Hence $d^n \in \mathbf{A}S_n$, hence $d \in \prod_{n=1}^{N} \mathbf{A}S_n$. Thus, $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$. □

沿原论证展开与批注 设 $C = S_1 \times \cdots \times S_K$ 非空，$d \in C_\infty$。取 $x_q = (x_{1,q}, \ldots, x_{K,q}) \in C$、$t_q \to \infty$ 使 $x_q/t_q \to d = (d_1, \ldots, d_K)$。向量收敛蕴含每个坐标块收敛，所以 $x_{k,q}/t_q \to d_k$，即 $d_k \in (S_k)_\infty$。因此

$$C_\infty \subset \prod_{k=1}^{K} (S_k)_\infty.$$

反向不自动成立：各个方向的见证序列可能只在互不匹配的尺度出现。对凸集，可由第105–108页的渐近正则性解决尺度匹配问题。

11.13 交集有界：用单调性证明

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 35 页。

对应已有 gpt-109.pdf 页脚：41。

原课件第 35 页

Vectors in $\mathbb{R}^N$

(8) Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

Proof. As we have that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$", then, for an arbitrary index set Γ, let $C_\gamma \subseteq \mathbb{R}^N$, $\gamma \in \Gamma$, we have

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \mathbf{A}C_\gamma$$

for all $\gamma \in \Gamma$. Hence

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma$$

If $\bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma = \{0\}$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) = \{0\}$. By Proposition 2.3, we have $\bigcap_{\gamma\in\Gamma} C_\gamma$ is bounded. □

沿原论证展开与批注 令 $D = \bigcap_{\gamma\in I} C_\gamma$。若 $D = \varnothing$，有界性直接成立。若非空，则 $D \subset C_\gamma$ 给 $D_\infty \subset (C_\gamma)_\infty$ 对每个 γ 成立，因此

$$D_\infty \subset \bigcap_{\gamma} (C_\gamma)_\infty = \{0\}.$$

因为 D 非空，$0 \in D_\infty$，于是 $D_\infty = \{0\}$，第29页即给 D 有界。这就是“先处理空集，再使用非空定理”的标准写法。

11.14 不能从递减交直接跳到某个截尾为空

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 36 页。

对应已有 gpt-109.pdf 页脚：42。

原课件第 36 页

Vectors in $\mathbb{R}^N$

(8) For all $\gamma \in \Gamma$, $S_\gamma \subseteq \mathbb{R}^N$, if $\bigcap_{\gamma\in\Gamma} \mathbf{A}(S_\gamma) = \{0\}$, then $\bigcap_{\gamma\in\Gamma} S_\gamma$ is bounded.

Proof. For all $\gamma \in \Gamma$, $\bigcap_{\gamma\in\Gamma} S_\gamma \subseteq S_\gamma$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) \subseteq \mathbf{A}(S_\gamma)$.

Whence $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}(S_\gamma) = \{0\}$. There exists $M_0 \geqq 0$ such that $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)^M = \varnothing$ for $M \geqq M_0$, thus $\bigcap_{\gamma\in\Gamma} S_\gamma$ is bounded.

□

沿原论证展开与批注 本页用截尾集合重述第35页。结论正确，但中间“存在 M_0 使截尾为空”需要第29页的有界性判别，不能只由递减集合交为空或交为 $\{0\}$ 直接断言。一般递减非空集可以有空交，例如 $[n,\infty)$。在这里，可反证：若交集无界，就抽出单位极限方向，同时属于全部渐近锥，与仅含零矛盾。

11.15 线性像证明的核心：归一化反证

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 38–41 页。

对应已有 gpt-109.pdf 页脚：44, 45, 46, 47。

原课件第 38 页

Vectors in $\mathbb{R}^N$

Proof. The condition will be shown first to imply that L carries unbounded sequences in C into unbounded sequences in $\mathbb{R}^K$. If this were not true, there would exist a sequence of points $\{x_k; k \in \mathbb{N}\} \subseteq C$ such that $\|x_k\| \to \infty$, but such that the sequence of images $y_k = L(x_k)$ is bounded. As $\left\{\frac{x_k}{\|x_k\|}; k \in \mathbb{N}\right\}$ is bounded, so we can extract a subsequence of $\left\{\frac{x_k}{\|x_k\|}; k \in \mathbb{K}\right\}$ such that $\lim\limits_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} = x$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|x\| = 1$. Hence $x \in C_\infty$, but also (because linear mappings are continuous)

$$L(x) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{1}{\|x_k\|} L(x_k) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{y_k}{\|x_k\|} = 0$$

which the condition prohibits.

原课件第 39 页

Vectors in $\mathbb{R}^N$

Assuming henceforth that the condition holds, we prove now that $L(C)$ is a closed set. Suppose $y_k \to y$ with $y_k \in L(C)$, or in other words, $y_k = L(x_k)$ for some $x_k \in C$. The sequence $\{L(x_k)\}$ is bounded, so the sequence $\{x_k; k \in \mathbb{N}\}$ must be bounded as well, for the reasons just given, and it therefore has a cluster point x. We have $x \in C$ (because C is closed) and $y = L(x)$ (again because linear mappings are continuous), and therefore $y \in L(C)$.

原课件第 40 页

Vectors in $\mathbb{R}^N$

The next task is to show that $L(C_\infty) \subseteq L(C)_\infty$. This is trivial when $C = \varnothing$, so we assume $C \neq \varnothing$. Any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. The points $y_k = L(x_k)$ in $L(C)$ and

$$\lim_{k\to\infty} \frac{y_k}{t_k} = \lim_{k\to\infty} \frac{L(x_k)}{t_k} = L\left(\lim_{k\to\infty} \frac{x_k}{t_k}\right) = L(d)$$

hence $L(d) \in L(C)_\infty$. Hence $L(C_\infty) \subseteq L(C)_\infty$.

原课件第 41 页

Vectors in $\mathbb{R}^N$

For the inclusion $L(C_\infty) \supseteq L(C)_\infty$, we consider any $y \in L(C)_\infty$, there exist $\{y_k; k \in \mathbb{N}\} \subseteq L(C)$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{y_k}{t_k} = y$. For any choice of $x_k \in C$ with $L(x_k) = y_k$, we have $\lim_{k\to\infty} L\left(\frac{x_k}{t_k}\right) = y$. The boundedness of the sequence $\left\{L\left(\frac{x_k}{t_k}\right); k \in \mathbb{N}\right\}$ implies the boundedness of the sequence $\left\{\frac{x_k}{t_k}; k \in \mathbb{N}\right\}$, which therefore must have a cluster point x. Then $x \in C_\infty$ and $L(x) = y$, and we conclude that $y \in L(C_\infty)$. □

沿原论证展开与批注 若存在 $x_k \in C$，$\|x_k\| \to \infty$，而 $L(x_k)$ 有界，则单位向量取子列 $x_k/\|x_k\| \to u$，$\|u\| = 1$。按定义 $u \in C_\infty$；连续性和线性给

$$L(u) = \lim_k \frac{L(x_k)}{\|x_k\|} = 0.$$

于是 $u \in \ker L \cap C_\infty$ 且非零，与条件矛盾。更准确的可用结论是：C 中任何像有界的点列自身必有界；若自身无界，先抽取范数趋无穷的子列，再运用本证明。

任取 $y_k \in L(C)$ 且 $y_k \to y$，每次选一个原像 $x_k \in C$ 满足 $L(x_k) = y_k$。(y_k) 有界，第38页使 (x_k) 有界；抽取 $x_{k_j} \to x$。C 闭给 $x \in C$，连续性给 $L(x) = \lim_j y_{k_j} = y$，故 $y \in L(C)$。注意 L 可能不单射，原像无须唯一；任意选取的原像都受这里的有界性论证控制。

任取 $d \in C_\infty$，选 $x_k \in C$、$t_k \to \infty$ 使 $x_k/t_k \to d$。由于 $L(x_k) \in L(C)$ 且

$$\frac{L(x_k)}{t_k} = L\left(\frac{x_k}{t_k}\right) \longrightarrow L(d),$$

得 $L(d) \in (L(C))_\infty$。这一步既不使用 C 闭，也不使用核条件。考试中指出哪些假设用在哪里，能避免把强条件当成所有步骤的理由。

须补严谨步骤，不能直接把第38页用于不在 C 内的 x_n/t_n。若 $Lx_n/t_n \to y$，假设 x_n/t_n 无界，则抽子列使 $r_n = \|x_n\|/t_n \to \infty$。归一化后再抽 $x_n/\|x_n\| \to d$，其中 $\|d\| = 1$，$d \in C_\infty$ 且 $Ld = \lim(Lx_n/t_n)/r_n = 0$，违反核条件。因此 x_n/t_n 有界，抽子列趋于 d，定义给 $d \in C_\infty$ 且 $Ld = y$。

11.16 闭集和与正半独立

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 43–44 页。

对应已有 gpt-109.pdf 页脚：49, 50。

原课件第 43 页

Vectors in $\mathbb{R}^N$

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.3 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}$$

so by Theorem 3.5, we have $L(C)$ is closed. As $L(C) = \sum_{n=1}^{N} C_n$, we have that $\sum_{n=1}^{N} C_n$ is closed.

原课件第 44 页

Vectors in $\mathbb{R}^N$

And again by Theorem 3.5, $L(C_\infty) = L(C)_\infty$, i.e.

$$L\left(\left(\prod_{n=1}^{N} C_n\right)_\infty\right) = \left(\sum_{n=1}^{N} C_n\right)_\infty$$

We also have

$$L\left(\prod_{n=1}^{N} (C_n)_\infty\right) = \sum_{n=1}^{N} (C_n)_\infty$$

and

$$\left(\prod_{n=1}^{N} C_n\right)_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$$

hence

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$$

□

沿原论证展开与批注 正半独立指若 $d_i \in (C_i)_\infty$ 且 $\sum_i d_i = 0$，则每个 $d_i = 0$。设积集 $C = \prod_i C_i$，线性映射 $L(x_1, \dots, x_m) = \sum_i x_i$。逐坐标极限给 $C_\infty \subseteq \prod_i (C_i)_\infty$，所以核条件成立。前面的线性像定理给 $L(C) = \sum_i C_i$ 闭，且 $(\sum_i C_i)_\infty = L(C_\infty) \subseteq \sum_i (C_i)_\infty$。这说明渐近方向不能互相抵消是保证闭和的重要条件；闭集的和一般不自动闭。

11.17 闭集和与正半独立

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 45 页。

对应已有 gpt-109.pdf 页脚：51。

原课件第 45 页

Vectors in $\mathbb{R}^N$

(9) Given K closed subsets of $\mathbb{R}^N$, if their asymptotic cones are positively semi-independent, then their sum is closed.

Proof. Let $C = \prod_{n=1}^{N} C_n$. Then $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$ from Theorem 3.3. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n, \qquad \text{for} \quad x_n \in C_n,$$

then L is a linear mapping.

If $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then

$$x_n \in (C_n)_\infty \text{ for } n = 1, \cdots, N \quad \text{and} \quad \sum_{n=1}^{N} x_n = 0 \quad \text{implies} \quad x_n = 0 \text{ for } n = 1, \cdots, N$$

$$(x_1, \cdots, x_N) \in \prod_{n=1}^{N} (C_n)_\infty \quad \text{and} \quad L((x_1, \cdots, x_N)) = 0 \quad \text{implies} \quad (x_1, \cdots, x_N) = \mathbf{0}$$

$$(x_1, \cdots, x_N) \in L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty \quad \text{implies} \quad (x_1, \cdots, x_N) = \mathbf{0}$$

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}.$$

For closed sets $C_n \in \mathbb{R}^L$, $n = 1, \cdots, N$, C is closed. By Theorem 3.5, we have $L(C) = \sum_{n=1}^{N} C_n$ is closed.

沿原论证展开与批注 正半独立指若 $d_i \in (C_i)_\infty$ 且 $\sum_i d_i = 0$，则每个 $d_i = 0$。设积集 $C = \prod_i C_i$，线性映射 $L(x_1, \ldots, x_m) = \sum_i x_i$。逐坐标极限给 $C_\infty \subseteq \prod_i (C_i)_\infty$，所以核条件成立。前面的线性像定理给 $L(C) = \sum_i C_i$ 闭，且 $(\sum_i C_i)_\infty = L(C_\infty) \subseteq \sum_i (C_i)_\infty$。这说明渐近方向不能互相抵消是保证闭和的重要条件；闭集的和一般不自动闭。

11.18 凸性的基本运算及连通性

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 49 页。

对应已有 gpt-109.pdf 页脚：55。

原课件第 49 页



(10) the intersection of a set of convex sets is convex.

Proof of (10). If $\{X_\gamma; \gamma \in \Gamma\}$ is a family of convex subsets of X. Then for any x_1 and x_2 in $\bigcap_{\gamma\in\Gamma} X_\gamma$, then for any $\gamma \in \Gamma$, we have x_1 and x_2 in X_γ. For any $0 \leqslant t \leqslant 1$, we have $tx_1 + (1-t)x_2 \in X_\gamma$, hence $tx_1 + (1-t)x_2 \in \bigcap_{\gamma\in\Gamma} X_\gamma$. So $\bigcap_{\gamma\in\Gamma} X_\gamma$ is convex. □

沿原论证展开与批注 交：两端点属于每个凸因子，线段逐个属于它们，故属于交。和：写 $x = \sum x_i, y = \sum y_i$，则 $tx + (1-t)y = \sum(tx_i + (1-t)y_i)$ 属于和；积逐坐标同理。闭包：选 $x_n, y_n \in C$ 趋于闭包两点，凸组合在 C，极限在闭包。凸集连通：若被相对开集分离，连接两侧点的连续路径 $t \mapsto tx + (1-t)y$ 把分离拉回 $[0,1]$，与区间连通矛盾。最后若 C 闭凸且含0，$x_n/t_n \to d$ 且 $t_n \to \infty$，充分大时 $x_n/t_n = (1 - 1/t_n)0 + x_n/t_n \in C$，闭性给 $d \in C$。

11.19 凸性的基本运算及连通性

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 50 页。

对应已有 gpt-109.pdf 页脚：56。

原课件第 50 页

Vectors in $\mathbb{R}^N$

(11) the sum and the product of K convex sets are convex.

Proof of (11). Let $X_1, \cdots, X_K$ be convex. For any x^1 and x^2 in $\sum_{k=1}^{K} X_k$, there exist $x_k^1 \in X_k$ and $x_k^2 \in X_k$ for $k = 1, \cdots, K$ such that $x^1 = \sum_{k=1}^{K} x_k^1$ and $x^2 = \sum_{k=1}^{K} x_k^2$. For any $t \in [0,1]$, we have

$$tx^1 + (1-t)\,x^2 = t\sum_{k=1}^{K} x_k^1 + (1-t)\sum_{k=1}^{K} x_k^2 = \sum_{k=1}^{K}\left[tx_k^1 + (1-t)\,x_k^2\right] \in \sum_{k=1}^{K} X_k$$

hence $\sum_{k=1}^{K} X_k$ is convex.

Let $x^i \in \prod_{k=1}^{K} X_k$ with $x^i = \left(x_1^i, \cdots, x_k^i, \cdots, x_K^i\right)$, $i =, 2$. For any $t \in [0,1]$, we have

$$\begin{aligned} & tx^1 + (1-t)\,x^2 = t\left(x_1^1, \cdots, x_k^1, \cdots, x_K^1\right) + (1-t)\left(x_1^2, \cdots, x_k^2, \cdots, x_K^2\right) \\ = \quad & \left(tx_1^1 + (1-t)x_1^2, \cdots, tx_k^1 + (1-t)x_k^2, \cdots, tx_K^1 + (1-t)x_K^2\right) \in \prod_{k=1}^{K} X_k \end{aligned}$$

hence $\prod_{k=1}^{K} X_k$ is convex. □

沿原论证展开与批注 交：两端点属于每个凸因子，线段逐个属于它们，故属于交。和：写 $x = \sum x_i, y = \sum y_i$，则 $tx + (1-t)y = \sum(tx_i + (1-t)y_i)$ 属于和；积逐坐标同理。闭包：选 $x_n, y_n \in C$ 趋于闭包两点，凸组合在 C，极限在闭包。凸集连通：若被相对开集分离，连接两侧点的连续路径 $t \mapsto tx + (1-t)y$ 把分离拉回 $[0,1]$，与区间连通矛盾。最后若 C 闭凸且含0，$x_n/t_n \to d$ 且 $t_n \to \infty$，充分大时 $x_n/t_n = (1 - 1/t_n)0 + x_n/t_n \in C$，闭性给 $d \in C$。

11.20 凸性的基本运算及连通性

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 51 页。

对应已有 gpt-109.pdf 页脚：57。

原课件第 51 页

Vectors in $\mathbb{R}^N$

(12) the adherence of a convex set is convex.

Proof of (12). Let x^1 and x^2 in convex set $\overline{X}$. By the definition of closure, there exist sequences $\left\{x_q^1\right\} \subseteq X$ and $\left\{x_q^2\right\} \subseteq X$ with limits x^1 and x^2, respectively. Now, because X is convex, we know that for each x_q^1 and x_q^2, and for all $t \in [0,1]$, we have $tx_q^1 + (1-t)x_q^2 \in X$. This establishes the sequence $\left\{tx_q^1 + (1-t)x_q^2\right\} \subseteq X$ which is convergent by the convergence of $\left\{x_q^1\right\}$ and $\left\{x_q^2\right\}$. So we have that $tx^1 + (1-t)\,x^2 \in \overline{X}$, as $\overline{X}$ includes all limit points of X. This establishes that $\overline{X}$ is convex. □

沿原论证展开与批注 交：两端点属于每个凸因子，线段逐个属于它们，故属于交。和：写 $x = \sum x_i, y = \sum y_i$，则 $tx + (1-t)y = \sum(tx_i + (1-t)y_i)$ 属于和；积逐坐标同理。闭包：选 $x_n, y_n \in C$ 趋于闭包两点，凸组合在 C，极限在闭包。凸集连通：若被相对开集分离，连接两侧点的连续路径 $t \mapsto tx + (1-t)y$ 把分离拉回 $[0,1]$，与区间连通矛盾。最后若 C 闭凸且含0，$x_n/t_n \to d$ 且 $t_n \to \infty$，充分大时 $x_n/t_n = (1 - 1/t_n)0 + x_n/t_n \in C$，闭性给 $d \in C$。

11.21 凸性的基本运算及连通性

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 52 页。

对应已有 gpt-109.pdf 页脚：58。

原课件第 52 页

Vectors in $\mathbb{R}^N$

(13) a convex set is connected.

Proof of (13). Suppose $E \subseteq \mathbb{R}^N$ is convex but not connected, then there exist non-empty open sets U_1 and U_2 in E such that $U_1 \cup U_2 = E$ and $U_1 \cap U_2 = \varnothing$. Let $x_1 \in U_1$ and $x_2 \in U_2$, We define

$$V_1 = \left\{t \in [0,1] : tx^1 + (1-t)\,x^2 \in U_1\right\}$$

and

$$V_2 = \left\{t \in [0,1] : tx^1 + (1-t)\,x^2 \in U_2\right\}$$

then $V_1 \neq \varnothing$ and $V_2 \neq \varnothing$ and $V_1 \cup V_2 = [0,1]$ and $V_1 \cap V_2 = \varnothing$. As U_1 and U_2 are open, hence V_1 and V_2 are also open on $[0,1]$. Hence $[0,1]$ is disconnected, contradicting (8) of 1.6.u. Hence E is connected. □

沿原论证展开与批注 交：两端点属于每个凸因子，线段逐个属于它们，故属于交。和：写 $x = \sum x_i, y = \sum y_i$，则 $tx + (1-t)y = \sum(tx_i + (1-t)y_i)$ 属于和；积逐坐标同理。闭包：选 $x_n, y_n \in C$ 趋于闭包两点，凸组合在 C，极限在闭包。凸集连通：若被相对开集分离，连接两侧点的连续路径 $t \mapsto tx + (1-t)y$ 把分离拉回 $[0,1]$，与区间连通矛盾。最后若 C 闭凸且含0，$x_n/t_n \to d$ 且 $t_n \to \infty$，充分大时 $x_n/t_n = (1 - 1/t_n)0 + x_n/t_n \in C$，闭性给 $d \in C$。

11.22 凸性的基本运算及连通性

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 53 页。

对应已有 gpt-109.pdf 页脚：59。

原课件第 53 页

Vectors in $\mathbb{R}^N$

(14) a closed, convex set owning 0, contains its asymptotic cone.

Proof of (14). Let $y \in \mathbf{A}C$. For any $M \geqq 0$, $\mathbb{R}_+ y \in \Gamma\left(C^M\right)$.

(1) $y \in C^M$, then $y \in C$.

(2) $y \notin C^M$, then there exists $\lambda < 1$ such that $\frac{1}{\lambda} y \in C^M \subseteq C$,

hence $y = (1-\lambda)0 + \lambda \frac{1}{\lambda} y \in C$.

Therefore $y \in \mathbf{A}C$ implies $y \in C$. That is, $\mathbf{A}C \subseteq C$. □

沿原论证展开与批注 交：两端点属于每个凸因子，线段逐个属于它们，故属于交。和：写 $x = \sum x_i, y = \sum y_i$，则 $tx + (1-t)y = \sum (tx_i + (1-t)y_i)$ 属于和；积逐坐标同理。闭包：选 $x_n, y_n \in C$ 趋于闭包两点，凸组合在 C，极限在闭包。凸集连通：若被相对开集分离，连接两侧点的连续路径 $t \mapsto tx + (1-t)y$ 把分离拉回 $[0,1]$，与区间连通矛盾。最后若 C 闭凸且含0，$x_n/t_n \to d$ 且 $t_n \to \infty$，充分大时 $x_n/t_n = (1 - 1/t_n)0 + x_n/t_n \in C$，闭性给 $d \in C$。

11.23 凸性的基本运算及连通性

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 54 页。

对应已有 gpt-109.pdf 页脚：60。

原课件第 54 页

Vectors in $\mathbb{R}^N$

(14) a closed, convex set owning 0, contains its asymptotic cone.

Proof of (14). Assume $d \in C_\infty$: By definition of the asymptotic cone, there exists a sequence $\{x_k\} \subseteq C$ and a sequence $\{t_k\} \subseteq \mathbb{R}$ with $t_k \to \infty$, such that $\dfrac{x_k}{t_k} \to d$.

- Since C is convex and contains 0, for each k, the point $\lambda_k x_k$ lies in C for $\lambda_k \in [0,1]$. Choose $\lambda_k = \dfrac{1}{t_k}$, then $\lambda_k x_k = \dfrac{x_k}{t_k}$.
- As $t_k \to \infty$, $\lambda_k \to 0$, and $\dfrac{x_k}{t_k} \to d$.
- Since C is closed and $\dfrac{x_k}{t_k} \in C$ for all k, the limit d must also belong to C, $d \in C$.

Since $d \in C_\infty$ implies $d \in C$, we have $C_\infty \subseteq C$.

沿原论证展开与批注 交：两端点属于每个凸因子，线段逐个属于它们，故属于交。和：写 $x = \sum x_i, y = \sum y_i$，则 $tx + (1-t)y = \sum(tx_i + (1-t)y_i)$ 属于和；积逐坐标同理。闭包：选 $x_n, y_n \in C$ 趋于闭包两点，凸组合在 C，极限在闭包。凸集连通：若被相对开集分离，连接两侧点的连续路径 $t \mapsto tx + (1-t)y$ 把分离拉回 $[0,1]$，与区间连通矛盾。最后若 C 闭凸且含0，$x_n/t_n \to d$ 且 $t_n \to \infty$，充分大时 $x_n/t_n = (1 - 1/t_n)0 + x_n/t_n \in C$，闭性给 $d \in C$。

11.24 凸包与和的交换及证明纠错

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 57–59 页。

对应已有 gpt-109.pdf 页脚：63, 64, 65。

原课件第 57 页

Vectors in $\mathbb{R}^N$

Proof of (15). Prove $\dot{S} \subseteq \sum_{k=1}^{K} \dot{S}_k$.

Let $x \in \dot{S}$. By definition, x is a convex combination of points in $S = \sum_{k=1}^{K} S_k$. $x = \sum_{i=1}^{I} \lambda_i \sum_{k=1}^{K} x_{ki}$, where $x_{ki} \in S_k$, $\lambda_i \geqq 0$, and $\sum_{i=1}^{I} \lambda_i = 1$. Rearranging the sum, we have $x = \sum_{k=1}^{K} \sum_{i=1}^{I} \lambda_i x_{ki}$. Each term $\sum_{i=1}^{I} \lambda_i x_{ki}$ is a convex combination of points in S_k, so $\sum_{i=1}^{I} \lambda_i x_{ki} \in \dot{S}_k$. Therefore $x \in \sum_{k=1}^{K} \dot{S}_k$. This proves $\dot{S} \subseteq \sum_{k=1}^{K} \dot{S}_k$.

原课件第 58 页

Vectors in $\mathbb{R}^N$

Proof of (15). Prove $\sum_{k=1}^{K} \dot{S}_k \subseteq \dot{S}$.

Let $x \in \sum_{k=1}^{K} \dot{S}_k$. By definition, $x = \sum_{k=1}^{K} x_k$, where $x_k \in \dot{S}_k$ for $k = 1, \cdots, K$. Each x_k is a convex combination of points in S_k. $x_k = \sum_{i_k=1}^{I_k} \lambda_{ki_k} x_{ki_k}$, where $x_{ki_k} \in S_k$, $\lambda_{ki_k} \geqq 0$, and $\sum_{i_k=1}^{I_k} \lambda_{ki_k} = 1$.

Substituting into x, we have $x = \sum_{k=1}^{K} \sum_{i_k=1}^{I_k} \lambda_{ki_k} x_{ki_k}$. This is a convex combination of points in $S = \sum_{k=1}^{K} S_k$, because $\sum_{k=1}^{K} \sum_{i=1}^{I_k} \lambda_{ki} = \sum_{k=1}^{K} 1 = K$ and we can normalize the coefficients to form a convex combination. Therefore $x \in \dot{S}$. This proves $\sum_{k=1}^{K} \dot{S}_k \subseteq \dot{S}$.

原课件第 59 页

Vectors in $\mathbb{R}^N$

$$
\begin{aligned}
x &= \sum_{k=1}^{K}\sum_{i_k=1}^{I_k}\lambda_{ki_k}x_{ki_k} = \sum_{i_1=1}^{I_1}\lambda_{1i_1}x_{1i_1} + \cdots + \sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{Ki_K} \\
&= \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{1i_1} + \cdots + \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}x_{Ki_K} \\
&= \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}\left(x_{1i_1}+\cdots+x_{Ki_K}\right) = \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K}\sum_{k=1}^{K}x_{ki_k} \\
&= \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\left[\lambda_{1i_1}\cdots\lambda_{Ki_K}\right]\sum_{k=1}^{K}x_{ki_k} = \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\left[\prod_{k=1}^{K}\lambda_{ki_k}\right]\left[\sum_{k=1}^{K}x_{ki_k}\right]
\end{aligned}
$$

where $\sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\prod_{k=1}^{K}\lambda_{ki_k} = \sum_{i_1=1}^{I_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{1i_1}\cdots\lambda_{Ki_K} = \sum_{i_1=1}^{I_1}\lambda_{1i_1}\cdots\sum_{i_K=1}^{I_K}\lambda_{Ki_K} = 1$, then $x \in \dot{S}$. □

沿原论证展开与批注 凸包是所有包含 S 的凸集的交，等价于有限凸组合全体。闭凸包是它的闭包。证明 $\operatorname{co}(A+B)=\operatorname{co}A+\operatorname{co}B$：一边把 $\sum\lambda_i(a_i+b_i)$ 分成两凸组合；另一边若 $a=\sum_i\alpha_i a_i, b=\sum_j\beta_j b_j$，则 $a+b=\sum_{i,j}\alpha_i\beta_j(a_i+b_j)$，积权重非负且总和1。有限多集合用权重 $\prod_k\lambda_{ki_k}$。第58页直接把总权重 K 归一化会把目标点也除以 K，因此不成立；第59页积权重论证才正确。

11.25 三种无穷远方向及量词

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 69–70 页。

对应已有 gpt-109.pdf 页脚：75, 76。

原课件第 69 页

Definitions

Remark. If $M_1 \leqq M_2$, then $S^{M_1} \supseteq S^{M_2}$ and $\Gamma\left(S^{M_1}\right) \supseteq \Gamma\left(S^{M_2}\right)$. $\Gamma\left(S^M\right)$ is a monotonically decreasing set-sequence. Hence

$$\mathbf{A}S = \bigcap_{M \geqq 0} \Gamma\left(S^M\right) = \lim_{M \to \infty} \Gamma\left(S^M\right).$$

Therefore $\mathbf{A}S$ is the cone generated by the points in S which are at infinity.

As $\Gamma\left(S^M\right)$ is the least closed cone with vertex 0 containing S_M, then we can see that

$$\begin{aligned} \Gamma\left(S^M\right) &= \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\}} \\ &= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S^M}\right\} \\ &= \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S}^M\right\} \end{aligned}$$

原课件第 70 页

Definitions

We just take the third definition from Alfred Auslender and Marc Teboulle (2003) *Asymptotic Cones and Functions in Optimization and Variational Inequalities*.

- Let $\{x_k; k \in \mathbb{N}\}$ be a sequence in $\mathbb{R}^N$. We are interested in knowing how to handle situations when the sequence $\{x_k; k \in \mathbb{N}\} \subseteq \mathbb{R}^N$ is unbounded.
- To derive some convergence properties, we are led to consider directions $d_k = \dfrac{x_k}{\|x_k\|}$ with $x_k \neq 0$ for $k \in \mathbb{N}$.
- From classical analysis, the Bolzano-Weierstrass theorem implies that we can extract a convergent subsequence $d = \lim\limits_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k$ with $\mathbb{K} \subseteq \mathbb{N}$ and $d \neq 0$.
- Now suppose that the sequence $\{x_k; k \in \mathbb{N}\} \subseteq \mathbb{R}^N$ is such that $\|x_k\| \to \infty$. Then there exists $t_k = \|x_k\|$, $k \in \mathbb{K} \subseteq \mathbb{N}$, such that $\lim\limits_{k\to\infty} t_k = \infty$ and $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$.

沿原论证展开与批注 截尾定义先保留 $\|x\| \geq M$ 的点，再取所生成最小闭锥，最后对全部 M 取交。序列定义为 $d \in C_\infty \iff \exists x_n \in C,\ t_n \to +\infty : x_n/t_n \to d$。衰退锥为 $\operatorname{rec} C = \{d : \forall x \in C, \forall t \geq 0, x + td \in C\}$，要求所有起点都可以一直沿该方向走；只在非空闭凸情形才保证与渐近锥一致。单位方向归一化必须选范数趋于无穷的子列，而不仅是声称任意无界序列范数都趋无穷。截尾最小闭锥不是凸锥，闭包横线不能漏。

11.26 渐近锥闭性及闭包不变

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 77–78 页。

对应已有 gpt-109.pdf 页脚：83, 84。

原课件第 77 页

Proposition of Definition 2

Proof of (1). $0 \in C_\infty$ because if we fix $x_k = x \in C$, for any $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$, we have $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = 0$.

If $d \in C_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. For any $\lambda > 0$, we have $\left\{\dfrac{1}{\lambda}t_k; k \in \mathbb{N}\right\} \subseteq \mathbb{R}$ with $\dfrac{1}{\lambda}t_k \to \infty$, we have $\lim\limits_{k\to\infty} \dfrac{x_k}{\frac{1}{\lambda}t_k} = \lambda d$, i.e. $\lambda d \in C_\infty$.

Hence C_∞ is a cone.

原课件第 78 页

Proposition of Definition 2

Next, we need to show C_∞ is closed.

For any $d \in \overline{C_\infty}$, there exists $\{d^q; q \in \mathbb{N}\} \subseteq C_\infty$ such that $d^q \to d$. For any d^q, we know there exist $\left\{x_k^q; k \in \mathbb{N}\right\} \subseteq C$ and $\left\{t_k^q; k \in \mathbb{N}\right\} \subseteq \mathbb{R}$ with $\lim\limits_{k\to\infty} t_k^q = \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k^q}{t_k^q} = d^q$.

Then there exists $K_q \in \mathbb{N}$ such that for all $k \geqq K_q$, we have $t_k^q > q$ and $\left\|\dfrac{x_k^q}{t_k^q} - d^q\right\| < \dfrac{1}{q}$. Let $\left\{x_q = x_{K_q}^q\right\} \subseteq C$ and $\left\{t'_q = t_{K_q}^q\right\} \subseteq \mathbb{R}$, we have $t'_q = t_{K_q}^q > q$ hence $\lim\limits_{q\to\infty} t'_q = \infty$.

$$\left\|\frac{x_q}{t'_q} - d\right\| \leqq \left\|\frac{x_q}{t'_q} - d^q\right\| + \|d^q - d\| < \frac{1}{q} + \|d^q - d\| \to 0$$

hence $\lim\limits_{q\to\infty} \dfrac{x_q}{t'_q} = d$, which implies $d \in C_\infty$. Hence C_∞ is a closed cone.

沿原论证展开与批注 非空 C 时固定 $x \in C$ 并取 $t_n = n$ 得0属于渐近锥。正倍数通过把 t_n 换为 t_n/λ 得到。闭性用对角选取：若 $d_j \to d$，对每个 j 从其见证序列选一点使 $t_j > j$ 且 $\|x_j/t_j - d_j\| < 1/j$；三角不等式给 $x_j/t_j \to d$。闭包不改变锥：用 $x'_j \in C$ 逼近 $x_j \in \overline{C}$ 至误差 $1/j$，除以 t_j 后误差更趋 0。若 C 本身是闭锥，取 $x_n = nd, t_n = n$ 得 $C \subset C_\infty$，反向用每个 $x_n/t_n \in C$ 与闭性；非闭锥的结论为 $C_\infty = \overline{C}$。

11.27 渐近锥闭性及闭包不变

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 79 页。

对应已有 gpt-109.pdf 页脚：85。

原课件第 79 页

Proposition of Definition 2

Proof of (2). It's trivial that $C_\infty \subseteq \left(\overline{C}\right)_\infty$. Conversely, let $d \in \left(\overline{C}\right)_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \overline{C}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. For any $x_k \in \overline{C}$, there exists $x_k' \in C$ such that $\|x_k' - x_k\| < \dfrac{1}{k}$. Hence

$$\begin{aligned}\lim_{k\to\infty} \left\| \frac{x_k'}{t_k} - d \right\| &\leqq \lim_{k\to\infty} \left[\left\| \frac{x_k'}{t_k} - \frac{x_k}{t_k} \right\| + \left\| \frac{x_k}{t_k} - d \right\| \right] \\ &< \lim_{k\to\infty} \frac{1}{k\,|t_k|} + \lim_{k\to\infty} \left\| \frac{x_k}{t_k} - d \right\| \longrightarrow 0\end{aligned}$$

hence $\lim\limits_{k\to\infty} \dfrac{x_k'}{t_k} = d$, i.e. $d \in C_\infty$. Hence $\left(\overline{C}\right)_\infty = C_\infty$.

沿原论证展开与批注 非空 C 时固定 $x \in C$ 并取 $t_n = n$ 得0属于渐近锥。正倍数通过把 t_n 换为 t_n/λ 得到。闭性用对角选取：若 $d_j \to d$，对每个 j 从其见证序列选一点使 $t_j > j$ 且 $\|x_j/t_j - d_j\| < 1/j$；三角不等式给 $x_j/t_j \to d$。闭包不改变锥：用 $x_j' \in C$ 逼近 $x_j \in \overline{C}$ 至误差 $1/j$，除以 t_j 后误差更趋 0。若 C 本身是闭锥，取 $x_n = nd, t_n = n$ 得 $C \subset C_\infty$，反向用每个 $x_n/t_n \in C$ 与闭性；非闭锥的结论为 $C_\infty = \overline{C}$。

11.28 渐近锥闭性及闭包不变

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 80 页。

对应已有 gpt-109.pdf 页脚：86。

原课件第 80 页

Proposition of Definition 2

Proof of (3). We just need to prove that if C is a closed cone, then $C_\infty = C$. Then for any cone C, we have $C_\infty = \left(\overline{C}\right)_\infty = \overline{C}$ as $\overline{C}$ is also a cone.

If C is a closed cone, then for any $x \in C$, for any $k > 0$, we have $kx \in C$. Let $x_k = kx$ and $t_k = k$, we have $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = x$, hence $x \in C_\infty$, i.e. $C \subseteq C_\infty$.

For any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Note that C is a cone, $\dfrac{x_k}{t_k} \in C$, then $d \in \overline{C} = C$. Hence $C_\infty = C$. $\square$

沿原论证展开与批注 非空 C 时固定 $x \in C$ 并取 $t_n = n$ 得0属于渐近锥。正倍数通过把 t_n 换为 t_n/λ 得到。闭性用对角选取：若 $d_j \to d$，对每个 j 从其见证序列选一点使 $t_j > j$ 且 $\|x_j/t_j - d_j\| < 1/j$；三角不等式给 $x_j/t_j \to d$。闭包不改变锥：用 $x_j' \in C$ 逼近 $x_j \in \overline{C}$ 至误差 $1/j$，除以 t_j 后误差更趋0。若 C 本身是闭锥，取 $x_n = nd, t_n = n$ 得 $C \subset C_\infty$，反向用每个 $x_n/t_n \in C$ 与闭性；非闭锥的结论为 $C_\infty = \overline{C}$。

11.29 截尾与序列定义等价

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 82 页。

对应已有 gpt-109.pdf 页脚：88。

原课件第 82 页

Proposition of Definition 2

Proof. For any $d \in S_\infty$. If $d = 0$, then $d \in \mathbf{A}S$ is trivial. If $d \neq 0$, then there exist $\{x_n\} \subseteq S$ and $\{t_n\} \subseteq \mathbb{R}$ with $t_n \to \infty$ such that $\lim_{n\to\infty} \frac{x_n}{t_n} = d$. There exists $N \in \mathbb{N}$ such that for all $n \geqq N$, we have $\left\| \frac{x_n}{t_n} \right\| > \frac{\|d\|}{2} > 0$. Then for any $M \geqq 0$, as $t_n \to \infty$, there exists $N_M > 0$ such that for all $n \geqq N_M$, we have $\|x_n\| \geqq M$. Hence $x_n \in S^M$. Hence $\frac{x_n}{t_n} \in \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\right\}$ for all $n \geqq N_M$. As $\lim_{n\to\infty} \frac{x_n}{t_n} = d$, we have $d \in \Gamma\left(S^M\right)$ for any $M \geqq 0$. Thus,

$$S_\infty \subseteq \bigcap_{M\geqq 0} \Gamma\left(S^M\right) = \mathbf{A}S$$

Conversely, if $d \in \mathbf{A}S$ with $d \neq 0$, then for all $M \geqq 0$, $d \in \Gamma\left(S^M\right) = \overline{\left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S^M\right\}}$, so there exists $\alpha_M x_M$ where $\alpha_M > 0$ and $x_M \in S^M$ such that $\|d - \alpha_M x_M\| < \frac{1}{M}$. Then let $t_M = \frac{1}{\alpha_M}$, if we can prove $t_M \to \infty$, i.e. $\alpha_M \to 0$, then we are done. As $\|\alpha_M x_M\| \leqq \|d\| + \|d - \alpha_M x_M\|$, then

$$|\alpha_M| \leqq \frac{\|d\| + \|d - \alpha_M x_M\|}{\|x_M\|} < \frac{\|d\| + \frac{1}{M}}{M} \to 0.$$

We are done! □

沿原论证展开与批注 非零 $d = x_n/t_n$ 的极限使 $\|x_n\| \to \infty$，所以对任意 M，最终原点均属截尾集，其缩放极限落入截尾闭锥，故属于交。反向对非零 d，每个整数 M 选 $\alpha_M x_M$ 距 d 小于 $1/M$，且 $\|x_M\| \geq M$。由 $\alpha_M \leq (\|d\| + 1/M)/M \to 0$，取 $t_M = 1/\alpha_M \to \infty$，得到见证序列。零向量对非空集合单独处理。归一化方向全体再取非负倍数得到相同锥；有界集方向集为空时需单独按约定加入0。

11.30 截尾与序列定义等价

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 83 页。

对应已有 gpt-109.pdf 页脚：89。

原课件第 83 页

Proposition of Definition 2

Proposition 2.2. Let $C \subseteq \mathbb{R}^N$ be non-empty and define the normalized set

$$C_N = \left\{ d \in \mathbb{R}^N \mid \text{there exists } \{x_k; k \in \mathbb{N}\} \subseteq C \text{ with } \|x_k\| \to \infty \text{ such that } d = \lim_{k\to\infty} \frac{x_k}{\|x_k\|} \right\}$$

then $C_\infty = \text{Cone}(C_N)$, where for any set C, $\text{Cone}(C_N) = \left\{ \lambda x \in \mathbb{R}^N \mid x \in C_N, \lambda \geqq 0 \right\}$.

Proof. Clearly, one always has $\text{Cone}(C_N) \subseteq C_\infty$ as $C_N \subseteq C_\infty$.

Conversely, let $0 \neq d \in C_\infty$. Then there exist $x_k \in C$ and $t_k \to \infty$, such that

$$d = \lim_{k\to\infty} \frac{x_k}{t_k} = \lim_{k\to\infty} \frac{\|x_k\|}{t_k} \frac{x_k}{\|x_k\|}$$

with $\|x_k\| \to \infty$. Thus the sequence $\left\{ \frac{\|x_k\|}{t_k}; k \in \mathbb{N} \right\}$ is a positive bounded sequence, and by the Bolzano–Weierstrass theorem, there exists a subsequence $\left\{ \frac{\|x_k\|}{t_k}; k \in \mathbb{K} \right\}$ with $\mathbb{K} \subseteq \mathbb{N}$ such that

$$\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{\|x_k\|}{t_k} = \lambda \geqq 0.$$

we also have

$$d = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{t_k} = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{\|x_k\|}{t_k} \frac{x_k}{\|x_k\|},$$

which means that $d = \lambda d_N$ with $d_N = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} \in C_N$, namely $d \in \text{Cone}(C_N)$. □

沿原论证展开与批注 非零 $d = x_n/t_n$ 的极限使 $\|x_n\| \to \infty$，所以对任意 M，最终原点均属截尾集，其缩放极限落入截尾闭锥，故属于交。反向对非零 d，每个整数 M 选 $\alpha_M x_M$ 距 d 小于 $1/M$，且 $\|x_M\| \geq M$。由 $\alpha_M \leq (\|d\| + 1/M)/M \to 0$，取 $t_M = 1/\alpha_M \to \infty$，得到见证序列。零向量对非空集合单独处理。归一化方向全体再取非负倍数得到相同锥；有界集方向集为空时需单独按约定加入0。

11.31 有界判据与平移不变

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 84 页。

对应已有 gpt-109.pdf 页脚：90。

原课件第 84 页

Proposition of Definition 2

Proposition 2.3. A set $C \subseteq \mathbb{R}^N$ is bounded if and only if $C_\infty = \{0\}$.

Proof. It is clear that C_∞ cannot contain any non-zero direction if C is bounded. As C is bounded, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = 0$. Hence $C_\infty = \{0\}$.

Conversely, if C is unbounded, then there exists a sequence $\{x_k; k \in \mathbb{N}\} \subseteq C$ with $x_k \neq 0$ for all $k \in \mathbb{N}$ such that $\|x_k\| \to \infty$ and thus the vectors $d_k = \frac{x_k}{\|x_k\|} \in \{d \in \mathbb{R}^N : \|d\| = 1\}$.
Therefore, we can extract a subsequence of $\{d_k\}$ such that $\lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} d_k = d$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|d\| = 1$. This non-zero vector d is an element of C_∞, a contradiction! □

沿原论证展开与批注 非空有界 C 的任意 x_n/t_n 趋于0，故渐近锥只有0。若无界，选 $\|x_n\| \to \infty$，单位向量抽子列趋于单位 d，令 $t_n = \|x_n\|$ 得非零渐近方向，矛盾。平移 $C + a$ 不改变方向，因为 $(x_n + a)/t_n - x_n/t_n = a/t_n \to 0$。若 E 非空，固定 $a \in E$，由 $C + a \subset C + E$ 及单调性得 $C_\infty \subset (C + E)_\infty$。

11.32 有界判据与平移不变

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 85 页。

对应已有 gpt-109.pdf 页脚：91。

原课件第 85 页

Asymptotic Cone

Theorem 3.1. If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S+\{x\}) = \mathbf{A}S$.

Proof. We just need to prove $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$, then to prove $\mathbf{A}S \subseteq \mathbf{A}(S+\{x\})$, we just need to let $S = (S+\{x\})+\{-x\}$ and we arrive at the conclusion. Let $T = S+\{x\}$.

For any $d \in \mathbf{A}(S+\{x\})$, $d \in (S+\{x\})_\infty$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq S+\{x\}$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. Let $\{x'_k = x_k - x\} \subseteq S$ and

$$\left\|\frac{x'_k}{t_k} - d\right\| \leqq \left\|\frac{x'_k}{t_k} - \frac{x_k}{t_k}\right\| + \left\|\frac{x_k}{t_k} - d\right\| = \left\|\frac{x}{t_k}\right\| + \left\|\frac{x_k}{t_k} - d\right\| \to 0$$

hence $\lim\limits_{k\to\infty} \dfrac{x'_k}{t_k} = d$, i.e., $d \in S_\infty$, hence $d \in \mathbf{A}S$. Hence $\mathbf{A}(S+\{x\}) \subseteq \mathbf{A}S$. □

沿原论证展开与批注 非空有界 C 的任意 x_n/t_n 趋于0，故渐近锥只有0。若无界，选 $\|x_n\| \to \infty$，单位向量抽子列趋于单位 d，令 $t_n = \|x_n\|$ 得非零渐近方向，矛盾。平移 $C+a$ 不改变方向，因为 $(x_n+a)/t_n - x_n/t_n = a/t_n \to 0$。若 E 非空，固定 $a \in E$，由 $C+a \subset C+E$ 及单调性得 $C_\infty \subset (C+E)_\infty$。

11.33 有界判据与平移不变

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 86 页。

对应已有 gpt-109.pdf 页脚：92。

原课件第 86 页

Asymptotic Cone

Theorem 3.1. If S is a subset of $\mathbb{R}^N$, and x is a vector in $\mathbb{R}^N$, then $\mathbf{A}(S + \{x\}) = \mathbf{A}S$.

Proof.

$$
\begin{aligned}
& d \in \mathbf{A}(S) \Longleftrightarrow d \in S_\infty \\
\Longleftrightarrow \quad & \text{there exist } \{x_k; k \in \mathbb{N}\} \subseteq S \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k}{t_k} = d \\
\Longleftrightarrow \quad & \text{there exist } \{x_k + x; k \in \mathbb{N}\} \subseteq S + \{x\} \text{ and } \{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R} \\
& \qquad \text{with } t_k \to \infty \text{ such that } \lim_{k\to\infty} \frac{x_k + x}{t_k} = d \\
\Longleftrightarrow \quad & d \in (S + \{x\})_\infty \Longleftrightarrow d \in \mathbf{A}(S + \{x\})
\end{aligned}
$$

□

In other words, a translation of S does not alter its asymptotic cone.

沿原论证展开与批注 非空有界 C 的任意 x_n/t_n 趋于0，故渐近锥只有0。若无界，选 $\|x_n\| \to \infty$，单位向量抽子列趋于单位 d，令 $t_n = \|x_n\|$ 得非零渐近方向，矛盾。平移 $C + a$ 不改变方向，因为 $(x_n + a)/t_n - x_n/t_n = a/t_n \to 0$。若 E 非空，固定 $a \in E$，由 $C + a \subset C + E$ 及单调性得 $C_\infty \subset (C + E)_\infty$。

11.34 有界判据与平移不变

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 87 页。

对应已有 gpt-109.pdf 页脚：93。

原课件第 87 页

Asymptotic Cone

Theorem 3.2. If $E \neq \varnothing$ and S are two subsets of $\mathbb{R}^N$, then $\mathbf{A}S \subseteq \mathbf{A}(S+E)$.

Proof. Let $x \in E$. Then $\mathbf{A}S = \mathbf{A}(S+\{x\}) \subseteq \mathbf{A}(S+E)$. □

沿原论证展开与批注 非空有界 C 的任意 x_n/t_n 趋于0，故渐近锥只有0。若无界，选 $\|x_n\| \to \infty$，单位向量抽子列趋于单位 d，令 $t_n = \|x_n\|$ 得非零渐近方向，矛盾。平移 $C+a$ 不改变方向，因为 $(x_n+a)/t_n - x_n/t_n = a/t_n \to 0$。若 E 非空，固定 $a \in E$，由 $C+a \subset C+E$ 及单调性得 $C_\infty \subset (C+E)_\infty$。

11.35 乘积与交集的渐近方向

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 88 页。

对应已有 gpt-109.pdf 页脚：94。

原课件第 88 页

Asymptotic Cone

Theorem 3.3. If, for every $n = 1, \cdots, N$, S_n is a subset of $\mathbb{R}^{K_n}$, then $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$.

Proof. For $d \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$. If $d = 0$, then $d \in \prod_{n=1}^{N} \mathbf{A}S_n$ is trivial. If $d \neq 0$, then there exist $\{x_k; k \in \mathbb{N}\} \subseteq \prod_{n=1}^{N} S_n$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let

$$x_k = \left(x_k^1, \cdots, x_k^n, \cdots, x_k^N\right), d = \left(d^1, \cdots, d^n, \cdots, d^N\right)$$

where $x_k^n \in S_n$ and $d^n \in \mathbb{R}$ for $n = 1, \cdots, N$. For each $n = 1, \cdots, N$, we have $\lim_{k\to\infty} \frac{x_k^n}{t_k} = d^n$. Hence $d^n \in \mathbf{A}S_n$, hence $d \in \prod_{n=1}^{N} \mathbf{A}S_n$. Thus, $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A}S_n$. □

沿原论证展开与批注 若 $(x_{1n}, \ldots, x_{mn})/t_n \to (d_1, \ldots, d_m)$，逐坐标就是 $x_{in}/t_n \to d_i$，所以积的渐近锥包含于渐近锥之积。反向一般不能单靠各自见证序列，因为缩放速度未必相同。第89页把总范数截尾当成各分量都截尾的等式错误，例如 $(M, 0)$ 总范数大但第二分量不大。交集包含于每个集合，锥的单调性给交集锥包含于各锥交；若右边只有0，交集有界（空交集本身有界，非空时用第84页）。

11.36 乘积与交集的渐近方向

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 89 页。

对应已有 gpt-109.pdf 页脚：95。

原课件第 89 页

Asymptotic Cone

Theorem 3.3. If, for every $n = 1, \cdots, N$, S_n is a subset of $\mathbb{R}^{K_n}$, then $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A} S_n$.

Proof. $y \in \mathbf{A}\left(\prod_{n=1}^{N} S_n\right)$, then, for any $M \geqq 0$, $y \in \Gamma\left(\left(\prod_{n=1}^{N} S_n\right)^M\right)$, then

$$
\begin{aligned}
y \quad \in \quad & \overline{\left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \left(\prod_{n=1}^{N} S_n\right)^M\right\}} = \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{\prod_{n=1}^{N} S_n^M}\right\} \\
\subseteq \quad & \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \prod_{n=1}^{N} \overline{S_n^M}\right\} = \prod_{n=1}^{N} \left\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in \overline{S_n^M}\right\} \\
= \quad & \prod_{n=1}^{N} \overline{\{\alpha x \in \mathbb{R}^N \mid \alpha \in \mathbb{R}_+, x \in S_n^M\}} = \prod_{n=1}^{N} \Gamma\left(S_n^M\right)
\end{aligned}
$$

hence $y \in \bigcap_{M \geqq 0}\left[\prod_{n=1}^{N} \Gamma\left(S_n^M\right)\right] = \prod_{n=1}^{N}\left[\bigcap_{M \geqq 0} \Gamma\left(S_n^M\right)\right] = \prod_{n=1}^{N} \mathbf{A}(S_n)$. Therefore $\mathbf{A}\left(\prod_{n=1}^{N} S_n\right) \subseteq \prod_{n=1}^{N} \mathbf{A} S_n$.

□

沿原论证展开与批注 若 $(x_{1n}, \ldots, x_{mn})/t_n \to (d_1, \ldots, d_m)$，逐坐标就是 $x_{in}/t_n \to d_i$，所以积的渐近锥包含于渐近锥之积。反向一般不能单靠各自见证序列，因为缩放速度未必相同。第89页把总范数截尾当成各分量都截尾的等式错误，例如 $(M, 0)$ 总范数大但第二分量不大。交集包含于每个集合，锥的单调性给交集锥包含于各锥交；若右边只有0，交集有界（空交集本身有界，非空时用第84页）。

11.37 乘积与交集的渐近方向

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 90 页。

对应已有 gpt-109.pdf 页脚：96。

原课件第 90 页

Asymptotic Cone

Theorem 3.4. Given a collection of subsets of $\mathbb{R}^N$, if the intersection of their asymptotic cones is $\{0\}$, then their intersection is bounded.

Proof. As we have that "$S_1 \subseteq S_2$" implies "$\mathbf{A}S_1 \subseteq \mathbf{A}S_2$", then, for an arbitrary index set Γ, let $C_\gamma \subseteq \mathbb{R}^N$, $\gamma \in \Gamma$, we have

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \mathbf{A}C_\gamma$$

for all $\gamma \in \Gamma$. Hence

$$\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) \subseteq \bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma$$

If $\bigcap_{\gamma\in\Gamma} \mathbf{A}C_\gamma = \{0\}$, then $\mathbf{A}\left(\bigcap_{\gamma\in\Gamma} C_\gamma\right) = \{0\}$. By Proposition 2.3, we have $\bigcap_{\gamma\in\Gamma} C_\gamma$ is bounded. □

沿原论证展开与批注 若 $(x_{1n},\dots,x_{mn})/t_n \to (d_1,\dots,d_m)$，逐坐标就是 $x_{in}/t_n \to d_i$，所以积的渐近锥包含于渐近锥之积。反向一般不能单靠各自见证序列，因为缩放速度未必相同。第89页把总范数截尾当成各分量都截尾的等式错误，例如 $(M,0)$ 总范数大但第二分量不大。交集包含于每个集合，锥的单调性给交集锥包含于各锥交；若右边只有0，交集有界（空交集本身有界，非空时用第84页）。

11.38 乘积与交集的渐近方向

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 91 页。

对应已有 gpt-109.pdf 页脚：97。

原课件第 91 页

Asymptotic Cone

Theorem 3.4. For all $j \in \mathcal{J}$, $S_j \subseteq \mathbb{R}^N$, if $\bigcap_{j\in\mathcal{J}} \mathbf{A}(S_j) = \{0\}$, then $\bigcap_{j\in\mathcal{J}} S_j$ is bounded.

Proof. For all $j \in \mathcal{J}$, $\bigcap_{j\in\mathcal{J}} S_j \subseteq S_j$, then $\mathbf{A}\left(\bigcap_{j\in\mathcal{J}} S_j\right) \subseteq \mathbf{A}(S_j)$. Whence $\mathbf{A}\left(\bigcap_{j\in\mathcal{J}} S_j\right) \subseteq \bigcap_{j\in\mathcal{J}} \mathbf{A}(S_j) = \{0\}$. There exists $M_0 \geqq 0$ such that $\left(\bigcap_{j\in\mathcal{J}} S_j\right)^M = \varnothing$ for $M \geqq M_0$, thus $\bigcap_{j\in\mathcal{J}} S_j$ is bounded. □

沿原论证展开与批注 若 $(x_{1n},\dots,x_{mn})/t_n \to (d_1,\dots,d_m)$，逐坐标就是 $x_{in}/t_n \to d_i$，所以积的渐近锥包含于渐近锥之积。反向一般不能单靠各自见证序列，因为缩放速度未必相同。第89页把总范数截尾当成各分量都截尾的等式错误，例如 $(M,0)$ 总范数大但第二分量不大。交集包含于每个集合，锥的单调性给交集锥包含于各锥交；若右边只有0，交集有界（空交集本身有界，非空时用第84页）。

11.39 线性像定理再证明

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 93–96 页。

对应已有 gpt-109.pdf 页脚：99, 100, 101, 102。

原课件第 93 页

Asymptotic Cone

Proof. The condition will be shown first to imply that L carries unbounded sequences in C into unbounded sequences in $\mathbb{R}^K$. If this were not true, there would exist a sequence of points $\{x_k; k \in \mathbb{N}\} \subseteq C$ such that $\|x_k\| \to \infty$, but such that the sequence of images $y_k = L(x_k)$ is bounded. As $\left\{\frac{x_k}{\|x_k\|}; k \in \mathbb{N}\right\}$ is bounded, so we can extract a subsequence of $\left\{\frac{x_k}{\|x_k\|}; k \in \mathbb{K}\right\}$ such that $\lim\limits_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{x_k}{\|x_k\|} = x$, $\mathbb{K} \subseteq \mathbb{N}$, and with $\|x\| = 1$. Hence $x \in C_\infty$, but also (because linear mappings are continuous)

$$L(x) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{1}{\|x_k\|} L(x_k) = \lim_{\substack{k\to\infty \\ k\in\mathbb{K}}} \frac{y_k}{\|x_k\|} = 0$$

which the condition prohibits.

原课件第 94 页

Asymptotic Cone

Assuming henceforth that the condition holds, we prove now that $L(C)$ is a closed set. Suppose $y_k \to y$ with $y_k \in L(C)$, or in other words, $y_k = L(x_k)$ for some $x_k \in C$. The sequence $\{L(x_k)\}$ is bounded, so the sequence $\{x_k; k \in \mathbb{N}\}$ must be bounded as well, for the reasons just given, and it therefore has a cluster point x. We have $x \in C$ (because C is closed) and $y = L(x)$ (again because linear mappings are continuous), and therefore $y \in L(C)$.

原课件第 95 页

Asymptotic Cone

The next task is to show that $L(C_\infty) \subseteq L(C)_\infty$. This is trivial when $C = \varnothing$, so we assume $C \neq \varnothing$. Any $d \in C_\infty$, there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim\limits_{k\to\infty} \dfrac{x_k}{t_k} = d$. The points $y_k = L(x_k)$ in $L(C)$ and

$$\lim_{k\to\infty} \frac{y_k}{t_k} = \lim_{k\to\infty} \frac{L(x_k)}{t_k} = L\left(\lim_{k\to\infty} \frac{x_k}{t_k}\right) = L(d)$$

hence $L(d) \in L(C)_\infty$. Hence $L(C_\infty) \subseteq L(C)_\infty$.

原课件第 96 页

Asymptotic Cone

For the inclusion $L(C_\infty) \supseteq L(C)_\infty$, we consider any $y \in L(C)_\infty$, there exist $\{y_k; k \in \mathbb{N}\} \subseteq L(C)$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{y_k}{t_k} = y$. For any choice of $x_k \in C$ with $L(x_k) = y_k$, we have $\lim_{k\to\infty} L\left(\frac{x_k}{t_k}\right) = y$. The boundedness of the sequence $\left\{L\left(\frac{x_k}{t_k}\right); k \in \mathbb{N}\right\}$ implies the boundedness of the sequence $\left\{\frac{x_k}{t_k}; k \in \mathbb{N}\right\}$, which therefore must have a cluster point x. Then $x \in C_\infty$ and $L(x) = y$, and we conclude that $y \in L(C_\infty)$. □

沿原论证展开与批注 本段重复第37–41页的重要定理。核心核条件为 $\ker L \cap C_\infty = \{0\}$。若 Lx_n 有界而 x_n 无界，取范数趋无穷子列并单位化，极限是非零渐近方向且被 L 送到0，矛盾。因此像序列收敛时原像可抽收敛子列，闭 C 与连续 L 给像闭。锥像的容易包含是对 x_n/t_n 直接取 L；困难包含须用第41页的缩放反证，不可仅引用原集合上的有界性。

11.40 闭和定理的两个证明

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 98–100 页。

对应已有 gpt-109.pdf 页脚：104, 105, 106。

原课件第 98 页

Asymptotic Cone

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.5 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}$$

hence

$$L^{-1}(0) \cap C_\infty = \{0\}$$

so by Theorem 3.5, we have $L(C)$ is closed. As $L(C) = \sum_{n=1}^{N} C_n$, we have that $\sum_{n=1}^{N} C_n$ is closed.

原课件第 99 页

Asymptotic Cone

And again by Theorem 3.5, $L(C_\infty) = L(C)_\infty$, i.e.

$$L\left(\left(\prod_{n=1}^{N} C_n\right)_\infty\right) = \left(\sum_{n=1}^{N} C_n\right)_\infty$$

We also have

$$L\left(\prod_{n=1}^{N} (C_n)_\infty\right) = \sum_{n=1}^{N} (C_n)_\infty$$

and

$$\left(\prod_{n=1}^{N} C_n\right)_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$$

hence

$$\left(\sum_{n=1}^{N} C_n\right)_\infty \subseteq \sum_{n=1}^{N} (C_n)_\infty$$

□

原课件第 100 页

Asymptotic Cone

Proof. Let $C = \prod_{n=1}^{N} C_n$. We define $L : \mathbb{R}^K \to \mathbb{R}^L$ as

$$L : (x_1, \cdots, x_n, \cdots, x_N) \longrightarrow \sum_{n=1}^{N} x_n$$

for $x_n \in C_n$. Then L is a linear mapping. As Theorem 3.5 shows, $C_\infty \subseteq \prod_{n=1}^{N} (C_n)_\infty$. And by the condition, we have

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}.$$

$$(x_1, \cdots, x_N) \in L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty$$

$$L((x_1, \cdots, x_N)) = 0 \quad \text{and} \quad x_n \in (C_n)_\infty \ \text{ for } n = 1, \cdots, N$$

$$x_n \in (C_n)_\infty \ \text{ for } n = 1, \cdots, N \quad \text{and} \quad \sum_{n=1}^{N} x_n = 0$$

If $(C_1)_\infty, \cdots, (C_N)_\infty$ are positively semi-independent (P.S.I.), then $x_n = 0$ for $n = 1, \cdots, N$, and hence

$$L^{-1}(0) \cap \prod_{n=1}^{N} (C_n)_\infty = \{0\}.$$

沿原论证展开与批注 积集加求和映射的证明已在第42页完整给出。两集合的直接证明：若 $z_n = a_n + b_n \to z$ 而 a_n 无界，选 $r_n = \|a_n\| \to \infty$ 及 $a_n/r_n \to d$，$\|d\| = 1$。由 $z_n/r_n \to 0$ 得 $b_n/r_n \to -d$，于是 $d \in A_\infty, -d \in B_\infty$，违反正半独立。故 a_n 有界，抽子列趋于 $a \in A$，再由 $b_n = z_n - a_n \to z - a \in B$，得 $z \in A + B$。有限多集合用积方法最直接，避免在归纳时漏检中间和的渐近锥条件。

11.41 闭和定理的两个证明

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 101–103 页。

对应已有 gpt-109.pdf 页脚：107, 108, 109。

原课件第 101 页

Asymptotic Cone

Theorem 3.6. $S_1, \cdots, S_j, \cdots, S_J$ are closed subsets of $\mathbb{R}^N$, if their asymptotic cones $\mathbf{A}(S_1), \cdots, \mathbf{A}(S_j), \cdots, \mathbf{A}(S_J)$ are positively semi-independent, then their sum $\sum_{j=1}^{J} S_j$ is closed.

Proof. We first verify the setting of $J = 2$. $\mathbf{A}(S_1)$ and $\mathbf{A}(S_2)$ are positively semi-independent if and only if $\mathbf{A}(S_1) \cap \mathbf{A}(-S_2) = \{0\}$.

Suppose $\mathbf{A}(S_1) \neq \{0\} \neq \mathbf{A}(S_2)$, then S_1 and S_2 are unbounded. We assume that $\{z^K; K \in \mathbb{R}_+\} \subseteq S_1 + S_2$ and $z^K \to z^0$. According to the definition, for any $K \in \mathbb{R}_+$, there exist $x_1^K \in S_1$ and $x_2^K \in S_2$ such that $z^K = x_1^K + x_2^K$.

原课件第 102 页

Asymptotic Cone

If $\{x_1^K; K \in \mathbb{R}_+\}$ is unbounded in S_1, then for any $K \in \mathbb{R}_+$, $x_1^K \in S_1^K$, $x_1^K \in \Gamma(S_1^K)$, $\|x_1^K\| \to \infty$, then $x_2^K = z^K - x_1^K$ and $\|x_2^K\| \to \infty$. For any $K \in \mathbb{R}_+$, $-x_2^K \in -S_2^K$, $-x_2^K \in \Gamma(-S_2^K)$, and

$$\frac{x_1^K}{\|x_1^K\|} = \frac{z^K}{\|x_1^K\|} - \frac{x_2^K}{\|x_1^K\|}, \quad \frac{x_1^K}{\|x_1^K\|} \in \Gamma(S_1^K) \quad \text{and} \quad -\frac{x_2^K}{\|x_1^K\|} \in \Gamma(-S_2^K)$$

then

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} = \lim_{K\to\infty} \frac{z^K}{\|x_1^K\|} + \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|},$$

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} \in \Gamma(S_1^K) \qquad \text{and} \qquad \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|} \in \Gamma(-S_2^K)$$

that is,

$$\lim_{K\to\infty} \frac{x_1^K}{\|x_1^K\|} = \lim_{K\to\infty} -\frac{x_2^K}{\|x_1^K\|} = e \neq 0, \quad e \in \Gamma(S_1^K) \quad \text{and} \quad e \in \Gamma(-S_2^K)$$

Hence $0 \neq e \in \Gamma(S_1^K) \cap \Gamma(-S_2^K) = \{0\}$, contradiction!

原课件第 103 页

Asymptotic Cone

Therefore $\{x_1^K; K \in \mathbb{R}_+\}$ id bounded in S_1, then there exists a convergent subsequence $\{x_1^{K_n}; K_n \in \mathbb{R}_+\} \subseteq \{x_1^K; K \in \mathbb{R}_+\}$ such that $x_1^{K_n} \to x_1^0 \in S_1$. On the other hand, for any $K_n \in \mathbb{R}_+$, $x_2^{K_n} = z^{K_n} - x_1^{K_n}$, then $x_2^{K_n} \to x_2^0 \in S_2$, and hence $z^{K_n} \to x_1^0 + x_2^0 \in S_1 + S_2$. Thus $z^0 = x_1^0 + x_2^0 \in S_1 + S_2$, and hence $S_1 + S_2$ is closed.

The general conclusion holds from mathematical induction. □

沿原论证展开与批注 积集加求和映射的证明已在第42页完整给出。两集合的直接证明：若 $z_n = a_n + b_n \to z$ 而 a_n 无界，选 $r_n = \|a_n\| \to \infty$ 及 $a_n/r_n \to d$，$\|d\| = 1$。由 $z_n/r_n \to 0$ 得 $b_n/r_n \to -d$，于是 $d \in A_\infty, -d \in B_\infty$，违反正半独立。故 a_n 有界，抽子列趋于 $a \in A$，再由 $b_n = z_n - a_n \to z - a \in B$，得 $z \in A + B$。有限多集合用积方法最直接，避免在归纳时漏检中间和的渐近锥条件。

11.42 渐近正则性与凸渐近锥

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 106–107 页。

对应已有 gpt-109.pdf 页脚：112, 113。

原课件第 106 页

Proposition of Definition 3

Proof. The inclusion $C_\infty^1 \subseteq C_\infty$ clearly holds from the definitions of C_∞^1 and C_∞, respectively.

Let $d \in C_\infty$. Then there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k'; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k' \to \infty$ such that $d = \lim\limits_{k\to\infty} \dfrac{x_k}{t_k'}$. Let $x \in C$ and define

$$d_k = \frac{1}{t_k'}(x_k - x),$$

then we have

$$d = \lim_{k\to\infty} d_k$$

and $x + t_k' d_k = x_k \in C$.

原课件第 107 页

Proposition of Definition 3

Now let $\{t_k; k \in \mathbb{N}\}$ be an arbitrary sequence such that $\lim_{k\to\infty} t_k = \infty$. For any fixed $N \in \mathbb{N}$, there exists $k(N)$ with $\lim_{N\to\infty} k(N) = \infty$ such that $t_N \leqq t'_{k(N)}$, and since C is convex, we have

$$z_N = x + t_N d_{k(N)} \in C.$$

As z_N is on the closed segment $\left[x, x + t'_{k(N)} d_{k(N)}\right]$. Hence,

$$d = \lim_{N\to\infty} \frac{z_N}{t_N},$$

showing that $d \in C^1_\infty$. □

沿原论证展开与批注 正则性把“存在缩放速度”增强为“每一种趋无穷速度都可匹配”。设 $x_j/s_j \to d$，固定 $a \in C$，给任意 $t_n \to \infty$ 选 $j(n) \to \infty$ 且 $s_{j(n)} \geq t_n$。由凸性 $z_n = a + (t_n/s_{j(n)})(x_{j(n)} - a) \in C$，且 $z_n/t_n = a/t_n + (x_{j(n)} - a)/s_{j(n)} \to d$。因此任意非空凸集渐近正则。对两方向使用共同 $t_n = n$ 的见证点，再作相同权重凸组合，就证明渐近锥凸；其闭锥性质已由第76页给出。

11.43 渐近正则性与凸渐近锥

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 108 页。

对应已有 gpt-109.pdf 页脚：114。

原课件第 108 页

Proposition of Definition 3

Proposition 4.2. Let C be a non-empty convex set in $\mathbb{R}^N$. Then the asymptotic cone C_∞ is a closed convex cone.

Proof. By Proposition 2.1, we have shown that C_∞ is a closed cone. We just need to verify C_∞ is convex.

By Proposition 4.1, we know that for any d_1 and d_2 in C_∞^1, for any $t_k \to \infty$, there exists x_k^1 and x_k^2 in C such that

$$\lim_{k\to\infty} \frac{x_k^1}{t_k} = d_1 \qquad \text{and} \qquad \lim_{k\to\infty} \frac{x_k^2}{t_k} = d_2$$

Then for any $\lambda \in [0,1]$, we have $\lambda x_k^1 + (1-\lambda)\, x_k^2 \in C$ and

$$\lim_{k\to\infty} \frac{\lambda x_k^1 + (1-\lambda)\, x_k^2}{t_k} = \lambda d_1 + (1-\lambda)\, d_2$$

hence $\lambda d_1 + (1-\lambda)\, d_2 \in C_\infty^1$. Hence C_∞^1 is convex, hence C_∞ is convex. □

沿原论证展开与批注 正则性把“存在缩放速度”增强为“每一种趋无穷速度都可匹配”。设 $x_j/s_j \to d$，固定 $a \in C$，给任意 $t_n \to \infty$ 选 $j(n) \to \infty$ 且 $s_{j(n)} \geq t_n$。由凸性 $z_n = a + (t_n/s_{j(n)})(x_{j(n)} - a) \in C$，且 $z_n/t_n = a/t_n + (x_{j(n)} - a)/s_{j(n)} \to d$。因此任意非空凸集渐近正则。对两方向使用共同 $t_n = n$ 的见证点，再作相同权重凸组合，就证明渐近锥凸；其闭锥性质已由第76页给出。

11.44 衰退锥相等与非闭反例

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 110–111 页。

对应已有 gpt-109.pdf 页脚：116, 117。

原课件第 110 页

Proposition of Definition 3

Proof. For any $y \in \mathbf{0}^+C$, for any $k > 0$, let $t_k = k \to \infty$ and $x_k = x + ky$ for any $x \in C$. Then

$$\left\| \frac{x_k}{t_k} - y \right\| = \left\| \frac{x}{k} \right\| \to 0$$

hence $\lim_{k\to\infty} \frac{x_k}{t_k} = y$, which implies that $\mathbf{0}^+C \subseteq C_\infty$.

If C is non-empty, closed and convex, then for any $d \in C_\infty$, we need to show that for all $x \in C$ and $t > 0$, we have $x + td \in C$. As there exist $\{x_k; k \in \mathbb{N}\} \subseteq C$ and $\{t_k; k \in \mathbb{N}\} \subseteq \mathbb{R}$ with $t_k \to \infty$ such that $\lim_{k\to\infty} \frac{x_k}{t_k} = d$. Let $d_k = \frac{x_k - x}{t_k}$, then we have $x_k = x + t_k d_k \in C$ and

$$\lim_{k\to\infty} \frac{x_k}{t_k} = \lim_{k\to\infty} \frac{x + t_k d_k}{t_k} = \lim_{k\to\infty} \left(d_k + \frac{x}{t_k} \right) = d$$

hence

$$\lim_{k\to\infty} d_k = d.$$

原课件第 111 页

Proposition of Definition 3

Take k sufficiently large such that $t \leqq t_k$, then since C is convex, we have

$$x + td_k = \left(1 - \frac{t}{t_k}\right) x + \frac{t}{t_k}\left(x + t_k d_k\right) \in C.$$

Passing to the limit, we thus have $x + td = \lim_{k\to\infty}\left(x + td_k\right) \in C$ as C is closed. So $d \in \mathbf{0}^+ C$. Hence $\mathbf{0}^+ C = C_\infty$.

So if C is non-empty convex, we have $\overline{C}$ is non-empty, closed and convex

$$\mathbf{0}^+ \overline{C} = \left(\overline{C}\right)_\infty = C_\infty$$

by Proposition 2.1. □

沿原论证展开与批注 任意非空集合有 $\operatorname{rec} C \subset C_\infty$：固定 $a \in C$，用 $x_n = a + nd$。反向需要闭凸：若 $x_n/t_n \to d$，对任意 $a \in C$ 和固定 $t > 0$，充分大时 $a + (t/t_n)(x_n - a) \in C$，取极限得 $a + td \in C$。所以 $\operatorname{rec} C = C_\infty$。第112页反例 $C = ([0,1) \times [0,\infty)) \cup \{(1,0)\}$ 中，衰退方向必须让 $(1,0)$ 及任意 $(0,y)$ 都留在集合，迫使两个方向分量均0，故衰退锥是 $\{0\}$，不是空集；渐近锥是 $\{(0,v) : v \geq 0\}$，不是整条竖直线。闭凸时 $C + C_\infty = C$ 由衰退定义及0在锥中立即得到。

11.45 衰退锥相等与非闭反例

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 112 页。

对应已有 gpt-109.pdf 页脚：118。

原课件第 112 页

Proposition of Definition 3

If C is not closed, the theorem may fail. Consider

$$C = \{(x_1, x_2) : 0 \leqq x_1 < 1, x_2 \geqq 0\} \cup \{(1, 0)\}$$

By using the definition of direction of recession, it can be verified that C has no nonzero directions of recession, i.e.

$$\mathbf{0}^+ C = \varnothing$$

but we can see that

$$C_\infty = \mathbf{0}^+ \overline{C} = \{(0, x) : x \in \mathbb{R}\}.$$

沿原论证展开与批注 任意非空集合有 $\operatorname{rec} C \subset C_\infty$：固定 $a \in C$，用 $x_n = a + nd$。反向需要闭凸：若 $x_n/t_n \to d$，对任意 $a \in C$ 和固定 $t > 0$，充分大时 $a + (t/t_n)(x_n - a) \in C$，取极限得 $a + td \in C$。所以 $\operatorname{rec} C = C_\infty$。第112页反例 $C = ([0, 1) \times [0, \infty)) \cup \{(1, 0)\}$ 中，衰退方向必须让 $(1, 0)$ 及任意 $(0, y)$ 都留在集合，迫使两个方向分量均0，故衰退锥是 $\{0\}$，不是空集；渐近锥是 $\{(0, v) : v \geq 0\}$，不是整条竖直线。闭凸时 $C + C_\infty = C$ 由衰退定义及0在锥中立即得到。

11.46 衰退锥相等与非闭反例

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 113 页。

对应已有 gpt-109.pdf 页脚：119。

原课件第 113 页

Proposition of Definition 3

Proposition 4.4. For any non-empty closed convex set $C \subseteq \mathbb{R}^N$, one has $C = C + C_\infty$.

Proof. The inclusion $C \subseteq C + C_\infty$ is clear because $0 \in C_\infty$.

Let $x \in C + C_\infty$. Then there exists $c \in C$, $d \in C_\infty$ with $x = c + d$. Thus, there exists $d_k \in C$, $t_k \to \infty$ such that $\dfrac{d_k}{t_k} \to d$, which implies that for k sufficiently large

$$\left(1 - \frac{1}{t_k}\right) c + \frac{d_k}{t_k} \in C$$

and

$$\left(1 - \frac{1}{t_k}\right) c + \frac{d_k}{t_k} \to c + d = x \in C$$

as C is closed. $\square$

沿原论证展开与批注 任意非空集合有 $\operatorname{rec} C \subset C_\infty$：固定 $a \in C$，用 $x_n = a + nd$。反向需要闭凸：若 $x_n/t_n \to d$，对任意 $a \in C$ 和固定 $t > 0$，充分大时 $a + (t/t_n)(x_n - a) \in C$，取极限得 $a + td \in C$。所以 $\operatorname{rec} C = C_\infty$。第112页反例 $C = ([0,1) \times [0,\infty)) \cup \{(1,0)\}$ 中，衰退方向必须让 $(1,0)$ 及任意 $(0,y)$ 都留在集合，迫使两个方向分量均0，故衰退锥是 $\{0\}$，不是空集；渐近锥是 $\{(0,v) : v \geq 0\}$，不是整条竖直线。闭凸时 $C + C_\infty = C$ 由衰退定义及0在锥中立即得到。

11.47 最大可含锥与基点无关

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 114 页。

对应已有 gpt-109.pdf 页脚：120。

原课件第 114 页

Proposition of Definition 3

Corollary A closed, convex set $C \subseteq \mathbb{R}^N$ owning 0, contains its asymptotic cone.

Proof. As $0 \in C$ and $\{0\} + C_\infty = C_\infty$. As $C = C + C_\infty$, we have

$$C_\infty \subseteq C.$$

□

沿原论证展开与批注 闭凸 S、$a \in S$ 下，令 $D_a = \bigcap_{\lambda>0} \lambda(S-a)$。$d \in D_a$ 等价于任意 $t > 0$ 有 $a + td \in S$，因此它是从 a 出发可完整放入 S 的最大锥。若 $b \in S$，取 $T \to \infty$，由凸性 $(1 - t/T)b + (t/T)(a + Td) \in S$，极限为 $b + td$，闭性给后者在 S，所以基点可以任意换，$D_a = \operatorname{rec} S = S_\infty$。当0在 S 中，取 $a = 0, t = 1$ 就得到渐近锥包含于原集。本段所有闭性使用均在最后取极限处，不是多余假设。

11.48 最大可含锥与基点无关

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 115 页。

对应已有 gpt-109.pdf 页脚：121。

原课件第 115 页

Proposition of Definition 3

Proposition 4.5. A closed, convex set $C \subseteq \mathbb{R}^N$ owning 0, contains its asymptotic cone.

Proof. Let $y \in \mathbf{A}C$. For any $M \geqq 0$, $\mathbb{R}_+ y \in \Gamma\left(C^M\right)$.

(1) $y \in C^M$, then $y \in C$.

(2) $y \notin C^M$, then there exists $\lambda < 1$ such that $\frac{1}{\lambda} y \in C^M \subseteq C$,

hence $y = (1-\lambda)0 + \lambda\frac{1}{\lambda} y \in C$.

Therefore $y \in \mathbf{A}C$ implies $y \in C$. That is, $\mathbf{A}C \subseteq C$. □

沿原论证展开与批注 闭凸 S、$a \in S$ 下，令 $D_a = \bigcap_{\lambda>0} \lambda(S-a)$。$d \in D_a$ 等价于任意 $t>0$ 有 $a+td \in S$，因此它是从 a 出发可完整放入 S 的最大锥。若 $b \in S$，取 $T \to \infty$，由凸性 $(1-t/T)b + (t/T)(a+Td) \in S$，极限为 $b+td$，闭性给后者在 S，所以基点可以任意换，$D_a = \operatorname{rec} S = S_\infty$。当0在 S 中，取 $a=0, t=1$ 就得到渐近锥包含于原集。本段所有闭性使用均在最后取极限处，不是多余假设。

11.49 最大可含锥与基点无关

来源： 109-ZSM-PPT.pdf，原课件 PDF 第 117–118 页。

对应已有 gpt-109.pdf 页脚：123, 124。

原课件第 117 页

Proposition of Definition 3

Proof of (1). $C \subseteq S - x_0 = S'$ and $0 \in S'$.

$$
\begin{aligned}
y \in C &\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq C \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq S' \\
&\implies \text{for all } \lambda > 0, \{\alpha y \mid \alpha \geqq 0\} \subseteq \lambda S' \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq \bigcap_{\lambda>0} \lambda S' \\
&\implies y \in \bigcap_{\lambda>0} \lambda S'
\end{aligned}
$$

then $C \subseteq \bigcap_{\lambda>0} \lambda(S - x_0)$. Conversely,

$$
\begin{aligned}
y \in \bigcap_{\lambda>0} \lambda S' &\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq \bigcap_{\lambda>0} \lambda S' \\
&\implies \text{for all } \lambda > 0, \{\alpha y \mid \alpha \geqq 0\} \subseteq \lambda S' \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq S' \\
&\implies \{\alpha y \mid \alpha \geqq 0\} \subseteq C \\
&\implies y \in C
\end{aligned}
$$

then $\bigcap_{\lambda>0} \lambda(S - x_0) \subseteq C$. □

原课件第 118 页

Proposition of Definition 3

Proof of (2). Suppose $x_0 \in S$, $y \in S \Longrightarrow \mathbb{R}_+ y \subseteq C \subseteq S'$. For any $x \in S$,

$$x_0 + \mathbb{R}_+ y \subseteq S \Longleftrightarrow x + \mathbb{R}_+ y \subseteq S$$

then C is independent of x_0.

Proof of (3).

$$\mathbb{R}_+ y \subseteq \bigcap_{\lambda>0} \lambda S' \Longrightarrow \text{for any } \lambda > 0, \mathbb{R}_+ y \subseteq \lambda S' \Longrightarrow \mathbb{R}_+ y \in \mathbf{A}S' = \mathbf{A}S$$

then $\bigcap_{\lambda>0} \lambda S' \subseteq \mathbf{A}S$. Conversely,

$$\begin{aligned} \mathbb{R}_+ y \in \mathbf{A}S = \mathbf{A}S' &\Longrightarrow \text{for any } M > 0, \mathbb{R}_+ y \subseteq \Gamma(S'^M) \\ &\Longrightarrow \text{for any } M > 0, [\alpha_M, \infty) y \subseteq S'^M \\ &\Longrightarrow \mathbb{R}_+ y \subseteq S' \\ &\Longrightarrow \text{for any } \lambda > 0, \mathbb{R}_+ y \subseteq \lambda S' \\ &\Longrightarrow \mathbb{R}_+ y \subseteq \bigcap_{\lambda>0} \lambda S' \end{aligned}$$

then $\mathbf{A}S \subseteq \bigcap_{\lambda>0} \lambda S'$. □

沿原论证展开与批注 闭凸 S、$a \in S$ 下，令 $D_a = \bigcap_{\lambda>0} \lambda(S-a)$。$d \in D_a$ 等价于任意 $t > 0$ 有 $a + td \in S$，因此它是从 a 出发可完整放入 S 的最大锥。若 $b \in S$，取 $T \to \infty$，由凸性 $(1 - t/T)b + (t/T)(a + Td) \in S$，极限为 $b + td$，闭性给后者在 S，所以基点可以任意换，$D_a = \operatorname{rec} S = S_\infty$。当0在 S 中，取 $a = 0, t = 1$ 就得到渐近锥包含于原集。本段所有闭性使用均在最后取极限处，不是多余假设。

12 110-Proof

12.1 球与一般紧凸集的归约纠错

来源： 110-ZSM-PPT.pdf，原课件 PDF 第 5 页。

对应已有 gpt-110.pdf 页脚：7。

原课件第 5 页

Fixed Points

(0) (Brouwer's Theorem) Let $D = \{x \in \mathbb{R}^N \mid \|x\| \leqslant 1\}$. If function $h : D \to D$ is continuous, then h has a fixed point.

Define a homeomorphic mapping $\phi : D \to S$, we prove that (0) and (1) are equivalent.

(0) $\Longrightarrow$ (1): Define function $h = \phi^{-1} \circ f \circ \phi : D \to D$, then h is continuous and has a fixed point x_0. That is, $h(x_0) = x_0$ or $\phi^{-1}(f(\phi(x_0))) = (\phi^{-1} \circ f \circ \phi)(x_0) = x_0$, then $f(\phi(x_0)) = \phi(x_0)$ and hence $\phi(x_0)$ is a fixed point of f.

(1) $\Longrightarrow$ (0): Define function $f = \phi \circ h \circ \phi^{-1} : S \to S$, then f is continuous and has a fixed point x_0. That is, $f(x_0) = x_0$ or $\phi(f(\phi^{-1}(x_0))) = (\phi \circ h \circ \phi^{-1})(x_0) = x_0$, then $h(\phi^{-1}(x_0)) = \phi^{-1}(x_0)$ and hence $\phi^{-1}(x_0)$ is a fixed point of h.

沿原论证展开与批注　老师证明的是同胚变换下的不动点转移，不是高维 Brouwer 本身。先明确此页论证的适用前提：确实存在同胚 $\varphi : D \to S$，即双射且 φ, φ^{-1} 都连续。由球版到 S 版：定义 $h = \varphi^{-1} \circ f \circ \varphi : D \to D$。各步的定义域与值域匹配，连续映射复合仍连续。若引用球版得到 $h(u) = u$，则 $\varphi^{-1}(f(\varphi(u))) = u$。两边施加 φ 得 $f(\varphi(u)) = \varphi(u)$，故 $\varphi(u)$ 是固定点。反向定义 $f = \varphi \circ h \circ \varphi^{-1} : S \to S$，若 $f(v) = v$，两边施加 φ^{-1}，得 $h(\varphi^{-1}(v)) = \varphi^{-1}(v)$，完成转移。原页反向中多写一层 φ 的公式应按此复合定义修正。**必要批注：**一般非空紧凸 $S \subset \mathbb{R}^N$ 可能是单点或低维线段，不一定与同维闭球同胚。故不能不经说明就假定该 φ 存在。上述推导在同胚前提下正确，但仅凭这一页不能覆盖所有紧凸集；本汇编保留老师的转移思路，不另外补入课件没有给出的完整高维定理证明。

12.2 一维证明与值域纠错

来源： 110-ZSM-PPT.pdf，原课件 PDF 第 6 页。

对应已有 gpt-110.pdf 页脚：8。

原课件第 6 页

Fixed Points

(0) (Brouwer's Theorem) Let $D^N = \{x \in \mathbb{R}^N \mid \|x\| \leqslant 1\}$. If function $h : D^N \to D^N$ is continuous, then h has a fixed point.

(0) (Brouwer's Theorem) Let $D = \{x \in \mathbb{R} \mid \|x\| \leqslant 1\} = [-1, 1]$. If function $h : D \to D$ is continuous, then h has a fixed point.

Define function $h' = h - Id : D \to D$, then h' is continuous with $h'(x) = h(x) - x$ and hence $h'(-1) = h(-1) + 1 \geqslant 0$ and $h'(1) = h(1) - 1 \leqslant 0$, thus h' has a solution to $h'(x) = 0$, that is, h has a fixed point.

沿原论证展开与批注 设 $h : [-1, 1] \to [-1, 1]$ 连续，$g(x) = h(x) - x$ 为实值连续函数，值域不一定在 $[-1, 1]$，原页此处应改为 $\mathbb{R}$。端点有 $g(-1) \geq 0, g(1) \leq 0$，若等于0已得到不动点，否则介值定理给内点零点。$h'$ 在课件只是新函数符号，不能误读为导数。